

First recirculation of SA ballot for Draft 3.1 of the

IEC/IEEE 60802 Time-Sensitive Networking Profile for Industrial Automation

This is an unapproved draft prepared by the IEC/IEEE 60802 Joint Project.

NOTE – This page is not subject to ballot comments.

CONTENTS

1		
2		
3	FOREWORD.....	9
4	INTRODUCTION.....	11
5	1 Scope.....	12
6	2 Normative references	12
7	3 Terms, definitions, symbols, abbreviated terms and conventions	16
8	3.1 General.....	16
9	3.2 List of terms, abbreviated terms and definitions given in various standards	16
10	3.3 Terms defined in this document	18
11	3.4 Abbreviated terms.....	20
12	3.5 Conventions.....	23
13	3.5.1 Convention for capitalizations	23
14	3.5.2 Unit conventions	23
15	3.5.3 Conventions for YANG contents	23
16	3.5.4 Conventions for YANG selection and Digital Data Sheet	23
17	4 Overview of TSN in industrial automation	24
18	4.1 Industrial application operation	24
19	4.2 Industrial applications	26
20	4.2.1 General	26
21	4.2.2 Control loop tasks.....	28
22	4.2.3 Start of control loop tasks	28
23	4.3 IA-stations	29
24	4.4 Ethernet interface	29
25	4.5 Mechanisms to meet control loop latency requirements.....	30
26	4.6 Translation between middleware and network provisioning	31
27	4.6.1 Interfaces of type I2vlan.....	31
28	4.6.2 PTP Instances	32
29	4.7 Industrial traffic types.....	34
30	4.7.1 General	34
31	4.7.2 Traffic type characteristics.....	34
32	4.7.3 Traffic type categories	34
33	4.7.4 Traffic types.....	36
34	4.8 Security for TSN-IA.....	38
35	4.8.1 General	38
36	4.8.2 Security configuration model.....	38
37	4.8.3 NETCONF/YANG processing.....	38
38	4.8.4 NETCONF/YANG access control	40
39	4.8.5 Identity checking.....	40
40	4.8.6 Secure device identity	41
41	5 Conformance	43
42	5.1 General.....	43
43	5.2 Requirements terminology	44
44	5.3 Profile conformance statement (PCS)	44
45	5.4 Conformance classes.....	44
46	5.5 IA-station requirements.....	45
47	5.5.1 IA-station PHY and MAC requirements for external ports.....	45

48	5.5.2	IA-station topology discovery requirements.....	46
49	5.5.3	IA-station requirements for time synchronization.....	46
50	5.5.4	Management requirements for non-proxied IA-stations	47
51	5.6	IA-station options.....	49
52	5.6.1	IA-station PHY and MAC options for external ports	49
53	5.6.2	IA-station options for time synchronization.....	49
54	5.6.3	IA-station options for management	49
55	5.6.4	Proxy target options.....	50
56	5.7	Bridge component requirements.....	50
57	5.7.1	Common Bridge component requirements.....	50
58	5.7.2	ccA Bridge component requirements.....	51
59	5.7.3	ccB Bridge component requirements.....	51
60	5.8	Bridge component options.....	51
61	5.8.1	Common Bridge component options.....	51
62	5.8.2	ccA Bridge component options.....	52
63	5.8.3	ccB Bridge component options.....	52
64	5.9	End station component requirements	53
65	5.9.1	Common end station Component requirements.....	53
66	5.9.2	ccA end station component requirements.....	53
67	5.9.3	ccB end station component requirements.....	53
68	5.10	End station component options	54
69	5.10.1	Common end station component options	54
70	5.10.2	ccA end station component options.....	54
71	5.10.3	ccB end station component options.....	54
72	6	Required functions for an industrial network	55
73	6.1	General.....	55
74	6.2	Synchronization	55
75	6.2.1	General	55
76	6.2.2	PTP Instance requirements.....	55
77	6.2.3	PTP protocol requirements	56
78	6.2.4	Clock control system requirements for PTP End Instances	57
79	6.2.5	Error generation limits	57
80	6.2.6	PTP Instance synchronization status	63
81	6.2.7	Application framework	63
82	6.2.8	Working Clock domain framework	64
83	6.2.9	Global Time domain framework.....	65
84	6.2.10	IA-station model for clocks.....	65
85	6.2.11	Clock usage for the Ethernet interface	66
86	6.2.12	Error model	66
87	6.2.13	gPTP domains and PTP Instances.....	67
88	6.3	Security model.....	68
89	6.3.1	General	68
90	6.3.2	Security functionality	68
91	6.3.3	IDeVID Profile	71
92	6.3.4	Security setup based on IDeVID.....	76
93	6.3.5	Secure configuration based on LDeVID-NETCONF	80
94	6.4	Management.....	81
95	6.4.1	General	81
96	6.4.2	IA-station management model	81

97	6.4.3	Discovery of IA-station internal structure	85
98	6.4.4	Network engineering model	85
99	6.4.5	Operation	88
100	6.4.6	Engineered time-synchronization spanning tree	95
101	6.4.7	Diagnostics.....	96
102	6.4.8	Data sheet.....	99
103	6.4.9	YANG representation of managed objects and nodes	100
104	6.4.10	YANG Data Model	121
105	6.5	Topology discovery and verification	164
106	6.5.1	Topology discovery and verification requirements	164
107	6.5.2	Topology discovery overview	165
108	6.5.3	Topology verification overview.....	167
109	6.6	CNC.....	168
110	6.6.1	General	168
111	6.6.2	Stream destination MAC address range	168
112	Annex A (normative)	PCS proforma – Time-sensitive networking profile for industrial	
113		automation	170
114	A.1	General.....	170
115	A.2	Abbreviated terms and special symbols	170
116	A.2.1	Status symbols	170
117	A.2.2	General abbreviated terms	171
118	A.3	Instructions for completing the PCS proforma	171
119	A.3.1	General structure of the PCS proforma	171
120	A.3.2	Additional information	171
121	A.3.3	Exception information	172
122	A.3.4	Conditional status	172
123	A.4	Common requirements	172
124	A.4.1	Instructions.....	172
125	A.4.2	Implementation identification	173
126	A.4.3	Profile summary.....	173
127	A.4.4	Implementation summary	173
128	A.5	IA-station requirements and options	174
129	A.5.1	Instructions.....	174
130	A.5.2	IA-station requirements.....	174
131	A.5.3	Management Requirements for non-proxied IA-stations	174
132	A.5.4	IA-station PHY and MAC options for external ports	174
133	A.5.5	IA-station options for time synchronization.....	174
134	A.5.6	IA-station secure management exchange options	175
135	A.5.7	Proxy target requirements.....	175
136	A.6	Bridge component.....	175
137	A.6.1	Bridge component requirements	175
138	A.6.2	Instructions.....	176
139	A.6.3	Bridge Common bridge component options	176
140	A.6.4	Conformance class A bridge component options	176
141	A.6.5	Conformance class B bridge component options	176
142	A.7	End station component	177
143	A.7.1	Instructions.....	177
144	A.7.2	Common end station component requirements	177
145	A.7.3	Common end station component options	177

146	A.7.4	Conformance class A end station component options	177
147	A.7.5	Conformance class B end station component options	178
148	Annex B (informative)	Representative Configuration Domain	179
149	Annex C (informative)	Description of clock control system	180
150	C.1	Clock control system introduction.....	180
151	C.2	Transfer function for control system	181
152	C.3	Frequency response for control system.....	182
153	C.4	Example.....	187
154	Annex D (informative)	Time synchronization annex.....	189
155	D.1	Overview.....	189
156	D.2	Principles of operation	190
157	D.2.1	General	190
158	D.2.2	Grandmaster PTP Instance implementation	192
159	D.2.3	Splitting, joining and aligning time domains	192
160	D.2.4	PTP link characteristics	193
161	D.3	Notes on normative requirements.....	194
162	D.3.1	Oscillator requirements.....	194
163	D.3.2	Timestamp Granularity Error.....	194
164	D.3.3	Dynamic Timestamp Error	194
165	D.3.4	Grandmaster PTP Instance error generation.....	195
166	D.3.5	PTP Relay Instance error generation	195
167	D.3.6	PTP End Instance error generation.....	197
168	D.3.7	Explanation for the asymmetric normative requirements for the allowable range of dTE in Table 14, rows 2 and 3.....	197
170	D.4	Approach to testing normative requirements	199
171	D.4.1	General	199
172	D.4.2	Testing Grandmaster PTP Instance	199
173	D.4.3	Testing PTP Relay Instance.....	200
174	D.4.4	Testing PTP End Instance	202
175	D.4.5	Testing meanLinkDelay Accuracy	203
176	D.5	Example algorithms	203
177	D.5.1	General	203
178	D.5.2	Algorithm for tracking NRR drift	203
179	D.5.3	Algorithm to compensate for errors in measured NRR due to clock drift.....	205
180	D.5.4	Algorithm for tracking RR Drift.....	208
181	D.5.5	Algorithm to compensate for errors in measured RR due to clock drift	209
182	D.5.6	Algorithm to compensate for errors in measured RR due to clock drift at PTP End Instance.....	211
184	D.5.7	Mean Link Delay averaging.....	212
185	Annex E (normative)	60802 constrained device proxy TLV definition.....	215
186	E.1	60802 constrained device proxy TLV.....	215
187	E.1.1	General	215
188	E.1.2	Device flags.....	215
189	E.1.3	Proxy type	215
190	Annex F (informative)	MAC and PHY delay.....	216
191	F.1	Accumulated delays in industrial automation.....	216
192	F.1.1	General	216
193	F.1.2	Forwarding delay	216
194	F.1.3	Time-triggered transmission	216

195	F.1.4	Synchronization	216
196	F.1.5	Recommended delays	217
197	Bibliography	218	
198			
199	Figure 1 – Data flow in a control loop	25	
200	Figure 2 – IA-station interaction with CNC – Transmit path	27	
201	Figure 3 – IA-station interaction with CNC – Receive path	28	
202	Figure 4 – IA-station example	29	
203	Figure 5 – Model for cycles	30	
204	Figure 6 – Traffic type translation example	32	
205	Figure 7 – PTP instance translation example	33	
206	Figure 8 – descriptionDS.userDescription used for PTP instance translation	33	
207	Figure 9 – NETCONF/YANG security processing steps	39	
208	Figure 10 – IA-station conformance model	45	
209	Figure 11 – Clock model	64	
210	Figure 12 – Example clock usage principles for PTP End Instances	65	
211	Figure 13 – Example clock usage principles for Grandmaster PTP Instances	66	
212	Figure 14 – Error budget scheme	67	
213	Figure 15 – Generic IEEE 802.1Q YANG Bridge management model	81	
214	Figure 16 – Internal LAN connection management model	82	
215	Figure 17 – IA-station example with IETF interfaces	83	
216	Figure 18 – VID/FID/MSTID example	84	
217	Figure 19 – Structure and interfaces of a CNC	86	
218	Figure 20 – IA-station structure example	87	
219	Figure 21 – CNC interaction	87	
220	Figure 22 – Operational management model	88	
221	Figure 23 – UNI service model	89	
222	Figure 24 – CNC southbound	89	
223	Figure 25 – NETCONF usage in a Configuration Domain	90	
224	Figure 26 – Boundary port model	91	
225	Figure 27 – Observer model	97	
226	Figure 28 – Creation of the digital data sheet of an IA-station	100	
227	Figure 29 – Module iecieee60802-ethernet-interface	129	
228	Figure 30 – Module iecieee60802-bridge	130	
229	Figure 31 – Module iecieee60802-dot1-sched-bridge	131	
230	Figure 32 – Module iecieee60802-subscribed-notifications	131	
231	Figure 33 – Module iecieee60802-ia-station	131	
232	Figure 34 – Module iecieee60802-cnc-config	132	
233	Figure 35 – Module iecieee60802-tt-mapping	132	
234	Figure 36 – Module iecieee60802-proxy	132	
235	Figure 37 – iecieee60802-proxy-target	132	
236	Figure 38 – Usage example of LLDP	165	
237	Figure 39 – Stream destination MAC Address	169	

238	Figure C.1 – Reference model for clock control system.....	180
239	Figure C.2 – Frequency response for the control system of Figure C.1	183
240	Figure C.3 – Detail of frequency response for the control system of Figure C.1 for	
241	dimensionless frequency in the range 0,1 to 10	184
242	Figure C.4 – Gain peaking (pure fraction) as a function of damping ratio	186
243	Figure C.5 – Gain peaking in dB as a function of damping ratio	186
244	Figure C.6 – Example frequency response.....	188
245	Figure D.1 – Approach to testing normative requirements for Grandmaster PTP	
246	Instance.....	199
247	Figure D.2 – Approach to testing normative requirements for PTP Relay Instance - 1	200
248	Figure D.3 – Approach to testing normative requirements for PTP Relay Instance – 2	201
249	Figure D.4 – Approach to testing normative requirements for PTP Relay Instance – 3	201
250	Figure D.5 – Approach to testing normative requirements for PTP End Instance	202
251	Figure D.6 – RR Drift tracking and error compensation calculations – PTP Relay	
252	Instance.....	209
253	Figure D.7 – RR Drift tracking and error compensation calculations – PTP End	
254	Instance.....	211
255	Figure D.8 – Signals and timestamps to measure path delay	213
256	Figure E.1 – 60802 constrained device proxy TLV	215
257		
258	Table 1 – List of terms	16
259	Table 2 – Traffic type characteristics	34
260	Table 3 – IA time-aware stream characteristics.....	35
261	Table 4 – IA stream characteristics	35
262	Table 5 – IA traffic engineered non-stream characteristics	35
263	Table 6 – IA non-stream characteristics	35
264	Table 7 – Industrial automation traffic types summary.....	36
265	Table 8 – Example traffic class to traffic type mapping.....	37
266	Table 9 – Required values	56
267	Table 10 – Protocol settings.....	56
268	Table 11 – Clock control system requirements	57
269	Table 12 – Error generation limits for Grandmaster PTP Instance	57
270	Table 13 – Error generation limits for PTP Relay Instance	58
271	Table 14 – Error generation limits for PTP End Instance	62
272	Table 15 – Error budget.....	67
273	Table 16 – descriptionDS.userDescription of gPTP Domains	68
274	Table 17 – Summary of the YANG modules	133
275	Table A.1 – Implementation identification template	173
276	Table A.2 – Profile summary template.....	173
277	Table A.3 – Implementation type.....	173
278	Table A.4 – IA-station requirements	174
279	Table A.5 – Additional requirements for non-proxied IA-stations	174
280	Table A.6 – IA-station PHY and MAC options.....	174
281	Table A.7 – IA-station time synchronization options	175

282	Table A.8 – IA-station secure management exchange options.....	175
283	Table A.9 – Proxy target requirements	175
284	Table A.10 –Bridge component requirements	176
285	Table A.11 – Common bridge component options	176
286	Table A.12 – Conformance class A bridge component options	176
287	Table A.13 – Conformance class B bridge component options	176
288	Table A.14 – Common end station component requirements	177
289	Table A.15 – Common end station component options	177
290	Table A.16 – Conformance class A end station component options.....	177
291	Table A.17 – Conformance class B end station component options.....	178
292	Table D.1 – Time synchronization error budget.....	189
293	Table D.2 – Protocol configurations & other measures to achieve dTE budget	190
294	Table D.3 – Main sources of error affecting normative requirements for Grandmaster	
295	PTP Instance	195
296	Table D.4 – Main sources of error affecting normative requirements for PTP Relay	
297	Instance.....	195
298	Table D.5 – Main sources of error affecting normative requirements for PTP End	
299	Instance.....	197
300	Table E.1 – Device flags	215
301	Table F.1 – Recommended PHY delay.....	217

302

303

TIME-SENSITIVE NETWORKING PROFILE FOR INDUSTRIAL AUTOMATION

FOREWORD

1) The International Electrotechnical Commission (IEC) is a worldwide organization for standardization comprising all national electrotechnical committees (IEC National Committees). The object of IEC is to promote international co-operation on all questions concerning standardization in the electrical and electronic fields. To this end and in addition to other activities, IEC publishes International Standards, Technical Specifications, Technical Reports, Publicly Available Specifications (PAS) and Guides (hereafter referred to as "IEC document(s)"). Their preparation is entrusted to technical committees; any IEC National Committee interested in the subject dealt with may participate in this preparatory work. International, governmental and non-governmental organizations liaising with the IEC also participate in this preparation.

IEEE Standards documents are developed within IEEE Societies and Standards Coordinating Committees of the IEEE Standards Association (IEEE SA) Standards Board. IEEE develops its standards through a consensus development process, approved by the American National Standards Institute, which brings together volunteers representing varied viewpoints and interests to achieve the final product. Volunteers are not necessarily members of IEEE and serve without compensation. While IEEE administers the process and establishes rules to promote fairness in the consensus development process, IEEE does not independently evaluate, test, or verify the accuracy of any of the information contained in its standards. Use of IEEE Standards documents is wholly voluntary. *IEEE documents are made available for use subject to important notices and legal disclaimers (see <https://standards.ieee.org/ipr/disclaimers.html> for more information).*

IEC collaborates closely with IEEE in accordance with conditions determined by agreement between the two organizations. This Dual Logo International Standard was jointly developed by the IEC and IEEE under the terms of that agreement.

2) The formal decisions of IEC on technical matters express, as nearly as possible, an international consensus of opinion on the relevant subjects since each technical committee has representation from all interested IEC National Committees. The formal decisions of IEEE on technical matters, once consensus within IEEE Societies and Standards Coordinating Committees has been reached, is determined by a balanced ballot of materially interested parties who indicate interest in reviewing the proposed standard. Final approval of the IEEE standards document is given by the IEEE Standards Association (IEEE SA) Standards Board.

3) IEC/IEEE Publications have the form of recommendations for international use and are accepted by IEC National Committees/IEEE Societies in that sense. While all reasonable efforts are made to ensure that the technical content of IEC/IEEE Publications is accurate, IEC or IEEE cannot be held responsible for the way in which they are used or for any misinterpretation by any end user.

4) In order to promote international uniformity, IEC National Committees undertake to apply IEC Publications (including IEC/IEEE Publications) transparently to the maximum extent possible in their national and regional publications. Any divergence between any IEC/IEEE Publication and the corresponding national or regional publication shall be clearly indicated in the latter.

5) IEC and IEEE do not provide any attestation of conformity. Independent certification bodies provide conformity assessment services and, in some areas, access to IEC marks of conformity. IEC and IEEE are not responsible for any services carried out by independent certification bodies.

6) All users should ensure that they have the latest edition of this publication.

7) No liability shall attach to IEC or IEEE or their directors, employees, servants or agents including individual experts and members of technical committees and IEC National Committees, or volunteers of IEEE Societies and the Standards Coordinating Committees of the IEEE Standards Association (IEEE SA) Standards Board, for any personal injury, property damage or other damage of any nature whatsoever, whether direct or indirect, or for costs (including legal fees) and expenses arising out of the publication, use of, or reliance upon, this IEC/IEEE Publication or any other IEC or IEEE Publications.

8) Attention is drawn to the normative references cited in this publication. Use of the referenced publications is indispensable for the correct application of this publication.

9) Attention is drawn to the possibility that implementation of this IEC/IEEE Publication may require use of material covered by patent rights. By publication of this standard, no position is taken with respect to the existence or validity of any patent rights in connection therewith. IEC or IEEE shall not be held responsible for identifying Essential Patent Claims for which a license may be required, for conducting inquiries into the legal validity or scope of Patent Claims or determining whether any licensing terms or conditions provided in connection with submission of a Letter of Assurance, if any, or in any licensing agreements are reasonable or non-discriminatory. Users of this standard are expressly advised that determination of the validity of any patent rights, and the risk of infringement of such rights, is entirely their own responsibility.

International Standard IEC/IEEE 60802 was prepared by subcommittee 65C: Industrial networks, of IEC technical committee 65: Industrial-process measurement, control and automation, in cooperation with IEEE 802.1: Higher Layer LAN Protocols Working Group of IEEE 802: LAN/MAN Standards Committee of the IEEE computer society, under the IEC/IEEE Dual Logo Agreement between IEC and IEEE. It is an International Standard.

369 This document is published as an IEC/IEEE Dual Logo standard.

370 The text of this International Standard is based on the following IEC documents:

Draft	Report on voting
65C/XX/FDIS	65C/XX/RVD

371
372 Full information on the voting for its approval can be found in the report on voting indicated in
373 the above table.

374 The language used for the development of this International Standard is English.

375 This document was drafted in accordance with the rules given in the ISO/IEC Directives, Part 2,
376 available at www.iec.ch/members_experts/refdocs. The main document types developed by IEC
377 are described in greater detail at www.iec.ch/publications/.

378 The IEC Technical Committee and IEEE Technical Committee have decided that the contents
379 of this document will remain unchanged until the stability date indicated on the IEC website
380 under webstore.iec.ch in the data related to the specific document. At this date, the document
381 will be

- 382 • reconfirmed,
- 383 • withdrawn, or
- 384 • revised.

385

IMPORTANT – The "colour inside" logo on the cover page of this document indicates that it contains colours which are considered to be useful for the correct understanding of its contents. Users should therefore print this document using a colour printer.

386

387

INTRODUCTION

This document defines time-sensitive networking profiles for industrial automation. The profile selects features, options, configurations, defaults, protocols, and procedures of bridges, end stations, and LANs to build industrial automation networks. This document also specifies YANG modules for the digital data sheet and for remote procedure calls.

The profile meets the industrial automation market objective of converging Operations Technology (OT) and Information Technology (IT) networks by defining a common, standardized network infrastructure. This objective is accomplished by taking advantage of the improvements that Time-Sensitive Networking provides to IEEE 802.1 and IEEE 802.3 standard Ethernet networks by providing guaranteed data transport with bounded low latency, low latency variation, zero congestion loss for critical traffic, and high availability.

The profile helps the convergence of industrial communication networks by referring only to international standards to build the lower layers of the communication stack and their management.

Ethernet extended with Time-Sensitive Networking technology provides the features required in the area of industrial communication networks, such as:

- meeting low latency and latency variation requirements concerning data transmission;
- efficient exchange of data records on a frequent time period;
- reliable communications with calculable downtime;
- high availability meeting application requirements;
- efficient mechanisms for bandwidth utilization of exchanges of data records, with zero congestion loss;
- improved clock synchronization mechanisms, including support of multiple gPTP domains.

TIME-SENSITIVE NETWORKING PROFILE FOR INDUSTRIAL AUTOMATION

1 Scope

This document defines time-sensitive networking profiles for industrial automation. The profiles select features, options, configurations, defaults, protocols, and procedures of bridges, end stations, and LANs to build industrial automation networks. This document also specifies YANG modules defining read-only information available online and offline as a digital data sheet. This document also specifies YANG modules for remote procedure calls and actions to address requirements arising from industrial automation networks.

2 Normative references

The following documents are referred to in the text in such a way that some or all of their content constitutes the requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

ISO/IEC 9594-8:2020, *Information technology — Open systems interconnection — Part 8: The Directory: Public-key and attribute certificate frameworks*, available at: <https://www.iso.org/obp/ui/#iso:std:iso-iec:9594:-8:en>

IEEE Std 1588-2019, *IEEE Standard for a Precision Clock Synchronization Protocol for Networked Measurement and Control Systems*¹

IEEE Std 1588e-2024, *Standard for a Precision Clock Synchronization Protocol for Networked Measurement and Control Systems Amendment: MIB and YANG Data Models*

IEEE Std 802.1AB-2016, *IEEE Standard for Local and metropolitan area networks – Station and Media Access Control Connectivity Discovery*

IEEE Std 802.1ABcu-2021, *IEEE Standard for Local and metropolitan networks – Station and Media Access Control Connectivity Discovery Amendment 1: YANG Data Model*

IEEE Std 802.1AR-2018, *IEEE Standard for Local and Metropolitan Area Networks – Secure Device Identity*

IEEE Std 802.1AS-2020, *IEEE Standard for Local and Metropolitan Area Networks – Timing and Synchronization for Time-Sensitive Applications*

IEEE Std 802.1ASdm-2024, *IEEE Standard for Local and Metropolitan Area Networks: Timing and Synchronization for Time-Sensitive Applications Amendment: Hot Standby*

IEEE Std 802.1ASdn-2024, *IEEE Standard for Local and Metropolitan Area Networks: Timing and Synchronization for Time-Sensitive Applications Amendment: YANG Data Model*

IEEE Std 802.1ASdr-2024, *IEEE Standard for Local and Metropolitan Area Networks – Timing and Synchronization for Time-Sensitive Applications Amendment: Inclusive Terminology*

¹ The IEEE standards or products referred to in Clause 2 are trademarks of The Institute of Electrical and Electronics Engineers, Incorporated

- 449 IEEE Std 802.1CB-2017, *IEEE Standard for Local and metropolitan area networks – Frame*
450 *Replication and Elimination for Reliability*
- 451 IEEE Std 802.1CBcv-2021, *IEEE Standard for Local and metropolitan area networks – Frame*
452 *Replication and Elimination for Reliability – Amendment 1: Information Model, YANG Data*
453 *Model and Management Information Base Module*
- 454 IEEE Std 802.1CBdb-2021, *IEEE Standard for Local and metropolitan area networks – Frame*
455 *Replication and Elimination for Reliability – Amendment 2: Extended Stream Identification*
456 *Functions*
- 457 IEEE Std 802.1Q-2022, *IEEE Standard for Local and Metropolitan Area Network – Bridges and*
458 *Bridged Networks*
- 459 IEEE Std 802.1Qcw-2023, *IEEE Standard for Local and Metropolitan Area Networks – Bridges*
460 *and Bridged Networks, Amendment: YANG Data Models for Scheduled Traffic, Frame*
461 *Preemption, and Per-Stream Filtering and Policing*
- 462 IEEE Std 802.1Qdj-2024, *Standard for Local and Metropolitan Area Networks – Bridges and*
463 *Bridged Networks, Amendment: Configuration Enhancements for Time-Sensitive Networking*
- 464 IEEE Std 802.1Qdx-2024, *Standard for Local and Metropolitan Area Networks – Bridges and*
465 *Bridged Networks, Amendment: YANG Data Models for the Credit-Based Shaper*
- 466 IEEE Std 802.1Qdy-2024, *Standard for Local and Metropolitan Area Networks – Bridges and*
467 *Bridged Networks, Amendment: YANG Data Models for the Multiple Spanning Tree Protocol*
- 468 IEEE Std 802.3-2022, *IEEE Standard for Ethernet*
- 469 IEEE Std 802.3.2-2019, *IEEE Standard for Ethernet – YANG Data Model Definitions*
- 470 IEEE Draft Std P802.3.2a, *Standard for Ethernet – YANG Data Model Definitions*
- 471 IEEE Std 802.3cx-2023, *IEEE Standard for Ethernet Amendment 6: Media Access Control*
472 *(MAC) Service Interface and Management Parameters to Support Improved Precision Time*
473 *Protocol (PTP) Timestamping Accuracy*
- 474 IEEE Std 802.3de-2022, *Standard for Ethernet Amendment 6: Enhancements to the MAC Merge*
475 *and Time Synchronization Service Interface for Point-to-Point 10 Mb/s Single Pair Ethernet*
- 476 IETF RFC 2131, Droms, R., *Dynamic Host Configuration Protocol*, March 1997, available at
477 <https://www.rfc-editor.org/info/rfc2131>
- 478 IETF RFC 2986, Nystrom, M. and Kaliski, B., *PKCS #10: Certification Request Syntax*
479 *Specification Version 1.7*, November 2000, available at <https://www.rfc-editor.org/info/rfc2986>
- 480 IETF RFC 3986, Berners-Lee, T., Fielding, R., and Masinter, L., *Uniform Resource Identifier*
481 *(URI): Generic Syntax*, January 2005, available at <https://www.rfc-editor.org/info/rfc3986>
- 482 IETF RFC 4836, Beili, E., *Definitions of Managed Objects for IEEE 802.3 Medium Attachment*
483 *Units (MAUs)*, April 2007, available at <https://www.rfc-editor.org/info/rfc4836>
- 484 IETF RFC 5246, Dierks, T. and Rescorla, E., *The Transport Layer Security (TLS) Protocol*,
485 August 2008, available at <https://www.rfc-editor.org/info/rfc5246>

- 486 IETF RFC 5277, Chisholm, S. and Trevino, H., *NETCONF Event Notification*, July 2008,
487 available at <https://www.rfc-editor.org/info/rfc5277>
- 488 IETF RFC 5280, Turner, D., Santesson, S., Farrell, S., Boeyen, S., Housley, R., and Polk, W.,
489 *Internet X.509 Public Key Infrastructure Certificate and Certificate Revocation List (CRL)*
490 *Profile*, May 2008, available at <https://www.rfc-editor.org/info/rfc5280>
- 491 IETF RFC 5289, Rescorla, E., *TLS Elliptic Curve Cipher Suites with SHA-256/384 and AES*
492 *Galois Counter Mode (GCM)*, August 2008, available at <https://www.rfc-editor.org/info/rfc5289>
- 493 IETF RFC 5480, Cooper, S., Brown, D., Yiu, K., Housley, R., and Polk, T., *Elliptic Curve*
494 *Cryptography Subject Public Key Information*, March 2009, available at [https://www.rfc-](https://www.rfc-editor.org/info/rfc5480)
495 [editor.org/info/rfc5480](https://www.rfc-editor.org/info/rfc5480)
- 496 IETF RFC 6022, Scott, M. and Bjorklund, M., *YANG Module for NETCONF Monitoring*, October
497 2010, available at <https://www.rfc-editor.org/info/rfc6022>
- 498 IETF RFC 6024, Reddy, R. and Wallace, C., *Trust Anchor Management Requirements*, October
499 2010, available at <https://www.rfc-editor.org/info/rfc6024>
- 500 IETF RFC 6125, Saint-Andre, P. and Hodges, J., *Representation and Verification of Domain-*
501 *Based Application Service Identity within Internet Public Key Infrastructure Using X.509 (PKIX)*
502 *Certificates in the Context of Transport Layer Security (TLS)*, March 2011, available at
503 <https://www.rfc-editor.org/info/rfc6125>
- 504 IETF RFC 6241, Enns, R., Bjorklund, M., Schoenwaelder, J. and Bierman, A., *Network*
505 *Configuration Protocol (NETCONF)*, June 2011, available at [https://www.rfc-](https://www.rfc-editor.org/info/rfc6241)
506 [editor.org/info/rfc6241](https://www.rfc-editor.org/info/rfc6241)
- 507 IETF RFC 7317, Bierman, A. and Bjorklund, M., *A YANG Data Model for System Management*,
508 August 2014, available at <https://www.rfc-editor.org/info/rfc7317>
- 509 IETF RFC 7589, Badra, M., Luchuk, A. and Schoenwaelder, J., *Using the NETCONF Protocol*
510 *over Transport Layer Security (TLS) with Mutual X.509 Authentication*, June 2015, available at
511 <https://www.rfc-editor.org/info/rfc7589>
- 512 IETF RFC 7748, Langley, A., Hamburg, M., and Turner, S., *Elliptic Curves for Security*, January
513 2016, available at <https://www.rfc-editor.org/info/rfc7748>
- 514 IETF RFC 7905, Langley, A., Chang, W., Mavrogiannopoulos, N., Strombergson, J., and
515 Josefsson, S., *ChaCha20-Poly1305 Cipher Suites for Transport Layer Security (TLS)*, June
516 2016, available at <https://www.rfc-editor.org/info/rfc7905>
- 517 IETF RFC 7950, Bjorklund, M., *The YANG 1.1 Data Modeling Language*, August 2016, available
518 at <https://www.rfc-editor.org/info/rfc7950>
- 519 IETF RFC 8032, Josefsson, S., and Liusvaara, I., *Edwards-Curve Digital Signature Algorithm*
520 *(EdDSA)*, January 2017, available at <https://www.rfc-editor.org/info/rfc8032>
- 521 IETF RFC 8069, Thomas, A., *URN Namespace for IEEE*, February 2017, available at
522 <https://www.rfc-editor.org/info/rfc8069>
- 523 IETF RFC 8141, Saint-Andre, P., and Klensin, J., *Uniform Resource Names (URNs)*, April 2017,
524 available at <https://www.rfc-editor.org/info/rfc8141>
- 525 IETF RFC 8341, Bierman, A. and Bjorklund, M., *Network Configuration Access Control Model*,
526 March 2018, available at <https://www.rfc-editor.org/info/rfc8341>

- 527 IETF RFC 8342, Bjorklund, M., Schoenwaelder, J., Shafer, P., Watsen, K. and Wilton, R.,
528 *Network Management Datastore Architecture (NMDA)*, March 2018, available at
529 <https://www.rfc-editor.org/info/rfc8342>
- 530 IETF RFC 8343, Bjorklund, M., *YANG Data Model for Interface Management*, March 2018,
531 available at <https://www.rfc-editor.org/info/rfc8343>
- 532 IETF RFC 8348, Bierman, A., Bjorklund, M., Dong, J., and Romascanu, D., *A YANG Data Model*
533 *for Hardware Management*, March 2018, available at <https://www.rfc-editor.org/info/rfc8348>
- 534 IETF RFC 8410, Josefsson, S., and Schaad, J., *Algorithm Identifiers for Ed25519, Ed448,*
535 *X25519, and X448 for Use in the Internet X.509 Public Key Infrastructure*, August 2018,
536 available at <https://www.rfc-editor.org/info/rfc8410>
- 537 IETF RFC 8446, Rescorla, E., *The Transport Layer Security (TLS) Protocol Version 1.3*, August
538 2018, available at <https://www.rfc-editor.org/info/rfc8446>
- 539 IETF RFC 8525, Bierman, A., Bjorklund, M., Schoenwaelder, J., Watsen, K. and Wilton, R.,
540 *YANG Library*, March 2019, available at <https://www.rfc-editor.org/info/rfc8525>
- 541 IETF RFC 8526, Bierman, A., Bjorklund, M., Schoenwaelder, J., Shafer, P., Watsen, K. and
542 Wilton, R., *NETCONF Extensions to Support the Network Management Datastore Architecture*,
543 March 2019, available at <https://www.rfc-editor.org/info/rfc8526>
- 544 IETF RFC 8639, Voit, E., Clemm, A., Gonzalez Prieto, A., Nilsen-Nygaard, E., and Tripathy, A.,
545 *Subscription to YANG Notifications*, September 2019, available at [https://www.rfc-](https://www.rfc-editor.org/info/rfc8639)
546 [editor.org/info/rfc8639](https://www.rfc-editor.org/info/rfc8639)
- 547 IETF RFC 8640, Voit, E., Clemm, A., Gonzalez Prieto, A., Nilsen-Nygaard, E. and Tripathy, A.,
548 *Dynamic Subscription to YANG Events and Datastores over NETCONF*, September 2019,
549 available at <https://www.rfc-editor.org/info/rfc8640>
- 550 IETF RFC 8641, Clemm, A. and Voit, E., *Subscription to YANG Notifications for Datastore*
551 *Updates*, September 2019, available at <https://www.rfc-editor.org/info/rfc8641>
- 552 IETF RFC 9195, Lengyel, B. and Claise, B., *A File Format for YANG Instance Data*, February
553 2022, available at <https://www.rfc-editor.org/info/rfc9195>
- 554 IETF RFC 9196, Lengyel, B., Clemm, A. and Claise, B., *YANG Modules Describing Capabilities*
555 *for Systems and Datastore Update Notifications*, February 2022, available at [https://www.rfc-](https://www.rfc-editor.org/info/rfc9196)
556 [editor.org/info/rfc9196](https://www.rfc-editor.org/info/rfc9196)
- 557 IETF RFC 9640, Watsen, K., *YANG Data Types and Groupings for Cryptography*, October,
558 available at <https://datatracker.ietf.org/doc/rfc9640/>
- 559 IETF RFC 9641, Watsen, K., *A YANG Data Model for a Truststore*, October 2024, available at
560 <https://datatracker.ietf.org/doc/rfc9641/>
- 561 IETF RFC 9642, Watsen, K., *A YANG Data Model for a Keystore*, October 2024, available at
562 <https://datatracker.ietf.org/doc/rfc9642/>
- 563 IETF RFC „Internet-Draft (I-D)“, Turner, S., and Housley, R., *Updates to Using the NETCONF*
564 *Protocol over Transport Layer Security (TLS) with Mutual X.509 Authentication* (draft-ietf-
565 netconf-over-tls13), Internet Draft, Work in Progress by NETCONF WG, available at
566 <https://datatracker.ietf.org/doc/draft-ietf-netconf-over-tls13/>

IETF RFC Internet-Draft (I-D)", Watsen, K., *NETCONF Client and Server Models* (draft-ietf-netconf-netconf-client-server), Internet Draft, Work in Progress by NETCONF WG, available at <https://datatracker.ietf.org/doc/html/draft-ietf-netconf-netconf-client-server-31>

NIST FIPS 180-4, *Secure Hash Standard (SHS)*, August 2015, available at <https://csrc.nist.gov/publications/detail/fips/180/4/final>

NIST FIPS 186-5, *Digital Signature Standard (DSS)*, February 2023, available at <https://csrc.nist.gov/publications/detail/fips/186/5/final>

NIST SP 800-186, *Recommendations for Discrete Logarithm-based Cryptography: Elliptic Curve Domain Parameters*, February 2023, available at <https://nvlpubs.nist.gov/nistpubs/SpecialPublications/NIST.SP.800-186.pdf>

Editor's note: Any draft standards will be removed prior to completion of SA Ballot

3 Terms, definitions, symbols, abbreviated terms and conventions

3.1 General

For the purposes of this document, the terms and definitions given in ITU-T G.8260, IEEE Std 802-2014, IEEE Std 802.3™-2022, IEEE Std 802.1Q™-2022, IEEE Std 802.1AS™-2020 and the following apply.

ISO and IEC maintain terminology databases for use in standardization at the following addresses:

- IEC Electropedia: available at <https://www.electropedia.org/>
- ISO Online browsing platform: available at <https://www.iso.org/obp>
- IEEE Standards Dictionary Online: available at <https://dictionary.ieee.org>
- ITU-T Terms and Definitions database: available at https://www.itu.int/br_tsb_terms/#/

3.2 List of terms, abbreviated terms and definitions given in various standards

For the purposes of this document, the terms and definitions given in Table 1 apply.

For ease of understanding, the most important terms used within this document are listed in Table 1 but the definitions are not repeated.

Table 1 – List of terms

Term	Source
BTCA	IEEE Std 802.1AS-2020 as amended by IEEE Std 802.1ASdr™-2024
Bridge	IEEE Std 802.1Q-2022
Bridge Port	IEEE Std 802.1Q-2022
CFM	IEEE Std 802.1Q-2022
Clock	IEEE Std 802.1AS-2020
ClockTimeTransmitter	IEEE Std 802.1AS-2020 as amended by IEEE Std 802.1ASdr-2024
ClockTimeReceiver	IEEE Std 802.1AS-2020 as amended by IEEE Std 802.1ASdr-2024
ClockSource	IEEE Std 802.1AS-2020
ClockTarget	IEEE Std 802.1AS-2020
CNC	IEEE Std 802.1Q-2022
Configuration Domain	IEEE Std 802.1Qdj™-2024

Term	Source
constant time error (cTE)	ITU-T G.8260
Customer Virtual Local Area Network (C-VLAN) component	IEEE Std 802.1Q-2022
CUC	IEEE Std 802.1Q-2022
device	IEEE Std 802.1AR™-2018
DLL	IEEE Std 802-2014
DTE	IEEE Std 802.3-2022
dynamic time error (dTE)	ITU-T G.8260
end entity (EE)	NIST Special Publication 800-57 Part 2, Revision 1
end station	IEEE Std 802-2014
Ethernet	IEEE Std 802.3-2022
FDB	IEEE Std 802.1Q-2022
FID	IEEE Std 802.1Q-2022
fingerprint	IETF RFC 7589
FQTSS	IEEE Std 802.1Q-2022
fractional frequency offset	IEEE Std 802.1AS-2020
frame	IEEE Std 802.1Q-2022
frame preemption	IEEE Std 802.1Q-2022
FRER	IEEE Std 802.1CB™-2017
gating cycle	IEEE Std 802.1Q-2022
gPTP communication path	IEEE Std 802.1AS-2020
gPTP domain	IEEE Std 802.1AS-2020
Grandmaster Clock	IEEE Std 802.1AS-2020
Grandmaster PTP Instance	IEEE Std 802.1AS-2020
IST	IEEE Std 802.1Q-2022
LAN	IEEE Std 802-2014
latency	IEEE Std 802.1Q-2022
Listener	IEEE Std 802.1Q-2022
LLDP	IEEE Std 802.1AB™-2016
LLDPDU	IEEE Std 802.1AB-2016
local clock	IEEE Std 802.1AS-2020
LocalClock	IEEE Std 802.1AS-2020
logical link	IEEE Std 802-2014
LPI	IEEE Std 802.3-2022
MAC	IEEE Std 802-2014
MMRP	IEEE Std 802.1Q-2022
MST	IEEE Std 802.1Q-2022
MVRP	IEEE Std 802.1Q-2022
NETCONF	IETF RFC 6241
PCP	IEEE Std 802.1Q-2022
PDU	IEEE Std 802.1Q-2022
PHY	IEEE Std 802.3-2022
PLS	IEEE Std 802.3-2022
Port	IEEE Std 802.1Q-2022

Term	Source
preciseOriginTimestamp	IEEE Std 802.1AS-2020
primary domain	IEEE Std 802.1ASdm™-2024
PTP End Instance	IEEE Std 802.1AS-2020
PTP Instance	IEEE Std 802.1AS-2020
PTP Link	IEEE Std 802.1AS-2020
PTP Port	IEEE Std 802.1AS-2020
PTP Relay Instance	IEEE Std 802.1AS-2020
PVID	IEEE Std 802.1Q-2022
redundancy	IEC 60050-192
residence time	IEEE Std 802.1AS-2020
secondary domain	IEEE Std 802.1ASdm-2024
signature suite	IEEE Std 802.1AR-2018
station	IEEE Std 802-2014
stream	IEEE Std 802.1Q-2022
synchronized time	IEEE Std 802.1AS-2020
Talker	IEEE Std 802.1Q-2022
time error	ITU-T G.8260
time-sensitive stream	IEEE Std 802.1Q-2022
traffic class	IEEE Std 802.1Q-2022
TLV	IEEE Std 802.3-2022
UNI	IEEE Std 802.1Q-2022
VID	IEEE Std 802.1Q-2022
VLAN	IEEE Std 802.1Q-2022
X.509	ISO/IEC 9594-8:2020
YANG	IETF RFC 6020

594

595 3.3 Terms defined in this document

596 3.3.1

597 application clock

598 clock used by the application to time events

599 Note 1 to entry: Events can be periodic or aperiodic.

600 3.3.2

601 Bridge component

602 Customer Virtual Local Area Network (C-VLAN) Bridge

603 3.3.3

604 control latency

605 time delay between the input to a sensor application and the output from an actuator application

606 Note 1 to entry: For the purposes of this document, control latency does not include latencies in the sensor,
607 actuator, or the physical system in a process.

608 3.3.4

609 deadline

610 application defined reference point that represents a time when data is required by the
611 application

3.3.5**digital data sheet**

information about the capabilities of an IA-station

3.3.6**end station component**

end station entity

3.3.7**Global Time**

synchronized time, derived from a gPTP domain, that is traceable to the PTP timescale

3.3.8**IA-controller**

industrial automation function, consisting of a comparing element and a controlling element, that performs a specified control function

Note 1 to entry: An IA-controller exchanges data with several IA-devices or other IA-controllers for the purpose of control of a system.

Note 2 to entry: The primary categories of IA-controllers are distributed control systems (DCS), programmable logic controllers (PLCs), and programmable automation controllers (PACs).

3.3.9**IA-device**

industrial automation function, consisting of sensor or actuator elements to read or write process data

Note 1 to entry: An IA-device exchanges data with an IA-controller or other IA-devices for the purpose of control of a system.

3.3.10**IA-station**

material element or assembly of one or more end station components, and zero, one or more bridge components

Note 1 to entry: IA-controllers and IA-devices are industrial automation functions of IA-stations.

3.3.11**imprinting**

<security> equipping IA-stations with an LDevID credential, corresponding trust anchor and certificate-to-name mapping instructions

3.3.12**management entity**

IA-station function responsible for configuration of Bridge components, end station components and ports

Note 1 to entry: The management entity interacts with remote management.

3.3.13**network diameter**

number of links in the longest of all the calculated shortest paths between each pair of nodes in the network

3.3.14**network provisioning**

process of defining a consistent network configuration, which is applied to all stations

3.3.15

nominal frequency

ideal frequency with zero uncertainty

Note 1 to entry: The nominal frequency of the PTP timescale is further explained in IEEE Std 1588TM-2019, 7.2.1, 7.2.2, and Annex B.

3.3.16

non-proxied IA-station

IA-station which contains a ccA Bridge component or a ccA end station component

3.3.17

ppm

μHz/Hz

Note 1 to entry: The term "ppm" refers to a pure multiplier of 0,000 001 and is used in the context of this document as an SI unit term to allow readable terms conformant to various rules related to expressions.

3.3.18

proxy

IA-station whose Netconf server can be utilized by other IA-stations for the purpose of remote management

3.3.19

proxy target

IA-station which contains only ccB Bridge or end station components

3.3.20

Working Clock

synchronized time, derived from a gPTP domain, that is traceable to the PTP timescale, or to an ARB timescale that is continuous

Note 1 to entry: In general, the Working Clock is traceable to an ARB timescale; however, the Working Clock time can be correlated to a recognized timing standard.

3.4 Abbreviated terms

AEAD	Authenticated Encryption with Associated Data
AES	Advanced Encryption Standard
ARB	Arbitrary
ASCII	American Standard Code for Information Interchange
ASN	Abstract Syntax Notation
BTCA	Best time Transmitter Clock Algorithm
CA	Certification Authority
CBC	Cipher Block Chaining
ccA	Conformance Class A
ccB	Conformance Class B
CFM	Connectivity Fault Management
CMLDS	Common Mean Link Delay Service
CMS	Cryptographic Message Syntax
CN	Common Name
CNC	Centralized Network Configuration
CRL	Certificate Revocation List
CRUDX	Create Read Update Delete eXecute
CSR	Certificate Signing Request

701	CUC	Centralized User Configuration
702	C-VLAN	Customer VLAN
703	DAC	Discretionary Access Control
704	DER	Distinguished Encoding Rules
705	DH	Diffie-Hellman
706	DHE	Diffie-Hellman Ephemeral
707	DLL	Data Link Layer
708	DMAC	Destination MAC Address
709	DNS	Domain Name Service
710	DSA	Digital Signature Algorithm
711	DTE	Data Terminal Equipment
712	EC	Elliptic Curve
713	ECC	Elliptic Curve Cryptography
714	ECDSA	Elliptic Curve Digital Signature Algorithm
715	EdDSA	Edwards-Curve Digital Signature Algorithm
716	EE	End Entity
717	FDB	Filtering Database
718	FID	Filtering Identifier
719	FQDN	Fully Qualified Domain Name
720	FQTSS	Forwarding and Queuing Enhancements for Time-Sensitive Streams
721	FRER	Frame Replication and Elimination for Reliability
722	GCM	Galois Counter Mode
723	gPTP	generalized Precision Time Protocol
724	HMAC	Keyed-Hashing for Message Authentication Code
725	HW	Hardware
726	IA	Industrial Automation
727	IDeVID	Initial Secure Device Identifier
728	IEC	International Electrotechnical Commission
729	IEEE	Institute of Electrical and Electronics Engineers
730	I-LAN	Internal Local Area Network
731	ISO	International Organization for Standardization
732	ISS	Internal Sublayer Service
733	IST	Internal Spanning Tree
734	IT	Information Technology
735	ITU	International Telecommunication Union
736	LDeVID	Locally Significant Secure Device Identifier
737	LLDP	Link Layer Discovery Protocol
738	LLDPDU	Link Layer Discovery Protocol Data Unit
739	LPI	Low Power Idle
740	LRP	Link-local Registration Protocol
741	MAC	Media Access Control
742	MD	Media-Dependent
743	MDI	Media Dependent Interface

744	MMRP	Multiple MAC Registration Protocol
745	MST	Multiple Spanning Tree
746	MVRP	Multiple VLAN Registration Protocol
747	N/A	Not applicable
748	NACM	Network configuration Access Control Model
749	NETCONF	Network Configuration Protocol
750	NMDA	Network Management Datastore Architecture
751	NPE	Network Provisioning Entity
752	NRR	Neighbor Rate Ratio
753	OCSP	Online Certificate Status Protocol
754	OID	Object Identifier
755	OMG®	Object Management Group ²
756	OT	Operations Technology
757	OUI	Organizationally Unique Identifier
758	PCP	Priority Code Point
759	PCS	Profile Conformance Statement
760	PDU	Protocol Data Unit
761	PE	Path Entity
762	PEM	Privacy Enhanced Mail
763	PFS	Perfect Forward Secrecy
764	PHY	Physical Layer Device
765	PII	Personally Identifiable Information
766	PKCS	Public Key Cryptography Standards
767	PKI	Public Key Infrastructure
768	PKIX	Public Key Infrastructure X.509
769	PLS	Physical Signaling Sublayer
770	PPS	Pulse Per Second
771	PTP	Precision Time Protocol
772	PVID	Port VLAN Identifier
773	QoS	Quality of Service
774	RBAC	Role-Based Access Control
775	RFC	Request for Comments
776	RPC	Remote Procedure Call
777	RSA	Rivest-Shamir-Adleman
778	RAE	Resource Allocation Entity
779	SAN	Subject Alternative Name
780	SHA	Secure Hash Algorithm
781	STE	Sync Tree Entity
782	TDE	Topology Discovery Entity

² OMG®, UML® and Unified Modeling Language™ are either registered trademarks or trademarks of a product supplied by the Object Management Group, Inc. in the United States and/or other countries. This information is given for the convenience of users of this document and does not constitute an endorsement by IEEE or IEC of the product named. Equivalent products may be used if they can be shown to lead to the same results.

783	TLS	Transport Layer Security
784	TLV	Type, Length, Value
785	TOFU	Trust On First Use
786	TSN	Time-Sensitive Networking
787	TSN-IA	Time-Sensitive Networking for Industrial Automation
788	TTP	Trusted Third Party
789	UML®	Unified Modeling Language™
790	UNI	User/Network Interface
791	URL	Uniform Resource Locator
792	URN	Uniform Resource Name
793	VID	VLAN Identifier
794	VLAN	Virtual Local Area Network
795	YANG	Yet Another Next Generation data modeling language

796 3.5 Conventions

797 3.5.1 Convention for capitalizations

798 Capitalized terms are either based on the rules given in the ISO/IEC Directives Part 2 or
799 emphasize that these terms have a specific meaning throughout this document.

800 Throughout this document "bridge" can be used instead of "Bridge", except when

- 801 • it occurs at the beginning of a sentence or
- 802 • it is being used as (or part of) a specific term such as "VLAN Bridge" rather than being used
803 to identify bridges (potentially of any type) in general. If "VLAN Bridge" is meant where only
804 "Bridge" is written, a change to "VLAN Bridge" would be appropriate.

805 3.5.2 Unit conventions

806 This document uses:

- 807 • Gb/s for gigabits per second,
- 808 • Mb/s for megabits per second and,
- 809 • kb/s for kilobits per second.

810 3.5.3 Conventions for YANG contents

811 YANG modules and XML instance data for YANG shown in this document use the following
812 style:

813 Text style `higher-layer-if` text style

814 Contents of a YANG module use the following style:

```
815 <ieee802-dot1q-bridge xmlns:dot1q="urn:ieee:std:802.1Q:yang:ieee802-dot1q-bridge">
816   <bridges>
817     <bridge> <!-- list -->
818       <name>functional-unit-x</name>
819     ...
```

820 3.5.4 Conventions for YANG selection and Digital Data Sheet

821 The digital data sheet expresses device capabilities and therefore, it is not required that all
822 nodes in a YANG module be included in the digital data sheet. YANG nodes in 6.4.9, marked
823 with [m], are mandatory nodes in the digital data sheet, nodes marked with [c] are conditional

mandatory if the IA-station supports the corresponding optional functionality. Nodes marked with [o] are optional nodes in the digital data sheet. These marking in no way affect whether the feature and associated YANG module is required for the IA-station. Please refer to Clause 5 for conformance criteria for the IA-station.

YANG node selections in 6.4.9 of parent nodes implicitly include all subsidiary child nodes.

4 Overview of TSN in industrial automation

4.1 Industrial application operation

Industrial network applications are based on three main types of building blocks, which can be combined in one IA-controller or provided as a combination of an IA-controller and IA-devices interconnected through a suitable communication network.

These basic building blocks are:

- IA-device Sensor subsystems, which provide input signals indicating the value of the parameter or state being monitored, such as temperature, pressure, or discrete input information;
- IA-controller subsystems, which operate on combinations of measurements and external demand settings to develop output requests, such as position corrections in a motion application;
- IA-device Actuator subsystems, which implement output requests that result in physical changes to the process or machine under control, such as a level in a storage tank, the speed of a printing press, or movement of a robot.

NOTE 1 In general, all subsystems have an internal state, based upon initial settings, and derived from execution; therefore, the application inputs are combined with the internal state to develop an updated internal state and associated outputs.

A control loop is formed when the process or machine responds to the IA-device Actuator output and produces a new measured value at the IA-device Sensor. The complete loop is shown in Figure 1 where an IA-controller and IA-devices are connected as end stations in the network.

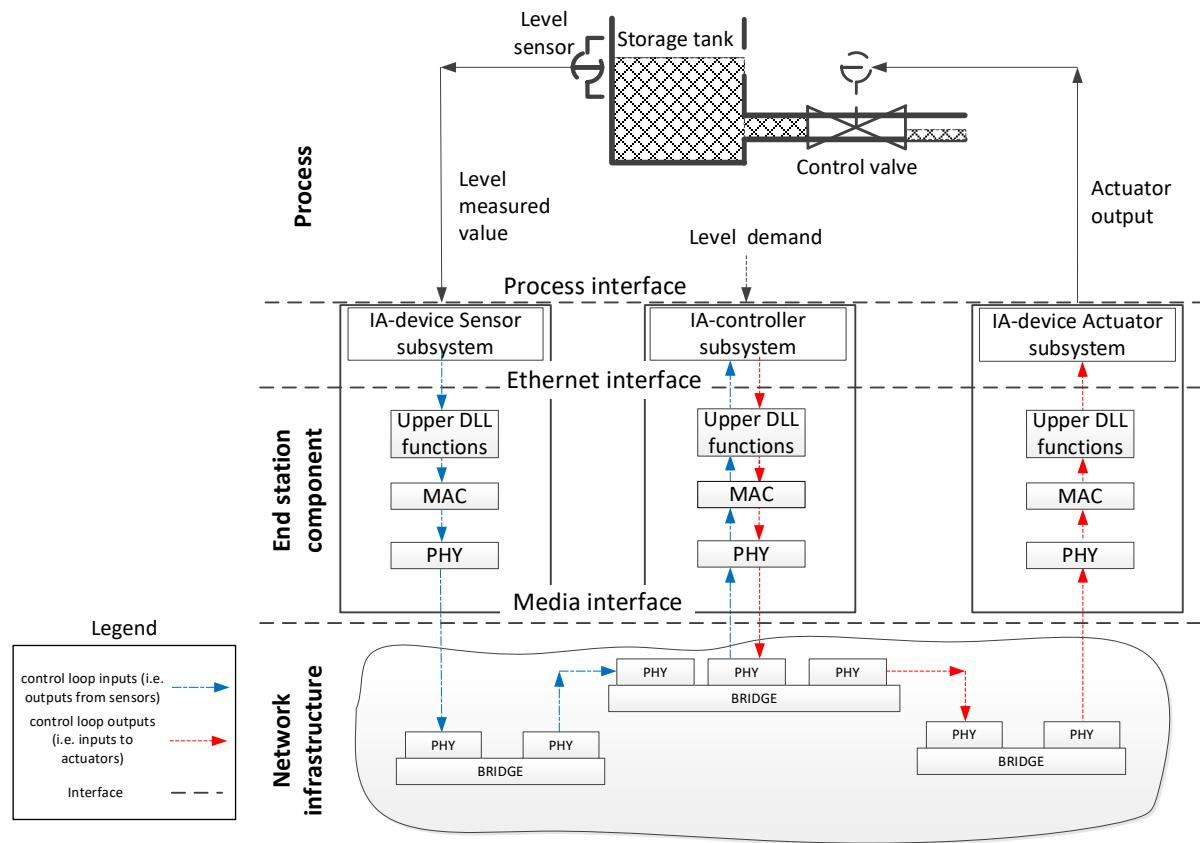


Figure 1 – Data flow in a control loop

In operation, the IA-device Sensor subsystem samples the measured value and the sampled values are transferred through the network as data packets for the IA-controller subsystem to compare with the demand value. After some computation time, the output is transferred from the IA-controller subsystem to the IA-device Actuator subsystem for implementation as a change in the external process.

This sequence repeats continuously as a regular operation using a Working Clock. The Working Clock is traceable to an ARB timescale or to the PTP timescale. Traceability to the PTP timescale is not required by all applications. For stability, the sequence repetition time (i.e., sampling time) is typically chosen to be between one fifth to one tenth of the time constant of the process response although shorter times can also be used.

NOTE 2 The actual ratio of the process response time constant to sampling time required for stability depends on the implementation.

Control latency is a critical factor in all types of control and is bounded. Components contributing to the control latency time (as shown in Figure 1) are:

- application time for sampling, computation, and processing within each IA-controller and IA-device. These are specific to the IA-device and IA-controller and known to the IA-device or IA-controller makers,

- the time for data transfer through the upper DLL functions, MAC and PHY layers within each IA-controller and IA-device. This time depends on the implementation of these components, their situation-dependent load and performance, and configuration elements related to QoS supported by these components, and
- the End Station and Bridge schedule and transfer time through the network. These are influenced by the configuration process, which allocates available bandwidth and priorities to various types of application messages.

Offline engineering of the network is possible, including the calculation of the control latency time. During system operation, management services are provided for diagnostics and checking the performance indicators of an installed network.

4.2 Industrial applications

4.2.1 General

Industrial applications can contain multiple tasks. These tasks are executed based upon time or other events. Thus, an industrial application can have multiple tasks executing on different cycles as shown in Figure 2 and Figure 3.

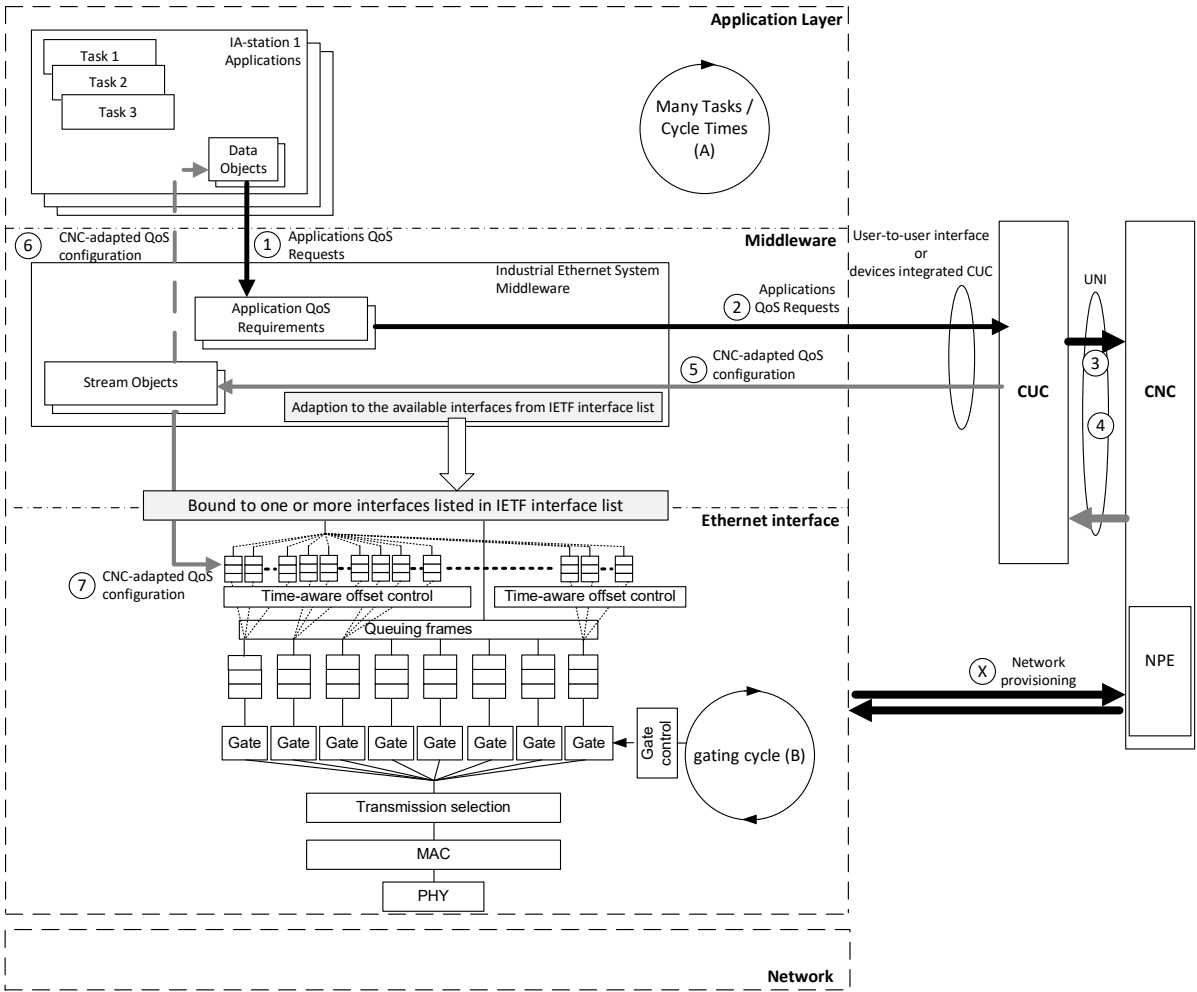
Examples of these tasks include the following.

- Background tasks, which are executed when no other task is running. There can be zero, one, or more such tasks in an industrial application.
- Main task which executes periodically. The start and execution of this task is often based upon the ARB timescale. There can be zero or one such task, in an industrial application.
- Global Time tasks. The start and execution of these tasks is often based upon Global Time (for example, at noon every day or at noon every Friday). There can be zero, one or more such tasks in an industrial application.
- Process driven tasks which are started by an event (for example, a sensor value reaches a defined point, or a process fault occurs). There can be zero, one or more such tasks in an industrial application.
- Control loop tasks which are bound to the Working Clock and started periodically. There can be zero, one or more such tasks in an industrial application.

A user defines the required automation tasks along with the data objects required as output and input for these tasks and the end station which hosts these tasks. Thus, these tasks are bound to data objects, which are exchanged between end stations per the user's definition. Many of these tasks have timing requirements, which are added as attributes to the assigned data objects. Examples of these attributes include:

- An update interval: [DataObject_Update_Interval] (time between two consecutive updates at the transmitting end station);
- A deadline: [DataObject_Deadline] (latest receive time at the end station, relative to the start of the DataObject Update Interval);
- The size of the DataObject: [DataObject_Data_Size];
- Other attributes to form a stream-list request according to IEEE Std 802.1Qdj-2024, 46.1.5.

NOTE These attributes are provided for illustration purposes. The list is not representative of all industrial applications. These are not network attributes.



910 Time-aware offset control Queues are shared between multiple middleware components.

911

Figure 2 – IA-station interaction with CNC – Transmit path

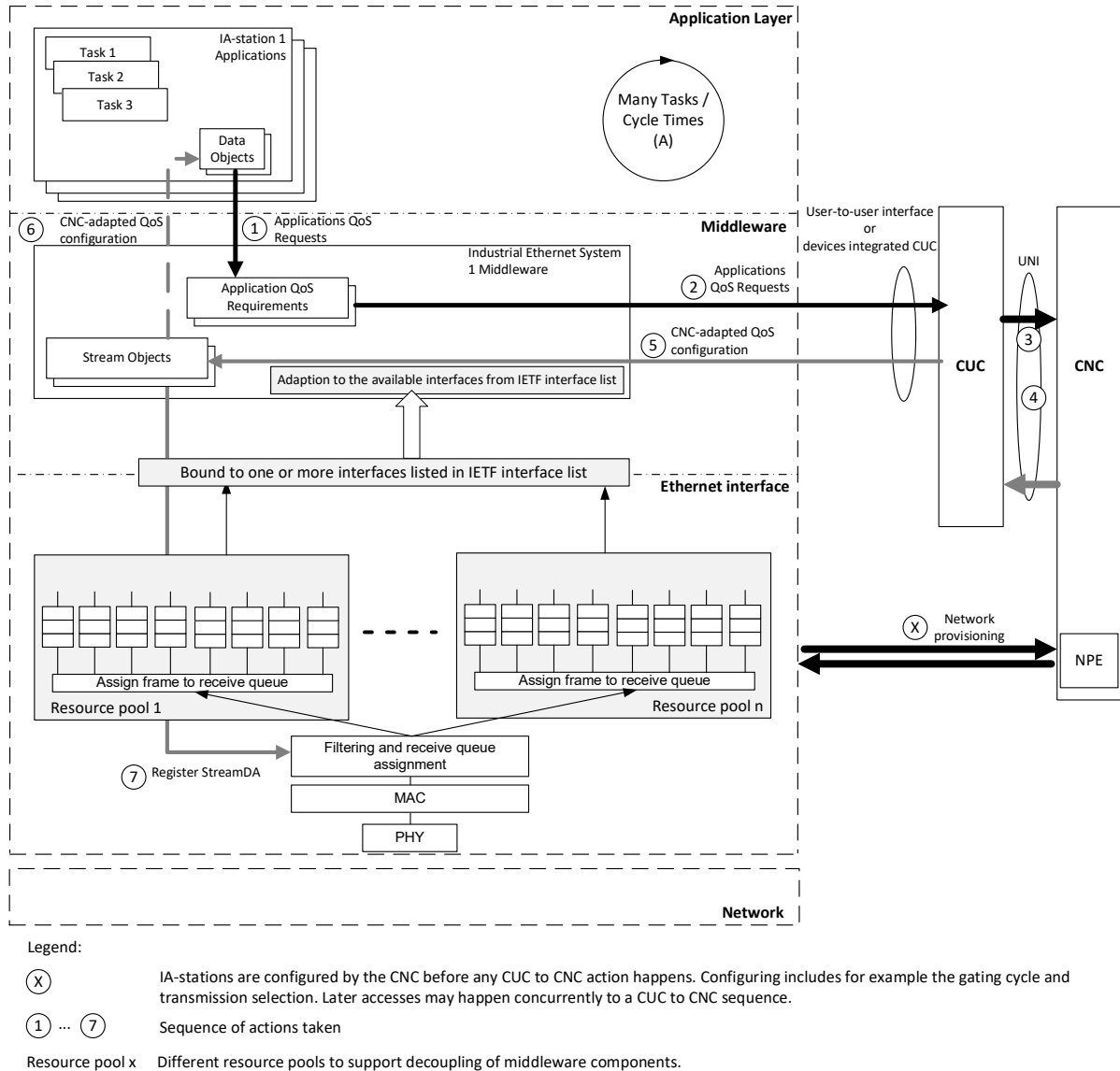


Figure 3 – IA-station interaction with CNC – Receive path

4.2.2 Control loop tasks

Multiple control loop tasks can be implemented and run in parallel in their automation devices. For example, this behavior can be implemented by using a common Working Clock, a common starting point relative to the Working Clock and a common duration for this control loop task at the involved IA-devices and IA-controllers. The data objects associated with the control loop share common values for some attributes (for example, the same values for DataObject_Update_Interval and DataObject_Deadline). Multiple control loop tasks can be implemented and run in parallel in their automation devices.

4.2.3 Start of control loop tasks

The calculation of the starting point for a control loop task is independent from the time when the device is powered up or connected to the Configuration Domain. The start of a control loop task, which is based on the Working Clock, can be calculated in the following manner: divide the Working Clock value, expressed as an integer, by the duration of the control loop task, expressed as an integer, whenever the Working Clock value increases by one. A remainder of zero provides the basis for the start of the control loop task.

NOTE The units of the Working Clock value and the units of the duration of the control loop task are the same.

Stations in the network associated with the control loop synchronize to a Working Clock using IEEE Std 802.1AS-2020.

4.3 IA-stations

An IA-station can be a simple end station acting as source or destination for control data traffic. In addition, an IA-station can be a combined functional unit that includes an end station component together with a Bridge component in one chassis. IA-stations, incorporating multiple functional units with several end station components and Bridge components within one chassis, can also be found in industrial automation. Within this kind of combined IA-station various components can be connected by internal ports and internal LANs. All components utilize a common management entity as shown in Figure 4.

Figure 4 shows an example IA-station incorporating four functional units in one chassis. Functional unit 1 and functional unit 2 each consist of a Bridge component and an end station component. The end station components are connected by internal ports via internal LANs to the Bridge components. The Bridge components include two external ports each. Functional unit 3 includes only a single end station component with one external port. Functional unit 4 includes a single end station component with two external ports.

IA-controllers and IA-devices as well as the management entity are IA-station functions acting as source of or destination for link layer data traffic. Thus, each IA-station incorporates at least one end station component where these functions can be located. Figure 4 shows that IA-station functions can either reside in a single end station component (IA-device 1, IA-controller 1, IA-device 2, IA-device 3, IA-controller 3) or in multiple end station components (IA-controller 2, management entity).

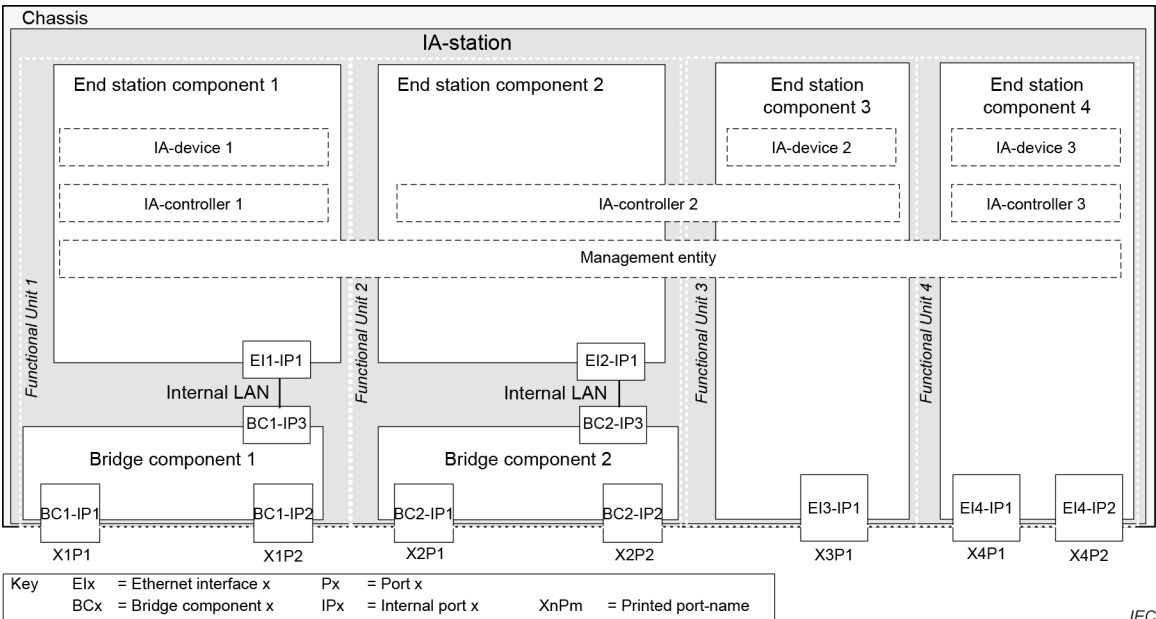


Figure 4 – IA-station example

4.4 Ethernet interface

One or more middleware components act as a layer between applications and the Ethernet interface. Figure 2 and Figure 3 show the relation between applications, middleware, Ethernet interface and the network. Various applications can run in parallel on an automation device. Data objects represent the information exchanged between applications running in different end stations. The application requirements contained in these data objects are translated by the

middleware into stream requirements for use by the CUC. This translation can be accomplished in one or both of the following ways.

- a) The user defines the data objects and translates them into stream requirements and end-station communication-configurations. A user-specific mechanism is used to configure the network components, establish paths, and the time-aware offset control.
- b) The user defines the data objects and associates them with QoS requirements for each stream (application QoS requirements). These can be forwarded as stream requirement requests by a CUC to a CNC. The CNC responds by providing a stream configuration response. The request and response are specified in IEEE Std 802.1Qdj-2024. This information is used to configure the time-aware offset control, which utilizes per-stream queues. The CUC can be integrated into the end station or can be accessed via a user-to-user protocol. The middleware uses this information for configuring Talkers and Listeners. This information is also used to add additional timing information to the data objects for application usage.

Time-aware offset control utilizes per-stream queues (see IEEE Std 802.1Q-2022, Figure 34-1) and the traffic specification of the streams, including transmission offsets, provided by the CNC to ensure the order of stream transmission.

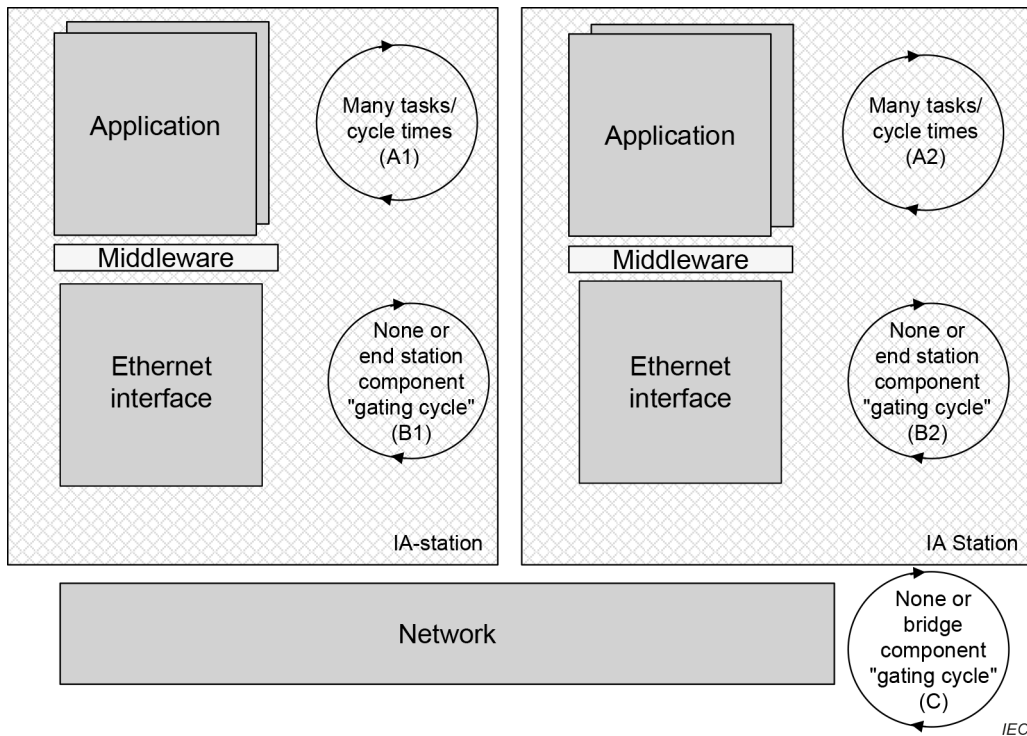


Figure 5 – Model for cycles

These automation systems, which are built from various end stations and connected via bridges, can share a common gating cycle or each station can have its own gating cycle. Alternatively, a bridge or end station can have no gating cycle (expressed as "none" in Figure 5).

4.5 Mechanisms to meet control loop latency requirements

Meeting latency requirements on a network can be accomplished using one or more combinations of the mechanisms enumerated below. The choice of a mechanism or a subset of the mechanisms listed below depends on the nature of the application(s) and the corresponding latency requirements:

- a) defining, testing, and simulating all possible application combinations and associated traffic patterns,

- b) overprovisioning the network,
- c) providing scheduled time slots for each application to transmit on the network,
- d) preempting lower priority traffic,
- e) providing scheduled time slots for certain traffic classes,
- f) time-aware offset control, and
- g) enforcing deterministic queuing delays in bridges.

NOTE This list is not comprehensive and not all mechanisms mentioned here are part of this specification. For specific mechanisms covered by this document please refer to Clause 5.

Frame preemption is specified in IEEE Std 802.1Q-2022 and IEEE Std 802.3-2022.

Reserving time on the network for certain traffic types can be done through enhancements for scheduled traffic according to IEEE Std 802.1Q-2022, 8.6.8.4. An aligned gating cycle is defined. Once a gating cycle is defined, portions of a cycle time can either be allocated to streams or classes of streams.

Multiple Talker-Listener pairs can be used for streams between end stations. Engineered time-triggered transmit can be used to coordinate transmission of all the traffic that shares a network to meet application requirements.

Creating a traffic load model in advance allows analysis of resulting traffic. It can be used to select and implement appropriate mechanisms to achieve latency requirements.

4.6 Translation between middleware and network provisioning

4.6.1 Interfaces of type I2vlan

Application engineering can be done without knowledge of the network provisioning. Since the application is not aware of the network provisioning, it cannot directly map to the network configuration, for example, the use of PCP or VID as configured in the network. This problem is solved by providing a translation table, in the form of a YANG module definition, to the middleware. The IA-station's local YANG datastore contains this information.

Figure 6 show examples of the translation models.

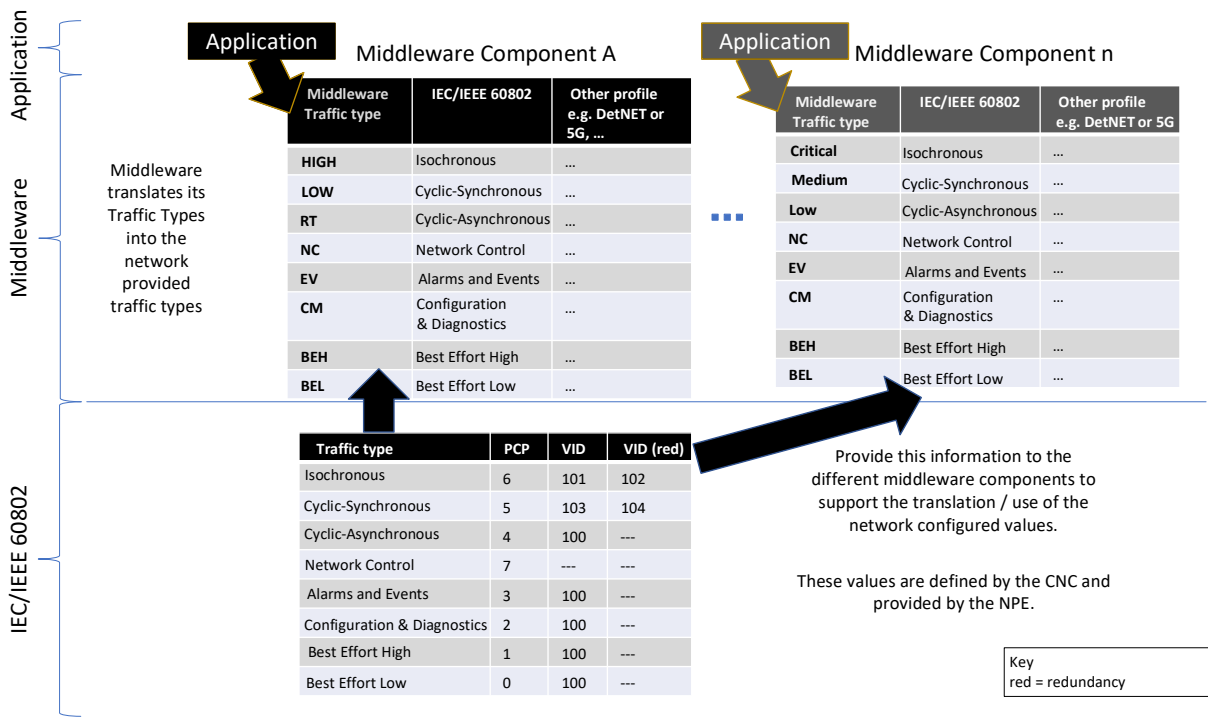


Figure 6 – Traffic type translation example

The YANG module `iecieeee60802-tt-mapping.yang` is used to provide the required mapping information to all installed middleware and applications. The information in the YANG module provides the VLAN-ID, the assigned traffic types with their PCP values and redundancy information (6.4.10.8.7).

4.6.2 PTP Instances

PTP domain numbers are also configured during network provisioning. The middleware is informed about which PTP domain is assigned to which target clock by providing `descriptionDS.userDescription` names according to IEEE Std 1588-2019, 8.2.5.5, in a translation table.

`descriptionDS.userDescription` names allow the support of multiple middleware components at one IA-station using the same PTP Instances (see 6.2.13). An IA-station's local database stores this information.

Figure 7 show examples of the translation models.

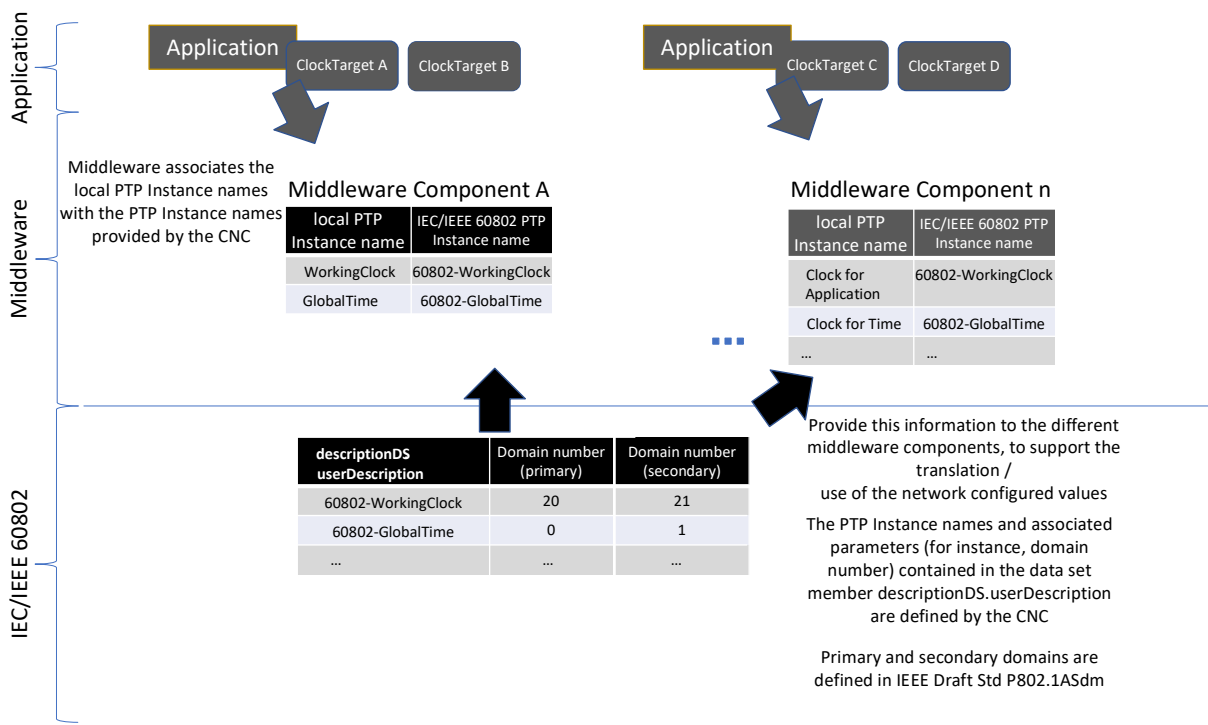


Figure 7 – PTP instance translation example

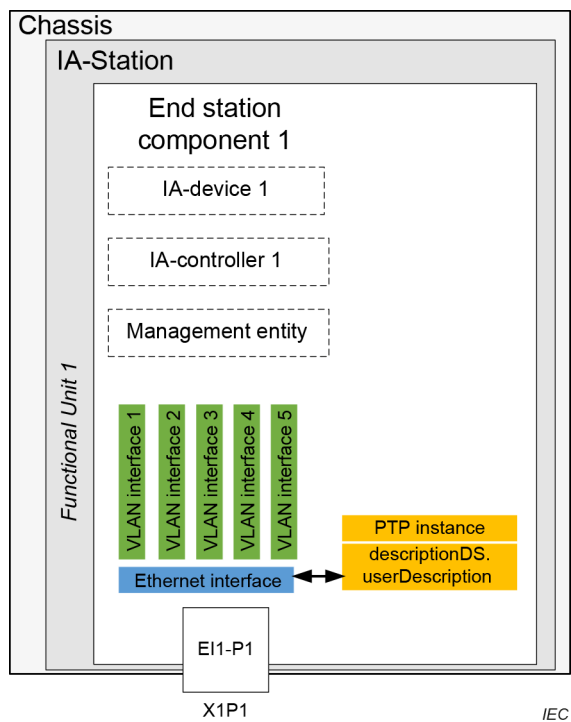


Figure 8 – descriptionDS.userDescription used for PTP instance translation

The userDescription contains the clock type (i.e., WorkingClock, GlobalTime, or both). This information is used by the middleware to align to the intended ClockTarget or ClockSource (see 6.2.13).

4.7 Industrial traffic types

4.7.1 General

Industrial automation applications make use of different traffic schemes/types for different functionalities (for example, parameterization, control, alarming). The various traffic patterns have different characteristics and thus impose different requirements on a network. To specify these traffic types, a two-step approach is used.

- a) First, define characteristics of generic traffic types (traffic type categories).
- b) Second, define instances of the generic traffic types (i.e. the traffic types).

4.7.2 Traffic type characteristics

The traffic type characteristics in Table 2 enable the identification of several distinct traffic types that are shared among sets of industrial applications.

Table 2 – Traffic type characteristics

Characteristic	Description
Cyclic	Traffic types consist of frames that can either be transmitted on a reoccurring time period (cyclic) or at no set period (acyclic). Available selections are: - required: traffic frames are transmitted cyclically, or - optional: implementation of cyclic traffic is at the discretion of the user.
Data delivery requirements	Denotes the delivery constraints for the traffic. Four options are specified. - Frame Latency: data delivery of a frame for a given Talker-Listener pair occurs within a bounded timespan. - Bandwidth: data delivery up to a certain number of frames or data size (including bursts of frames) occurs over a defined period. - Deadline: data delivery of a frame to a given Listener occurs at or before a specific point in time. - No: Denotes the case of traffic types with no special data delivery requirements
Time-triggered transmission	Talker data transmission occurs at a specific point in time based upon the Working Clock. Available selections are: - required, or - optional: implementation of time-triggered transmission is at the discretion of the user. Enhancements of scheduled traffic is only one means of achieving time-triggered transmission. Other, application-based, methods are possible
Stream-based	Indicates whether a traffic type requires registering a stream, i.e., allocating dedicated resources, for the traffic, before a given Talker can start transmitting data. - True: Data can only be transmitted after a Stream has been set up. - False: Data can be transmitted at any time.

4.7.3 Traffic type categories

4.7.3.1 General

The two-step approach described in 4.7.1 allows a clear differentiation between characteristics as seen from the "network" point of view and "application" point of view. Traffic type categories allow different IEEE 802 feature selections to achieve the goals of a specific network deployment. Four traffic type categories are identified in industrial automation systems:

- a) IA time-aware stream,
- b) IA stream,
- c) IA traffic engineered non-stream, and

d) IA non-stream.

4.7.3.2 IA time-aware stream

The characteristics of this traffic type category are shown in Table 3.

Table 3 – IA time-aware stream characteristics

Characteristics	
Cyclic	Required
Data delivery requirement	Deadline or Frame Latency
Time-triggered transmission	Required
Stream-based	True

4.7.3.3 IA stream

The characteristics of this traffic type category are shown in Table 4.

Table 4 – IA stream characteristics

Characteristics	
Cyclic	Required
Data delivery requirement	Frame Latency
Time-triggered transmission	Optional
Stream-based	True

4.7.3.4 IA traffic engineered non-stream

The characteristics of this traffic type category are shown in Table 5.

Table 5 – IA traffic engineered non-stream characteristics

Characteristics	
Cyclic	Optional
Data delivery requirement	Bandwidth
Time-triggered transmission	Optional
Stream-based	False

4.7.3.5 IA non-stream

The characteristics of this traffic type category are shown in Table 6.

Table 6 – IA non-stream characteristics

Characteristics	
Cyclic	Optional
Data delivery requirement	No
Time-triggered transmission	Optional
Stream-based	False

4.7.4 Traffic types

4.7.4.1 General

Table 7 summarizes relevant industrial automation traffic types and their associated characteristics. In an industrial automation system, other applications, such as audio or video, utilizes one of these traffic types.

Table 7 – Industrial automation traffic types summary

Traffic type name	Cyclic	Data delivery requirements	Time-triggered transmission	Stream based	Traffic type category
Isochronous	Required	Deadline	Required	True	IA time-aware stream
Cyclic-synchronous	Required	Frame Latency	Required	True	IA time-aware stream
Cyclic-asynchronous	Required	Frame Latency	Optional	True	IA stream
Alarms & events	Optional	Bandwidth	Optional	False	IA traffic engineered non-stream
Configuration & Diagnostics	Optional	Bandwidth	Optional	False	IA traffic engineered non-stream
Network Control	Optional	Bandwidth	Optional	False	IA traffic engineered non-stream
Best Effort High	Optional	No	Optional	False	IA non-stream
Best Effort Low	Optional	No	Optional	False	IA non-stream

4.7.4.2 Isochronous

A type of IA time-aware stream traffic. This type of traffic is transmitted cyclically using time-triggered transmission. Listeners have individual deadline requirements. Cycle times are typically in the range of microseconds to tens of milliseconds. Frame size is typically below 500 octets. Talker-Listener pairs are synchronized to the Working Clock. The network is configured by the CNC to provide zero congestion loss for this traffic type. This type of traffic is normally used in control loop tasks.

4.7.4.3 Cyclic-synchronous

Cyclic-synchronous is a type of IA time-aware stream traffic. This type of traffic is transmitted cyclically using time-triggered transmission. Talker-Listener pairs have individual latency requirements. Cycle times are typically in the range of hundreds of microseconds to hundreds of milliseconds. Frame size is unconstrained except as indicated in 5.5.1. Talker-Listener pairs are synchronized to the Working Clock. The network is configured by the CNC to provide zero congestion loss for this traffic type. This type of traffic is normally used in control loop tasks.

4.7.4.4 Cyclic-asynchronous

Cyclic-asynchronous is a type of IA stream traffic. This type of traffic is transmitted cyclically with latency requirements bounded by the interval as specified in IEEE Std 802.1Q-2022, 46.2.3.5.1. Talker-Listener pairs have individual latency requirements. Cycle times are typically in the range of milliseconds to seconds. Frame size is unconstrained except as indicated in 5.5.1. Data exchanges between Talker-Listener pairs are typically not dependent on the Working Clock. This traffic type typically tolerates limited congestion loss. The network is

configured by the CNC to handle this traffic type without loss, up to a certain number of frames or data size.

4.7.4.5 Alarms and events

Alarms and events are a type of IA traffic engineered non-stream. This type of traffic is transmitted cyclically or acyclically. This traffic expects bounded latency including time for retransmission in the range of milliseconds to hundreds of milliseconds. The source of the alarm or event typically limits the bandwidth allocated to this traffic. Frame size is unconstrained except as indicated in 5.5.1. Congestion loss can occur. Retransmission to mitigate frame loss is expected. The network is configured by the CNC to handle these frames, including bursts of frames, up to a certain number of frames or data size over a defined period.

4.7.4.6 Configuration and diagnostics

Configurations and dagnostics are a type of IA traffic engineered non-stream. This type of traffic is transmitted cyclically or acyclically. This traffic expects bounded latency, up to seconds, including time for retransmission. The source of configuration or diagnostics frames typically limits the bandwidth allocated to this traffic. Frame size is unconstrained except as indicated in 5.5.1. Congestion loss can occur. Retransmission to mitigate frame loss is expected. The network is configured by the CNC to handle these frames, including bursts of frames, up to a certain number of frames or data size over a defined period.

4.7.4.7 Network control

Network control is a type of IA traffic engineered non-stream. This type of traffic can be transmitted cyclically or acyclically. This traffic expects bounded latency including time for retransmission. Frame size is unconstrained except as indicated in 5.5.1. The network is configured by the CNC to handle these frames, including bursts of frames, up to a certain number of frames or data size over a defined period. If these limits are exceeded congestion loss can occur. Network control is comprised of services required to maintain network operation. Examples include time synchronization, loop prevention, and topology detection.

4.7.4.8 Best effort

Best effort is a type of IA non-stream. The network is configured by the CNC so that these frames do not interfere with other traffic types. These frames are forwarded when resources are available. Congestion loss resulting in frame drop can occur. It is sometimes desirable to have more than one traffic class for best effort traffic (see Table 8).

4.7.4.9 Traffic class to traffic type mapping

Table 8 provides an example for the usage of traffic classes based on the traffic type:

Table 8 – Example traffic class to traffic type mapping

Traffic class	Queue Assignment (8 queues)	Queue Assignment (4 queues)	Traffic type
7	6	2	Isochronous
6	5	1	Cyclic-Synchronous
5	4	1	Cyclic-Asynchronous
4	7	3	Network Control
3	3	0	Alarms and events
2	2	0	Configuration & Diagnostics
1	1	0	Best Effort High
0	0	0	Best Effort Low

NOTE An example of mapping of PCP and traffic type to an application is provided in Figure 6.

The traffic type categories definition allows different IEEE 802 feature selections to achieve specified goals. Moreover it helps in identification of the traffic protection mechanisms. Adherence to this example of a common mapping helps minimize potential conflicts between traffic types.

4.8 Security for TSN-IA

4.8.1 General

Subclause 4.8 describes selected aspects of TSN-IA security. Protecting the management of industrial communication is the main objective of TSN-IA security. The protection of communications that use industrial traffic types is not addressed by this document.

4.8.2 Security configuration model

Security configuration is a part of system engineering and configuration. The security configuration in this document does not encompass the supply of configuration objects for middleware and application security. Security configuration settles the prerequisites for protecting the establishment and management of communications that use industrial traffic types (see 4.7). It ensures that the security features of IA-stations (including CNCs) can be used for protecting message exchanges and authorizing the resource accesses during stream establishment and management. This security configuration supplies deployment-specific configuration objects to IA-stations. They encompass:

- instructions about cryptographic algorithms,
- credentials and trust anchors,
- instructions to interpret the outcome of peer entity authentication while enforcing resource access controls, and
- access control rules and permissions.

This security configuration uses NETCONF/YANG request/response exchanges, as follows.

- The to-be-configured IA-stations act in NETCONF server role with respect to their security configuration.
- A NETCONF client is responsible for setting-up IA-stations for security. This NETCONF client possesses information about the security relationship to be established during security configuration or about the expectations on the IA-stations in a Configuration Domain. It can be implemented as part of an interactive or automated process (for example an engineering tool, or CNC operation). As an implication, the security configuration includes options for interactive and automated setup, i.e. security configuration is done by human or non-human actors.

NOTE NETCONF notifications can also be used to recognize events such as a near-term end-of-life of certificate objects, especially EE certificate objects (see IETF RFC 4210, 3.1.1).

- The security configuration exchanges supply deployment-specific objects (trust anchors, credentials, etc.) to IA-stations and manages them. IA-stations that are in factory default state can only possess manufacturer-specific security objects (trust anchors, credentials, etc.) when booting initially. The protected NETCONF/YANG exchanges with IA-stations that are in factory default state are outlined in 4.8.3 to 4.8.6.

4.8.3 NETCONF/YANG processing

Securing NETCONF/YANG resources (for example, NETCONF sessions or managed objects) on NETCONF servers is specified by IETF RFC 6241 (NETCONF). Therefore, message exchange protection between NETCONF clients and servers as well as resource access authorization by NETCONF servers are utilized.

- IETF RFC 7589 and draft-ietf-netconf-over-tls13 (NETCONF-over-TLS) specify a solution to protect NETCONF message exchanges by TLS.
- IETF RFC 8341 (NACM) specifies three access control points, covering the request/response and notification model in NETCONF according to IETF RFC 8341, 2.1.

NETCONF servers enforce security as shown in Figure 9. The processing steps are executed upon the current configuration of the NETCONF server's YANG modules.

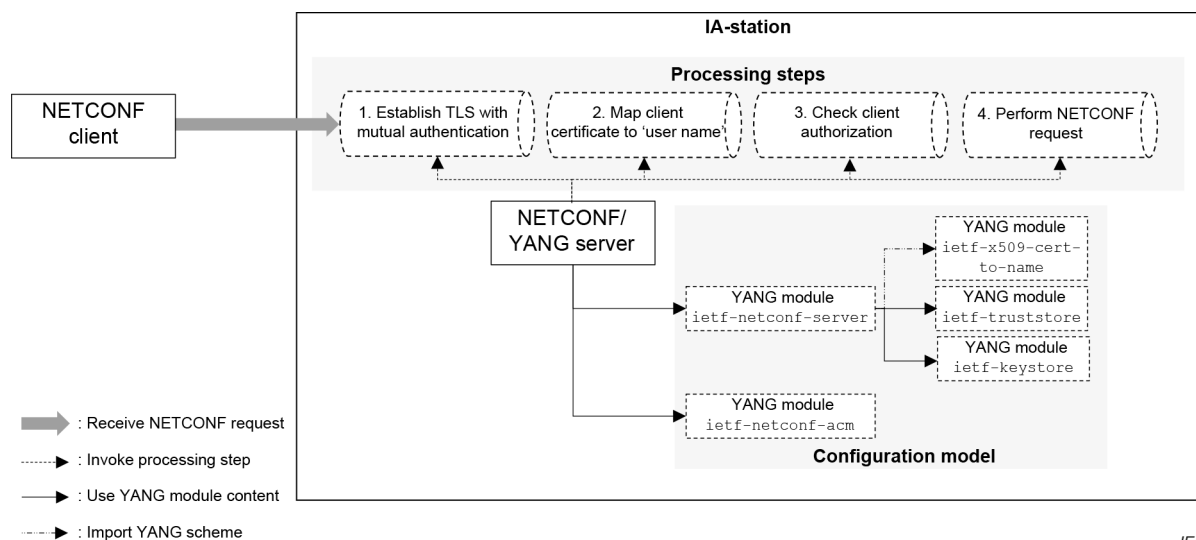


Figure 9 – NETCONF/YANG security processing steps

The processing steps on the side of NETCONF servers are as follows.

- 1) Establish a TLS connection with mutual authentication: the NETCONF server acts as TLS server and awaits connection requests of NETCONF clients (TLS clients). At the beginning of the TLS handshake, the TLS client and server negotiate the TLS protocol version to be used. During the TLS handshake the NETCONF server authenticates itself towards the NETCONF client by a credential from its ietf-keystore YANG module. In addition, the NETCONF server challenges the NETCONF client for authentication and verifies its authentication by trust anchors in its ietf-truststore YANG module according to 6.3.4. A successful mutual authentication is a prerequisite for proceeding to the next step.
- 2) Map the client certificate to a username: the NETCONF server maps the authenticated TLS client certificate to a "NETCONF username"³ by applying an ordered list of mapping instructions. These instructions are provided in its ietf-x509-cert-to-name YANG module. The applicable list item is identified by matching its configured fingerprint (according to IETF RFC 7589, Clause 7) against the certification path that was used for TLS client authentication (an end entity certificate or a CA certificate). According to the map type of the identified list item, the NETCONF server determines the "NETCONF username". This can be done by extracting information from the end entity certificate of the NETCONF client. A successful certificate-to-"NETCONF username" mapping is a prerequisite for proceeding to the next step.
- 3) Check client authorization: the NETCONF server checks if the NETCONF client has the permission to access the requested NETCONF/YANG resource based on its "NETCONF username" and the access control rules available in its ietf-netconf-acm YANG module. See 4.8.4 for more information about NETCONF/YANG access control. A successful authorization is a prerequisite for proceeding to the next step.
- 4) Perform NETCONF request: if all preceding steps succeeded, the NETCONF server performs the NETCONF request.

³ In this document and, in almost all cases, NETCONF username values do not represent human users.

4.8.4 NETCONF/YANG access control

NACM defines a YANG information model for describing permitted/denied access operations. NETCONF servers are responsible for enforcing access control to their resources according to the information in their ietf-netconf-acm YANG modules. NACM allows the description of access-controlled resources in terms of NETCONF protocol operations, nodes in YANG datastores or types of notification events. NACM uses character strings to represent the subject actors, i.e. NETCONF clients. These character strings are known as "NETCONF username". The NACM access control information of a NETCONF server is created, updated, and deleted per IA-station. The management of this information happens along the IA-station lifecycle, for example, manufacturing, bootstrapping, operation, maintaining, re-owning, destructing. Moreover, the management of the NACM access control information itself is subject to NACM access control.

This document employs multiple YANG data models for fulfilling its purposes. This extends beyond the above identified YANG modules (see 4.8.3). The NETCONF server on an IA-station enforces access control for NETCONF/YANG resources. To meet this objective, the NETCONF server on an IA-station is supplied with access control information for the used NETCONF/YANG resources. NACM is employed for this purpose and profiles default access control information for the NETCONF/YANG resources (see 6.3.2.2). This relieves other organizations or individuals for example, manufacturers, integrators, operators, owners from being responsible to create NACM access control information for the respective NETCONF/YANG resources.

NACM relies on character strings (known as "NETCONF username") to refer to clients. NACM access control information as specified in this document, populates the "NETCONF username" character strings in NACM with role names specified in 6.3.2.1.4 c). This allows to create default NACM information without knowing actual names of individual entities. A role name can refer to 0, 1 or more individual entities. It is the responsibility of users to assign role names to individual entities. This happens by binding the assigned role names to the credentials of individual entities. The current form to express this binding is a role extension in the identity certificates of end entities defined in this document. These are NETCONF clients, i.e. these role extensions appear in the end entity certificates of LDevID credentials for NETCONF clients.

As initial step NACM maps the NETCONF username to a set of groups. The set of groups determines the set of rules to be applied for access-controlled resources.

4.8.5 Identity checking

IETF RFC 7589 (NETCONF-over-TLS) specifies that NETCONF clients check the identity of NETCONF servers (IETF RFC 7589, Clause 6) and that NETCONF servers verify the identity of NETCONF clients (IETF RFC 7589, Clause 7).

The NETCONF server identity check happens inside NETCONF clients. It matches an actual against an expectation.

- The actual server identity is established by the end entity certificate of the NETCONF server (authenticated by means of TLS).
- The expectations on server identity are established by the information that is used to connect to the NETCONF server.

IETF RFC 7589 refers to IETF RFC 6125, Clause 6, for the details of retrieving the actual and comparing it against the expected.

The NETCONF client identity check happens inside NETCONF servers. It also matches an actual against an expectation.

- The actual client identity is established by the end entity certificate of the NETCONF client (authenticated by means of TLS).

- The expectations on client identity are established by the contents of the ietf-netconf-acm and ietf-x509-cert-to-name YANG modules.

The details of this check are subject to the requested NETCONF operation. IETF RFC 7589, Clause 7, specifies the mapping of an authenticated client certificate to a "NETCONF username" whose permissions are then enforced by IETF RFC 8341 (NACM). More information is provided in 4.8.3, steps 2 and 3.

1272 **4.8.6 Secure device identity**

1273 **4.8.6.1 Device identity**

The term 'device' originates from IEEE Std 802.1AR-2018. It matches the term IA-station in this document.

The device identity refers to a set of information items about a device that:

- describes a device as a physical or virtual entity in a distributed system (identifier and attribute information);
- is used by a device to describe itself as such entity (identifier and attribute information);
- allows to interact with this device (addressing information, i.e. a specific identifier class).

The targeted use case, for example application data exchanges, configuration exchanges, inventory, or ordering, determines the required amount of identity information about a device.

The device identity of any single IA-station encompasses:

- MAC addresses, IP addresses, TCP ports, DNS names;
- ietf-hardware YANG module contents (IETF RFC 8348, Clause 3 and 7.1).

1286 **4.8.6.2 Verifiable device identity**

Certain aspects of device identity are verified before relying on them during online interactions. These are examples.

- DNS names or IP addresses are used to call the management entity of an IA-station, i.e. its NETCONF/YANG server. Their value represents the caller's expectation on the identity of their responder in network communications. Verification of the responder's identity helps defeat DNS spoofing, component impersonation and man-in-the-middle attacks. This is specified by IETF RFC 7589 and described in IETF RFC 6125, Clause 6. Passing this check is a prerequisite before NETCONF application exchanges can happen.
- mfg-name values in instances of the ietf-hardware YANG module. These values make claims about the IA-station manufacturer. Their verification is a means to protect against counterfeiting.

The verification of IA-station identity happens according to a model that is fully specified by this document. That verification can be done in a manufacturer-agnostic manner. This verification is important before supplying locally significant credentials especially LDevID to IA-stations that are in factory-default state.

1302 **4.8.6.3 Verification support mechanisms**

1303 **4.8.6.3.1 General**

Subclause 4.8.6.3 considers mechanisms that support device identity verification during online interactions with IA-stations.

4.8.6.3.2 Secure transports

Sending information in plain form over a protected channel, e.g. ietf-hardware YANG module contents via NETCONF-over-TLS protects the transferred information during its transit through the network but does not vouch for the correctness of the received information, e.g. the mfg-name value.

4.8.6.3.3 Secure information

Protecting information objects by means of a cryptographic authentication code or digital signature enables verification of the authenticity and integrity of that information. These cryptographic authentication codes can use symmetric or asymmetric schemes. In case of asymmetric schemes, raw and self-signed public keys are distinguished from CA-signed public keys.

Asymmetric schemes with CA-signed public keys are preferable for the verifiable device identity use case: claimants and verifiers share a public key; the claimant possesses the corresponding private key. The establishment and storage of shared public keys uses public key certificates. For this approach self-signed CA certificates are to be established in an authentic manner. The number of self-signed CA certificates is independent from the number of verifiers (CNCs) as well as claimants (IA-stations).

4.8.6.3.4 IDevID and LDevID credentials

IDeVID and LDeVID credentials are specified by IEEE Std 802.1AR-2018. These objects are comprised of a certification path and a private key. The certification path encompasses an end entity certificate which contains verifiable device identity in a CA-signed form. The device identity verification happens after validating the certification path (IETF RFC 5280, Clause 6) and checking the proof-of-possession for the private key. The certification path validation demands trust anchors as input arguments (IETF RFC 5280, 6.1.1 input argument (d)).

Two types of credentials are distinguished by IEEE Std 802.1AR-2018:

- IDeVIDs are issued by device manufacturers. They represent an initial identity as it is known at device production-time. The initial device identity is not locally significant: it cannot contain deployment-specific information such as DNS names or IP addresses.
- LDeVIDs are issued by other actors, e.g. a device user. They represent a locally significant device identity: they can contain deployment-specific information, e.g. DNS names or IP addresses.

IEEE Std 802.1AR-2018, Clause 6, uses signature suites to describe the subject public key and the signature fields in IDeVID and LDeVID certification paths. This notion is different from TLS cipher suites.

NOTE IDeVID and LDeVID credentials also serve purposes beyond secure device identity, for instance the realization of secure transports. This facilitates the use case of NETCONF/YANG security setup from factory default state.

4.8.6.3.5 IDevID items beyond IEEE Std 802.1AR-2018

IEEE Std 802.1AR-2018 allows verification of the following identity items:

- certificate issuer (not necessarily: manufacturer) by issuer field (data type: ASN.1 Name), and
- if present: device instance by serialNumber value (data type: ASN.1 PrintableString).

NOTE 1 IEEE Std 802.1AR-2018 represents the initial device identity as an optional serialNumber attribute (OID 2.5.4.5) in the subject field of the EE certificate. This value is unique within the domain of significance of the EE certificate issuer.

NOTE 2 This verification can happen after certification path validation and the proof-of-possession checking for the private key.

The following bullet points describe options beyond IEEE Std 802.1AR-2018 for verifying the device identity of IA-stations in factory default state. It also identifies informational items for the corresponding checks:

- IA-station manufacturer check: using names that identify IA-station manufacturers, e.g. mfg-name in ietf-hardware YANG module,
- IA-station type check: using attributes that identify IA-station types, e.g. model-name, hw-revision, description in ietf-hardware YANG module, and
- IA-station instance check: using values that identify IA-station instances, e.g. serial-num in ietf-hardware YANG module.

The following model described in the bullet points applies to the verification of the initial device identity of IA-stations:

- the set of to-be-conducted checks is determined by IA-station and CNC users,
- an IA-station uses IDevID credentials to prove its device identity. The checking happens by means of online interactions in the operational network. It happens automatically and is done by CNCs. This does not depend on Configuration Domain external repositories, and
- other stakeholders, e.g. middleware/application consortia or individual manufactures are allowed to additionally express information items in IDevID credentials to reflect their device identity model. CNCs do not assess such additional information.

4.8.6.3.6 Device identity representation in IDevID and LDevID credentials

The best practices for representing verifiable device identity information in IDevID and LDevID credentials (see 6.3.3.2.2 for more information) are as follows.

- Corresponding information (actual values or references to them) appears in EE certificates:
 - IDevID EE certificates bind initial device identity items that are known by the device manufacturer at production time, e.g. mfg-name.
 - LDevID EE certificates bind locally significant device identity items that are known by other actors such as device users, e.g. DNS names or IP addresses. They can also bind initial device identity information.
 - Items that encode device naming information appear in the subjectAltName extension.
- NOTE This is specified in IETF RFC 5280, 4.2.1.6. It is further explained in IETF RFC 6125, 2.3.
- A binding can take one of following forms. Multiple forms can appear in one EE certificate:
 - by-value: the verifiable device identity information is represented by its value inside the IDevID resp. LDevID EE certificate. Examples are:
 - the product serialNumber in IDevID credentials (IEEE Std 802.1AR-2018), and
 - the hostname of the NETCONF/YANG server in LDevID credentials (IETF RFC 6125, Clause 6).
 - by-ref: the verifiable device identity information is represented by a reference inside the IDevID resp. LDevID EE certificate, not by its value.
 - The actual value can be provided by the device itself or by a device-external source.
 - If it is provided in form of an unprotected information object, then the reference object that is embedded to EE certificates includes a digest value.

5 Conformance

5.1 General

A claim of conformance to this document is a claim that the behavior of an implementation of an IA-station (see 5.5, 5.6) with its Bridge components (see 5.7, 5.8) and end station

1397 components (see 5.9, 5.10) meets the mandatory requirements of this document and may
1398 support options identified in this document.

1399 **5.2 Requirements terminology**

1400 The verbal forms for required expressions of provisions follow the conventions below.

- 1401 a) Requirements terminology is provided in the ISO/IEC Directives Part 2:2021, Clause 7. This
1402 document can be found at www.iec.ch/members_experts/refdocs.
- 1403 b) The Profile Conformance Statement (PCS) proformas (see Annex A) reflect the occurrences
1404 of the words "shall," "may," and "should" within this document.
- 1405 c) This document avoids needless repetition and apparent duplication of its formal
1406 requirements by using "is", "is not", "are", and "are not" for definitions and the logical
1407 consequences of conformant behavior. Behavior that is permitted but is neither always
1408 required nor directly controlled by an implementer or administrator, or whose conformance
1409 requirement is detailed elsewhere, is described by "can". Behavior that never occurs in a
1410 conformant implementation or system of conformant implementations is described by
1411 "cannot". The word "allow" is used as a replacement for the phrase "support the ability for",
1412 and the word "capability" means "can be configured to".

1413 **5.3 Profile conformance statement (PCS)⁴**

1414 The supplier of an implementation that is claimed to conform to this document shall provide the
1415 information necessary to identify both the supplier and the implementation and shall complete
1416 a copy of the PCS proforma provided in Annex A.

1417 **5.4 Conformance classes**

1418 This document includes conformance requirements and options that are related to an entire
1419 station, as well as conformance requirements and options that are related to single Bridge or
1420 end station components within an IA-station. Figure 10 illustrates this conformance model.

⁴ Copyright release for the PCS: Users of this document may freely reproduce the PCS contained in this document so that it can be used for its intended purpose.

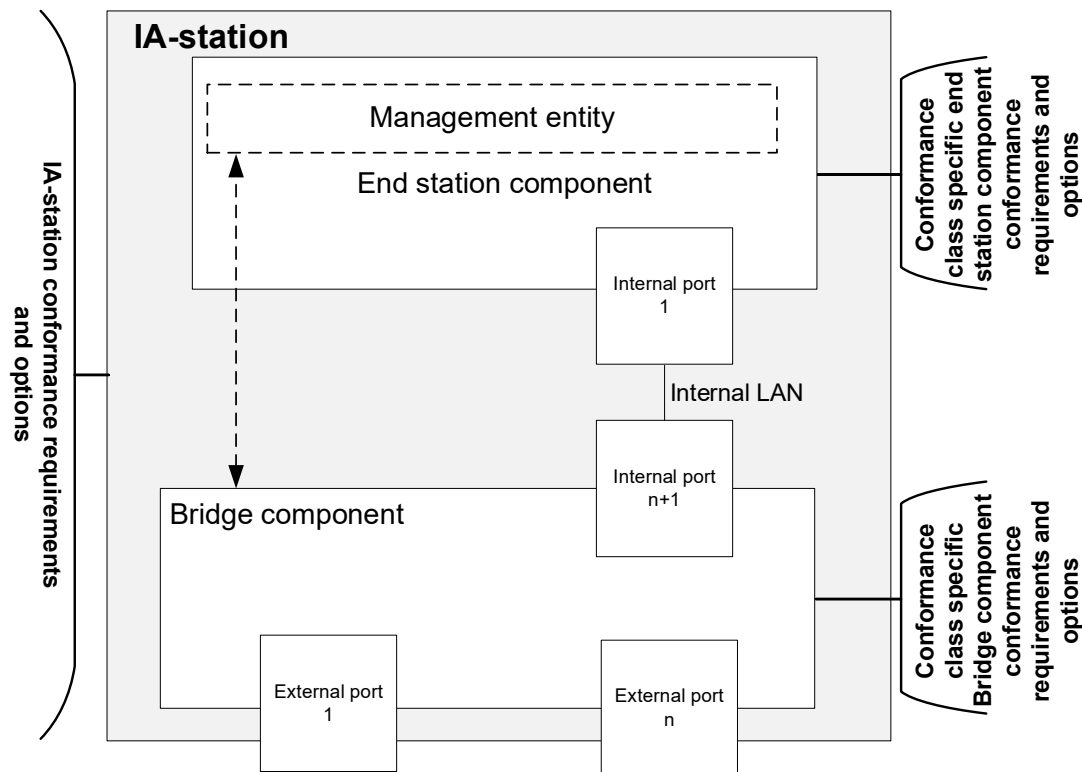


Figure 10 – IA-station conformance model

This document supports a variety of industrial use cases. In some of these use cases, support of certain TSN features can be mandatory, while in others, supporting these features can lead to non-optimal implementations. Therefore, this document defines two conformance classes that are applicable both to Bridge components and end station components. Conformance Class A (ccA) is feature-rich, i.e. tailored to use cases requiring support of many TSN-IA features. Conformance Class B (ccB) targets implementations that are more resource constrained. The details for the conformance classes are specified in 5.7 and 5.8 for Bridge components, and in 5.9 and 5.10 for end station components.

NOTE 1 It is the responsibility of the IA-station manufacturer to carefully consider the implications of mixing ccA and ccB Bridge components and end station components in a single IA-station.

NOTE 2 It is the responsibility of the user to carefully consider the implications of mixing ccA and ccB Bridge components and end station components in a single Configuration Domain.

NOTE 3 Any Bridge compliant to this document is an IA-station. Any IA-station contains a management entity (i.e. an end station component).

5.5 IA-station requirements

5.5.1 IA-station PHY and MAC requirements for external ports

IA-stations for which a claim of conformance to this document is made shall support the following list of requirements for external ports.

- Support the Media Access Control (MAC) service specification according to IEEE Std 802.3-2022, Clause 2.
- Support the Media Access Control (MAC) frame and packet specifications according to IEEE Std 802.3-2022, Clause 3, especially the MAC Client Data field size according to IEEE Std 802.3-2022, 3.2.7, item c).
- Support Layer Management according to IEEE Std 802.3-2022, 5.2.4.

- d) Implement at least one IEEE Std 802.3-2022 MAC that shall operate in full-duplex mode, and associated IEEE Std 802.3-2022 PHY with a data rate of at least one of the following speeds: 10 Mb/s, 100 Mb/s, 1 000 Mb/s, 2.5 Gb/s, 5 Gb/s, or 10 Gb/s together with the corresponding managed objects:
 - 1) 10BASE-T1L MAU type according to IEEE Std 802.3-2022, Clauses 22 and 146,
 - 2) 100BASE-TX and 100BASE-FX MAU types according to IEEE Std 802.3-2022, Clauses 21, 22, 24, 25, 26, 30, 31 and IEEE Std 802.3-2022, Annexes 23A, 28A, 28B, 28C, 28D, 31A, 31B, 31C, and 31D,
 - 3) 1000BASE-T and 1000BASE-SX MAU types according to IEEE Std 802.3-2022, Clauses 28, 34, 35, 36, 37, 38, and 40,
 - 4) 2.5GBASE-T and 5GBASE-T MAU types according to IEEE Std 802.3-2022, Clauses 28, 125, and 126,
 - 5) 2.5GBASE-T1 and 5GBASE-T1 MAU types according to IEEE Std 802.3-2022, Clause 149,
 - 6) 10GBASE-T and 10GBASE-SR MAU types according to IEEE Std 802.3-2022, Clauses 44, 46, 47, 49, 51, 52, 55, and IEEE Std 802.3-2022, Annexes 48A and 55A,
 - 7) 10GBASE-T1 MAU type according to IEEE Std 802.3-2022, Clause 149,
 - 8) 100BASE-T1 MAU type according to IEEE Std 802.3-2022, Clause 96, and
 - 9) 1000BASE-T1 MAU type according to IEEE Std 802.3-2022, Clause 97.
- e) Support the YANG features and nodes of the ieee802-ethernet-interface module according to 6.4.9.2.1.
- f) Implement Ethernet support for time synchronization protocols according to IEEE Std 802.3-2022, Clause 90 as amended by IEEE Std 802.3cxTM-2023 with the aTimeSyncSelectionDmnp set to 'symbol-after-sfd' according to IEEE Std 802.3cx-2023, 30.13.1.14.

Clauses and subclauses not mentioned can be implemented but are not obligatory.

5.5.2 IA-station topology discovery requirements

IA-stations for which a claim of conformance to this document is made shall support the following list of requirements.

- a) The required capabilities according to IEEE Std 802.1AB-2016, 5.3 and IEEE Std 802.1ABcuTM-2021, 5.3.
- b) Topology discovery and verification according to 6.5.
- c) The YANG features and nodes of the ieee802-dot1ab-lldp module according to 6.4.9.2.2.

5.5.3 IA-station requirements for time synchronization

These requirements are related to the entire IA-station with all its PTP Instances and PTP Ports. IA-stations for which a claim of conformance to this document is made shall support the following list of requirements.

- a) PTP Instance requirements according to IEEE Std 802.1AS-2020 as amended by IEEE Std 802.1AS-2020/Cor 1-2021, IEEE Std 802.1ASdr-2024, IEEE Std 802.1ASdnTM-2024, and IEEE Std 802.1ASdm-2024, 5.4.1 items a) through h).

NOTE A gPTP domain in a PTP End Instance can be used for Global Time, Working Clock, or both.

- b) PTP Instance defaults for BTCA according to IEEE Std 802.1AS-2020 as amended by IEEE Std 802.1AS-2020/Cor 1-2021, IEEE Std 802.1ASdr-2024, IEEE Std 802.1ASdn-2024, and IEEE Std 802.1ASdm-2024, 5.4.2a.
- c) Timing and synchronization management according to IEEE Std 802.1AS-2020, 5.4.2 items j) and k).
- d) PTP Instance requirements according to 6.2.2.

- 1494 e) PTP Protocol requirements according to 6.2.3.
 - 1495 f) Clock control system requirements according to 6.2.4.
 - 1496 g) Error generation limits according to 6.2.5.
 - 1497 h) PTP Instance synchronization status according to 6.2.6.
 - 1498 i) The transmission of the Drift_Tracking TLV according to IEEE Std 802.1AS-2020 as
1499 amended by IEEE Std 802.1AS-2020/Cor 1-2021, IEEE Std 802.1ASdr-2024, IEEE Std
1500 802.1ASdn-2024, and IEEE Std 802.1ASdm-2024, 5.4.2 item n).
 - 1501 j) External port configuration capability according to IEEE Std 802.1AS-2020 as amended by
1502 IEEE Std 802.1AS-2020/Cor 1-2021, IEEE Std 802.1ASdr-2024, IEEE Std 802.1ASdn-2024,
1503 and IEEE Std 802.1ASdm-2024, 5.4.2 f).
 - 1504 k) MAC-specific timing and synchronization methods for IEEE Std 802.3 full-duplex links
1505 according to IEEE Std 802.1AS-2020 as amended by IEEE Std 802.1AS-2020/Cor 1-2021,
1506 IEEE Std 802.1ASdr-2024, IEEE Std 802.1ASdn-2024, and IEEE Std 802.1ASdm-2024, 5.5
1507 items a) through e) and item j), and 11.2.17.
 - 1508 l) The YANG features and nodes of the:
 - 1509 i) ieee1588-ptp-tt module according to 6.4.9.2.3.1,
 - 1510 ii) ieee802-dot1as-gptp module according to 6.4.9.2.3.2, and
 - 1511 iii) ieee802-dot1as-hs module according to 6.4.9.2.3.3.
 - 1512 m) The message timestamp point according to IEEE Std 802.1AS-2020, 11.3.9.
 - 1513 n) The Common Mean Link Delay Service (CMLDS) according to IEEE Std 802.1AS-2020,
1514 11.2.17.
 - 1515 o) The descriptionDS according to 6.2.13 and IEEE Std 1588-2019, 8.2.5.
 - 1516 p) gptpCapableStateMachinesEnabled is FALSE according to IEEE Std 802.1ASdm- 2024,
1517 5.4.2a and 14.8.54a.
- 1518 NOTE Setting gptpCapableStateMachinesEnabled to FALSE allows transmission of signaling message to be
1519 disabled by design.

1520 **5.5.4 Management requirements for non-proxied IA-stations**

1521 **5.5.4.1 Secure management exchanges**

1522 IA-stations which contain a ccA Bridge component or a ccA end station component and for
1523 which a claim of conformance to this document is made shall support the following list of
1524 requirements.

- 1525 a) NETCONF server functionality according to IETF RFC 6241 including:
 - 1526 1) candidate configuration capability as described in IETF RFC 6241, 8.3,
 - 1527 2) rollback-on-error capability as described in IETF RFC 6241, 8.5, and
 - 1528 3) validate capability as described in IETF RFC 6241, 8.6.
- 1529 b) NETCONF-over-TLS server supporting TLS version 1.2 according to 6.3.2.1.1 and 6.3.2.1.4.
- 1530 c) Secure Device Identity according to 6.3.3 and IEEE Std 802.1AR-2018, 5.3 a), using the
1531 signature suite in IEEE Std 802.1AR-2018 9.2, 5.3 d), and 5.3 i).
- 1532 d) PKIX according to 6.3.2.1.5 and IETF RFC 5280, 4.1, 4.2.1.1-3, 4.2.1.6, 6.1 and 6.2.
- 1533 e) NACM (IETF RFC 8341) supporting four different roles according to 6.3.2.1.5 c).
- 1534 f) The YANG features and nodes of the:
 - 1535 1) ietf-keystore module according to 6.4.9.2.4.1,
 - 1536 2) ietf-netconf-acm module according to 6.4.9.2.4.2 and,
 - 1537 3) ietf-truststore according to 6.4.9.2.4.3.
- 1538 g) NETCONF Event Notifications according to IETF RFC 5277 including operations according
1539 to IETF RFC 5277, Clause 2.

- 1540 h) Dynamic Subscriptions to YANG Events and Datastores over NETCONF according to
1541 6.4.7.7.
- 1542 i) NETCONF Extensions to support the Network Management Datastore Architecture (NMDA)
1543 as described in IETF RFC 8526.
- 1544 j) DHCP client according to IETF RFC 2131, 4.1, 4.2, and 4.4.
- 1545 k) At least one of the following asymmetric key pair generation methods:
1546 1) component-internal generation according to 6.3.4.3;
1547 2) component-external generation according to 6.3.4.3;
- 1548 l) Storage of at least one IDevID credential according to 6.3.3.4.2.5 and one LDevID-
1549 NETCONF credential according to 6.3.4.1.
- 1550 m) If the IA-station supports CNC functionality, that IA-station shall also support the following
1551 list of requirements.
1552 1) TSN CNC station requirements according to IEEE Std 802.1Q-2022, 5.29.
1553 2) NETCONF-over-TLS server and related client functionality 5.5.4.2.
1554 3) The common YANG modules, features, and nodes according to 6.4.9.2.
1555 4) The optional YANG modules, features, and nodes according to 6.4.9.3.
1556 5) Support the use of at least one customer VLAN Identifier for an isolation VLAN and one
1557 VLAN identifier for a default VLAN.
1558 6) Support CUC/CNC YANG modules, features and nodes according to 6.4.9.4.
- 1559 n) If the IA-station supports CUC functionality, that IA-station shall also support the following
1560 list of requirements.
1561 1) Support NETCONF-over-TLS client functionality with client related security requirements
1562 according to 5.5.4.2.
1563 2) The TSN UNI YANG module, features, and nodes according to 6.4.9.4.1.
1564 3) The ietf-netconf-client module according to 6.4.9.4.1.
- 1565 o) If the IA-station supports proxy functionality, support the iecieee60802-proxy YANG module
1566 according to 6.4.10.8.8.
- 1567 IA-stations which contain a ccA Bridge component or a ccA end station component and for
1568 which a claim of conformance to this document is made should support internal key generation
1569 according to 6.3.4.3.2.

1570 **5.5.4.2 IA-station management YANG modules**

- 1571 IA-stations which contain a ccA Bridge component or a ccA end station component and for
1572 which a claim of conformance to this document is made shall support the YANG features and
1573 nodes for IA-station management of the:
- 1574 a) ietf-system-capabilities module according to 6.4.9.2.5.1,
- 1575 b) ietf-yang-library module according to 6.4.9.2.5.2,
- 1576 c) ietf-yang-push module according to 6.4.9.2.5.3,
- 1577 d) ietf-notification-capabilities module according to 6.4.9.2.5.4,
- 1578 e) ietf-subscribed-notifications module according to 6.4.9.2.5.5,
- 1579 f) diagnostics using YANG-Push subscriptions according to 6.4.7,
- 1580 g) ietf-netconf-monitoring module according to 6.4.9.2.5.6,
- 1581 h) ietf-system module according to 6.4.9.2.5.7,
- 1582 i) ietf-hardware module according to 6.4.9.2.5.8,
- 1583 j) ietf-interfaces module according to 6.4.9.2.5.9,
- 1584 k) ieee802-dot1q-bridge module according to 6.4.9.2.5.10,

- l) iecieeee60802-ethernet-interface module according to 6.4.9.2.5.11,
- m) ietf-netconf-server according to 6.4.9.2.5.12,
- n) iecieeee60802-bridge according to 6.4.9.2.5.11,
- o) ietf-subscribed-notifications according to 6.4.9.2.5.13,
- p) iecieeee60802-subscribed-notifications according to 6.4.9.2.5.13, and
- q) iecieeee60802-ia-station according to 6.4.9.2.5.11.
- r) ieee802-dot1q-rstp-bridge according to 6.4.9.2.5.14.

5.5.4.3 Digital data sheet

IA-stations which contain a ccA Bridge component or a ccA end station component and for which a claim of conformance to this document is made shall provide a 60802 instance data file according to 6.4.8. The instance data file shall contain at least the YANG nodes of 6.4.9 that are marked with [m]. Nodes marked with [c] shall be included if the associated feature is supported.

NOTE It is the user's responsibility to ensure that the filename is unique by using a standardized mechanism (for example, GUID, URL, or ReverseDomainName).

5.6 IA-station options

5.6.1 IA-station PHY and MAC options for external ports

IA-stations for which a claim of conformance to this document is made may support the following list of requirements.

- a) Power over Ethernet (PoE) over 2 Pairs according to IEEE Std 802.3-2022, Clause 33.
- b) Power Interfaces according to IEEE Std 802.3-2022, Clause 104.
- c) Power over Ethernet according to IEEE Std 802.3-2022 Clause 145.

5.6.2 IA-station options for time synchronization

IA-stations for which a claim of conformance to this document is made may support the following list of requirements.

- a) The media-independent TimeTransmitter capability according to IEEE Std 802.1AS-2020, 5.4.2 item b) as amended by IEEE Std 802.1ASdr-2024.
- b) Grandmaster PTP Instance capability according to IEEE Std 802.1AS-2020 as amended by IEEE Std 802.1ASdr-2024 and IEEE Std 802.1ASdm-2024, 5.4.2 item c).
- c) More than one PTP port as a PTP Relay Instance according to IEEE Std 802.1AS-2020, 5.4.2 item d).
- d) Hot standby redundancy requirements according to IEEE Std 802.1AS-2020 as amended by IEEE Std 802.1ASdm-2024, 5.4.2, item m).

5.6.3 IA-station options for management

IA-stations for which a claim of conformance to this document is made may support the following list of requirements.

- a) Writable-Running capability according to IETF RFC 6241, 8.2.
- b) Confirmed Commit capability according to IETF RFC 6241, 8.4.
- c) Distinct Startup capability according to IETF RFC 6241, 8.7.
- d) URL capability according to IETF RFC 6241, 8.8.
- e) XPath capability according to IETF RFC 6241, 8.9.
- f) NETCONF-over-TLS server supporting TLS version 1.3, according to 6.3.2.1.3.
- g) PKIX according to IETF RFC 5280, 4.2.1.13, Clause 5 and 6.3.

5.6.4 Proxy target options

IA-stations which contain only ccB Bridge or end station components and for which a claim of conformance to this document is made may support the following list of requirements.

- a) Secure management exchanges according to 5.5.4.1, a) through l) including support of IA-station management YANG modules according to 5.5.4.2.
- b) Secure management exchanges using a management proxy according to 6.4.5.5 including the following list of requirements.
 - 1) Provide a digital datasheet according to 5.5.4.3 including the YANG features and nodes of the `iecieee60802-proxy-target` module according to 6.4.10.8.9.
 - 2) Support the 60802 constrained device proxy TLV described in Annex E.

5.7 Bridge component requirements

5.7.1 Common Bridge component requirements

A Bridge component implementation of any conformance class for which a claim of conformance to this document is made shall support the following list of requirements.

- a) C-VLAN component requirements according to IEEE Std 802.1Q-2022, 5.5 and 5.4 except item o) in IEEE Std 802.1Q-2022, 5.4.
- b) The use of Customer VLAN Identifiers (C-VID).
- c) FDB to contain Static and Dynamic VLAN Registration Entries for a minimum of 10 VIDs according to IEEE Std 802.1Q-2022, 8.8.

NOTE 1 An example use case for 10 VIDs: 2 VIDs for IA time-aware stream or IA stream traffic, 2 VIDs for IA time-aware stream or IA stream redundancy, 4 VIDs for IA traffic engineered non-stream or IA non-stream traffic, 1 VID for the isolation VLAN, and 1 VID for the Default VLAN (see 6.4.5.2).

- d) Translation of VIDs through support of the VID Translation Table or through support of both the VID Translation Table and Egress VID translation table on one or more Bridge Ports according to IEEE Std 802.1Q-2022, 6.9.
- e) The strict priority algorithm for transmission selection on each port for each traffic class according to IEEE Std 802.1Q-2022, 8.6.8.1.
- f) The capability to disable Priority-based flow control according to IEEE Std 802.1Q-2022, Clause 36, if it is implemented.
- g) The Priority Regeneration requirements according to IEEE Std 802.1Q-2022, 5.4.1, item o).
- h) MST according to IEEE Std 802.1Q-2022, 5.4.1.1 a) to i) and k) to o) and 6.4.2.4.
- i) TE-MSTID according to IEEE Std 802.1Q-2022, 5.5.2, 8.6, and 8.8.
- j) Spanning tree, VLAN, and TE-MSTID configuration according to 6.4.2.4.
- k) YANG features and nodes of the `iecieee60802-tt-mapping` module according to 6.4.10.8.7.
- l) PSFP according to IEEE Std 802.1Q-2022, 8.6.5.2.1, with support for:
 - 1) at least 3 stream filters per port according to IEEE Std 802.1Q-2022, 8.6.5.3,
 - 2) at least one stream gate according to IEEE Std 802.1Q-2022, 8.6.5.4 with at least zero gate control list entries according to IEEE Std 802.1Q-2022, 12.31.1.4, and
 - 3) at least 3 flow meters per port according to IEEE Std 802.1Q-2022, 8.6.5.5.

NOTE 2 The flow meter parameter values Excess Information Rate (EIR) = 0, Excess burst size (EBS) = 0 and Color Mode (CM) = `color_blind`, as configurable by the CNC, are applicable to many industrial automation use cases. When CM = `color_blind`, `DropOnYellow` (IEEE Std 802.1Q-2022, 8.6.5.5, item h), `MarkAllFramesRed` (IEEE Std 802.1Q-2022, 8.6.5.1.3, item j), and `MarkAllFramesRedEnable` (IEEE Std 802.1Q-2022, 8.6.5.5, item i) are not used.

NOTE 3 For example, an implementation could contain one flow meter for broadcast traffic, one flow meter for multicast traffic and one flow meter for unicast traffic. This example can be realized using 49 mask and match stream identification functions per Port and 49 associated stream filter instances (one-to-one relationship) that identify and map unicast (1 stream identification function for identifying all frame with the group address bit cleared in the DA), broadcast (1 stream identification function for identifying all frames with all DA bits set) and

1677 multicast (47 stream identification function with the DA group address bit set and one other DA bit cleared for
1678 all 47 combinations).

1679 m) The YANG features and nodes for flow meter configuration according to 6.4.9.2.5.14.

1680 n) Stream identification component behaviors according to IEEE Std 802.1CB-2017, 5.3 and
1681 IEEE Std 802.1CBdbTM-2021, 5.5 d).

1682 **5.7.2 ccA Bridge component requirements**

1683 A Bridge component implementation for which a claim of conformance to ccA of this document
1684 is made shall support the following list of requirements.

1685 a) Common Bridge component requirements according to 5.7.1.

1686 b) At least 2 PTP Instances according to 5.5.3.

1687 c) Eight queues according to IEEE Std 802.1Q-2022, 8.6.6.

1688 d) Enhancements for scheduled traffic for data rates of 100 Mb/s and 1 Gb/s according to IEEE
1689 Std 802.1Q-2022, 5.4.1 items ac) and ad) including:

1690 1) TickGranularity of less than or equal to 10 ns according to IEEE Std 802.1Q-2022,
1691 8.6.9.4.16 and Table 12-32,

1692 2) the allowable delay between the transmission selection timing point and the on-the-wire
1693 timing point, less any delay for the PHY (IEEE Std 802.1Q-2022, Figure 12-6), of less
1694 than or equal to 10 ns,

1695 NOTE Transmission selection timing points have a granularity of 1 ns; however, operation is determined by the
1696 precision of the "tick" event.

1697 3) support of the YANG features and nodes of the ieee802-dot1q-sched module according
1698 to 6.4.9.3.2, and

1699 4) support of the YANG features and nodes of the icieeee60802-sched-bridge module
1700 according to 6.4.9.3.3.

1701 e) Frame preemption according to IEEE Std 802.1Q-2022, 5.4.1 item ae), for data rates of
1702 100 Mb/s and 1 Gb/s, including:

1703 1) support of Interspersing Express Traffic with preemptable traffic according to IEEE
1704 Std 802.3-2022, Clause 99, including support of the Additional Ethernet Capabilities for
1705 TLV in an LLDPDU to indicate supported functions of frame preemption according to
1706 IEEE Std 802.3-2022, 79.3.7, and

1707 2) support of the YANG features and nodes of the ieee802-dot1q-preemption module
1708 according to 6.4.9.3.4.

1709 **5.7.3 ccB Bridge component requirements**

1710 A Bridge component implementation for which a claim of conformance to ccB of this document
1711 is made shall support the following list of requirements.

1712 a) Common Bridge component requirements according to 5.7.1.

1713 b) At least 1 PTP Instance according to 5.5.3.

1714 c) At least four queues according to IEEE Std 802.1Q-2022, 8.6.6.

1715 **5.8 Bridge component options**

1716 **5.8.1 Common Bridge component options**

1717 A Bridge component implementation of any conformance class for which a claim of conformance
1718 to this document is made may support the operation of the credit-based shaper algorithm
1719 according to IEEE Std 802.1Q-2022, 8.6.8.2 on all Ports as the transmission selection algorithm
1720 for at least 4 traffic classes including support of the YANG features and nodes of the < ieee802-
1721 dot1q-cbsa-bridge> module according to 6.4.9.3.5.

5.8.2 ccA Bridge component options

A Bridge component implementation for which a claim of conformance to ccA of this document is made may support the following list of requirements.

- a) Any or none of the common Bridge component options according to 5.8.1.
- b) More than 2 PTP Instances according to 5.5.3.
- c) Enhancements for scheduled traffic for data rates of 10 Mb/s, 2,5 Gb/s, 5 Gb/s, and 10 Gb/s according to IEEE Std 802.1Q-2022, 5.4.1 items ac) and ad) including:
 - 1) TickGranularity of less than or equal to 10 ns according to IEEE Std 802.1Q-2022, 8.6.9.4.16 and Table 12-32,
 - 2) the allowable delay between the transmission selection timing point and the on-the-wire timing point, less any delay for the PHY (IEEE Std 802.1Q-2022, Figure 12-6), of less than or equal to 10 ns,
 - 3) support of the YANG features and nodes of the `ieee802-dot1q-sched` module according to 6.4.9.3.2, and
 - 4) support of the YANG features and nodes of the `ieciieee60802-sched-bridge` module according to 6.4.9.3.3.
- d) Frame preemption according to IEEE Std 802.1Q-2022, 5.4.1 item ae), for data rates of 10 Mb/s, 2,5 Gb/s, 5 Gb/s, and 10 Gb/s, including:

NOTE IEEE Std 802.3de™-2022, 99.1, comprises 10 Mb/s.

- 1) support of Interspersing Express Traffic with preemptable traffic according to IEEE Std 802.3-2022, Clause 99, including support of the Additional Ethernet Capabilities for TLV in an LLDPDU to indicate supported functions of frame preemption according to IEEE Std 802.3-2022, 79.3.7, and
- 2) support of the YANG features and nodes of the `ieee802-dot1q-preemption` module according to 6.4.9.3.4.

5.8.3 ccB Bridge component options

A Bridge component implementation for which a claim of conformance to ccB of this document is made may support the following list of requirements.

- a) Any or none of the common Bridge component options according to 5.8.1.
- b) Up to eight queues according to IEEE Std 802.1Q-2022, 8.6.6.
- c) More than 1 PTP Instance according to 5.5.3.
- d) Enhancements for scheduled traffic according to IEEE Std 802.1Q-2022, 5.4.1, items ac) and ad) including:
 - 1) TickGranularity of less than or equal to 10 ns according to IEEE Std 802.1Q-2022, 8.6.9.4.16 and Table 12-32,
 - 2) the allowable delay between the transmission selection timing point and the on-the-wire timing point, less any delay for the PHY (IEEE Std 802.1Q-2022, Figure 12-6), of less than or equal to 10 ns,
 - 3) support of the YANG features and nodes of the `ieee802-dot1q-sched` module according to 6.4.9.3.2, and
 - 4) support of the YANG features and nodes of the `ieciieee60802-sched-bridge` module according to 6.4.9.3.3.
- e) Frame preemption according to IEEE Std 802.1Q-2022, 5.4.1, item ae), including:
 - 1) support of Interspersing Express Traffic with preemptable traffic according to IEEE Std 802.3-2022, Clause 99 including support of the Additional Ethernet Capabilities for TLV in an LLDPDU to indicate supported functions of frame preemption according to IEEE Std 802.3-2022, 79.3.7, and

- 1769 2) support of the YANG features and nodes of the ieee802-dot1q-preemption module
1770 according to 6.4.9.3.4.

1771 **5.9 End station component requirements**

1772 **5.9.1 Common end station Component requirements**

1773 An end station component implementation of any conformance class for which a claim of
1774 conformance to this document is made shall support the following list of requirements.

- 1775 a) The use of at least one customer VID for IA traffic engineered non-stream or IA non-stream
1776 traffic.
- 1777 b) The use of an additional customer VID for IA time-aware stream traffic if that traffic type
1778 category is supported.
- 1779 c) The use of an additional customer VID for IA stream traffic if that traffic type category is
1780 supported.
- 1781 d) The use of an additional customer VID for IA time-aware stream traffic if redundancy for that
1782 traffic type category is supported.
- 1783 e) The use of an additional customer VID for IA stream traffic if redundancy for that traffic type
1784 category is supported.
- 1785 f) Participate in only a single Configuration Domain.
- 1786 g) The use of an additional customer VID for an isolation VLAN.
- 1787 h) The use of an additional customer VID for a default VLAN

1788 **5.9.2 ccA end station component requirements**

1789 An end station component implementation for which a claim of conformance to ccA of this
1790 document is made shall support the following list of requirements.

- 1791 a) Common end station component requirements according to 5.9.1.
- 1792 b) At least 2 PTP Instances according to 5.5.3.
- 1793 c) End station requirements for enhancements for scheduled traffic according to IEEE Std
1794 802.1Q-2022, 5.25, for data rates of 100 Mb/s and 1 Gb/s including:
- 1795 1) TickGranularity of less than or equal to 10 ns according to IEEE Std 802.1Q-2022,
1796 8.6.9.4.16 and Table 12-32,
- 1797 2) the allowable delay between the transmission selection timing point and the on-the-wire
1798 timing point, less any delay for the PHY (IEEE Std 802.1Q-2022, Figure 12-6), of less
1799 than or equal to 10 ns,
- 1800 3) support of the YANG features and nodes of the ieee802-dot1q-sched module according
1801 to 6.4.9.3.2, and
- 1802 4) support of the YANG features and nodes of the icieeee60802-sched-bridge module
1803 according to 6.4.9.3.3.
- 1804 d) End station requirements for frame preemption according to IEEE Std 802.1Q-2022, 5.26,
1805 for data rates of 100 Mb/s, and 1 Gb/s, if the IA time-aware stream traffic or the IA stream
1806 traffic type categories are supported, including:
- 1807 1) support of Interspersing Express Traffic according to IEEE Std 802.3-2022, Clause 99,
1808 including support of the Additional Ethernet Capabilities TLV in an LLDPDU to indicate
1809 supported functions of frame preemption according to IEEE Std 802.3-2022, 79.3.7 and
1810 Table 79-8, and
- 1811 2) support of the YANG features and nodes of the ieee802-dot1q-preemption module
1812 according to 6.4.9.3.4.

1813 **5.9.3 ccB end station component requirements**

1814 An end station component implementation for which a claim of conformance to ccB of this
1815 document is made shall support the following list of requirements:

e) Common end station component requirements according to 5.9.1.

f) At least 1 PTP Instance according to 5.5.3.

5.10 End station component options

5.10.1 Common end station component options

An end station component implementation of any conformance class for which a claim of conformance to this document is made may support the following list of requirements.

a) The operation of the credit-based shaper algorithm according to IEEE Std 802.1Q-2022, 8.6.8.2 including support of the YANG features and nodes of the `ieee802-dot1q-cbs` module according to 6.4.9.3.5.

b) Talker end system behaviors according to IEEE Std 802.1CB-2017, as amended by IEEE Std 802.1CBdb-2021 and IEEE Std 802.1CBcv™-2021, 5.6, and 5.7 c) on one or more ports and for 1 or more Compound Streams, including support of the `ieee802-dot1cb-stream-identification` and `ieee802-dot1cb-frer` YANG modules according to 6.4.9.3.6.

c) Listener end system behaviors according to IEEE Std 802.1CB-2017, as amended by IEEE Std 802.1CBdb-2021 and IEEE Std 802.1CBcv-2021, 5.9 on one or more ports and for 1 or more Compound Streams, including support of the `ieee802-dot1cb-stream-identification` and `ieee802-dot1cb-frer` YANG modules according to 6.4.9.3.6.

5.10.2 ccA end station component options

An end station component implementation for which a claim of conformance to ccA of this document is made may support the following list of requirements.

a) Common end station options according to 5.10.1

b) More than 2 PTP Instances according to 5.5.3.

c) End station requirements for enhancements for scheduled traffic according to IEEE Std 802.1Q-2022, 5.25, for data rates of 10 Mb/s, 2,5 Gb/s, 5 Gb/s, and 10 Gb/s including:

1) TickGranularity of less than or equal to 10 ns according to IEEE Std 802.1Q-2022, 8.6.9.4.16 and Table 12-32,

2) the allowable delay between the transmission selection timing point and the on-the-wire timing point, less any delay for the PHY (IEEE Std 802.1Q-2022, Figure 12-6), of less than or equal to 10 ns,

3) support of the YANG features and nodes of the `ieee802-dot1q-sched` module according to 6.4.9.3.2, and

4) support of the YANG features and nodes of the `ieciieee60802-sched-bridge` module according to 6.4.9.3.3.

d) End station requirements for frame preemption according to IEEE Std 802.1Q-2022, 5.26, for data rates of 10 Mb/s, 2,5 Gb/s, 5 Gb/s, and 10 Gb/s including:

NOTE IEEE Std 802.3de-2022, 99.1, comprises 10 Mb/s.

1) support of Interspersing Express Traffic according to IEEE Std 802.3-2022, Clause 99, and IEEE Std 802.3de, 99.1, including support of the Additional Ethernet Capabilities TLV in an LLDPDU to indicate supported functions of frame preemption according to IEEE Std 802.3-2022, 79.3.7 and Table 79-8, and

2) support of the YANG features and nodes of the `ieee802-dot1q-preemption` module according to 6.4.9.3.4.

5.10.3 ccB end station component options

An end station component implementation for which a claim of conformance to ccB of this document is made may support the following list of requirements.

a) Common end station component options according to 5.10.1.

b) One or more PTP Instances according to 5.5.3.

- c) End station requirements for enhancements for scheduled traffic according to IEEE Std 802.1Q-2022, 5.25, including:
 - 1) TickGranularity of less than or equal to 10 ns according to IEEE Std 802.1Q-2022, 8.6.9.4.16 and Table 12-32,
 - 2) the allowable delay between the transmission selection timing point and the on-the-wire timing point, less any delay for the PHY (IEEE Std 802.1Q-2022, Figure 12-6), of less than or equal to 10 ns,
 - 3) support of the YANG features and nodes of the `ieee802-dot1q-sched` module according to 6.4.9.3.2, and
 - 4) support of the YANG features and nodes of the `ieciieee60802-sched-bridge` module according to 6.4.9.3.3.
- d) End station requirements for frame preemption according to IEEE Std 802.1Q-2022, 5.26, including:
 - 1) support of Interspersing Express Traffic according to IEEE Std 802.3-2022, Clause 99, and IEEE Std 802.3de, 99.1, including support of the Additional Ethernet Capabilities TLV in an LLDPDU to indicate supported functions of frame preemption according to IEEE Std 802.3-2022, 79.3.7 and Table 79-8, and
 - 2) support of the YANG features and nodes of the `ieee802-dot1q-preemption` module according to 6.4.9.3.4.

6 Required functions for an industrial network

6.1 General

Clause 6 provides requirements specific to this document and the industrial use case.

6.2 Synchronization

6.2.1 General

An IA-station can contain more than one Grandmaster PTP Instance and PTP End Instance to support:

- a) hot-standby use cases, or
- b) Working Clock or Global Time.

For further explanation of the requirements for time synchronization, refer to Annex D.

6.2.2 PTP Instance requirements

A Grandmaster PTP Instance, a PTP Relay Instance and a PTP End Instance, and the Working Clock or Global Time clocks connected to them, shall meet the following requirements under their allowed working conditions and for their lifetime.

- a) The fractional frequency offset of the LocalClock relative to the nominal frequency shall be according to Table 9.
- b) The range of the rate of change of fractional frequency offset of the LocalClock shall be according to Table 9.
- c) During operation, the Working Clock and Global Time at Grandmaster PTP Instances and PTP End Instances shall increase monotonically, where monotonic means that for a time y that occurs after time x , the ClockTarget's timestamp of y is greater than or equal to the ClockTarget's timestamp of x .
- d) The Working Clock and Global Time at a PTP End Instance can be controlled by applying a frequency change over a period of time. The frequency applied can have a fine resolution to speed up or slow down the clock smoothly, and it has a total range of frequency adjustment.

e) For the Global Time at a PTP End Instance, the maximum value of frequency adjustment shall be according to Table 9.

f) For the Working Clock at a PTP End Instance, the maximum value of frequency adjustment shall be according to Table 9.

For Working Clock or Global Time, decoupled from a ClockTarget, a higher maximum value of frequency adjustment and maximum rate of change of fractional frequency offset are allowed. As soon as it is coupled (or coupled again), a) to f) apply.

Table 9 – Required values

Topic	Value
Local Clock at non-Grandmaster PTP Instance, range of fractional frequency offset relative to the nominal frequency	± 50 ppm
Local Clock at non-Grandmaster PTP Instance, range of rate of change of fractional frequency offset with respect to the nominal frequency	± 1 ppm/s
Working Clock and Global Time (acting as ClockSource) and Local Clock at Grandmaster PTP Instance, range of fractional frequency offset with respect to the nominal frequency	± 25 ppm
Working Clock and Global Time (acting as ClockSource) and Local Clock at Grandmaster PTP Instance, range of rate of change of fractional frequency offset with respect to the nominal frequency (steady state, see Annex D.2.3)	± 1 ppm/s
Working Clock and Global Time (acting as ClockSource) at Grandmaster PTP Instance, range of rate of change of fractional frequency offset (transient, see Annex D.2.3)	± 3 ppm/s
Working Clock and Global Time at PTP End Instance, maximum value of frequency adjustment	± 250 ppm over any observation interval of 1 ms
NOTE 1 If the Grandmaster PTP Instance implementation is such that its Working Clock and Local Clock are the same or otherwise locked to the same frequency, the normative requirements on the Working Clock take priority over those on the Local Clock. NOTE 2 The Maximum value of frequency adjustment represents an upper bound that limits how much a PTP End Instance can change the frequency of its Working Clock or Global Time during a given period. However, the adjustment is expected to be gradual over the defined interval rather than instantaneous. NOTE 3 The example algorithms that track clock drift use up to 4 seconds of historical data and can take that length of time to respond to changes in clock drift. The example algorithm has been used for simulating cases where no fast changes in the observed frequency drift rate were observed. In most of the real-life situations, this condition can be satisfied.	

6.2.3 PTP protocol requirements

Table 10 shows the required protocol times.

Table 10 – Protocol settings

Topic	Value
Nominal time between successive Announce messages (announce interval)	1 s
Nominal time between successive Pdelay_Req messages (Pdelay_Req message transmission interval)	125 ms
Range of allowed time between successive Pdelay_Req messages	119 ms to 131 ms
Nominal time between successive Sync messages at the Grandmaster (Sync message transmission interval)	125 ms
Range of allowed time between successive Sync messages at the Grandmaster	119 ms to 131 ms
Time between reception of a Sync message and transmission of the subsequent Sync message (i.e. residence time) at a PTP Relay instance	Maximum: 15 ms Measured Mean: ≤ 5 ms
Maximum time between transmission of a Sync message and transmission of the related Follow_Up message	2,5 ms
Time between reception of a Pdelay_Req message and transmission of the subsequent Pdelay_Resp message (i.e. Pdelay turnaround time).	Maximum: 15 ms

Topic	Value
NOTE 1 A consequence of having a single allowed value of mean sync interval is that syncLocked mode is achieved. If the TimeTransmitter port sync interval is the same as that of the timeReceiver port, syncLocked mode is achieved.	
NOTE 2 The values contained in this table apply to both the Working Clock and Global Time.	

1920

1921 6.2.4 Clock control system requirements for PTP End Instances

1922 Table 11 shows the required clock control system characteristics at a PTP End Instance.

1923 **Table 11 – Clock control system requirements**

Topic	Value
Maximum Bandwidth	1,0 Hz
Minimum Bandwidth	0,9 Hz
Maximum Gain Peaking	2,2 dB
Minimum absolute value of Roll-off	20 dB/decade
NOTE 1 For more information regarding the clock control system, see Annex C.	
NOTE 2 The values contained in this table apply to both the Working Clock and Global Time.	

1924

1925 6.2.5 Error generation limits

1926 Table 12 shows the required limits on error generation at a Grandmaster PTP Instance. A limit
 1927 on error generation for a Grandmaster PTP Instance is a limit on the amount of error it generates
 1928 in the output Sync message compared to its Working Clock (acting as ClockSource) and Local
 1929 Clock. See D.3.4.

1930 **Table 12 – Error generation limits for Grandmaster PTP Instance**

Topic	Value
Accuracy of forwarded time information: (preciseOriginTimestamp + correctionField) in PTP timing message minus Working Clock at Grandmaster when Sync message is transmitted	Allowable range of the measured mean: - 10 ns to + 10 ns Range around the measured mean within which 90% of measurements fall: ± 7 ns Range around the measured mean within which 100% of measurements fall: ± 10 ns
Accuracy of forwarded Rate Ratio information: rateRatio field in Follow_Up information TLV minus true Rate Ratio between Working Clock at Grandmaster and Local Clock when Sync message is transmitted	Mean 0 ppm ± 0,1 ppm Standard deviation ≤ 0,1 ppm

Topic	Value
Accuracy of syncEgressTimestamp:	
syncEgressTimestamp in Drift_Tracking TLV	Range around the measured mean within which 90% of measurements fall: ± 7 ns
minus	Range around the measured mean within which 100% of measurements fall: ± 10 ns
Local Clock when Sync message is transmitted	
NOTE 1 “Allowable range of the measured mean” specifies limits on constant error. “Range around the measure mean” and “Standard deviation” specify limits on dynamic error. A limit on the constant error of syncEgressTimestamp is not specified because constant error in this characteristic is not a source of time synchronization error. NOTE 2 This document specifies statistical normative requirements without a test duration. It is the responsibility of a test plan to determine the test duration and pass/fail criteria, taking sampling error into consideration, and striking an acceptable balance between test duration and accuracy.	

1931

1932 Table 13 shows the required limits on error generation at a PTP Relay Instance. A limit on error
 1933 generation for a PTP Relay Instance is a limit on the amount of error it adds to the output Sync
 1934 message compared to the input Sync message. These requirements are written for the case
 1935 when errors due to change of fractional frequency offset of its Local Clock with respect to the
 1936 nominal frequency and errors in the input Sync message are negligible with respect to the
 1937 specified error generation limits. See D.3.5.

1938

Table 13 – Error generation limits for PTP Relay Instance

Topic	Value
Accuracy of residenceTime measurement (no clock drift):	
(preciseOriginTimestamp + correctionField) in the PTP timing message transmitted by PTP Relay Instance	Allowable range of the measured mean: - 2 ns to + 2 ns
minus	Range around the measured mean within which 90% of measurements fall: ± 10 ns
Working Clock at Grandmaster when the Sync message is transmitted	Range around the measured mean within which 100% of measurements fall: ± 20 ns
while...	
• Working Clock (acting as ClockSource) at Grandmaster is stable.	
• Local Clock at upstream PTP Instance is stable	
• meanLinkDelay between upstream PTP Instance and PTP Relay Instance is negligible	

Topic	Value
Accuracy of Neighbor Rate Ratio measurement and correct modification of Rate Ratio (no clock drift): rateRatio field in the Follow_Up information TLV transmitted by PTP Relay Instance minus Rate Ratio from the PTP Relay Instance's Local Clock to the Working Clock at the Grandmaster while... • Working Clock (acting as ClockSource) at Grandmaster is stable. • Local Clock at upstream PTP Instance is stable	Allowable range of the measured mean: - 0,1 ppm to + 0,1 ppm Allowable measured standard deviation around the measured mean: 0,02 ppm
Accuracy of Neighbor Rate Ratio measurement and correct modification of Rate Ratio (Rate Ratio drift): rateRatio field in the Follow_Up information TLV transmitted by PTP Relay Instance minus Rate Ratio from the PTP Relay Instance's Local Clock to the Working Clock at the Grandmaster while... • WorkingClock (acting as ClockSource) at Grandmaster PTP Instance, fractional frequency offset with respect to the nominal frequency is increasing at 1 ppm/s • Local Clock at upstream PTP Instance is stable.	Allowable range of the measured mean: - 0,1 ppm to + 0,1 ppm Allowable measured standard deviation around the measured mean: 0,08 ppm
Accuracy of Neighbor Rate Ratio measurement and correct modification of Rate Ratio (Rate Ratio drift and Neighbor Rate Ratio drift): rateRatio field in the Follow_Up information TLV transmitted by PTP Relay Instance minus Rate Ratio from the PTP Relay Instance's Local Clock to the Working Clock at the Grandmaster while... • WorkingClock (acting as ClockSource) at Grandmaster PTP Instance, fractional frequency offset with respect to the nominal frequency is increasing at 1 ppm/s • Local Clock at upstream PTP Instance, fractional frequency offset with respect to the nominal frequency is increasing at 1 ppm/s	Allowable range of the measured mean: - 0,1 ppm to + 0,1 ppm Allowable measured standard deviation around the measured mean: 0,08 ppm

Topic	Value
Accuracy of Neighbor Rate Ratio Drift measurement and correct modification of Rate Ratio Drift (no clock drift) rateRatioDrift field in the Drift_Tracking TLV transmitted by PTP Relay Instance minus Rate Ratio Drift from the PTP Relay Instance's Local Clock to the Working Clock at the Grandmaster while... • WorkingClock (acting as ClockSource) atGrandmaster is stable. • Local Clock at upstream PTP Instance is stable.	Allowable range of the measured mean: - 0,1 ppm to + 0,1 ppm Allowable measured standard deviation around the measured mean: 0,02 ppm
Accuracy of Neighbor Rate Ratio Drift measurement and correct modification of Rate Ratio Drift (Rate Ratio drift) rateRatioDrift field in the Drift_Tracking TLV transmitted by PTP Relay Instance minus Rate Ratio Drift from the PTP Relay Instance's Local Clock to the Working Clock at the Grandmaster while... • WorkingClock (acting as ClockSource) at Grandmaster PTP Instance, fractional frequency offset with respect to the nominal frequency is increasing at 1 ppm/s • Local Clock at upstream PTP Instance is stable.	Allowable range of the measured mean: - 0,1 ppm to + 0,1 ppm Allowable measured standard deviation around the measured mean: 0,08 ppm
Accuracy of Neighbor Rate Ratio Drift measurement and correct modification of Rate Ratio Drift (Rate Ratio drift and Neighbor Rate Ratio drift) rateRatioDrift field in the Drift_Tracking TLV transmitted by PTP Relay Instance minus Rate Ratio Drift from the PTP Relay Instance's Local Clock to the Working Clock at the Grandmaster while... • WorkingClock (acting as ClockSource) at Grandmaster PTP Instance, fractional frequency offset with respect to the nominal frequency is increasing at 1 ppm/s • Local Clock at upstream PTP Instance, fractional frequency offset with respect to the nominal frequency is increasing at 1 ppm/s	Allowable range of the measured mean: - 0,1 ppm to + 0,1 ppm Allowable measured standard deviation around the measured mean: 0,08 ppm

Topic	Value
Accuracy of syncEgressTimestamp: syncEgressTimestamp in Drift_Tracking TLV minus Local Clock when Sync message is transmitted	Range around the measured mean within which 90% of measurements fall: ± 7 ns Range around the measured mean within which 100% of measurements fall: ± 10 ns
Accuracy of meanLinkDelay: meanLinkDelay measured by the PTP Relay Instance minus actual path delay	± 3 ns
Accuracy of Pdelay turnaround time measurement: (responseOriginTimestamp – requestReceiptTimestamp) for transmitted Pdelay_Resp and associated Pdelay_Resp_Follow_Up messages minus the actual interval between reception of the associated Pdelay_Req and transmission of the Pdelay_Resp as measured with respect to the Local Clock	Allowable range of the measured mean: - 2 ns to + 2 ns Range around the measured mean within which 90% of measurements fall: ± 10 ns Range around the measured mean within which 100% of measurements fall: ± 20 ns
NOTE 1 “Allowable range of the measured mean” specifies limits on constant error. “Range around the measured mean” and “Standard deviation” specify limits on dynamic error. A limit on the constant error of syncEgressTimestamp is not specified because constant error in this characteristic is not a source of time synchronization error. NOTE 2 This document specifies statistical normative requirements without a test duration. It is the responsibility of a test plan to determine the test duration and pass/fail criteria, taking sampling error into consideration, and striking an acceptable balance between test duration and accuracy. NOTE 3 These requirements are for errors generated by the PTP Relay Instance. They assume input Sync messages with negligible errors. However, timestamps of Pdelay_Req reception and Pdelay_Resp transmission by the upstream PTP Instance should include expected timestamp errors when meanLinkDelay accuracy is assessed.	

1939

1940 Table 14 shows the required limits on error generation at a PTP End Instance. A limit on error
 1941 generation for a PTP End Instance is a limit on the amount of error it adds to its Working Clock
 1942 (acting as ClockTarget) compared to the input Sync message. These requirements are written
 1943 for the case when errors due to change of fractional frequency offset of its Local Clock with
 1944 respect to the nominal frequency and errors in the input Sync message are negligible with
 1945 respect to the specified error generation limits. See D.3.6.

1946

Table 14 – Error generation limits for PTP End Instance

Topic	Value
Accuracy of recovered clock (no clock drift): Working Clock (acting as ClockTarget) at PTP End Instance minus Working Clock (acting as Clock Source) at Grandmaster while... • WorkingClock (acting as ClockSource) at Grandmaster is stable. • Local Clock at upstream PTP Instance is stable. • meanLinkDelay between upstream PTP Instance and PTP End Instance is negligible	Allowable range of cTE: - 10 ns to + 10 ns Allowable range of dTE: - 15 ns to + 15 ns
Accuracy of recovered clock (Rate Ratio drift): Working Clock (acting as ClockTarget) at PTP End Instance minus Working Clock (acting as Clock Source) at Grandmaster while... • WorkingClock (acting as ClockSource) at Grandmaster PTP Instance, fractional frequency offset with respect to the nominal frequency is increasing at 1 ppm/s • Local Clock at upstream PTP Instance is stable. • meanLinkDelay between upstream PTP Instance and PTP Relay Instance is negligible	Allowable range of cTE: - 10 ns to + 10 ns Allowable range of dTE: - 15 ns to + 15 ns
Accuracy of recovered clock (Rate Ratio drift and Neighbor Rate Ratio drift): Working Clock (acting as ClockTarget) at PTP End Instance minus Working Clock (acting as Clock Source) at Grandmaster while... • WorkingClock (acting as ClockSource) at Grandmaster PTP Instance, fractional frequency offset with respect to the nominal frequency is increasing at 1 ppm/s • Local Clock at upstream PTP Instance, fractional frequency offset with respect to the nominal frequency is increasing at 1 ppm/s • meanLinkDelay between upstream PTP Instance and PTP Relay Instance is negligible	Allowable range of cTE: - 10 ns to + 10 ns Allowable range of dTE: - 15 ns to + 15 ns

1964 In the case of Working Clock, a time change that exceeds the user-defined threshold (for
1965 example 1 μ s) is avoided to protect assets and prevent damage. Thus, the ClockSource or
1966 ClockTarget can be decoupled (see Figure 13 and Figure 14) from the PTP-maintained clock
1967 when such a time change occurs.

1968 In Figure 13, two ClockTargets are traceable to a reliable source of time, which should be
1969 synchronized to Global Time or Working Clock.

1970 The status of a ClockSource, ClockTarget, ClockTimeTransmitter or ClockTimeReceiver is
1971 given by the status of the clock (see 6.2.6) as shown in Figure 11. When timestamps are
1972 provided to the application, the current ClockSource or ClockTarget state can also be provided
1973 to the application.

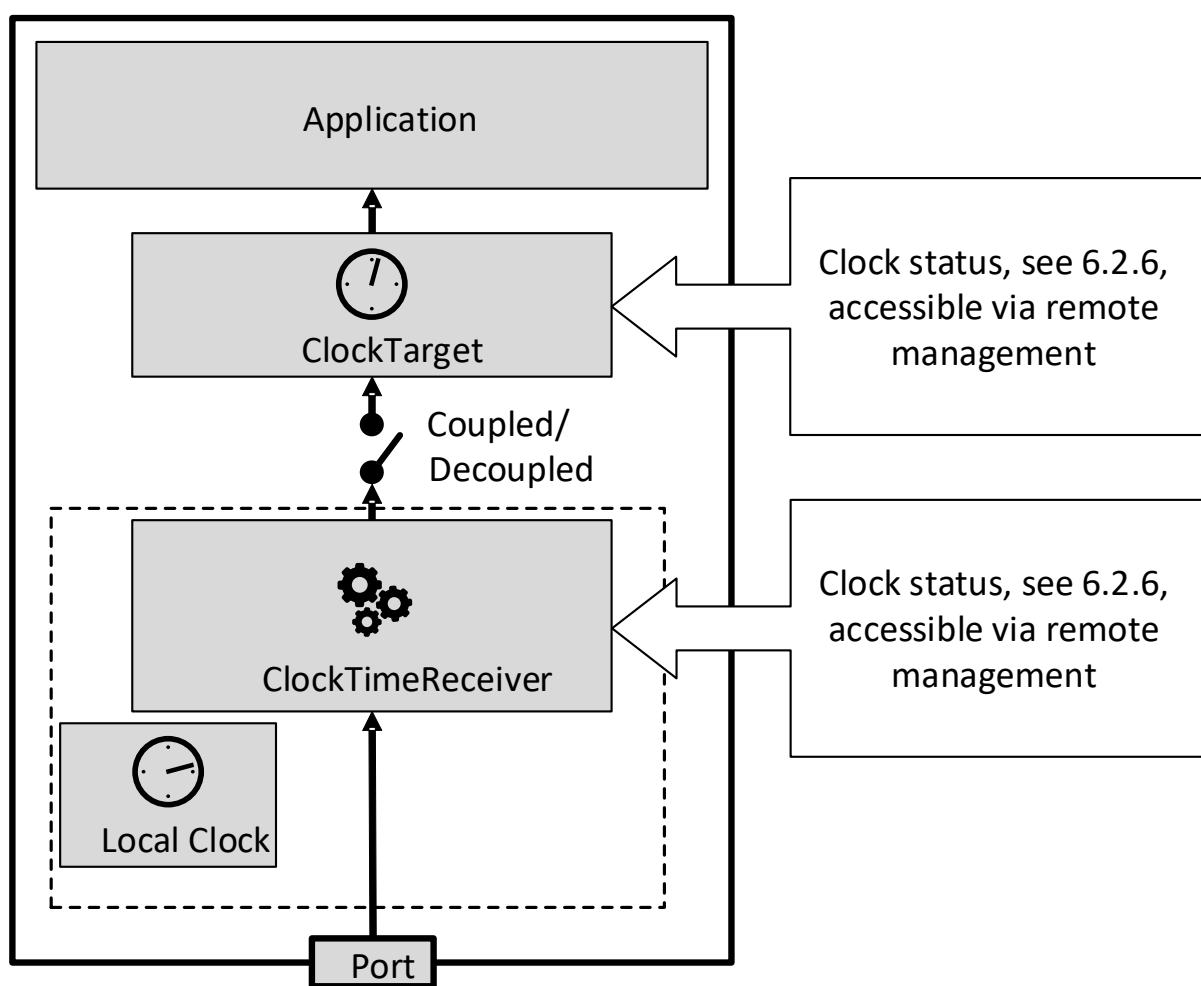


Figure 11 – Clock model

6.2.8 Working Clock domain framework

The gPTP domainNumber of a Working Clock domain is assigned by the CNC. In industrial applications, when the number of PTP Relay Instances between the Grandmaster PTP Instance and any PTP End Instance is less than or equal to 99, max|TER| of the synchronized time of any ClockTarget, relative to the Grandmaster ClockSource, is less than or equal to 1 μ s (see error budget A in Figure 14). Thus it is incumbent upon any PTP Instance to ensure that the requirements specified in 5.5.3, 6.2.2, 6.2.3, 6.2.4, and 6.2.5 are met.

6.2.9 Global Time domain framework

The gPTP domainNumber of a Global Time domain is assigned by the CNC. In industrial applications, when the number of PTP Relay Instances between the Grandmaster PTP Instance and any PTP End Instance is less than or equal to 99, $\max|TER|$ of the synchronized time of any ClockTarget, relative to the Grandmaster ClockSource, is less than or equal to 100 μ s (see error budget A in Figure 14). Thus it is incumbent upon any PTP Instance to ensure that the requirements specified in 5.5.3, 6.2.2, 6.2.3, 6.2.4, and 6.2.5 are met.

6.2.10 IA-station model for clocks

Industrial automation applications, as described in 4.1, require synchronized time that is traceable to a known source (i.e. Global Time) and a source of time synchronized to the Working Clock. Figure 12 and Figure 13 show examples of the IA-station internal model for clocks with the two PTP Instances. It is possible for the ClockSource or ClockTarget to start decoupled or become decoupled from the ClockTimeTransmitter or ClockTimeReceiver, respectively, of a PTP Instance; the ClockSource or ClockTarget runs independently of the availability of the network or a Grandmaster. For example, if the PTP Instance enters a state where isSynced is FALSE, the application might choose to decouple its clock from the PTP Instance and continue to run on its internal clock. If isSynced for the PTP Instance changes to TRUE, the application can choose to again synchronize to the PTP Instance.

Figure 13 shows the IA-station internal model for clocks, with the two PTP Instances used as ClockTimeReceiver/ClockTarget.

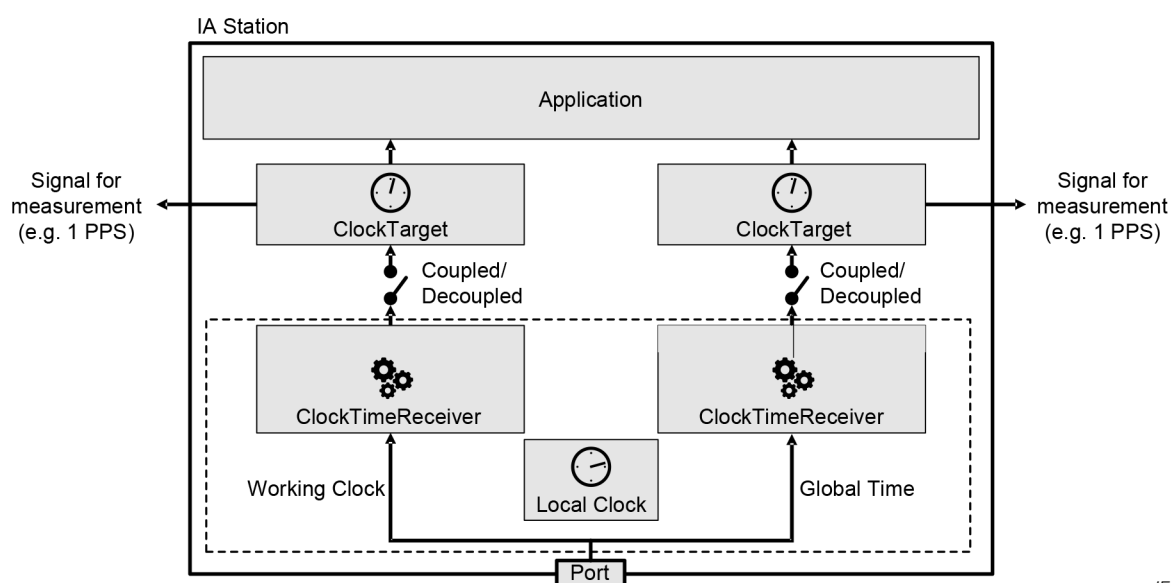


Figure 12 – Example clock usage principles for PTP End Instances

Figure 14 shows the IA-station internal model for clocks, with the two PTP Instances used as Grandmaster.

c) the providing of time to a ClockTarget entity (end application) via the ClockTimeReceiver entity.

NOTE Item a) includes time error introduced in a PTP End Instance between the timeReceiver port and the ClockTimeReceiver entity, and between the ClockTimeTransmitter entity and a TimeTransmitter port.

An output synchronization signal (for example, 1 pulse per second (PPS)) synchronized to the Working Clock as shown in Figure 13 and Figure 14, at any PTP Instance, is used to measure the time error between the ClockSource of the Grandmaster and the ClockTarget of a PTP Instance that is not the Grandmaster. The additional error introduced by implementation of the output synchronization signal is in the range of -10 ns to +10 ns. Figure 14 shows the error budget principle used. These budgets do not include any deviation from the PTP timescale. Representative budgets are provided in Annex D.

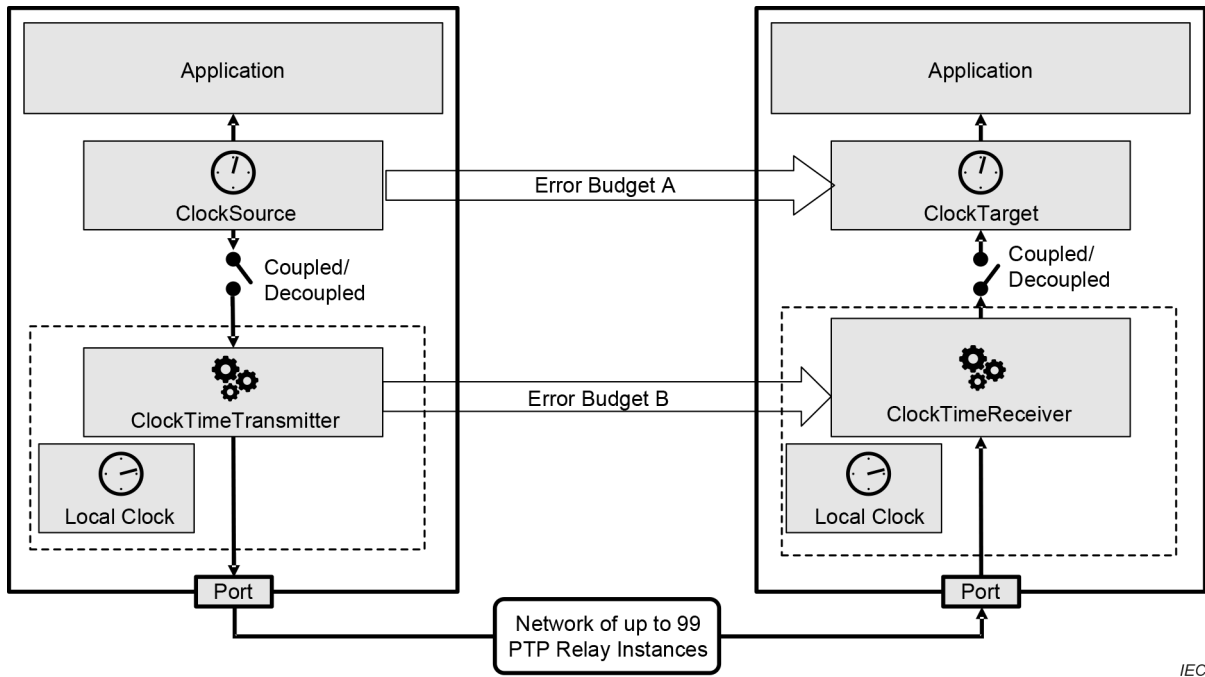


Figure 14 – Error budget scheme

Table 15 shows example values for the splitting of the available error budgets (see Figure 14).

Table 15 – Error budget

Domain	Error budget A	Error budget B
Working Clock	1 μ s	900 ns
Global Time	100 μ s	99,9 μ s

Global Time is often used for tracking events in industrial applications (i.e. sequence of events). Any usage of Global Time for time stamping of application events is allowed an error budget of 1 μ ms.

6.2.13 gPTP domains and PTP Instances

Any valid gPTP domain number as specified in IEEE Std 802.1AS-2020 can be used. The IEEE Std 1588-2019 attribute descriptionDS.userDescription shall be used according to Table 16 to support the translation of PTP Instances and middleware as described in 4.6.2. One gPTP domain can be used for both Working Clock and Global Time. If only one gPTP domain is used, then the requirements for the Working Clock apply (see 6.2.8).

2058

Table 16 – descriptionDS.userDescription of gPTP Domains

gPTP Domain	descriptionDS.userDescription
Working Clock (no hot standby configured)	"60802-WorkingClock"
Primary Working Clock (with configured hot standby)	"60802-Primary-WorkingClock"
Secondary Working Clock (with configured hot standby)	"60802-Secondary-WorkingClock"
Global Time (no hot standby configured)	"60802-GlobalTime"
Primary Global Time (with configured hot standby)	"60802-Primary-GlobalTime"
Secondary Global Time (with configured hot standby)	"60802-Secondary-GlobalTime"
GlobalTime and WorkingClock (no hot standby configured)	"60802-GlobalTime-WorkingClock"
Primary GlobalTime and WorkingClock (with configured hot standby)	"60802-Primary-GlobalTime-WorkingClock"
Secondary GlobalTime and WorkingClock (with hot standby configured)	"60802-Secondary-GlobalTime-WorkingClock"

2059

2060 The descriptionDS.userDescription attribute is represented in the ieee1588-ptp-tt YANG module
2061 by the user-description leaf in the description-ds container of a PTP Instance.

2062 The linking between a gPTP domain and the IETF interfaces is provided by the underlying-
2063 interface.

2064 **6.3 Security model**

2065 **6.3.1 General**

2066 Subclause 6.3 specifies the security model starting with NETCONF/YANG. It describes the
2067 security functionality, the security objects in factory default state, the imprinting of Configuration
2068 Domain-specific security objects and the secure configuration based on Configuration Domain-
2069 specific security objects.

2070 **6.3.2 Security functionality**

2071 **6.3.2.1 Message exchange protection**

2072 **6.3.2.1.1 NETCONF-over-TLSv1.2 profile**

2073 Network configuration with NETCONF/YANG is protected by NETCONF-over-TLS according to
2074 IETF RFC 7589, where the to-be-configured IA-stations act in the NETCONF server role. TLS
2075 protocol version 1.2 according to IETF RFC 5246, 6.2.3.3, 7.4.7.2 and 8.1.2 shall be supported
2076 according to the following list of requirements.

2077 a) Mutual authentication in conjunction with the IDevID and LDevID credentials according to
2078 6.3.4 and 6.3.5.

2079 b) The cipher suite TLS_ECDHE_ECDSA_WITH_AES_128_GCM_SHA256 according to IETF
2080 RFC 5289, 3.2 and Clause 5.

2081 NOTE IETF RFC 7589 implicitly mandates the cipher suite TLS_RSA_WITH_AES_128_CBC_SHA by referring to
2082 IETF RFC 5246. This cipher suite is not used in this document because it requires excessive asymmetric key lengths,
2083 it is not an Authenticated Encryption with Associated Data (AEAD) scheme, and it does not provide perfect forward
2084 secrecy.

2085 c) Signature algorithm ECDSA with SHA-256 and Curve P-256 according to NIST FIPS 186-5
2086 Digital Signature Standard (DSS).

2087 The TLS Certificate message from the TLS client and server contains the self-signed root
2088 certificate. This approach allows simplifying and flattening the PKI hierarchy on base of the

current TLS client certificate to NETCONF username mapping algorithm in IETF RFC 7589. Implementations shall support TLS Certificate message with at least 2 certificate objects.

NETCONF-over-SSH according to IETF RFC 6242 is not used in this document.

NOTE This document selects TLS as a secure transport for NETCONF since TLS is the better match for the case of configuration clients that rely upon unattended or automated operation. This case is dominant in industrial automation.

6.3.2.1.2 NETCONF-over-TLSv1.2 options

The following list of TLS protocol version 1.2 requirements may be supported.

a) The cipher suites TLS_ECDHE_ECDSA_WITH_AES_256_GCM_SHA384 according to IETF RFC 5289, 3.2 and Clause 5, and TLS_ECDHE_ECDSA_WITH_CHACHA20_POLY1305_SHA256 according to IETF RFC 7905, Clause 2.

b) Signature algorithms ECDSA with SHA-512 and Curve P-521 according to NIST FIPS 186-5, Ed25519 according to IETF RFC 8032, 5.1, and Ed448 according to IETF RFC 8032, 5.2.

6.3.2.1.3 NETCONF-over-TLSv1.3 profile

Network configuration with NETCONF/YANG is protected by NETCONF-over-TLS according to IETF RFC 7589 and IETF draft-ietf-netconf-over-tls13. TLS protocol version 1.3 according to IETF RFC 8446 may be used with mutual authentication for NETCONF/YANG as follows.

a) The cipher suites TLS_AES_128_GCM_SHA256, TLS_AES_256_GCM_SHA384 and TLS_CHACHA20_POLY1305_SHA256 according to IETF RFC 8446, 9.1.

b) One or more of the following signature schemes:

- ecdsa_secp256r1_sha256 according to NIST FIPS 186-5 and NIST SP 800-186, 3.2.1.3, ecdsa_secp521r1_sha512 according to NIST FIPS 186-5 and NIST SP 800-186, 3.2.1.5,
- ed25519 according to IETF RFC 8032, 5.1, and
- ed448 according to IETF RFC 8032, 5.2.

The TLS Certificate message from the TLS client and server contains the self-signed root certificate.

6.3.2.1.4 Certificate-to-name mapping

The IETF RFC 7589 based certificate-to-name mapping procedure is as follows.

NOTE IETF RFC 7589, Clause 7, specifies that NETCONF servers map client certificates to "NETCONF usernames" and specifies a concrete mapping procedure for this purpose. This mapping is represented by the YANG module ietf-x509-cert-to-name.

The list of mapping entries has a single element containing:

- fingerprint: the fingerprint of the trust anchor for the Configuration Domain, and
- map_type: ext-60802-roles.

The map-type ext-60802-roles maps the roles provided in the id-60802-pe-roles extension (defined in 6.3.2.1.4) of the end entity certificate presented by the NETCONF client to a NETCONF username. The UTF-8 string representation of each role is added to the NETCONF username in chronological order of the enumeration values, whereas multiple roles are separated by ':' character.

2129 **6.3.2.1.5 Role extension**

2130 The id-60802-pe-roles extension in LDevID-NETCONF end entity certificates shall be
2131 constructed as follows:

2132 a) Extension field extnID

2133 The extnID shall provide the following OBJECT IDENTIFIER to identify the id-60802-pe-roles
2134 extension:

2135 id-60802 OBJECT IDENTIFIER ::= { iso (1) iso-identified-organization (3) ieee (111)
2136 standards-association-numbered-series-standards (2) lan-man-stds (802) ieee802dot1(1)
2137 ieee802dot1stds(2) 1 }

2138
2139 id-60802-pe OBJECT IDENTIFIER ::= { id-60802 1 }

2140
2141 id-60802-pe-roles OBJECT IDENTIFIER ::= { id-60802-pe 1 }

2142

2143 b) Extension field critical

2144 The id-60802-pe-roles extension is marked as non-critical (critical:= FALSE).

2145 c) Extension field extnValue

2146 60802RoleNamesSyntax ::= SEQUENCE OF 60802RoleName

2147

2148 60802RoleName ::= ENUMERATED {
2149 SecurityAdminRole (0),
2150 ConfiguratorRole (1),
2151 StreamConfiguratorRole (2),
2152 SubscriberRole (3)}

2153 **6.3.2.2 Resource access authorization**

2154 Access control to NETCONF/YANG resources shall be protected by NACM according to IETF
2155 RFC 8341.

2156 NACM specifies a YANG data model (ietf-netconf-acm) for expressing rules to control access
2157 to NETCONF/YANG resources. This document profiles NACM to deliver role-based access
2158 control.

2159 NOTE 1 NACM does not natively deliver role-based access control but can be geared by profiling.

2160 This role-based model for security resources should be applied according to the following list
2161 of requirements.

- 2162 • The global switch enable-nacm is set to true.
- 2163 • The set of NETCONF/YANG resources of an IA-station is partitioned according to the YANG
2164 modules specified in 6.4.9 with a permission-to-role assignment as listed below. An access
2165 operation is allowed through the keyword "permitted" and not allowed through the keyword
2166 "denied".

2167 NOTE 2 NACM recognizes following "access-operations": create, read, update, delete, exec and uses the term write
2168 access for the access operations "create", "delete", and "update". This document uses the terms read, write and
2169 exec access.

- 2170 • All authenticated entities (default rules): All YANG modules: read access permitted, write
2171 access denied, exec-access denied.

2172 NOTE 3 The default rules apply for YANG modules that are listed in 6.4.9 but are not listed in the rules of the
2173 individual roles.

- 2174 • Rules for StreamConfiguratorRole: YANG module ieee802-dot1q-cnc-config: write and
2175 execute operations permitted.

- 2176 • Rules for SubscriberRole:
 - 2177 – YANG module ietf-subscribed-notifications: write and execute operations permitted, and
 - 2178 – YANG module ietf-yang-push: write and execute operations permitted.
- 2179 • Rules for ConfiguratorRole: All YANG modules except those listed below, write and execute
 2180 operations permitted:
 - 2181 – YANG modules for security configuration, i.e. ietf-truststore, ietf-keystore, path to cert-
 2182 to-name nodes of ietf-netconf-server, path to tls-server-parameters nodes of ietf-
 2183 netconf-server,
 - 2184 – YANG modules for stream configuration, i.e. ieee802-dot1q-cnc-config, and
 - 2185 – YANG modules for subscription configuration, i.e. ietf-subscribed-notifications, ietf-
 2186 yang-push.
- 2187 • Rules for SecurityAdminRole:
 - 2188 – YANG module ietf-truststore, path to certificate node of IDevID trust anchor: write and
 2189 execute operations denied, and
 - 2190 – YANG module ietf-truststore (besides path to certificate node of IDevID trust anchor):
 2191 write and execute operations permitted.
 - 2192 – YANG module ietf-keystore, path to asymmetric-key node of IDevID credential: write and
 2193 execute operations denied, and
 - 2194 – YANG module ietf-keystore (besides path to asymmetric-key node of IDevID credential):
 2195 write and execute operations permitted.
 - 2196 – YANG module ietf-netconf-server (besides path to cert-to-name nodes and path to tls-
 2197 server-parameters nodes): write and execute operations denied, and
 - 2198 – YANG module ietf-netconf-server, path to cert-to-name nodes: write and execute
 2199 operations permitted.
 - 2200 – YANG module ietf-netconf-server, path to tls-server-parameters nodes: write and
 2201 execute operations permitted.

2202 In addition, the following access control should be applied for NETCONF protocol operations:

- 2203 • <lock>, <unlock>: permitted for any role specified in this document,
- 2204 • <partial-lock>, <partial-unlock>: denied (not used in this document),
- 2205 • <get> and <get-config>: mapped to a "read" access operation to the target datastore,
- 2206 • <edit-config>: permitted for any role specified in this document,
- 2207 • <copy-config>: permitted for ConfiguratorRole,
- 2208 • <delete-config>: denied (not used in this document),
- 2209 • <commit>: permitted for any role specified in this document,
- 2210 • <discard-changes>: permitted for any role specified in this document,
- 2211 • <close-session>: permitted for any role specified in this document, and
- 2212 • <kill-session>: denied (not used in this document).

2213 This document does not specify the assignment of role names to actual system entities. This is
 2214 a duty of system owners or operators.

2215 6.3.3 IDevID Profile

2216 6.3.3.1 General

2217 IA-stations shall possess IDevID credentials according to 6.3.3. CNCs shall contain trust
 2218 anchors for validating IDevID credentials.

2219 6.3.3.2 Object Contents

2220 6.3.3.2.1 General

2221 The IDevID credential contents shall comply to 6.3.3.2.2, 6.3.3.2.3, and IEEE Std 802.1AR-
2222 2018, Clause 6.

2223 6.3.3.2.2 IA-station Identity

2224 Any IDevID EE certificate of an IA-station shall take one of the following forms:

- 2225 • raw form: the IDevID EE certificate complies to IEEE Std 802.1AR-2018, Clause 8, and
- 2226 • extended form: the IDevID EE certificate complies to requirements provided IEEE Std
2227 802.1AR-2018, Clause 8. The extended form of an IDevID EE certificate shall be constructed
2228 as follows:
 - 2229 – the verifiable device identity shall appear as a URN in a GeneralName of type
2230 uniformResourceIdentifier in the subjectAltName extension,
 - 2231 – the URN value shall be constructed according to IETF RFC 8141 and as follows:
 - 2232 • namespace identifier: ieee (see IETF RFC 8069),
 - 2233 • namespace-specific string: iec-ieee-60802#verifiable-device-identity, and
 - 2234 • q-component (see IETF RFC 8141, 2.3.2) to parameterize the named resource: an
2235 ampersand-separated list of keyword=value tuples with following keywords and
2236 values. These tuples can appear in any order inside the q-component:
 - 2237 – the keywords: hardware-rev, serial-num, mfg-name, model-name;
 - 2238 – their corresponding values from the single "chassis" component list entry in the
2239 ietf-hardware YANG module (see 6.4.9.2.5.8) that represents the management
2240 entity of the IA-station respectively from its pre-material form in percent-encoding
2241 according to IETF RFC 3986, 2.1.

2242 NOTE 1 These are the items with the YANG property config=false from the 'component' list entry that represents
2243 the management entity of the IA-station. The config=false items firmware-rev and software-rev are excluded to avoid
2244 IDevID credential updates in case of FW or SW updates.

2245 NOTE 2 An object looks like urn:ieee:iec-ieee-60802#verifiable-device-identity?=mfg-name=<mfg-name>&model-
2246 name=<model-name>&hardware-rev=<hardware-rev>&serial-num=<serial-num>.

2247 NOTE 3 One IDevID EE certificate can have one subjectAltName extension which can have one or more
2248 GeneralName entries. In particular, there can be one or more GeneralName entries of type
2249 uniformResourceIdentifier. This allows other organizations (e.g. middleware and application consortia or individual
2250 manufacturers) to also represent their perception of verifiable device identity in addition to the perception of this
2251 document.

2252 6.3.3.2.3 Signature Suites

2253 An IDevID shall utilize the signature suite: ECDSA P-256/SHA-256 according to NIST FIPS
2254 186-5, NIST FIPS 180-4 and NIST SP 800-186 using the algorithm identifiers according to IETF
2255 RFC 5480.

2256 An IDevID may utilize the following signature suites:

- 2257 • ECDSA P-521/SHA-512 according to NIST FIPS 186-5, NIST FIPS 180-4 and NIST SP 800-
2258 186 using the algorithm identifiers according to IETF RFC 5480,
- 2259 • EdDSA instance Ed25519 according to IETF RFC 8032 using Curve25519 according to IETF
2260 RFC 7748 and using the algorithm identifiers according to IETF RFC 8410, and
- 2261 • EdDSA instance Ed448 according to IETF RFC 8032 using Curve448 according to IETF
2262 RFC 7748 and using the algorithm identifiers according to IETF RFC 8410.

2263 **6.3.3.3 Information Model**

2264 **6.3.3.3.1 General**

2265 The information model for IDevID credentials and trust anchors shall comply to YANG and
2266 NMDA, in particular the YANG modules ietf-keystore and ietf-truststore, as well as subsequent
2267 subclauses of 6.3.3.3.

2268 **6.3.3.3.2 Entries**

2269 IDevID credentials shall be provided in form of built-in keys of an IA-station by its manufacturer.
2270 In YANG, they are modeled as config-false nodes and are represented in the 'keystore'
2271 container that is instantiated by the YANG module ietf-keystore. The private key shall use the
2272 private-key-type choice hidden-private-key, i.e. the IDevID private key is not presented in
2273 NETCONF/YANG. The details of storing and protecting IDevID private keys as well as using
2274 them for signing purposes are implementation specific.

2275 Trust anchors for IDevID credentials are CNC user-configured data objects; these objects shall
2276 be available as applied configuration (IETF RFC 8342) upon CNCs. In YANG, they are modeled
2277 as config-true nodes and are represented in the 'truststore' container that is instantiated by the
2278 YANG module ietf-truststore.

2279 NOTE IA-station built-in trust anchors for use cases such as firmware/software update are not addressed in this
2280 document.

2281 **6.3.3.3.3 Entry Manifolddness**

2282 An IA-station shall possess one IDevID credential with a certification path plus trust anchor
2283 information issued under the required signature suite according to 6.3.3.2.3 as part of its factory
2284 default state.

2285 If an IA-station supports an optional signature suite according to 6.3.3.2.3, it shall possess in
2286 addition one IDevID credential with a certification path plus trust anchor information issued
2287 under the optional signature suite as part of its factory default state.

2288 An IA-station can have additional IDevID credential(s) with a certification path plus trust anchor
2289 information issued under a combination of any required or any supported optional DevID
2290 signature suites.

2291 If an IA-station possesses multiple IDevID credentials, then they shall be issued by the same
2292 organization (the IA-station manufacturer). Their EE certificates shall contain the same device
2293 identity information.

2294 A CNC shall support at least one trust anchor for IDevID credentials per supported IA-station
2295 manufacturer.

2296 **6.3.3.3.4 Entry Naming**

2297 IDevID credentials shall be present in an 'asymmetric-key' entry that is identified as: /ietf-
2298 keystore:keystore/asymmetric-keys/asymmetric-key/name=
2299 IDevID-<SignatureSuiteName>-<CertificateSerialNumberOfEECertificate>.

2300 IDevID trust anchors shall be present in 'certificate' entries that are identified as: /ietf-
2301 truststore:truststore/certificate-bags/certificate-bag/certificate/name=
2302 IDevID-<SignatureSuiteName>-<CertificateSerialNumberOfCACertificate>.

2303 Such entries shall be present underneath a 'certificate-bag' entry that is identified as: /ietf-
2304 truststore:truststore/certificate-bags/certificate-bag/name=IDevID.

2305 **6.3.3.4 Processing model**

2306 **6.3.3.4.1 General**

2307 The processing model for IDevID credentials and trust anchors shall comply to IEEE Std
2308 802.1AR-2018 and 6.3.3.4.

2309 **6.3.3.4.2 Credentials**

2310 **6.3.3.4.2.1 General**

2311 IDevID credentials are used in following use cases:

- 2312 • NETCONF/YANG security setup from factory default; the number of such events scales with
2313 the number of factory resets, i.e. this use case is performed sporadically. It is conducted by
2314 CNCs and encompasses a device identity verification, and
- 2315 • device identity verification happens as a subtask during NETCONF/YANG security setup
2316 from factory default. It can also happen at the discretion of the CNC user. The details of
2317 device identity verification are also subject to given policy.

2318 In these use cases, IA-stations act in claimant role and CNCs act in verifier role:

- 2319 • IA-stations shall present the certification path of and prove private key possession for an
2320 IDevID credential, and
- 2321 • CNCs shall validate the certification path, check the proof-of-possession for the private key,
2322 and verify the obtained device identity information.

2323 **6.3.3.4.2.2 Creation**

2324 IA-station manufacturers select the form factor for representing verifiable device identity in
2325 IDevID credentials: raw or extended form. The details of the IDevID credential issuance process
2326 are manufacturer-specific and not addressed in this document.

2327 IA-station manufacturers are not required to offer an update feature for IDevID credentials.

2328 **6.3.3.4.2.3 Distribution**

2329 IA-stations shall supply IDevID credentials in form of built-in keys, see 6.3.3.3.

2330 **6.3.3.4.2.4 Use**

2331 Verifiers (CNCs) shall perform the following checks when they challenge claimants (IA-stations)
2332 to authenticate themselves by means of an IDevID credential.

- 2333 • IDevID certification path validation according to IETF RFC 5280, Clause 6. Whether this
2334 validation happens with or without revocation checks is at the discretion of the CNC user.
2335 It is the responsibility of the CNC user to supply a trust anchor configuration (set of trusted
2336 certificates or trusted public keys), a revocation check instruction (Boolean) and optionally,
2337 X.509 CRL objects according to IETF RFC 5280, Clause 5, to CNCs. The certification path
2338 validation is passed if and only if the IDevID EE certificate is the leaf of a valid certification
2339 path that ends with a CA certificate which is signed by a configured trust anchor and which
2340 is not revoked (if revocation check is enabled).
- 2341 • Proof-of-possession checking for the private key. The proof-of-possession check is passed
2342 if and only if the IA-station possesses the private key which matches the public key in the
2343 IDevID EE certificate.

- It is the responsibility of the CNC user to establish and supply to CNCs: a device identity verification policy which determines the verifiable device identity subset that shall be checked by the CNC for the IA-stations in a Configuration Domain. This is a subset of {hardware-rev, serial-num, mfg-name, model-name}. The empty subset ("no-identity-check") as well as the whole set are allowed. The device identity verification for an IA-station instance shall behave according to the following list of requirements.

- If this subset is empty, then the device identity check is passed. If the user chooses not to verify identity, information about the devices is considered unreliable. Tracking the unverified status of such devices is the responsibility of the user. It is the responsibility of the user to establish policies for the use of such devices.

- If this subset is non-empty, then the CNC performs the following expected vs. actual check for each verifiable device identity item in this subset:

- The check for any item in this subset is passed if the expected value (from ietf-hardware YANG module) matches the actual value (from the verifiable device identity URN value for this document in the subjectAltName extension of the IDevID EE certificate). This check fails if the IDevID has raw form.

- The device identity check is passed if it is passed for all items in the subset.

IDevIDs in raw form (without verifiable device identity URN) can be used if the device identity verification setting option "no-identity-check" is employed. This allows to perform the NETCONF/YANG security setup from factory default for IA-stations with IDevID credentials in raw form. From CNC perspective these IA-stations remain anonymous.

NOTE This document does not specify a mechanism for device identity verification for IDevIDs in raw form. Whether and how device identity checks for such IA-stations are done in an offline mode is at the discretion of CNC users.

6.3.3.4.2.5 Storage

Credentials shall be stored persistently upon an IA-station. The details for implementing this persistent storage are IA-station manufacturer-specific and not addressed in this document.

IA-stations shall support storage of at least one IDevID credential.

6.3.3.4.2.6 Revocation

It is the responsibility of IA-station manufacturers to report revocation for the IDevID credentials issued by them in form of X.509 CRL objects according to IETF RFC 5280, Clause 5. These objects are made available in a form that allows relying parties, i.e. CNC users to retrieve them at their own discretion.

CNC users decide whether they support IDevID certification path validation with or without revocation:

- if revocation checks are disabled, then certificate path validation shall be performed according to IETF RFC 5280, 6.1, or
- if revocation checks are enabled, then certificate path validation shall be performed according to IETF RFC 5280, 6.1 and 6.3.

NOTE It is the responsibility of CNC users to obtain up-to-date X.509 CRL objects from manufactures and make them locally available for verifiers.

6.3.3.4.3 Trust anchors

6.3.3.4.3.1 General

Trust anchors are input arguments for certification path validation according to IETF RFC 5280, 6.1.1, input argument (d). Relying parties decide about these input arguments in a discretionary fashion, i.e. these objects are not created and distributed as literal trust anchor objects but in a pre-material form of, for example, self-signed certificate objects.

2390 NOTE The digital signature in self-signed certificates do not vouch for authenticity of this object: Actor X can issue
2391 self-signed certificates featuring the name of actor A that cannot be distinguished from self-signed certificates issued
2392 by A. The mechanisms to verify the authenticity of self-signed certificates are not addressed in this document.

2393 The trust anchors for use cases where IA-stations act in claimant role are determined by CNC
2394 users.

2395 **6.3.3.4.3.2 Creation**

2396 The details of the issuance and update processes for trust anchors for validation of IDevID
2397 credentials are not addressed by this document.

2398 **6.3.3.4.3.3 Distribution**

2399 With respect to use cases where IA-stations act in claimant role, e.g. NETCONF/YANG security
2400 setup and device identity verification, the following model applies.

- 2401 • Issuers (IA-station manufacturers) create and distribute trust anchors. Issuers also provide
2402 out-of-band means that allow relying parties to check the authenticity of these objects, and
- 2403 • Relying parties (CNC users) check the authenticity of trust anchors and decide about their
2404 acceptance as trust anchors for certification path validation in a discretionary manner and
2405 configure their verifiers (CNCs) accordingly.

2406 The details of distribution and validation of trust anchors are not addressed by this document.

2407 **6.3.3.4.3.4 Use**

2408 Trust anchors for IDevID credentials are used for certification path validation according to IETF
2409 RFC 5280, 6.1.1 d). This concerns CNCs with respect to the use cases NETCONF/YANG
2410 security setup from factory default, device identity verification.

2411 **6.3.3.4.3.5 Storage**

2412 Trust anchors for IDevID credentials shall be stored persistently upon CNCs. The details for
2413 implementing this persistent storage are not addressed in this document.

2414 **6.3.3.4.3.6 Revocation**

2415 IA-station manufacturers are not required to support an authority revocation feature for IDevID
2416 credential certification authorities.

2417 **6.3.4 Security setup based on IDevID**

2418 **6.3.4.1 General**

2419 IA-stations in factory default state shall conduct a security setup sequence for the Configuration
2420 Domain. This sequence consists of the following steps, each step is described in 6.3.4.

- 2421 • imprintTrustAnchor: imprint of a Configuration Domain specific trust anchor to an IA-station
2422 that allows to validate LDevID-NETCONF certificates presented by communication partners.
- 2423 • imprintCredential: imprint of a Configuration Domain specific credential to an IA-station, i.e.
2424 a private key and the corresponding X.509 v3 end entity certificate according to ISO/IEC
2425 9594-8 as profiled in IETF RFC 5280, Clause 4, (plus intermediate CA certificates, if
2426 applicable) plus self-signed root CA certificate that serves as own LDevID credential.
- 2427 • imprintCertToNameMapping: imprint a Configuration Domain specific certificate-to-name
2428 mapping to an IA-station.

2429 Credentials shall be stored persistently upon an IA-station. The details for implementing this
2430 persistent storage are IA-station manufacturer-specific and not addressed in this document.

2431 IA-stations shall support storage of at least one LDevID-NETCONF credential.

2432 **6.3.4.2 imprintTrustAnchor**

2433 IA-stations in factory default state shall support the imprinting of a single Configuration Domain
2434 specific trust anchor according to IETF RFC 6024 via NETCONF-over-TLS according to a
2435 procedure called "provisional accept of client certificate", which uses an IDevID credential on
2436 NETCONF and TLS server side (IA-station) and a LDevID credential on NETCONF and TLS
2437 client side (for example, a CNC) and operates as follows at the NETCONF and TLS server.

- 2438 a) Challenge the client for TLS client authentication according to IETF RFC 7589 by sending
2439 a CertificateRequest message with an empty certificate_authorities entry.
- 2440 b) Perform certification path validation according to IETF RFC 5280, Clause 6, for the contents
2441 of the client's Certificate message. This certification path validation fails due to a missing
2442 trust anchor for the LDevID credential.
- 2443 c) Provisionally accept the failing certification path validation when the reason is "no matching
2444 trust anchor" (and only this reason) and proceed with the TLS exchange.
- 2445 d) Expect the client to send a trust anchor for LDevID over the provisionally accepted TLS
2446 session (no other object type).
- 2447 e) If the trust anchor in the NETCONF application payload was accepted, then redo the priorly
2448 failing certification path validation using this trust anchor, see step b).
- 2449 f) If this certification path revalidation is successful, then keep the TLS session alive and send
2450 an <rpc-reply> with success. The client then is expected to perform the NETCONF
2451 exchanges for imprintCredential (described in 6.3.4.3) and for imprintCertToNameMapping
2452 (described in 6.3.4.4) via the already established TLS session.
- 2453 g) If this certification path revalidation is not successful, then terminate the TLS session. The
2454 usual NETCONF/YANG hygiene applies. This is expected to remove the entry in the ietf-
2455 truststore that was created in step d).

2456 NOTE This "provisional accept of client certificate" is a mirrored version of the "provisional accept of server cert"
2457 in IETF RFC 8995.

2458 The "provisional accept of client cert" in factory default state shall skip the certificate-to-name
2459 mapping and shall use the NACM recovery session, i.e. skip permission checking. In this model
2460 all authenticated clients are accepted as authorized for doing the first imprinting of the LDevID
2461 credential and the corresponding trust anchor. Only contextual checks such as "once only when
2462 being in factory default state" are feasible. This model is also known as "trust on first use"
2463 (TOFU) and, e.g. also allows to read contents of the ietf-hardware module by the client for an
2464 extended identity check.

2465 The imprinting NETCONF client checks the actual server identity that is stated by the IA-station
2466 on TLS level.

2467 The NETCONF client checks the IDevID end entity certificate presented by the NETCONF
2468 server on TLS level for existence of subjectAltName extension with GeneralName entries of
2469 type uniformResourceIdentifier. If an entry contains the namespace identifier and the
2470 namespace-specific string as defined in 6.3.3.2.2, the presented server certificate is in
2471 extended form, otherwise it is in raw form.

2472 In case that the server certificate is in raw form, the following matching can be done:

- 2473 • Match a list of accepted (or blocked) manufacturers against the issuer or subject field entries
2474 of the certificate.
- 2475 • Match a list of accepted (or blocked) product instances against the product serial number
2476 from the subject field per accepted manufacturer.
- 2477 • Match the end entity certificate object as a whole against a list of pinned certificates.

2478 In case that the server certificate is in extended form, the following additional matching can be
 2479 done: match the q-components included in the verifiable device identity according to
 2480 6.3.3.2.2 against those that can be read out from the corresponding leaves of the YANG module
 2481 ietf-hardware or against reference values obtained by a method not addressed in this document.

2482 Details of how the matching happens depend on the implementation of the client that performs
 2483 this imprinting.

2484 The LDevID-NETCONF trust anchor certificate shall be imprinted using the truststore container
 2485 of the ietf-truststore module with:

- 2486 • /ts:truststore/ts:certificate-bags/ts:certificate-bag/ts:name = IEC60802,
- 2487 • /ts:truststore/ts:certificate-bags/ts:certificate-bag/[ts:name=IEC60802]/,
 - 2488 – ts:certificate/ts:name = IEC60802-LDevID,
 - 2489 – ts:certificate/ts:cert-data containing the IEC60802-LDevID trust anchor certificate data
 - 2490 object of type trust-anchor-cert-cms according to IETF RFC 9640, i.e., enveloped in
 - 2491 Base64-encoded CMS SignedData in degenerated form "certs-only" (no signature
 - 2492 value), and
 - 2493 – The imprintTrustAnchor step shall use the NETCONF operation <edit-config> according
 - 2494 to IETF RFC 6241 for the truststore container. The NETCONF operation <commit> is not
 - 2495 yet applied, but rather after successful completion of all security setup sequence steps.

2496 **6.3.4.3 imprintCredential**

2497 **6.3.4.3.1 General**

2498 The LDevID-NETCONF end entity certificate shall be provided as X.509 v3 public key certificate
 2499 according to ISO/IEC 9594-8 as profiled in IETF RFC 5280, Clause 4, with the following criteria:

- 2500 • contains the FQDN of the NETCONF server in its subjectAltName extension according to
- 2501 IETF RFC 7589, Clause 6, and IETF RFC 6125, 2.2 and B.7;
- 2502 • contains a public key and is signed by a signature suite according to 5.5.4.2 or 5.6.3;
- 2503 • contains a digitalSignature in its keyUsage extension;
- 2504 • has a finite validity period.

2505 NOTE The actual length of the validity period is at the discretion of the user of the Configuration Domain.

2506 Depending on the key generation capabilities, different steps are applied to this keystore
 2507 container.

2508 **6.3.4.3.2 Internal key generation**

2509 For IA-station with internal key generation capabilities, two NETCONF exchanges are
 2510 performed. Processing steps for the first NETCONF exchange shall be applied as follows at the
 2511 NETCONF server.

2512 a) Receive and process the NETCONF request message with action <generate-csr> and input
 2513 values as follows:

- 2514 • /ks:keystore/ks:asymmetric-keys/ks:asymmetric-key/[ks:name=LDevID_NETCONF]/ks:
- 2515 generate-csr/ks:input/ks:csr-format containing derived identity p10-csr-format according
- 2516 to IETF RFC 9640, and
- 2517 • /ks:keystore/ks:asymmetric-keys/ks:asymmetric-key/[ks:name=LDevID_NETCONF]/ks:
- 2518 generate-csr/ks:input/ks:csr-info containing a Base64-encoded PKCS#10
- 2519 CertificationRequestInfo according to IETF RFC 2986, Clause 4.

- 2520 b) Base64-decode the <csr-info> value and parse it as a PKCS#10 CertificationRequestInfo
2521 object.
- 2522 c) Extract the algorithm information from the child element SubjectPublicKeyInfo of
2523 CertificationRequestInfo and randomly generate a key pair for the specified algorithm.
- 2524 d) Internally store the private key together with its metadata for example, algorithm information,
2525 <name> value in a secure manner.
- 2526 e) Put the public key into the (parsed) PKCS#10 CertificationRequestInfo.
- 2527 f) Serialize the PKCS#10 CertificationRequestInfo (including the public key).
- 2528 g) Use the private key to create signature value for the (serialized) PKCS#10
2529 CertificationRequestInfo (including the public key).
- 2530 h) Create a NETCONF reply message with /ks:keystore/ks:asymmetric-keys/ks:asymmetric-
2531 key/[ks:name=LDevID-NETCONF]/ks:generate-csr/ks:output/ks:p10-csr containing the data
2532 object of the previous step.

2533 In the second NETCONF exchange, the LDevID-NETCONF end entity certificate (plus
2534 intermediate CA certificates) shall be imprinted using the keystore container of the ietf-keystore
2535 module with:

- 2536 • /ks:keystore/ks:asymmetric-keys/ks:asymmetric-key/ks:name = LDevID-NETCONF,
2537 • /ks:keystore/ks:asymmetric-keys/ks:asymmetric-key/[ks:name=LDevID-NETCONF]/,
2538 – ks:certificates/ks:certificate/ks:name = LDevID-NETCONF, and
2539 – ks:certificates/ks:certificate/ks:cert-data containing the certificate chain LDevID-
2540 NETCONF end entity certificate (plus intermediate CA certificates, if applicable) plus
2541 self-signed root CA certificate as data object of type end-entity-cert-cms according to
2542 IETF RFC 9640

2543 The imprintCredential step shall use the NETCONF operation <edit-config> according to IETF
2544 RFC 6241 for the keystore container. The NETCONF operation <commit> is not yet applied, but
2545 rather after successful completion of all security setup sequence steps.

2546 6.3.4.3.3 External key generation

2547 External key generation can be used for IA-stations without internal key generation capability.
2548 For external key generation, one NETCONF exchange is performed.

2549 The LDevID-NETCONF private key and end entity certificate (plus intermediate CA certificates)
2550 shall be imprinted using the keystore container of the ietf-keystore module with:

- 2551 • /ks:keystore/ks:asymmetric-keys/ks:asymmetric-key/ks:name = LDevID-NETCONF,
2552 • /ks:keystore/ks:asymmetric-keys/ks:asymmetric-key/[ks:name=LDevID-NETCONF]/,
2553 – ks:certificates/ks:certificate/ks:name = LDevID-NETCONF,
2554 – ks:certificates/ks:certificate/ks:public-key-format describing the encoding of the public
2555 key of the selected cryptographic algorithm according to IETF RFC 9640,
2556 – ks:certificates/ks:certificate/ks:public-key containing the public key value in the selected
2557 public-key-format,
2558 – ks:certificates/ks:certificate/ks:private-key-format describing the encoding of the private
2559 key of the selected cryptographic algorithm according to IETF RFC 9640,
2560 – ks:certificates/ks:certificate/ks:cleartext-private-key containing the private key value in
2561 the selected private-key-format,

2562 NOTE The private key is confidentiality protected by NETCONF-over-TLS even if the option <cleartext-private-key>
2563 is used.

- 2564 – ks:certificates/ks:certificate/ks:name = LDevID-NETCONF, and
2565 – ks:certificates/ks:certificate/ks:cert-data containing the certificate chain LDevID-
2566 NETCONF end entity certificate (plus intermediate CA certificates, if applicable) plus

2567 self-signed root CA certificate as data object of type end-entity-cert-cms according to
2568 IETF RFC 9640.

2569 The imprintCredential step shall use the NETCONF operation <edit-config> according to IETF
2570 RFC 6241 for the keystore container. The NETCONF operation <commit> is not yet applied, but
2571 rather after successful completion of all security setup sequence steps.

2572 External key generation can introduce security vulnerabilities during the generation and loading
2573 process. Ensuring those processes are secure is the responsibility of the user and not
2574 addressed in this document.

2575 **6.3.4.4 imprintCertToNameMapping**

2576 The Configuration Domain specific certificate-to-name mapping is imprinted in the ietf-netconf-
2577 server YANG module under the following node.

- 2578 • /ncs:netconf-server/ncs:listen/ncs:endpoint/ncs:tls/ncs:netconf-server-
2579 parameters/ncs:client-identity-mappings/ncs:cert-to-name, with the following leaves:
2580 – id = 1,
2581 – fingerprint = Configuration Domain specific fingerprint of the LDevID-NETCONF trust
2582 anchor using the hash algorithm sha256 according to IETF RFC 7589, Clause 7, and
2583 – map-type = ext-60802-roles.

2584 The application of this map-type is described in 6.3.5, steps e) and f).

2585 The imprintCertToNameMapping step uses the NETCONF operation <edit-config> according to
2586 IETF RFC 6241 for the certificate-to-name mapping. Afterwards the NETCONF operation
2587 <commit> is applied to finalize the security setup sequence steps and to leave the factory
2588 default state.

2589 **6.3.5 Secure configuration based on LDevID-NETCONF**

2590 Configuration by NETCONF/YANG is protected by NETCONF-over-TLS as described in 6.3.2.1
2591 and NACM as described in 6.3.2.2. The NETCONF/YANG servers and clients shall use LDevID
2592 credentials for authentication.

2593 The procedure called "provisional accept of client certificate" as described in 6.3.4.2 is not
2594 applied anymore if the IA-station has left the factory default state. Instead, after successful
2595 establishment of a TLS session according to IETF RFC 7589 and IETF draft-ietf-netconf-over-
2596 tls13, the NETCONF server shall perform a certificate-to-name mapping and authorization
2597 check as follows.

- 2598 a) Compare the fingerprint of the trust anchor of the NETCONF client's certification path with
2599 the fingerprint contained in cert-to-name list entries of the x509c2n container for equal
2600 values.
- 2601 b) If no cert-name list entry match is found, then terminate the TLS session.
- 2602 c) If a cert-to-name list entry match is found, then verify if the map-type is equal to ext-60802-
2603 roles.
- 2604 d) If the map-type does not match, then terminate the TLS session.
- 2605 e) If the map-type value matches, then extract the role values from the id-60802-pe-roles
2606 certificate extension of the NETCONF client's TLS-authenticated end entity certificate. The
2607 output is a list of string values from the enumeration of specified role names according to
2608 this document.
- 2609 f) The list of role name string values is provided as input to NACM for permission checking.
2610 The access to the requested resource is checked according to the rules configured in the
2611 nacm container of the ietf-netconf-acm YANG module.

The NETCONF client checks if the expected identity to address the NETCONF server (IP address or DNS name) matches to the actual server identity that is stated by the IA-station on TLS level. This shall be done by comparing the expected identity with the subjectAltName extension of the TLS authenticated LDevID-NETCONF end entity certificate of the NETCONF server.

6.4 Management

6.4.1 General

Subclause 6.4 describes a model for configuration, deployment, and management of an industrial automation network.

6.4.2 IA-station management model

6.4.2.1 General

The management model of IA-stations covers simple end station IA-stations as well as combined IA-stations as described in 4.3. The IA-station management model is applied for topology discovery, network provisioning and stream establishment.

6.4.2.2 IEEE 802.1Q management model

In industrial automation both Bridge and end station components make use of IEEE 802.1Q specified functionality (for example, traffic classes, gate control). Thus, the IEEE 802.1Q management model is the basic management model to be applied to all IA-stations. Figure 15 shows the implementation of the IEEE Std 802.1Q Bridge model in YANG as specified in IEEE Std 802.1Q-2022, Clause 48. The IETF Interface Management YANG data model is specified in IETF RFC 8343.

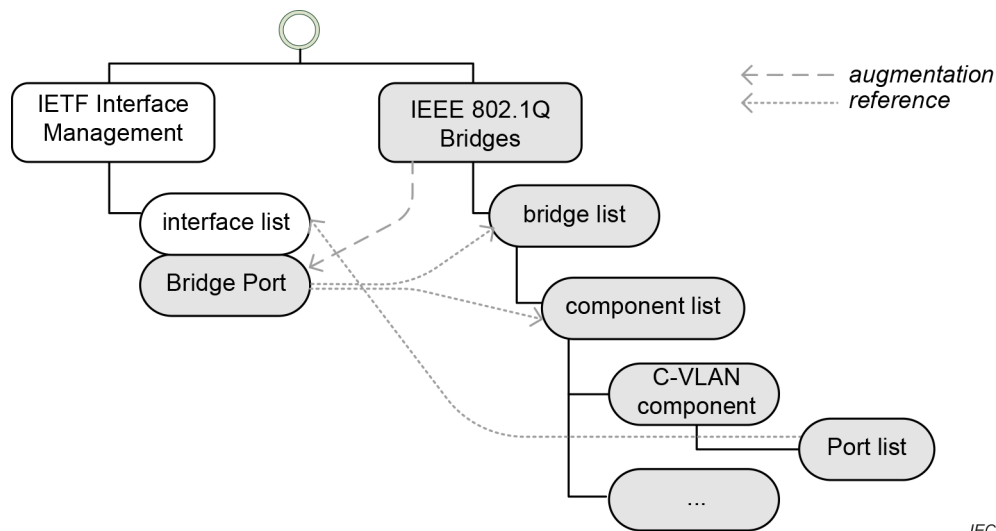


Figure 15 – Generic IEEE 802.1Q YANG Bridge management model

The IEEE 802.1Q Bridge model is organized as a bridge list where each bridge includes an underlying component list (for example, C-VLAN components). Each component has a Port list attached with references to the representation of the ports in the IETF interface list. The managed data of the ports is defined as Bridge Port augmentation to the IETF interface model. Each Bridge Port includes a reference to its bridge and component instances in the IEEE 802.1Q Bridge model.

The YANG data model for an IEEE 802.1Q Bridge is applied to IA-stations as follows.

- Each functional unit of an IA-station is modeled as bridge entry in the bridge list.

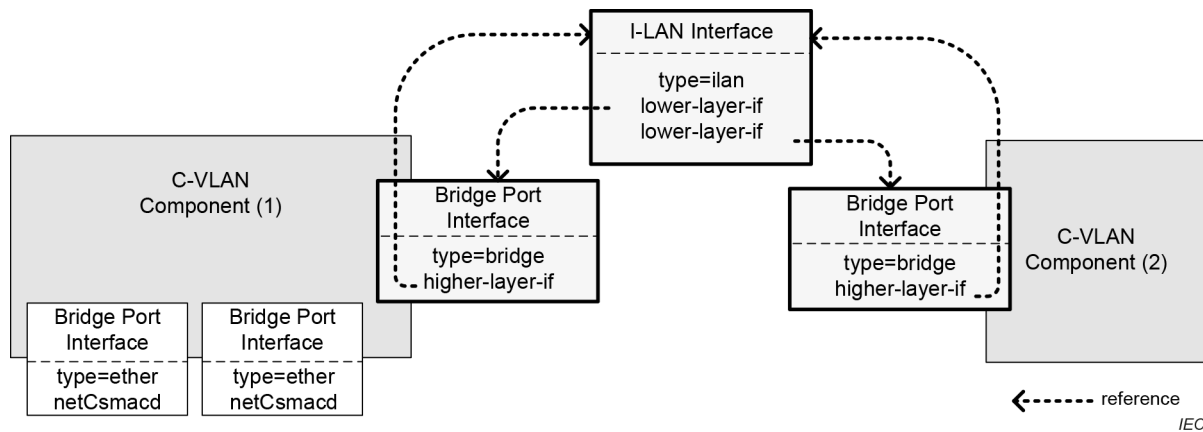
- 2643 • Each Bridge and end station component of an IA-station is modeled as C-VLAN component.
- 2644 • IA-station components belonging to the same functional unit are added to the component
- 2645 list of this functional unit's bridge entry.
- 2646 • Each IA-station external or internal port is modeled as Bridge Port.

2647 IA-station ports belonging to the same component are added to the Port list of the related
2648 component list entry.

2649 Further YANG data models which are relevant for IA-stations are described in 6.4.9.

2650 6.4.2.3 Internal LAN connection model

2651 The modeling of internal connections between C-VLAN components within an IA-station is
2652 aligned to IEEE Std 802.1Q-2022, 17.3.2.2, which includes an I-LAN interface. As shown in
2653 Figure 16, the I-LAN interface is modeled as an ilan IETF interface object (see IETF RFC 7224)
2654 together with appropriate `higher-layer-if` and `lower-layer-if` reference objects to
2655 describe the internal connection.



2657 **Figure 16 – Internal LAN connection management model**

2658 This internal LAN connection model comprises three configuration steps.

- 2659 • The internal Ports of the C-VLAN components are modelled as IETF interfaces of type bridge
- 2660 with Bridge Port augmentation.
- 2661 • An additional I-LAN interface of type ilan as described in IETF RFC 7224 is created.
- 2662 • The I-LAN interface references the internal Bridge Port interfaces of the connected C-VLAN
- 2663 components as lower-layer-if, and the internal Bridge Port interfaces of the connected C-
- 2664 VLAN components reference the I-LAN interface as higher-layer-if.

2665 Figure 17 shows the application of this model to the example IA-station of Figure 16.

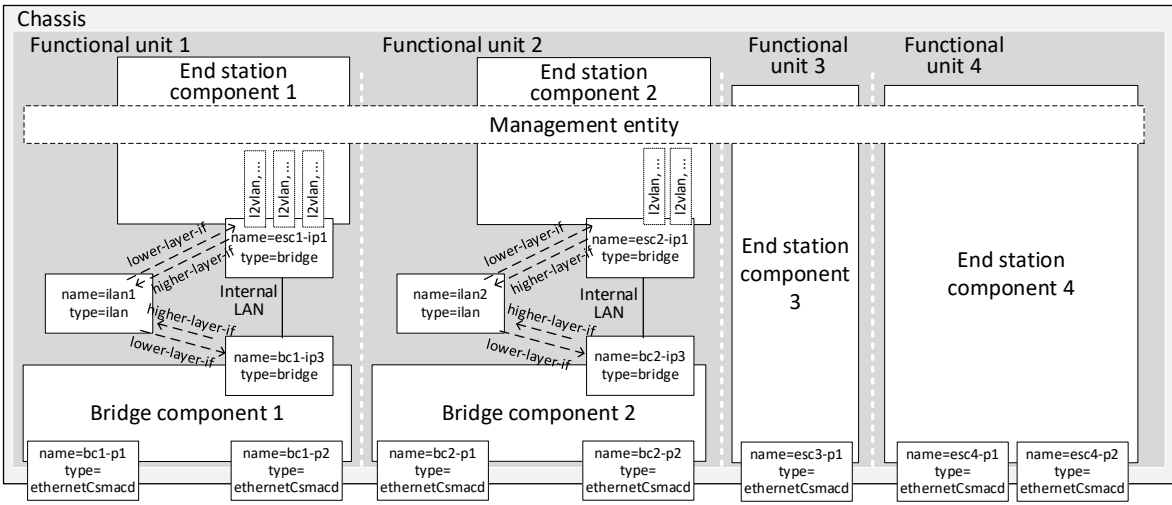


Figure 17 – IA-station example with IETF interfaces

NOTE Figure 17 represents an abstract model and is not intended to imply a particular implementation or partitioning.

Late binding of IA-station applications to the configured VLANs and priorities is performed using the `ieciieee60802-tt-mapping` YANG module in 6.4.10.8.7.

6.4.2.4 Spanning Tree, VLAN and TE-MSTID configuration

C-VLAN Bridge components of IA-stations shall support:

- the Common and Internal Spanning Tree (CIST) calculated by the Multiple Spanning Tree Algorithm and Protocol (MSTP), and
- the Traffic Engineering Multiple Spanning Tree Instance Identifier (TE-MSTID) as specified in IEEE Std 802.1Q-2022, 5.5.2.

The MSTP configuration is either default or accomplished by IA-station specific means.

CNCs configure VLANs in the `vlan` list in the `bridge-vlan` container of the `ieee802-dot1q-bridge` YANG module. Ports are assigned to a `vlan` as `static-filtering-entries` in a `filtering-database`.

NOTE `vlan`, in lowercase, refers to a YANG element.

VLANs are assigned to filtering databases in the `vid-to-fid` list of the `bridge-vlan` container. The filtering databases, and in consequence the VLANs, are by default assigned to the MSTP calculated Internal Spanning Tree and can be assigned to the TE-MSTID by management. IA-time-aware streams and IA-streams are assigned to the TE-MSTID.

TE-MSTID assignment is accomplished via the `bridge-mst` container of the `ieee802-dot1q-bridge` YANG module.

The following example shows the VID/FID/MSTID configuration of an IA-station's C-VLAN Bridge component, which supports three VLANs in three Forwarding Databases (VID 100 in FID 1, VID 101 in FID 2 and VID 102 in FID 3). FID 2 and FID 3 – and in consequence VID 101 and VID 102 – are assigned to the TE-MSTID. FID 1 – and in consequence VID 100 – is not assigned to a MSTID and thus, is implicitly assigned to the Internal Spanning Tree (IST).

Figure 18 shows the representation of this example configuration in the MST configuration table.

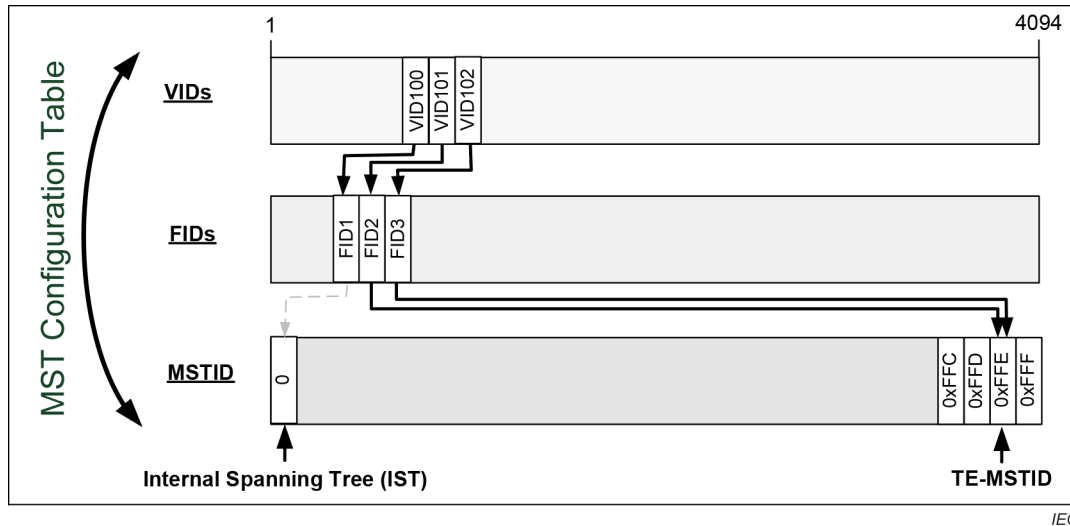


Figure 18 – VID/FID/MSTID example

The YANG-based configuration of this example is shown as YANG instance data snippet of the ieee802-dot1q-bridge YANG module. Herein the MST configuration table is included in component "bridge-component-x", which is part of bridge "functional-unit-x". It is the responsibility of the user to ensure that traffic type names are consistent with those provided in the YANG module iecieeee60802-tt-mapping (see 6.4.10.8.7).

```
<ieee802-dot1q-bridge xmlns:dot1q="urn:ieee:std:802.1Q:yang:ieee802-dot1q-bridge">
  <bridges>
    <bridge> <!-- list -->
      <name>functional-unit-x</name>
      ...
      <component> <!-- list -->
        <name>bridge-component-x</name>
        ...
        <bridge-vlan>
          <version>2</version> <!-- MST supported -->
          ...
          <vlan>
            <vid>100</vid>
            <name>60802-best-effort-hl-100</name> <!-- best effort high and
low -->
          </vlan>
          <vlan>
            <vid>101</vid>
            <name>60802-isochronous-101</name> <!-- isochronous -->
          </vlan>
          <vlan>
            <vid>102</vid>
            <name>60802-isochronous-102R</name> <!-- isochronous -->
          </vlan>
          ...
          <vid-to-fid>
            <vid>100</vid>
            <fid>1</fid>
          </vid-to-fid>
          <vid-to-fid>
            <vid>101</vid>
            <fid>2</fid>
          </vid-to-fid>
          <vid-to-fid>
            <vid>102</vid>
            <fid>3</fid>
          </vid-to-fid>
        </bridge-vlan>
      ...
    </bridge>
  </bridges>
</ieee802-dot1q-bridge>
```

```

2741         <bridge-mst>
2742             ...
2743             <fid-to-mstid> <!-- list -->
2744                 <!-- fid 1 is implicitly assigned to mstid 0 -->
2745                 <fid>2</fid>
2746                 <mstid>4094</mstid> <!-- TE-MSTID -->
2747             </fid-to-mstid>
2748             <fid-to-mstid> <!-- list -->
2749                 <fid>3</fid>
2750                 <mstid>4094</mstid> <!-- TE-MSTID -->
2751             </fid-to-mstid>
2752         </bridge-mst>
2753         ...
2754     </component>
2755 </bridge>
2756 </bridges>
2757 </ieee802-dot1q-bridge>

```

2758 6.4.3 Discovery of IA-station internal structure

2759 LLDP provides information about the external connectivity of IA-stations. To identify the internal
 2760 structure of complex IA-stations (see 4.3) the IEEE 802.1Q management model (see 6.4.2.2)
 2761 and the IETF Interface management model are applied.

- 2762 • The functional units of an IA-station are represented as bridge entries in the bridge-list.
- 2763 • The components of a functional unit are represented as component entries in the associated
 2764 bridge entry's component-list.
- 2765 • Internal LAN connections between components of a functional unit are identified by I-LAN
 2766 entries in the IETF interface list (6.4.2.3).

2767 6.4.4 Network engineering model

2768 To understand the requirements for network configuration, deployment and management, an
 2769 engineering model covering industrial use cases is required. The "fully centralized model"
 2770 described in IEEE Std 802.1Q-2022, 46.1.3.3 includes two functional entities: the CUC and the
 2771 CNC. The relationship between user and network configuration is described in IEEE Std
 2772 802.1Q-2022, Clause 46. This document further elaborates this relationship to address use
 2773 cases for industrial automation. A conceptual block diagram of a CNC is shown in Figure 19,
 2774 which adds further details to the CNC specified in IEEE Std 802.1Q-2022 to serve the industrial
 2775 automation use case. The following functional entities are introduced.

2776 a) The Topology Discovery Entity (TDE)

2777 The topology discovery entity is responsible for the topology discovery (i.e. Bridge
 2778 component and end station component discovery). The TDE also performs a topology
 2779 verification in cases where an expected topology is provided by the engineering tool. The
 2780 resulting topology information is used by the CNC. The TDE detects added or removed IA-
 2781 stations, including internal structure and connectivity. Thus, the CNC becomes aware of
 2782 them. Overall, the TDE discovers and maintains an inventory of the devices, including their
 2783 capabilities and the topology they form.

2784 b) The Path Entity (PE)

2785 The PE computes, establishes and maintains the forwarding paths for the IA time-aware
 2786 stream and IA stream traffic type categories according to 4.7.3.

2787 c) The Sync Tree Entity (STE)

2788 The STE computes, establishes and maintains the sync trees. For example, for Working
 2789 Clock and Global Time.

2790 d) The Resource Allocation Entity (RAE)

2791 The RAE is responsible for the allocation of the resources that are necessary for all traffic
 2792 type categories, according to 4.7.3, to meet their requirements via their forwarding paths.
 2793 For example, frame buffers at egress ports and FDB entries.

2794 e) The Network Provisioning Entity (NPE)

2795 The NPE applies a network policy provided by the Engineering Tool to the IA-stations within
2796 the Configuration Domain. It uses the information discovered by the TDE to create a network
2797 configuration based upon this policy which is then applied to all IA-stations. The CNC uses
2798 the chosen network configuration together with the discovered IA-stations and their
2799 capabilities as input for its stream calculation and deployment.

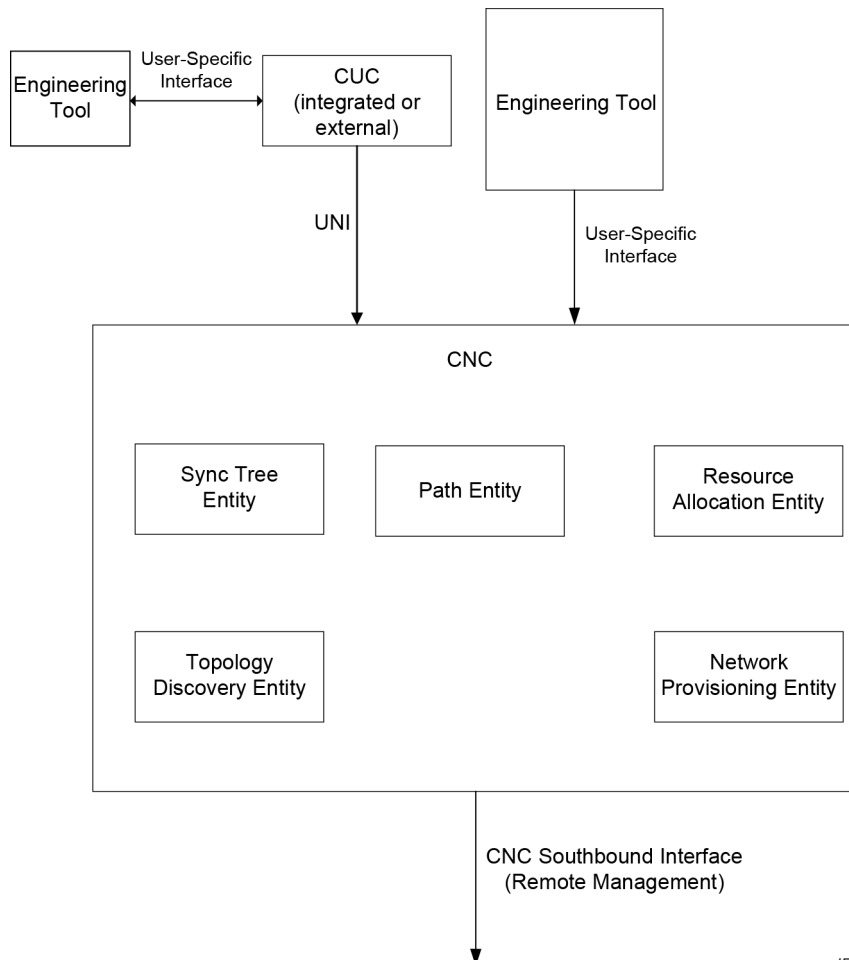
2800 A CNC includes these functional entities. The implementation of these functional entities and
2801 the CNC can vary. The means of communication among these functional entities is
2802 implementation dependent.

2803 If there are multiple CNCs in one Configuration Domain, then, by some means not addressed
2804 by this document, only a single CNC is in charge at any time in the given Configuration Domain.

2805 The CNC can be in a dedicated station or integrated into any IA-controller or IA-device.
2806 Generally, its engineering tool interface is user-specific and can only work with the compatible
2807 engineering tools. The definition of this interface is not addressed in this document.

2808 The CUC can be in a dedicated station or integrated into any IA-controller or IA-device.
2809 Generally, the CUC is user-specific. In industrial automation use cases, an IA-controller
2810 integrated CUC is very likely.

2811 For stream establishment, the UNI of the CNC component is exposed.



IEC

Figure 19 – Structure and interfaces of a CNC

2814 Figure 20 shows an example of the structure of an IA-station which the CNC can discover and
2815 manage.

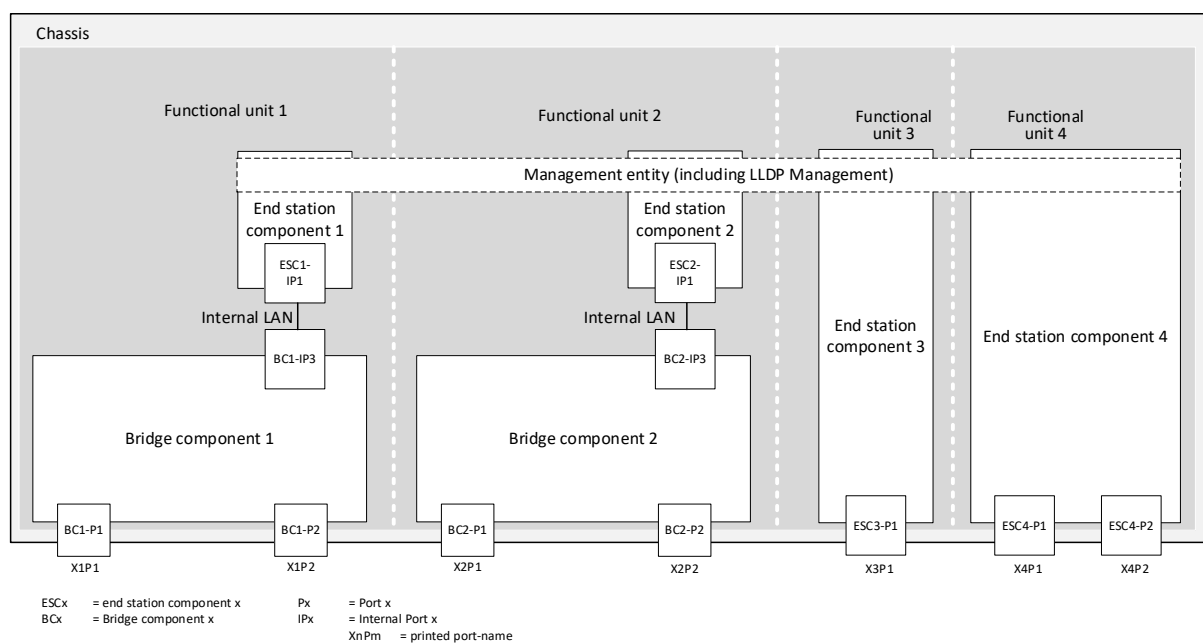


Figure 20 – IA-station structure example

Figure 21 shows the interaction of IA-stations with the CNC.

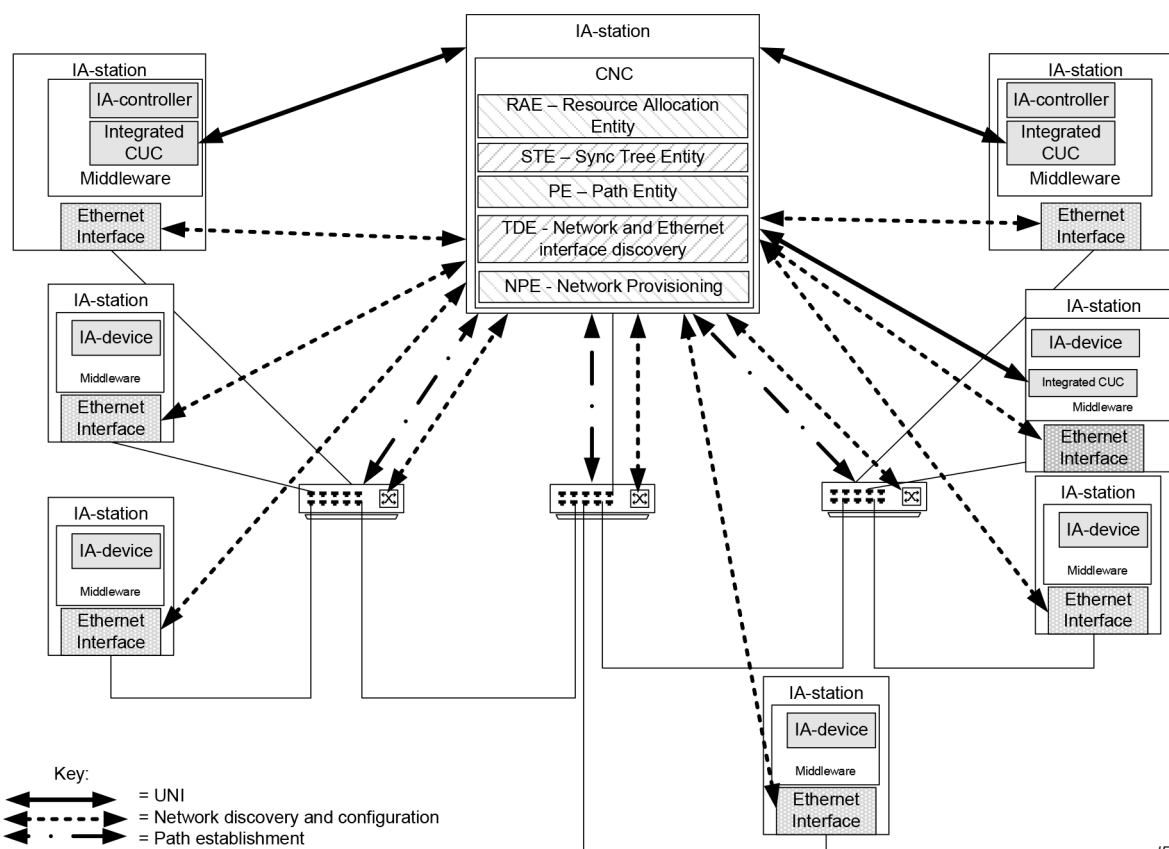


Figure 21 – CNC interaction

6.4.5 Operation

6.4.5.1 General

A representative model for network configuration is shown in Figure 22. This diagram maintains the traditional role of the IA-controller and the IA-device in an industrial automation network. IA-devices and IA-controllers require configuration from engineering tools (refer to engineering tools A, B, D, and E). These tools and associated interfaces are not addressed by this document. In this example, engineering tool C communicates directly with the CNC to provide traffic requirements for the network. The protocols that the engineering tool uses for communication with end stations are specific to the user application.

The UNI is the interface to the CNC which is serviced by NETCONF over TLS. The UNI service recognizes that industrial automation communications are typically connection oriented. There is a communication initiator, typically in an IA-controller, which is responsible for establishing those connections, determining what data is of interest and providing the required update rate. So, while an application/middleware of an IA-station (for example a Drive) understands what information it can produce and the maximum rate at which that information can be provided, until an IA-controller establishes a connection with that device, it does not know where that information goes and what update rate is required to close the control loop. The IA-controller gets this information from its engineering tool. There can be multiple IA-controllers in each Configuration Domain. The CNC uses the topology, the device capabilities, the device configuration, and the traffic specifications from the user to calculate a path for each Talker-Listener pair. The UNI then provides stream identification (VLAN, DMAC, etc.) to the middleware.

The operational management model, see Figure 22, reflects the model used in industrial automation. Figure 22 shows an active CNC managing multiple IA-stations. Each station can wholly incorporate a CUC and interact with the CNC directly.

Security requirements (see 6.3) are an important consideration for these networks and are integrated into the design, configuration, and deployment of any management model.

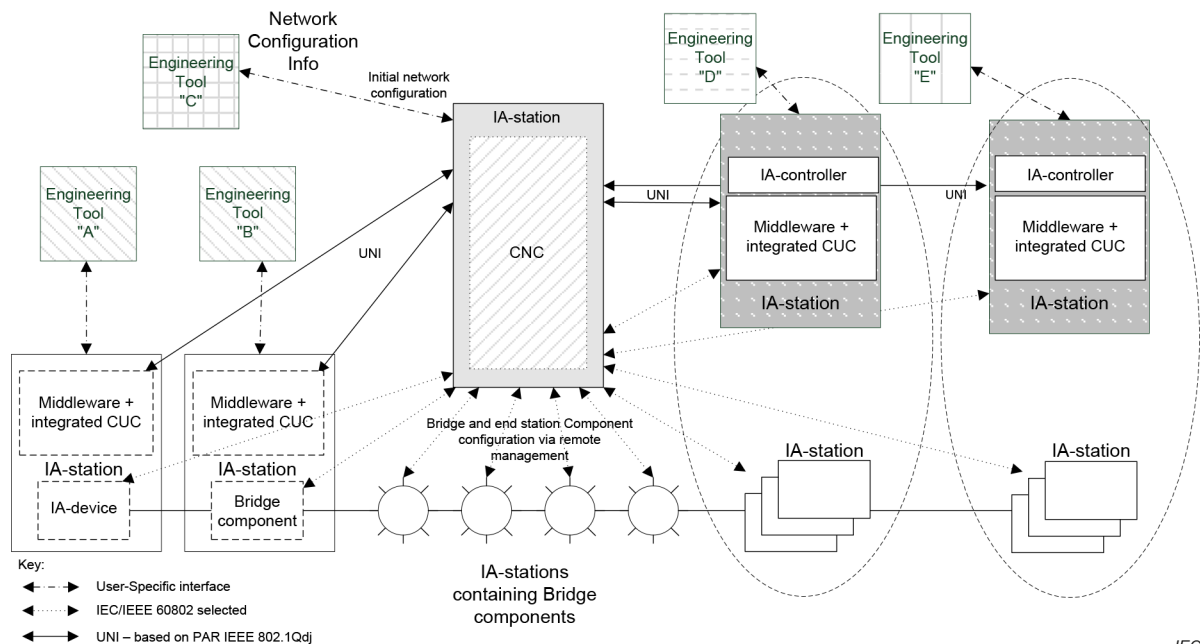


Figure 22 – Operational management model

Figure 23 shows the steps that are typically performed in the scope of the CUC-CNC interaction.

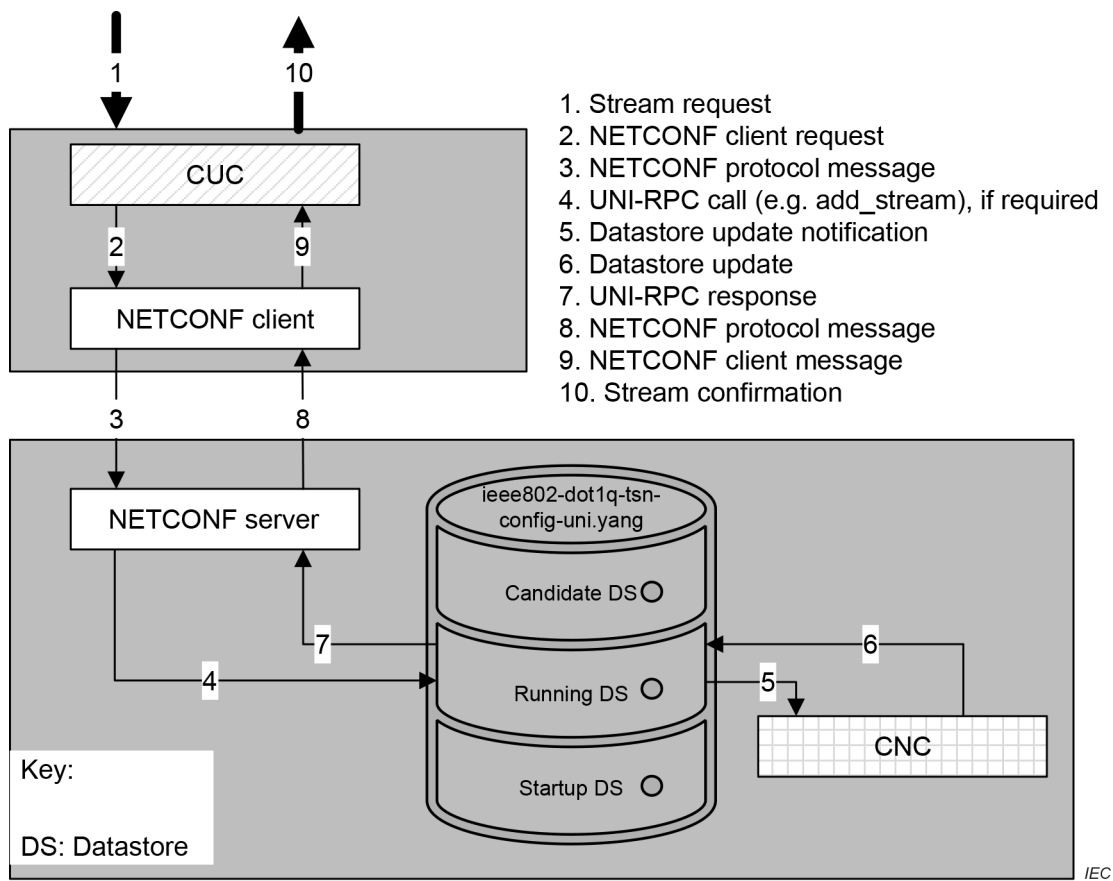


Figure 23 – UNI service model

After the computation of the paths and the scheduling or shaping configuration have been done, the CNC configures the IA-stations via NETCONF client. The typical steps that are performed in this process are shown in Figure 24 below.

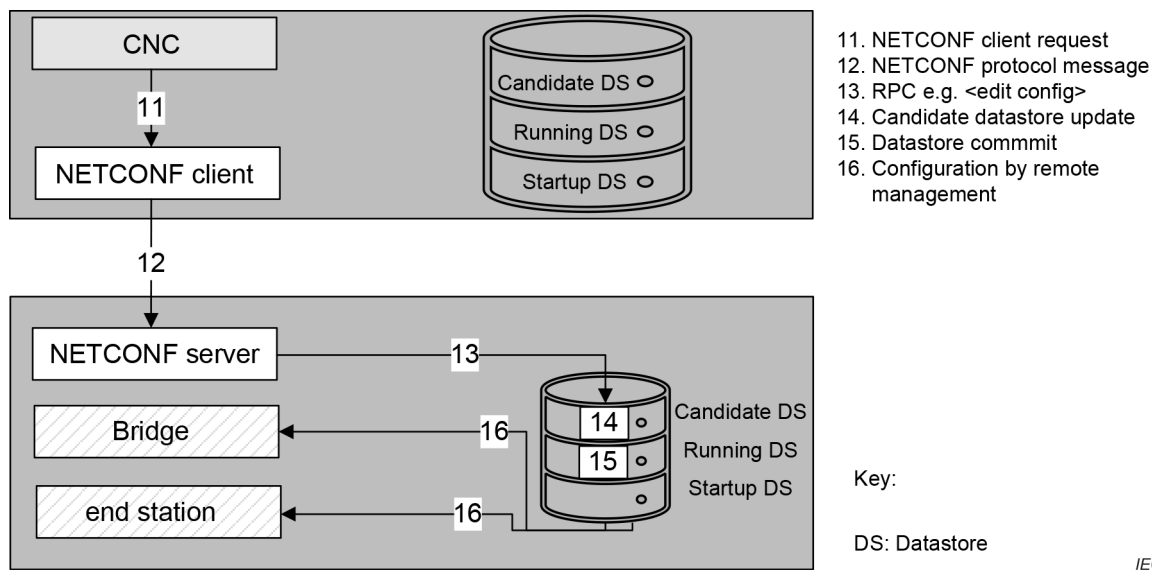


Figure 24 – CNC southbound

Instances of NETCONF servers and clients within a Configuration Domain are shown in Figure 25. IA-stations that contain a CNC or CUC entity contain both a NETCONF server and a NETCONF client. The CUC utilizes a NETCONF client for the UNI. The CNC utilizes a

NETCONF server to accommodate the UNI as well as remote network management of the end stations and bridges that are contained in the same chassis as the CNC entity. The CNC utilizes a NETCONF client for the southbound interface of the CNC, i.e. for the remote management of the bridges and end stations in the scope of stream configuration. All IA-stations have a NETCONF server to make remote management possible. The NETCONF server used by the CNC serves multiple NETCONF Clients (CUCs) within a single Configuration Domain whose requests clients can occur simultaneously.

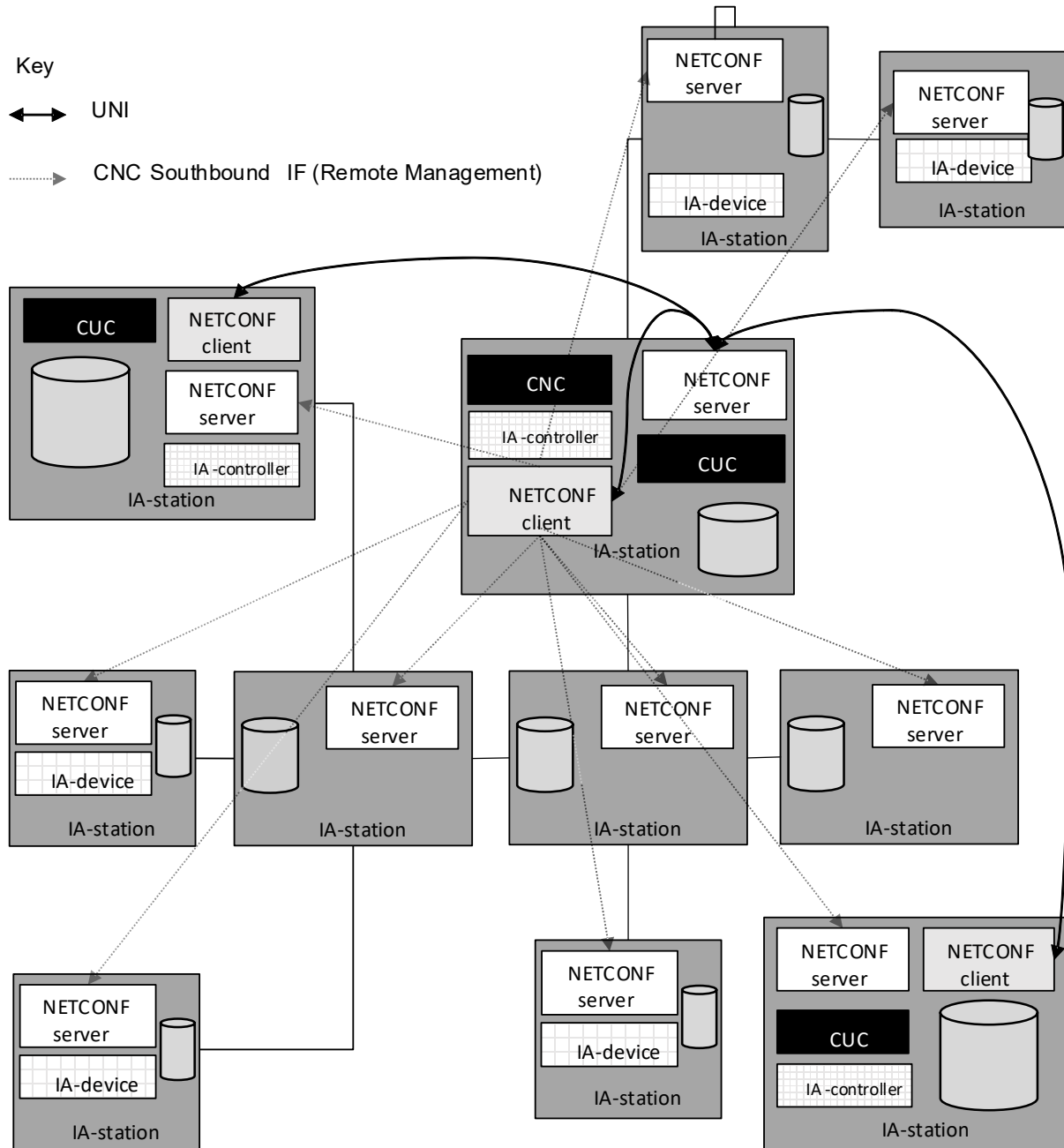


Figure 25 – NETCONF usage in a Configuration Domain

6.4.5.2 Domain port states

A CNC manages available network resources and assigns them to the IA-stations. Management of the network resources is only possible if the CNC owns these resources. Thus, no connected station is allowed to make use of network resources that are not granted by the CNC. The security configuration of a connected station allows remote access for the CNC.

Protection of the network resources is done by managing the ports (see Figure 26) at the boundary of the Configuration Domain. The state of any newly connected station is unknown. The CNC is responsible for determining if the newly connected station is added to the Configuration Domain and configuring the IA-station appropriately.

This port state model avoids any assumptions about configuration of added stations or network portions.

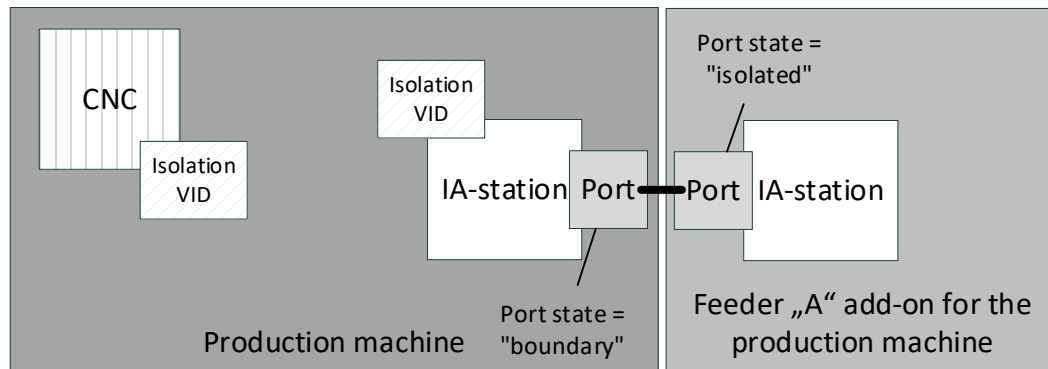


Figure 26 – Boundary port model

Ports of an IA-station that is a member of a Configuration Domain have different states:

- Isolated – a station connected via this port can only exchange information with a CNC. The CNC is responsible for establishing an isolation VID and for onboarding the station. In the isolated state:
 - the port gets to or remains in isolated state in case of a link down event, e.g. when nothing is connected, or no link is established;
 - the port gets to or remains in isolated state in case of a link up event;
 - the port stays in isolated state as long as the neighbor is unknown, not able to enter Boundary state.
- Boundary – a station connected via this port is not part of the Configuration Domain, but is allowed to access devices inside the Configuration Domain and to pass traffic through the Configuration Domain
- Inside – a station connected via this port is part of the Configuration Domain

The determination of whether a given port of an IA-station remains in the Isolated state or transitions to the Boundary or Inside state is performed by the CNC using remote management. A port acts as a domain boundary if it is in the Isolated or Boundary state.

For example, a port could be configured as follows:

- Isolated state:
 - port is IST boundary,
 - port is not part of a sync tree,
 - port uses VLAN stripping for egress,
 - port uses VLAN assignment and priority regeneration to assign all traffic to an isolated VLAN, and
 - port uses an ingress rate limiter to control the amount of traffic for the Configuration Domain.
- Boundary state:
 - port is part of IST,

- 2910 – port is part of a sync tree,
- 2911 – port uses VLAN stripping for egress,
- 2912 – port uses VLAN assignment and priority regeneration to assign all traffic to a default
- 2913 VLAN, and
- 2914 – port uses an ingress rate limiter to control the amount of traffic for the Configuration
- 2915 Domain.
- 2916 • Inside state:
 - 2917 – port is part of IST,
 - 2918 – port is part of a sync tree, and
 - 2919 – port is part of the active topology for stream and non-stream traffic.

2920 An example workflow includes the following steps executed by the CNC:

- 2921 a) Topology discovery:
 - 2922 1) Case A: Link down / Port not connected:
 - 2923 i) set port to isolated state;
 - 2924 ii) configure a NETCONF subscription "on data change" to the port state leaf
 - 2925 2) Case B: Neighbor is not a Configuration Domain member:
 - 2926 i) set port to boundary state;
 - 2927 ii) configure a NETCONF subscription "on data change" to the port state leaf.
 - 2928 3) Case C: Neighbor is not a Configuration Domain member – but part of expected topology:
 - 2929 i) set port to boundary state;
 - 2930 ii) configure the neighbor station as Configuration Domain member;
 - 2931 iii) set port to inside state.
- 2932 b) NETCONF subscription trigger:
 - 2933 Issued to the CNC upon change of subscribed YANG data.

2934 **6.4.5.3 Engineered network**

2935 For an offline engineered (based on the available digital data sheets of the used IA-stations)
 2936 centralized approach with fixed topology, fixed stations and fixed paths, the user provides traffic
 2937 requirements, path information, topology information and expected network configuration to the
 2938 CNC. The CNC then uses the TDE, RAE and the NPE to perform the calculation of paths,
 2939 resources, and stream schedules necessary to meet the specified traffic requirements and
 2940 deploys the result of these calculations via remote management. The CNC also provides the
 2941 relevant results to the CUC via the UNI. The CUC then configures the end stations using the
 2942 user-to-user interface (see Figure 3).

2943 The workflow for this example consists of the following steps:

- 2944 a) The user determines:
 - 2945 1) the expected network topology,
 - 2946 2) the expected stations and its capabilities, value ranges and quantities,
 - 2947 3) the expected paths and resources,
 - 2948 4) the required streams, and
 - 2949 5) the requirements for IA non-stream traffic.

2950 This step focuses on network capabilities including the Ethernet interface of the end stations.
 2951 For example, if the end station is a sensor, the user considers the Ethernet interface capabilities
 2952 of the sensor as they apply to the physical world.

- 2953 b) Engineering Tool provides this information to the CNC via a user-specific interface.

2954 Although the communication between the CNC and any Engineering Tool is user-specific, the
2955 CNC obtains all necessary information via the integrated TDE and NPE.

2956 c) The CNC uses the TDE to discover the topology and checks it against the expected
2957 topology. The NPE is used to configure the IA-stations of the Configuration Domain.

2958 d) The CNC uses STE and NPE to setup, validate, and monitor synchronization configuration
2959 in the Configuration Domain.

2960 e) The CNC uses the information from engineering item a), steps 1 to 5, above to respond to
2961 requests from middleware (with integrated CUC) using UNI. These requests are handled
2962 using the already established communication paths received from the user.

2963 If the CNC is not required after commissioning, then the CNC can be removed after setting up
2964 the IA-stations. That requires that all IA-stations have a persistent storage for the data provided
2965 by the CNC.

2966 **6.4.5.4 Dynamic topology**

2967 **6.4.5.4.1 General**

2968 For a centralized approach with a dynamic topology and dynamic paths, the user provides the
2969 network policy to the CNC. The TDE performs topology discovery including IA-station
2970 capabilities (YANG representation of the digital data sheet) and the NPE performs network
2971 configuration for the CNC. IA-stations then provide traffic requirements via the middleware to
2972 the CNC via the UNI. The CNC then uses the TDE, RAE, and NPE to perform the calculation of
2973 paths, resources, and stream schedules necessary to meet the specified traffic requirements
2974 and deploys the result of these calculations via remote management. The CNC also provides
2975 the relevant results to the CUC via the UNI. The CUC then configures the end stations using
2976 the user-to-user interface (see Figure 3).

2977 The workflow for this example consists of the following steps.

2978 a) The user determines the network policy and provides it to the CNC.

2979 b) The TDE continuously discovers the physical network topology and station capabilities of
2980 each station using remote management.

2981 c) The NPE uses the information gathered in steps a) to b) to configure the IA-stations in the
2982 Configuration Domain.

2983 d) The CNC uses STE and NPE to setup, validate and monitor time synchronization
2984 configuration in the Configuration Domain.

2985 The CNC uses the information from steps a) to d) to respond to requests from middleware using
2986 UNI. The CNC establishes streams in the bridges via a remote management protocol.

2987 **6.4.5.4.2 Adding an IA-station**

2988 Each IA-station added to the Configuration Domain is discovered by the TDE and receives the
2989 initial network configuration from the NPE. After this, the middleware can request stream
2990 establishment.

2991 When an IA-station is added to the network, it is isolated until the CNC determines that its traffic
2992 requirements can be accommodated without disrupting other traffic (see 6.4.5.2).

2993 **6.4.5.4.3 Removing an IA-station**

2994 Each IA-station removed from the Configuration Domain is discovered by the TDE. A
2995 neighboring station can receive an updated network configuration by the NPE. After this, the
2996 removed IA-station is no longer part of the Configuration Domain.

2997 **6.4.5.4.4 Replacing an IA-station**

2998 In the simplest case, replacing an IA-station is simply the sequence of removing an IA-station
2999 (6.4.5.4.3) and adding an IA-station (6.4.5.4.2). In more complex cases, the user can take other
3000 precautions or actions following deployment.

3001 **6.4.5.5 Management proxies**

3002 **6.4.5.5.1 General**

3003 Implementing NETCONF server functionality can be burdensome for resource constrained IA-
3004 stations. IA-stations which contain only ccB Bridge and end station components may utilize the
3005 NETCONF server of another IA-station as a management proxy. IA-stations supporting proxy
3006 functionality (proxies), maintain a list of the types of proxy functionality supported as well as
3007 the IA-stations for which they act as proxy (proxy targets). Proxy types are described in E.1.3.
3008 The information is maintained in the YANG module `iecieee60802-proxy` as described in 0.

3009 **6.4.5.5.2 Proxy operation**

3010 When onboarding a new proxy target, a different process is followed. Proxies communicate with
3011 proxy targets via the isolation VLAN. When a new IA station is connected, the following
3012 sequence is followed.

- 3013 a) The IA-station to which the proxy target is connected (neighbor), which is already part of
3014 the configuration domain, reports a change in link status to the CNC.
- 3015 b) The CNC reads the stored LLDP information related to the proxy target from the neighbor
3016 via NETCONF. The neighbor can be a proxied device in which case the LLDP information
3017 about the proxy target would be retrieved using the neighbor's proxy.
- 3018 c) Each proxy target identifies itself using the 60802 constrained device proxy TLV described
3019 in Annex E.
- 3020 d) Based on this TLV data, the CNC determines whether the connected station:
3021
 - natively speaks NETCONF/YANG (no information regarding proxy operation is included),
 - 3022 or
 - 3023 • requires a proxy, and if so, which one.
- 3024 e) The proxy onboards the new device via a user-specific protocol. This user-specific protocol
3025 is not described in this document. Ensuring the security of this user-specific mechanism is
3026 not described in this document.
- 3027 f) The proxy communicates the result of the Authentication to the CNC. For instance, the CNC
3028 can utilize dynamic subscriptions and the on-change feature of the `ietf-yang-push` module
3029 to automatically be notified when the new proxy target is added to the proxy's list of proxy
3030 targets. Based upon the results of the authentication, the CNC sets the boundary port state
3031 appropriately
- 3032 g) The CNC communicates with the appropriate proxy and configures the device accordingly.
3033 The proxy handles the NETCONF parameterization and translates it into the required
3034 organization-defined or vendor-specific protocol and configures the proxy target.

3035 **6.4.5.6 Engineered network extended by dynamic topology**

3036 The engineered and dynamic topology workflows can be used together. For instance, modular
3037 machines, robot tool changers or more general plug & produce can add or remove modules.
3038 The basic machine is handled as an engineered network. Additional modules or removed
3039 modules are handled dynamically.

6.4.6 Engineered time-synchronization spanning tree

6.4.6.1 General

Engineered time-synchronization spanning tree (sync tree) for a given gPTP domain refers to the usage of external port configuration instead of BTCA for the construction of a desired sync tree with the Grandmaster PTP Instance as the root (see IEEE Std 802.1AS-2020 as amended by IEEE Std 802.1ASdr-2024 and IEEE Std 802.1ASdm-2024, 10.3.1). The Grandmaster PTP Instance can reside in a dedicated grandmaster-capable IA-station.

One of the advantages of engineered sync trees is to enable a planned, deterministic, and stable configuration of the IEEE Std 802.1AS-2020 sync tree for a given gPTP domain. For example, this approach prevents sync tree changes in case of IA-station addition or removal from the network. Hot standby (see IEEE Std 802.1ASdm-2024) is a use case of an engineered sync tree.

6.4.6.2 Sync tree requirements

If an engineered synchronization spanning tree is used, the sync tree requirements for all participating PTP Instances in a gPTP domain are specified in 5.5.3, h).

6.4.6.3 STE phases

6.4.6.3.1 General

The STE follows the logical sequence described in 6.4.6.3 if an engineered sync tree is utilized in a gPTP domain. Each STE phase describes an externally observable behavior of the participating PTP Instances in a gPTP domain.

6.4.6.3.2 Discovery phase

In discovery phase, STE utilizes the topology discovered by the TDE to verify the capabilities and status of participating IA-stations via a diagnostics entity (see 6.4.7.1) by reading the following managed objects.

- The status of oper-status parameter is up (see IETF RFC 8343) for all participating Ethernet links.
- The status of isMeasuringDelay (see IEEE Std 802.1AS-2020 as amended by IEEE Std 802.1ASdm-2024, 14.16.4) is TRUE for all PTP Ports.
- The status of asCapable (see IEEE Std 802.1AS-2020, 14.8.7) is TRUE for all PTP Ports.
- The status of asCapableAcrossDomains (see IEEE Std 802.1AS-2020, 14.16.5) is TRUE for all LinkPorts.
- The status of gmCapable (see IEEE Std 802.1AS-2020, 14.2.7) is TRUE, only applicable to the Grandmaster PTP Instance.

STE should use the information collected via managed objects and the discovered topology to verify the constraints on the gPTP domain, for example:

- Verify that the number of PTP Relay Instances (hops) between the Grandmaster PTP Instance and any given timeReceiver PTP End Instance is within the limit prescribed by for example, CNC.

6.4.6.3.3 Provisioning phase

In provisioning phase, STE should apply the desired configuration to all participating PTP Instances, for example:

- externalPortConfigurationEnabled is set to true,

- 3082 • the desiredState of all PTP ports of the Grandmaster PTP Instance is set to
3083 TimeTransmitterPort,
- 3084 • the desiredState of exactly one PTP port of all the other PTP Instances is set to
3085 TimeReceiverPort,
- 3086 • the desiredState of remaining PTP ports that are part of sync tree in non-Grandmaster PTP
3087 Relay Instances is set to TimeTransmitterPort, and
- 3088 • the desiredState of all other PTP ports is set to PassivePort.

3089 Then STE should validate, for example, the syncLocked (see IEEE Std 802.1AS-2020, 14.8.52)
3090 parameter is TRUE for all PTP ports of PTP Relay Instances that are in TimeTransmitterPort
3091 state.

3092 **6.4.6.4 Adding an IA-station**

3093 Each IA-station added to the gPTP domain is discovered by STE via TDE. It is the responsibility
3094 of the CNC to on-board this newly added station. IA-stations can receive an updated gPTP
3095 configuration via STE.

3096 A newly installed IA-station can disrupt the operation of a gPTP domain. The extent of disruption
3097 is dependent on the location of the IA-station in the gPTP domain and the type of PTP Instance
3098 running on that IA-station. For example, if PTP Instances are arranged in a daisy-chain
3099 formation and if an IA-station with a non-Grandmaster Relay Instance is installed in the middle
3100 of a daisy-chain then this change can disrupt for example, the operation of downstream PTP
3101 Instances.

3102 **6.4.6.5 Removing an IA-station**

3103 The removal of a station from the gPTP domain is detected by STE via TDE. IA-stations can
3104 receive an updated gPTP configuration via STE.

3105 **6.4.6.6 Replacing an IA-station**

3106 An IA-station replacement follows the sequence of removing an IA-station according to 6.4.6.5
3107 and adding an IA-station according to 6.4.6.4.

3108 **6.4.7 Diagnostics**

3109 **6.4.7.1 General**

3110 Diagnosis for an IA-station is done by monitoring YANG representation of the IA-station's local
3111 database.

3112 A vendor can implement an observer in a diagnostics entity, which could reside in the CNC.
3113 This diagnostics entity uses the information provided by remote management to define the
3114 monitored objects and set up fitting notifications.

3115 **6.4.7.2 Observer model**

3116 A diagnostic entity can select any objects described via YANG and observe them via NETCONF.
3117 The NETCONF binding is specified in IETF RFC 8640, and the subscription model in IETF RFC
3118 8641. NETCONF messages can be pipelined, i.e. a client can invoke multiple RPCs without
3119 having to wait for RPC result messages first. RPC messages are specified in IETF RFC 6241,
3120 and notification messages are specified in IETF RFC 5277. To reduce the load on the diagnostic
3121 entity when many stations are providing notifications, the diagnostic objects can be monitored
3122 and notifications can be retrieved from individual IA-stations.

3123 Figure 27 shows the model of a diagnostic observer.

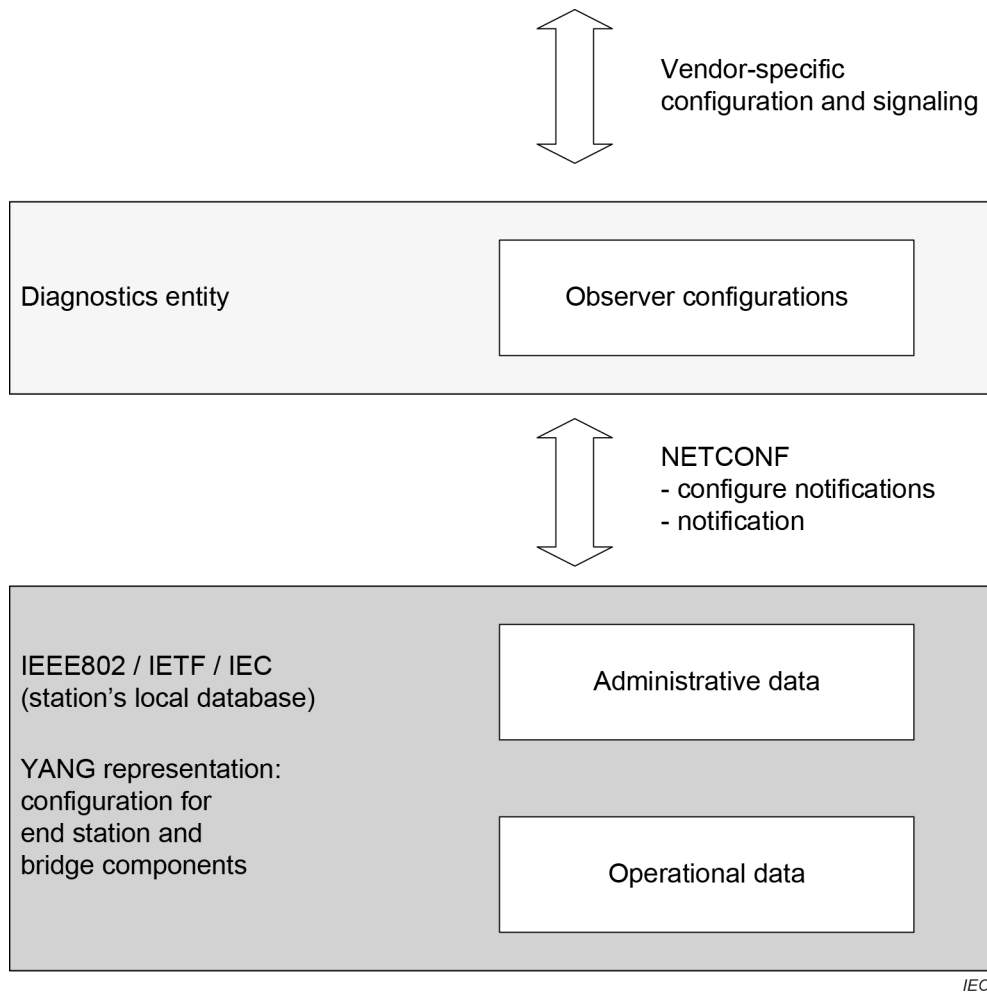


Figure 27 – Observer model

6.4.7.3 Usage of YANG Push

For diagnostics, an IA-station shall support YANG-Push subscriptions according to IETF RFC 8641 (YANG Push) and IETF RFC 8639 (Subscribed Notifications).

IA-stations shall support the "subtree" selection filter as specified in IETF RFC 8641, 3.6

6.4.7.4 Mandatory RPCs

An IA-station shall support following RPCs as specified in IETF RFC 8641:

- a) establish-subscription,
- b) modify-subscription,
- c) delete-subscription, and
- d) kill-subscription.

6.4.7.5 Mandatory notifications

An IA-station shall support following notifications as specified in IETF RFC 8641:

- a) subscription-resumed,
- b) subscription-modified,
- c) subscription-terminated,

- 3141 d) subscription-suspended,
- 3142 e) push-update, and
- 3143 f) push-change-update.

3144 **6.4.7.6 Mandatory diagnostics data nodes**

3145 An IA-station shall provide the following data nodes for NETCONF subscriptions.

- 3146 a) Change of link-status per Ethernet port:

3147 /ietf-interfaces/interface/oper-status

- 3148 b) Change of MAU-type per Ethernet port:

3149 /ieee802-ethernet-lldp/lldp/port/ operational-mau-type

- 3150 c) Change of sync-status:

- 3151 1) per PTP Instance:

3152 for every application-clock: /ieciieee60802-
3153 bridge/bridges/bridge/component/clock/is-synced

- 3154 2) per hot standby Instance:

3155 /dotlas-hs/ptp/common-services/hss/hot-standby-system-
3156 list/hot-standby-system-ds/hot-standby-system-state

- 3157 d) Data to be provided as periodic time-aligned subscriptions:

- 3158 1) Dropped frames statistic counters per Ethernet interface

- 3159 – /ietf-interfaces/interface/statistics/in-octets
- 3160 – /ietf-interfaces/interface/statistics/in-discards
- 3161 – /ietf-interfaces/interface/statistics/in-errors
- 3162 – /ietf-interfaces/interface/statistics/out-octets
- 3163 – /ietf-interfaces/interface/statistics/out-discards
- 3164 – /ietf-interfaces/interface/statistics/out-errors

- 3165 2) VLAN specific counters per Ethernet Interface and VLAN ID

- 3166 – /ieee802-dot1q-bridge/interfaces/interface/bridge-
3167 port/statistics/octets-rx
- 3168 – /ieee802-dot1q-bridge/interfaces/interface/bridge-
3169 port/statistics/octets-tx
- 3170 – /ieee802-dot1q-bridge/interfaces/interface/bridge-
3171 port/statistics/forward-outbound
- 3172 – /ieee802-dot1q-bridge/interfaces/interface/bridge-
3173 port/statistics/discard-inbound

3174 **6.4.7.7 Usage of NETCONF notifications**

3175 IA-stations shall implement the binding of a stream of events according to IETF RFC 8640
3176 (NETCONF Notifications) using the "encode-xml" feature and the "NETCONF" event stream of
3177 IETF RFC 8639.

3178 An IA-station shall support dynamic subscriptions according to 6.4.7.6, IETF RFC 8640 Clauses
3179 5, 6 and 7. The number of dynamic subscriptions shall be reported.

6.4.8 Data sheet

6.4.8.1 General

Data sheets containing the capabilities, value ranges and quantities of IA-stations allow a user to select appropriate IA-stations and enable users to configure a system using online and offline engineering. See Annex B for quantities in a representative Configuration Domain.

Online data sheets are modeled using YANG. YANG modeling is used for the offline data sheet to keep the offline (6.4.5.3) and online (6.4.5.4) format the same.

6.4.8.2 Digital data sheet of an IA-station

Both engineering models, offline via an engineering tool and online with plug & produce by the CNC, require information about the capabilities of an IA-station, for example, states, configurations, or supported features. An example depicting the creation of a digital data sheet is provided in Figure 28.

This data is extracted from the implemented YANG modules, which are available in the local database of the IA-station.

The data from the implemented YANG modules is also available offline in the form of a digital data sheet of an IA-station as a digital data sheet file.

The digital data sheet of an IA-station provides a collection of all instantiated data nodes of all YANG modules that are required by this document (see 6.4.9). A manufacturer may reduce the instance data set by removing statistical config-false YANG nodes from the digital data sheet.

The digital data sheet does not contain any additional information that is not modeled by the YANG modules that exist in the local database of the IA-station.

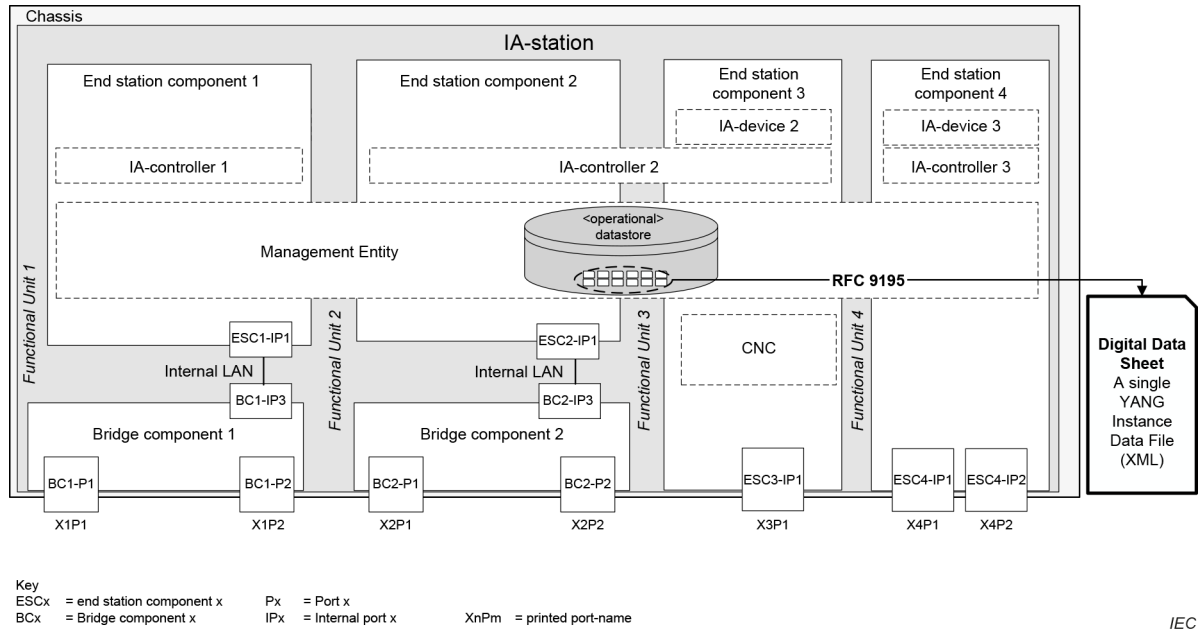
The data sheet contains a single instance data set. It carries complete configuration and state data of each YANG module that is present in the local database of the IA-station.

The identity of the datastore with which the instance data set is associated and the format of the YANG instance data set shall be reported according to IETF RFC 9195. The file format is based on the XML encoding. It is created by applying the respective XML encoding rules for the YANG structure of all YANG modules included in the digital data sheet.

A user uses the information from the digital data sheet to understand the quantities and capabilities of an IA-station, which is required for successful offline engineering of the network.

The features of a CNC are used for offline and online engineering or diagnostics. For this purpose, YANG modules are used that allow structured access to the local database of the CNC according to 6.4.9.2.5.11.

Any IA-station can include a CNC entity in which case the collection of YANG modules of such IA-station includes all CNC specific YANG modules for example, the ieee802-dot1q-cnc-config YANG module. Since all IA-stations meet the requirements from 5.5.4, the CNC related YANG instance data is automatically included in the digital data sheet of the IA-station that hosts the CNC as described in 6.4.9.2.



IEC

Figure 28 – Creation of the digital data sheet of an IA-station

6.4.9 YANG representation of managed objects and nodes

6.4.9.1 General

All managed objects shall be represented in the YANG 1.1 format as described in IETF RFC 7950. The markings (i.e. [m], [o], [c]) indicate whether the node is included in the digital data sheet (see 3.5.4). These markings are independent of the conformance criteria for an IA-station (see 5.2).

6.4.9.2 Common YANG modules, features, and nodes

6.4.9.2.1 IEEE standard for Ethernet

IA-stations shall support the `ieee802-ethernet-interface` YANG module according to IEEE Std 802.3.2™-2019 as modified by IEEE Draft Std P802.3.2a with the following nodes:

[o] `/ieee802-ethernet-interface/interfaces/interface/ethernet/duplex`

[o] `/ieee802-ethernet-interface/interfaces/interface/ethernet/speed`

[o] `/ieee802-ethernet-interface/interfaces/interface/ethernet/flow-control/pause/direction` (if the feature "ethernet-pause" is supported)

[o] `ieee802-ethernet-lldp/lldp/port/operational-mau-type`

[o] `ieee802-ethernet-lldp/lldp/port/preemption-supporated`

[o] `ieee802-ethernet-lldp/lldp/port/preemption-enabled`

[o] `ieee802-ethernet-lldp/lldp/port/preemption-active`

[o] `ieee802-ethernet-lldp/lldp/port/additional-fragment-size`

[o] `ieee802-ethernet-lldp/lldp/port/local-max-frame-size`

3239 6.4.9.2.2 Station and media access control connectivity discovery

3240 IA-stations shall support the following nodes from the ieee802-dot1ab-lldp YANG module
3241 according to IEEE Std 802.1ABcu-2021 with values and value ranges according to 6.5.

- 3242 [o] /ieee802-dot1ab-lldp/lldp/message-fast-tx
- 3243 [o] /ieee802-dot1ab-lldp/lldp/message-tx-hold-multiplier
- 3244 [o] /ieee802-dot1ab-lldp/lldp/message-tx-interval
- 3245 [o] /ieee802-dot1ab-lldp/lldp/reinit-delay
- 3246 [o] /ieee802-dot1ab-lldp/lldp/tx-credit-max
- 3247 [o] /ieee802-dot1ab-lldp/lldp/tx-fast-init
- 3248 [o] /ieee802-dot1ab-lldp/lldp/notification-interval
- 3249 [o] /ieee802-dot1ab-lldp/lldp/remote-statistics
- 3250 [o] /ieee802-dot1ab-lldp/lldp/remote-statistics/last-change-time
- 3251 [o] /ieee802-dot1ab-lldp/lldp/remote-statistics/remote-inserts
- 3252 [o] /ieee802-dot1ab-lldp/lldp/remote-statistics/remote-deletes
- 3253 [o] /ieee802-dot1ab-lldp/lldp/remote-statistics/remote-drops
- 3254 [o] /ieee802-dot1ab-lldp/lldp/remote-statistics/remote-ageouts
- 3255 [m] /ieee802-dot1ab-lldp/lldp/local-system-data
- 3256 [o] /ieee802-dot1ab-lldp/lldp/local-system-data/chassis-id-subtype
- 3257 [o] /ieee802-dot1ab-lldp/lldp/local-system-data/chassis-id
- 3258 [o] /ieee802-dot1ab-lldp/lldp/local-system-data/system-name
- 3259 [o] /ieee802-dot1ab-lldp/lldp/local-system-data/system-description
- 3260 [m] /ieee802-dot1ab-lldp/lldp/local-system-data/system-capabilities-
3261 supported
- 3262 [o] /ieee802-dot1ab-lldp/lldp/local-system-data/system-capabilities-
3263 enabled
- 3264 [o] /ieee802-dot1ab-lldp/lldp/port
- 3265 [o] /ieee802-dot1ab-lldp/lldp/port/name
- 3266 [o] /ieee802-dot1ab-lldp/lldp/port/dest-mac-address
- 3267 [o] /ieee802-dot1ab-lldp/lldp/port/admin-status

3268 [o] /ieee802-dot1ab-lldp/lldp/port/notification-enable

3269 [o] /ieee802-dot1ab-lldp/lldp/port/tlvs-tx-enable

3270 [o] /ieee802-dot1ab-lldp/lldp/port/message-fast-tx

3271 [o] /ieee802-dot1ab-lldp/lldp/port/message-tx-hold-multiplier

3272 [o] /ieee802-dot1ab-lldp/lldp/port/message-tx-interval

3273 [o] /ieee802-dot1ab-lldp/lldp/port/reinit-delay

3274 [o] /ieee802-dot1ab-lldp/lldp/port/tx-credit-max

3275 [o] /ieee802-dot1ab-lldp/lldp/port/tx-fast-init

3276 [o] /ieee802-dot1ab-lldp/lldp/port/notification-interval

3277 [o] /ieee802-dot1ab-lldp/lldp/port/management-address-tx-port

3278 [o] /ieee802-dot1ab-lldp/lldp/port/port-id-subtype

3279 [o] /ieee802-dot1ab-lldp/lldp/port/port-id

3280 [o] /ieee802-dot1ab-lldp/lldp/port/port-desc

3281 [o] /ieee802-dot1ab-lldp/lldp/port/remote-systems-data

3282 **6.4.9.2.3 Synchronization**

3283 **6.4.9.2.3.1 Timesync**

3284 IA-stations shall support the ieee1588-ptp-tt YANG module according to IEEE Std 1588e™-
3285 2024 with the following features:

- 3286 • cmls (Common Mean Link Delay Service), and
- 3287 • external-port-config.

3288 IA-stations shall support the ieee1588-ptp-tt YANG module according to IEEE Std 1588e-2024
3289 with the following nodes:

3290 [o] /ieee1588-ptp-tt/ptp/instances/instance/instance-index

3291 [o] /ieee1588-ptp-tt/ptp/instances/instance/default-ds/clock-identity

3292 [o] /ieee1588-ptp-tt/ptp/instances/instance/default-ds/number-ports

3293 [o] /ieee1588-ptp-tt/ptp/instances/instance/default-ds/priority1

3294 [o] /ieee1588-ptp-tt/ptp/instances/instance/default-ds/domain-number

3295 [o] /ieee1588-ptp-tt/ptp/instances/instance/default-ds/time-receiver-
3296 only

3297 [o] /ieee1588-ptp-tt/ptp/instances/instance/default-ds/sdo-id

3298 [o] /ieee1588-ntp-tt/ntp/instances/instance/default-ds/instance-enable

3299 [o] /ieee1588-ntp-tt/ntp/instances/instance/default-ds/external-port-
3300 config-enable

3301 [o] /ieee1588-ntp-tt/ntp/instances/instance/default-ds/instance-type

3302 [o] /ieee1588-ntp-tt/ntp/instances/instance/description-ds/user-
3303 description

3304 [o] /ieee1588-ntp-tt/ntp/instances/ports/port/port-index

3305 [o] /ieee1588-ntp-tt/ntp/instances/ports/port/underlying-interface

3306 [o] /ieee1588-ntp-tt/ntp/instances/ports/port/port-ds/port-state

3307 [o] /ieee1588-ntp-tt/ntp/instances/ports/port/port-ds/delay-mechanism

3308 [o] /ieee1588-ntp-tt/ntp/instances/ports/port/port-ds/port-enable

3309 [o] /ieee1588-ntp-tt/ntp/instances/ports/port/external-port-config-
3310 port-ds/desired-state

3311 [o] /ieee1588-ntp-tt/ntp/common-services/cmlds/default-ds/clock-
3312 identity

3313 [o] /ieee1588-ntp-tt/ntp/common-services/cmlds/default-ds/number-link-
3314 ports

3315 [o] /ieee1588-ntp-tt/ntp/common-services/cmlds/ports/port/port-index

3316 [o] /ieee1588-ntp-tt/ntp/common-services/cmlds/ports/port/underlying-
3317 interface

3318 [o] /ieee1588-ntp-tt/ntp/common-services/cmlds/ports/port/link-port-
3319 ds/port-identity/clock-identity

3320 [o] /ieee1588-ntp-tt/ntp/common-services/cmlds/ports/port/link-port-
3321 ds/port-identity/port-number

3322 [o] /ieee1588-ntp-tt/ntp/common-services/cmlds/ports/port/link-port-
3323 ds/domain-number

3324 [o] /ieee1588-ntp-tt/ntp/common-services/cmlds/ports/port/link-port-
3325 ds/service-measurement-valid

3326 [o] /ieee1588-ntp-tt/ntp/common-services/cmlds/ports/port/link-port-
3327 ds/mean-link-delay

3328 [o] /ieee1588-ntp-tt/ntp/common-services/cmlds/ports/port/link-port-
3329 ds/scaled-neighbor-rate-ratio

3330 [o] /ieee1588-ntp-tt/ntp/common-services/cmlds/ports/port/link-port-
3331 ds/log-min-pdelay-req-interval

3332 [m] /ieee1588-ptp-tt/ptp/common-services/cmlsds/ports/port/link-port-
3333 ds/version-number

3334 [m] /ieee1588-ptp-tt/ptp/common-services/cmlsds/ports/port/link-port-
3335 ds/minor-version-number

3336 [o] /ieee1588-ptp-tt/ptp/common-services/cmlsds/ports/port/link-port-
3337 ds/delay-asymmetry

3338 **6.4.9.2.3.2 Timesync (ieee802-dot1as-gptp)**

3339 IA-stations shall support the ieee802-dot1as-gptp YANG module according to EEE Std
3340 802.1ASdn-2024™ with the following nodes:

3341 [o] /ieee802-dot1as-gptp/ptp/instances/instance/default-ds/gm-capable

3342 [o] /ieee802-dot1as-gptp/ptp/instances/instance/default-ds/current-
3343 utc-offset-valid

3344 [o] /ieee802-dot1as-gptp/ptp/instances/instance/default-ds/ptp-
3345 timescale

3346 [o] /ieee802-dot1as-gptp/ptp/instances/ports/port/port-ds/sync-
3347 receipt-timeout

3348 [o] /ieee802-dot1as-gptp/ptp/instances/ports/port/port-ds/current-one-
3349 step-tx-oper

3350 [o] /ieee802-dot1as-gptp/ptp/instances/ports/port/port-ds/use-mgt-one-
3351 step-tx-oper

3352 [o] /ieee802-dot1as-gptp/ptp/instances/ports/port/port-ds/mgt-one-
3353 step-tx-oper

3354 [o] /ieee802-dot1as-gptp/ptp/instances/ports/port/port-ds/sync-locked

3355 [o] /ieee802-dot1as-gptp/ptp/common-services/cmlsds/ports/port/link-
3356 port-ds/cmlsds-link-port-enabled

3357 [o] /ieee802-dot1as-gptp/ptp/common-services/cmlsds/ports/port/link-
3358 port-ds/is-measuring-delay

3359 [o] /ieee802-dot1as-gptp/ptp/common-services/cmlsds/ports/port/link-
3360 port-ds/as-capable-across-domains

3361 [o] /ieee802-dot1as-gptp/ptp/common-services/cmlsds/ports/port/link-
3362 port-ds/mean-link-delay-thresh

3363 [o] /ieee802-dot1as-gptp/ptp/common-services/cmlsds/ports/port/link-
3364 port-ds/current-log-pdelay-req-interval

3365 [o] /ieee802-dot1as-gptp/ptp/common-services/cmlsds/ports/port/link-
3366 port-ds/use-mgt-log-pdelay-req-interval

3367 [o] /ieee802-dot1as-gptp/ptp/common-services/cmlsds/ports/port/link-
3368 port-ds/mgt-log-pdelay-req-interval

3369 [o] /ieee802-dot1as-gptp/ptp/common-services/cmls/ports/port/link-
 3370 port-ds/current-compute-rate-ratio

3371 [o] /ieee802-dot1as-gptp/ptp/common-services/cmls/ports/port/link-
 3372 port-ds/use-mgt-compute-rate-ratio

3373 [o] /ieee802-dot1as-gptp/ptp/common-services/cmls/ports/port/link-
 3374 port-ds/mgt-compute-rate-ratio

3375 [o] /ieee802-dot1as-gptp/ptp/common-services/cmls/ports/port/link-
 3376 port-ds/current-compute-mean-link-delay

3377 [o] /ieee802-dot1as-gptp/ptp/common-services/cmls/ports/port/link-
 3378 port-ds/use-mgt-compute-mean-link-delay

3379 [o] /ieee802-dot1as-gptp/ptp/common-services/cmls/ports/port/link-
 3380 port-ds/mgt-compute-mean-link-delay

3381 [o] /ieee802-dot1as-gptp/ptp/common-services/cmls/ports/port/link-
 3382 port-ds/allowed-lost-responses

3383 [o] /ieee802-dot1as-gptp/ptp/common-services/cmls/ports/port/link-
 3384 port-ds/allowed-faults

3385 **6.4.9.2.3 Timesync (ieee802-dot1as-hs)**

3386 IA-stations shall support the ieee802-dot1as-hs YANG module according to IEEE Std
 3387 802.1ASdm-2024 with the following nodes:

3388 [o] /ieee802-dot1as-hs/ptp/instances/instance/parent-ds/gmPresent

3389 [o] /ieee802-dot1as-hs/ptp/instances/instance/ptp-instance-ds/is-
 3390 synced

3391 [o] /ieee802-dot1as-hs/ptp/instances/instance/ptp-instance-ds/nrr-
 3392 pdelay

3393 [o] /ieee802-dot1as-hs/ptp/instances/instance/ptp-instance-ds/nrr-sync

3394 [o] /ieee802-dot1as-hs/ptp/instances/instance/ptp-instance-ds/nrr-
 3395 comp-method

3396 [o] /ieee802-dot1as-hs/ptp/instances/instance/ptp-instance-ds/as-
 3397 capable-across-domains

3398 [o] /ieee802-dot1as-hs/ptp/instances/instance/ptp-instance-ds/gptp-
 3399 capable-state-machines-enabled

3400 **6.4.9.2.4 Security configuration modules**

3401 **6.4.9.2.4.1 YANG module for a keystore**

3402 IA-stations shall support the ietf-keystore YANG module according to IETF RFC 9642 with the
 3403 following features:

- 3404 • central-truststore-supported, and
- 3405 • asymmetric-keys.

3406 IA-stations shall support the ietf-keystore YANG module according to IETF RFC 9642 with the
3407 following nodes:

- 3408 [o] /ietf-keystore/keystore/asymmetric-keys/asymmetric-key/name
- 3409 [o] /ietf-keystore/keystore/asymmetric-keys/asymmetric-key/public-key-
3410 format
- 3411 [o] /ietf-keystore/keystore/asymmetric-keys/asymmetric-key/public-key
- 3412 [o] /ietf-keystore/keystore/asymmetric-keys/asymmetric-key/private-
3413 key-format
- 3414 [o] /ietf-keystore/keystore/asymmetric-keys/asymmetric-key/hidden-
3415 private-key
- 3416 [o] /ietf-keystore/certificates/certificate/name
- 3417 [o] /ietf-keystore/certificates/certificate/cert-data
- 3418 [o] /ietf-keystore/certificates/certificate/expiration-date
- 3419 [o] /ietf-keystore/certificates/certificate/csr-info
- 3420 [o] /ietf-keystore/certificates/certificate/certificate-signing-
3421 request

3422 **6.4.9.2.4.2 Network configuration access control**

3423 IA-stations shall support the ietf-netconf-acm YANG module according to IETF RFC 8341 with
3424 the following nodes:

- 3425 [o] /ietf-netconf-acm/nacm/enable-nacm
- 3426 [o] /ietf-netconf-acm/nacm/read-default
- 3427 [o] /ietf-netconf-acm/nacm/write-default
- 3428 [o] /ietf-netconf-acm/nacm/exec-default
- 3429 [o] /ietf-netconf-acm/nacm/enable-external-groups
- 3430 [o] /ietf-netconf-acm/nacm/groups
- 3431 [o] /ietf-netconf-acm/nacm/rule-list

3432 **6.4.9.2.4.3 A YANG data module for a truststore**

3433 IA-stations shall support the ietf-truststore YANG module according to IETF RFC 9641 with the
3434 following features:

- 3435 • central-keystore-supported, and
- 3436 • certificates.

3437 IA-stations shall support the ietf-truststore YANG module according to IETF RFC 9641 with the
3438 following nodes:

3439 [o] /ietf-truststore/truststore/certificate-bags/certificate-bag/name

3440 [o] /ietf-truststore/truststore/certificate-bags/certificate-
3441 bag/certificate/name

3442 [o] /ietf-truststore/truststore/certificate-bags/certificate-
3443 bag/certificate/cert-data

3444 [o] /ietf-truststore/truststore/certificate-bags/certificate-
3445 bag/certificate/expiration-date

3446 **6.4.9.2.5 IA-station management**

3447 **6.4.9.2.5.1 System capabilities**

3448 IA-stations shall support the ietf-system-capabilities YANG modules according to IETF RFC
3449 9196 with the following nodes:

3450 [m] /ietf-system-capabilities/system-capabilities/datastore-
3451 capabilities/datastore

3452 [m] /ietf-system-capabilities/system-capabilities/datastore-
3453 capabilities/per-node-capabilities

3454 [m] /ietf-system-capabilities/system-capabilities/datastore-
3455 capabilities/on-change-supported

3456 **6.4.9.2.5.2 YANG library**

3457 IA-stations shall support the ietf-yang-library YANG module according to IETF RFC 8525 with
3458 the following nodes:

3459 [m] /ietf-yang-library/yang-library/module-set

3460 [m] /ietf-yang-library/yang-library/schema

3461 [m] /ietf-yang-library/yang-library/datastore

3462 [m] /ietf-yang-library/yang-library/content-id

3463 **6.4.9.2.5.3 YANG push**

3464 IA-stations shall support the ietf-yang-push YANG module according to IETF RFC 8641, 4.1,
3465 with the on-change feature.

3466 IA-stations shall support the ietf-yang-push YANG module according to IETF RFC 8641, 4.1,
3467 with the following nodes:

3468 [o] /ietf-subscribed-notifications/filters/selection-filter

3469 [o] /ietf-subscribed-
3470 notifications/subscriptions/subscription/target/datastore

3471 [o] /ietf-subscribed-notifications/subscriptions/subscription/update-
3472 trigger

3473 **6.4.9.2.5.4 YANG notification capabilities**

3474 IA-stations shall support the ietf-notification-capabilities YANG module according to IETF RFC
3475 9196 with the following nodes:

3476 [m] /ietf-notification-capabilities/system-capabilities/subscription-
3477 capabilities

3478 [m] /ietf-notification-capabilities/system-capabilities/datastore-
3479 capabilities/per-node-capabilities/subscription-capabilities

3480 **6.4.9.2.5.5 YANG notifications**

3481 IA-stations shall support the ietf-subscribed-notifications YANG module according to IETF RFC
3482 8639 with the following features:

- 3483 • Configured,
- 3484 • encode-xml, and
- 3485 • subtree.

3486 IA-stations shall support the ietf-subscribed-notifications YANG module according to IETF RFC
3487 8639 with the following nodes:

3488 [o] /ietf-subscribed-notifications/streams/stream/name

3489 [o] /ietf-subscribed-notifications/streams/stream/description

3490 [o] /ietf-subscribed-notifications/streams/stream/replay-support

3491 [o] /ietf-subscribed-notifications/streams/stream/replay-log-creation-
3492 time

3493 [o] /ietf-subscribed-notifications/streams/stream/replay-log-aged-time

3494 [o] /ietf-subscribed-notifications/filters/stream-filter/name

3495 [o] /ietf-subscribed-notifications/filters/stream-filter/filter-spec

3496 [o] /ietf-subscribed-notifications/subscriptions/subscription/id

3497 [o] /ietf-subscribed-notifications/subscriptions/subscription/target

3498 [o] /ietf-subscribed-notifications/subscriptions/subscription/stop-
3499 time

3500 [o] /ietf-subscribed-notifications/subscriptions/subscription/dscp

3501 [o] /ietf-subscribed-
3502 notifications/subscriptions/subscription/weighting

3503 [o] /ietf-subscribed-
3504 notifications/subscriptions/subscription/dependency

3505 [o] /ietf-subscribed-
3506 notifications/subscriptions/subscription/transport

3507 [o] /ietf-subscribed-notifications/subscriptions/subscription/encoding

3508 [o] /ietf-subscribed-notifications/subscriptions/subscription/purpose

3509 [o] /ietf-subscribed-
3510 notifications/subscriptions/subscription/notification-message-origin

3511 [o] /ietf-subscribed-
3512 notifications/subscriptions/subscription/configured-subscription-state

3513 [o] /ietf-subscribed-
3514 notifications/subscriptions/subscription/receivers

3515 **6.4.9.2.5.6 NETCONF monitoring**

3516 IA-stations shall support the ietf-netconf-monitoring YANG module according to IETF RFC 6022
3517 with the following nodes:

3518 [m] /ietf-netconf-monitoring/netconf-state/capabilities

3519 [m] /ietf-netconf-monitoring/netconf-state/datastores

3520 [m] /ietf-netconf-monitoring/netconf-state/schemas

3521 **6.4.9.2.5.7 System management**

3522 IA-stations shall support the ietf-system YANG module according to IETF RFC 7317 with the
3523 following nodes:

3524 [o] /ietf-system/system/contact

3525 [o] /ietf-system/system/hostname

3526 [o] /ietf-system/system/location

3527 **6.4.9.2.5.8 Hardware management**

3528 IA-stations shall support the ietf-hardware YANG module according to IETF RFC 8348 with the
3529 following nodes:

3530 [m] /ietf-hardware/hardware/component/name

3531 [m] /ietf-hardware/hardware/component/class

3532 [m] /ietf-hardware/hardware/component/description

3533 [m] /ietf-hardware/hardware/component/hardware-rev

3534 [m] /ietf-hardware/hardware/component/software-rev

3535 [o] /ietf-hardware/hardware/component/serial-num

3536 [m] /ietf-hardware/hardware/component/mfg-name

3537 [m] /ietf-hardware/hardware/component/model-name

3538 An IA-station shall provide exactly one /ietf-hardware/component with class "chassis" and may
3539 provide further components with other classes.

3540 **6.4.9.2.5.9 Interface management**

3541 IA-stations shall support the ietf-interfaces YANG module according to IETF RFC 8343 with the
3542 following nodes:

3543 [m] /ietf-interfaces/interfaces/interface/name

3544 [m] /ietf-interfaces/interfaces/interface/description

3545 [m] /ietf-interfaces/interfaces/interface/type

3546 [o] /ietf-interfaces/interfaces/interface/enabled

3547 [o] /ietf-interfaces/interfaces/interface/oper-status

3548 [o] /ietf-interfaces/interfaces/interface/phys-address

3549 [o] /ietf-interfaces/interfaces/interface/higher-layer-if

3550 [o] /ietf-interfaces/interfaces/interface/lower-layer-if

3551 [o] /ietf-interfaces/interfaces/interface/speed

3552 [o] /ietf-interfaces/interfaces/interface/statistics/discontinuity-
3553 time

3554 [o] /ietf-interfaces/interfaces/interface/statistics/in-octets

3555 [o] /ietf-interfaces/interfaces/interface/statistics/in-discards

3556 [o] /ietf-interfaces/interfaces/interface/statistics/in-errors

3557 [o] /ietf-interfaces/interfaces/interface/statistics/out-octets

3558 [o] /ietf-interfaces/interfaces/interface/statistics/out-discards

3559 [o] /ietf-interfaces/interfaces/interface/statistics/out-errors

3560 **6.4.9.2.5.10 Bridge and end station component management**

3561 **6.4.9.2.5.10.1 General**

3562 IA-stations shall support the ieee802-dot1q-bridge YANG module according to
3563 IEEE Std 802.1Q-2022, Clause 48, as amended by IEEE Std 802.1Qcw™-2023 with the
3564 following feature: ingress-filtering.

3565 IA-stations shall support the ieee802-dot1q-bridge YANG module according to
3566 IEEE Std 802.1Q-2022, Clause 48, as amended by IEEE Std 802.1Qcw-2023 with the following
3567 nodes. A distinction is made between nodes that shall be supported by bridge and end station
3568 components, or by bridge components only.

3569 6.4.9.2.5.10.2 Bridge nodes to be supported by bridge and end station components

3570 [m] /ieee802-dot1q-bridge/bridges/bridge/name

3571 [o] /ieee802-dot1q-bridge/bridges/bridge/address

3572 [m] /ieee802-dot1q-bridge/bridges/bridge/bridge-type

3573 [m] /ieee802-dot1q-bridge/bridges/bridge/ports

3574 [m] /ieee802-dot1q-bridge/bridges/bridge/components

3575 [m] /ieee802-dot1q-bridge/bridges/bridge/component/name

3576 [o] /ieee802-dot1q-bridge/bridges/bridge/component/id

3577 [m] /ieee802-dot1q-bridge/bridges/bridge/component/type

3578 [o] /ieee802-dot1q-bridge/bridges/bridge/component/traffic-class-
3579 enabled

3580 [m] /ieee802-dot1q-bridge/bridges/bridge/component/ports

3581 [o] /ieee802-dot1q-bridge/bridges/bridge/component/bridge-port

3582 [m] /ieee802-dot1q-bridge/bridges/bridge/component/capabilities

3583 [m] /ieee802-dot1q-bridge/bridges/bridge/component/filtering-
3584 database/size

3585 [o] /ieee802-dot1q-bridge/bridges/bridge/component/filtering-
3586 database/static-vlan-registration-entries

3587 [o] /ieee802-dot1q-bridge/bridges/bridge/component/filtering-
3588 database/vlan-registration-entry

3589 6.4.9.2.5.10.3 Filtering-database nodes to be supported by bridge components

3590 [o] /ieee802-dot1q-bridge/bridges/bridge/component/filtering-
3591 database/aging-time

3592 [o] /ieee802-dot1q-bridge/bridges/bridge/component/filtering-
3593 database/static-entries

3594 [o] /ieee802-dot1q-bridge/bridges/bridge/component/filtering-
3595 database/dynamic-entries

3596 [o] /ieee802-dot1q-bridge/bridges/bridge/component/filtering-
3597 database/dynamic-vlan-registration-entries

3598 [o] /ieee802-dot1q-bridge/bridges/bridge/component/filtering-
3599 database/mac-address-registration-entries

3600 [o] /ieee802-dot1q-bridge/bridges/bridge/component/filtering-
3601 database/filtering-entry

3602 **6.4.9.2.5.10.4 Permanent-database nodes to be supported by bridge components**

3603 [m] /ieee802-dot1q-bridge/bridges/bridge/component/permanent-
3604 database/size

3605 [o] /ieee802-dot1q-bridge/bridges/bridge/component/permanent-
3606 database/static-entries

3607 [o] /ieee802-dot1q-bridge/bridges/bridge/component/permanent-
3608 database/static-vlan-registration-entries

3609 [o] /ieee802-dot1q-bridge/bridges/bridge/component/permanent-
3610 database/filtering-entry

3611 **6.4.9.2.5.10.5 Bridge-vlan nodes to be supported by bridge and end station components**

3612 [m] /ieee802-dot1q-bridge/bridges/bridge/component/bridge-vlan/version

3613 [m] /ieee802-dot1q-bridge/bridges/bridge/component/bridge-vlan/max-
3614 vids

3615 [o] /ieee802-dot1q-bridge/bridges/bridge/component/bridge-
3616 vlan/override-default-pvid

3617 [o] /ieee802-dot1q-bridge/bridges/bridge/component/bridge-vlan/vlan

3618 **6.4.9.2.5.10.6 Bridge-vlan nodes to be supported by bridge components**

3619 [m] /ieee802-dot1q-bridge/bridges/bridge/component/bridge-vlan/max-
3620 msti

3621 [o] /ieee802-dot1q-bridge/bridges/bridge/component/bridge-vlan/vid-to-
3622 fid-allocation

3623 [o] /ieee802-dot1q-bridge/bridges/bridge/component/bridge-vlan/fid-to-
3624 vid-allocation

3625 [o] /ieee802-dot1q-bridge/bridges/bridge/component/bridge-vlan/vid-to-
3626 fid

3627 **6.4.9.2.5.10.7 Bridge-mst nodes to be supported by bridge components**

3628 [o] /ieee802-dot1q-bridge/bridges/bridge/component/bridge-mst

3629 **6.4.9.2.5.10.8 Bridge-port nodes to be supported by bridge and end station components**

3630 [m] /ietf-interfaces/interfaces/interface/bridge-port/bridge-name

3631 [m] /ietf-interfaces/interfaces/interface/bridge-port/component-name

3632 [m] /ietf-interfaces/interfaces/interface/bridge-port/port-type

3633 [o] /ietf-interfaces/interfaces/interface/bridge-port/pvid

3634 [o] /ietf-interfaces/interfaces/interface/bridge-port/default-priority

3635 [m] /ietf-interfaces/interfaces/interface/bridge-port/traffic-class

3636 [o] /ietf-interfaces/interfaces/interface/bridge-port/statistics
3637 [m] /ietf-interfaces/interfaces/interface/bridge-port/capabilities
3638 [m] /ietf-interfaces/interfaces/interface/bridge-port/type-capabilities
3639 [o] /ietf-interfaces/interfaces/interface/bridge-port/transmission-
3640 selection-algorithm-table

3641 **6.4.9.2.5.10.9 Bridge-port nodes to be supported by bridge component ports**

3642 [o] /ietf-interfaces/interfaces/interface/bridge-port/priority-
3643 regeneration
3644 [o] /ietf-interfaces/interfaces/interface/bridge-port/acceptable-frame
3645 [o] /ietf-interfaces/interfaces/interface/bridge-port/enable-ingress-
3646 filtering
3647 [o] /ietf-interfaces/interfaces/interface/bridge-port/enable-vid-
3648 translation-table
3649 [o] /ietf-interfaces/interfaces/interface/bridge-port/vid-translations
3650 [o] /ietf-interfaces/interfaces/interface/bridge-port/enable-egress-
3651 vid-translation-table
3652 [o] /ietf-interfaces/interfaces/interface/bridge-port/egress-vid-
3653 translations

3654 **6.4.9.2.5.11 IEC/IEEE 60802 YANG modules**

3655 IA-stations shall support the *ieciieee60802-ethernet-interface* YANG module according to this
3656 document with the following nodes:

3657 [m] /ieciieee60802-ethernet-
3658 interface/interfaces/interface/ethernet/supported-mau-types/mau-type
3659 [m] /ieciieee60802-ethernet-
3660 interface/interfaces/interface/ethernet/supported-mau-
3661 types/preemption-supported

3662 IA-stations shall support the *ieciieee60802-bridge* YANG module according to this document
3663 with the following nodes:

3664 [m] /ieciieee60802-bridge/interfaces/interface/bridge-port/max-burst-
3665 params

3666 [m] /ieciieee60802-bridge/interfaces/interface/bridge-port/committed-
3667 data-rates

3668 [m] /ieciieee60802-bridge/interfaces/interface/bridge-
3669 port/transmission-selection-algorithm

3670 [m] /ieciieee60802/interfaces/interface/bridge-port/supported-resource-
3671 pools

3672 [m] /iecieee60802-bridge/bridges/bridge/component/frer-supported

3673 [m] /iecieee60802-bridge/bridges/bridge/component/max-redundant-
3674 streams

3675 [m] /iecieee60802-bridge/bridges/bridge/component/max-fids

3676 [m] /iecieee60802-bridge/bridges/bridge/component/max-fdb-entries

3677 [m] /iecieee60802-bridge/bridges/bridge/component/delay-variance

3678 [m] /iecieee60802-bridge/bridges/bridge/component/max-ptp-instances

3679 [m] /iecieee60802-bridge/bridges/bridge/component/max-hot-standby-
3680 systems

3681 [m] /iecieee60802-bridge/bridges/bridge/component/clock

3682 [m] /iecieee60802/interfaces/interface/bridge-port/phy-latencies/mau-
3683 type

3684 [m] /iecieee60802/interfaces/interface/bridge-port/phy-
3685 latencies/latency

3686 [m] /iecieee60802/interfaces/interface/bridge-port/phy-
3687 latencies/latency-variation

3688 **IA-stations shall support the iecieee60802-ia-station YANG module according to this document**
3689 **with the following nodes:**

3690 [m] /iecieee60802-ia-station/ia-station-capabilities/capability-lldp

3691 [m] /iecieee60802-ia-station/ia-station-capabilities/capability-timesync

3692 [m] /iecieee60802-ia-station/ia-station-capabilities/capability-keystore

3693 [m] /iecieee60802-ia-station/ia-station-capabilities/capability-truststore

3694 [m] /iecieee60802-ia-station/ia-station-capabilities/capability-nacm

3695 [m] /iecieee60802-ia-station/ia-station-capabilities/capability-yang-library

3696 [m] /iecieee60802-ia-station/ia-station-capabilities/capability-yang-push

3697 [m] /iecieee60802-ia-station/ia-station-capabilities/capability-yang-
3698 notifications

3699 [m] /iecieee60802-ia-station/ia-station-capabilities/capability-netconf-
3700 monitoring

3701 [m] /iecieee60802-ia-station/ia-station-capabilities/capability-netconf-
3702 client

3703 [m] /iecieee60802-ia-station/ia-station-capabilities/capability-cnc-config

3704 [m] /iecieeee60802-ia-station/ia-station-capabilities/capability-sched-
3705 traffic

3706 [m] /iecieeee60802-ia-station/ia-station-capabilities/capability-frame-
3707 preemption

3708 IA-stations shall support the iecieeee60802-subscribed-notifications YANG module according to
3709 this document with the following nodes:

3710 [m] /iecieeee60802-subscribed-notifications/subscriptions/max-
3711 subscriptions

3712 [m] /iecieeee60802-subscribed-notifications/subscriptions/max-on-
3713 change-subscription-leaves

3714 [m] /iecieeee60802-subscribed-notifications/subscriptions/max-periodic-
3715 subscription-leaves

3716 [m] /iecieeee60802-subscribed-notifications/subscriptions/max-periodic-
3717 subscription-interval

3718 IA-stations shall support the iecieeee60802-proxy YANG module according to this document with
3719 the following nodes:

3720 [c] /iecieeee60802-proxy/proxy/max-num-proxy-targets

3721 [c] /iecieeee60802-proxy/proxy/proxy-types

3722 [o] /iecieeee60802-proxy/proxy/proxy-targets

3723 IA-stations shall support the iecieeee60802-proxy-target YANG module according to this
3724 document with the following nodes:

3725 [c] /iecieeee60802-proxy-target/proxy-target/required-proxy-type

3726 **6.4.9.2.5.12 NETCONF server**

3727 IA-stations shall support the ietf-netconf-server YANG module according to draft-ietf-netconf-
3728 netconf-client-server, 3.1.1, with the following features:

- 3729 • tls-call-home, and
3730 • central-netconf-server-supported.

3731 IA-stations shall support the ietf-netconf-server YANG module according to draft-ietf-netconf-
3732 netconf-client-server, 3.3, with the following nodes:

3733 [o] /ietf-netconf-server/netconf-server/listen/idle-timeout

3734 [o] /ietf-netconf-server/netconf-server/listen/endpoint/name

3735 [o] /ietf-netconf-server/netconf-
3736 server/listen/endpoint/transport/tls/netconf-server-parameters

3737 [o] /ietf-netconf-server/netconf-
3738 server/listen/endpoint/transport/tls/tls-server-parameters

3739 [o] /ietf-netconf-server/netconf-server/call-home/netconf-client/name

3740 [o] /ietf-netconf-server/netconf-server/call-home/netconf-
3741 client/endpoints/endpoint/name

3742 [o] /ietf-netconf-server/netconf-server/call-home/netconf-
3743 client/endpoints/endpoint/transport/tls/netconf-server-parameters

3744 [o] /ietf-netconf-server/netconf-server/call-home/netconf-
3745 client/endpoints/endpoint/transport/tls/tls-server-parameters

3746 **6.4.9.2.5.13 Flow Meter Management**

3747 IA-stations which incorporate a bridge component shall support the ieee802-dot1q-stream-
3748 filters-gates YANG module according to IEEE Std 802.1Qcz-2023 as amended by IEEE Std
3749 802.1Qcw-2023 with the following nodes:

3750 [o] /ieee802-dot1q-bridge/bridges/bridge/component/stream-
3751 filters/stream-filter-instance-table/stream-filter-instance-id

3752 [o] /ieee802-dot1q-bridge/bridges/bridge/component/stream-
3753 filters/stream-filter-instance-table/stream-handle

3754 [o] /ieee802-dot1q-bridge/bridges/bridge/component/stream-
3755 filters/stream-filter-instance-table/flow-meter-ref

3756 [o] /ieee802-dot1q-bridge/bridges/bridge/component/stream-
3757 filters/stream-filter-instance-table/flow-meter-enable

3758 [m] /ieee802-dot1q-bridge/bridges/bridge/component/stream-filters/max-
3759 stream-filter-instances

3760 [o] /ieee802-dot1q-psfp-bridge/bridges/bridge/component/stream-
3761 gates/stream-gate-instance-table

3762 [m] /ieee802-dot1q-psfp-bridge/bridges/bridge/component/stream-
3763 gates/max-stream-gate-instances

3764 [m] /ieee802-dot1q-psfp-bridge/bridges/bridge/component/stream-
3765 gates/supported-list-max

3766 IA-stations which incorporate a bridge component shall support the ieee802-dot1cb-stream-
3767 identification YANG module according to IEEE Std 802.1CBcv-2021 as amended by IEEE Std
3768 802.1CBdb-2021 with the following nodes:

3769 [o] /ieee802-dot1cb-stream-identification/stream-identity/index

3770 [o] /ieee802-dot1cb-stream-identification/stream-identity/handle

3771 [o] /ieee802-dot1cb-stream-identification/stream-identity/out-
3772 facing/input-port

3773 [o] /ieee802-dot1cb-stream-identification/stream-
3774 identity/parameters/mask-and-match-stream-identification/destination-
3775 mac-mask

3776 [o] /ieee802-dot1cb-stream-identification/stream-
 3777 identity/parameters/mask-and-match-stream-identification/destination-
 3778 mac-match

3779 NOTE For example, an implementation could contain per out-facing/input-port one mask and match stream
 3780 identification for broadcast traffic, one mask and match stream identification for multicast traffic and one mask and
 3781 match stream identification for unicast traffic.

3782 IA-stations which incorporate a bridge component shall support the ieee802-dot1q-psfp-bridge
 3783 YANG module according to IEEE Std 802.1Qcw-2023 with the following nodes:

3784 [o] /ieee802-dot1q-psfp-bridge/bridges/bridge/component/flow-
 3785 meters/flow-meter-instance-table/flow-meter-instance-id

3786 [o] /ieee802-dot1q-psfp-bridge/bridges/bridge/component/flow-
 3787 meters/flow-meter-instance-table/committed-information-rate

3788 [o] /ieee802-dot1q-psfp-bridge/bridges/bridge/component/flow-
 3789 meters/flow-meter-instance-table/committed-burst-size

3790 [o] /ieee802-dot1q-psfp-bridge/bridges/bridge/component/flow-
 3791 meters/flow-meter-instance-table/excess-information-rate

3792 [o] /ieee802-dot1q-psfp-bridge/bridges/bridge/component/flow-
 3793 meters/flow-meter-instance-table/excess-burst-size

3794 [o] /ieee802-dot1q-psfp-bridge/bridges/bridge/component/flow-
 3795 meters/flow-meter-instance-table/coupling-flag

3796 [o] /ieee802-dot1q-psfp-bridge/bridges/bridge/component/flow-
 3797 meters/flow-meter-instance-table/color-mode

3798 [o] /ieee802-dot1q-psfp-bridge/bridges/bridge/component/flow-
 3799 meters/flow-meter-instance-table/drop-on-yellow

3800 [m] /ieee802-dot1q-psfp-bridge/bridges/bridge/component/flow-
 3801 meters/max-flow-meter-instances

3802 **6.4.9.2.5.14 RSTP/MSTP**

3803 IA-stations shall support the <ieee802-dot1q-rstp-bridge> YANG module according to IEEE Std
 3804 802.1Qdy™-2024 with the following nodes.

3805 [o] /ieee802-dot1q-rstp-bridge/bridges/bridge/rstp

3806 [o] /ieee802-dot1q-rstp-bridge/interfaces/interface/bridge-port/rstp

3807 IA-stations shall support the <ieee802-dot1q-mstp-bridge> YANG module according to IEEE
 3808 Std 802.1Qdy-2024 with the following nodes.

3809 [o] /ieee802-dot1q-mstp-bridge/bridges/bridge/component/bridge-
 3810 mst/bridge-mstp

3811 [o] /ieee802-dot1q-mstp-bridge/interfaces/interface/bridge-port/port-
 3812 mstp

3813 **6.4.9.3 Optional YANG data models, features, and nodes**

3814 **6.4.9.3.1 General**

3815 The following YANG modules, features and nodes shall be supported by IA-stations if the
3816 functionality they describe is included.

3817 **6.4.9.3.2 Scheduled traffic**

3818 IA-stations supporting the enhancements for scheduled traffic shall support the ieee802-dot1q-
3819 sched-bridge YANG module according to IEEE Std 802.1Qcw-2023 with the following feature:
3820 `scheduled-traffic`.

3821 IA-stations supporting the enhancements for scheduled traffic shall support the ieee802-dot1q-
3822 sched-bridge YANG module according to IEEE Std 802.1Qcw-2023 with the following nodes:

3823 [o] `ieee802-dot1q-sched-bridge/interfaces/interface/bridge-port/gate-`
3824 `parameter-table/queue-max-sdu-table`

3825 [o] `ieee802-dot1q-sched-bridge/interfaces/interface/bridge-port/gate-`
3826 `parameter-table/gate-enabled`

3827 [o] `ieee802-dot1q-sched-bridge/interfaces/interface/bridge-port/gate-`
3828 `parameter-table/admin-gate-states`

3829 [o] `ieee802-dot1q-sched-bridge/interfaces/interface/bridge-port/gate-`
3830 `parameter-table/oper-gate-states`

3831 [o] `ieee802-dot1q-sched-bridge/interfaces/interface/bridge-port/gate-`
3832 `parameter-table/admin-control-list`

3833 [o] `ieee802-dot1q-sched-bridge/interfaces/interface/bridge-port/gate-`
3834 `parameter-table/oper-control-list`

3835 [o] `ieee802-dot1q-sched-bridge/interfaces/interface/bridge-port/gate-`
3836 `parameter-table/admin-cycle-time`

3837 [o] `ieee802-dot1q-sched-bridge/interfaces/interface/bridge-port/gate-`
3838 `parameter-table/oper-cycle-time`

3839 [o] `ieee802-dot1q-sched-bridge/interfaces/interface/bridge-port/gate-`
3840 `parameter-table/admin-cycle-time-extension`

3841 [o] `ieee802-dot1q-sched-bridge/interfaces/interface/bridge-port/gate-`
3842 `parameter-table/oper-cycle-time-extension`

3843 [o] `ieee802-dot1q-sched-bridge/interfaces/interface/bridge-port/gate-`
3844 `parameter-table/admin-base-time`

3845 [o] `ieee802-dot1q-sched-bridge/interfaces/interface/bridge-port/gate-`
3846 `parameter-table/oper-base-time`

3847 [o] `ieee802-dot1q-sched-bridge/interfaces/interface/bridge-port/gate-`
3848 `parameter-table/config-change`

3849 [o] `ieee802-dot1q-sched-bridge/interfaces/interface/bridge-port/gate-`
3850 `parameter-table/config-change-time`

3851 [o] ieee802-dot1q-sched-bridge/interfaces/interface/bridge-port/gate-
3852 parameter-table/tick-granularity

3853 [o] ieee802-dot1q-sched-bridge/interfaces/interface/bridge-port/gate-
3854 parameter-table/current-time

3855 [o] ieee802-dot1q-sched-bridge/interfaces/interface/bridge-port/gate-
3856 parameter-table/config-pending

3857 [o] ieee802-dot1q-sched-bridge/interfaces/interface/bridge-port/gate-
3858 parameter-table/config-change-error

3859 [c] ieee802-dot1q-sched-bridge/interfaces/interface/bridge-port/gate-
3860 parameter-table/supported-list-max

3861 [c] ieee802-dot1q-sched-bridge/interfaces/interface/bridge-port/gate-
3862 parameter-table/supported-cycle-max

3863 [c] ieee802-dot1q-sched-bridge/interfaces/interface/bridge-port/gate-
3864 parameter-table/supported-interval-max

3865 **6.4.9.3.3 IEC/IEEE 60802 YANG modules**

3866 IA-stations that support enhancements for scheduled traffic shall support the iecieee60802-
3867 sched-bridge YANG module according to this document with the following nodes:

3868 [c] /ieciieee60802-sched-bridge/interfaces/interface/bridge-port/gate-
3869 parameter-table/min-gating-times

3870 **6.4.9.3.4 Frame preemption**

3871 IA-stations supporting frame preemption according to IEEE Std 802.1Q-2022, 5.4.1, ae), shall
3872 support the ieee802-dot1q-preemption-bridge YANG module according to IEEE Std
3873 802.1Qcw-2023 with the following feature: frame-preemption.

3874 IA-stations supporting frame preemption according to IEEE Std 802.1Q-2022, 5.4.1, ae), shall
3875 support the ieee802-dot1q-preemption-bridge YANG module according to IEEE Std
3876 802.1Qcw-2023 with the following nodes:

3877 [o] ieee802-dot1q-preemption-bridge/interfaces/interface/bridge-
3878 port/frame-preemption-parameters/frame-preemption-status-table

3879 [o] ieee802-dot1q-preemption-bridge/interfaces/interface/bridge-
3880 port/frame-preemption-parameters/preemption-active

3881 **6.4.9.3.5 Credit-based shaper**

3882 IA-stations supporting the credit-based shaper according to IEEE Std 802.1Q-2022, 8.6.8.2,
3883 shall support the <ieee802-dot1q-cbsa-bridge> YANG module according to IEEE Std
3884 802.1QdxTM-2024 with the following nodes.

3885 [c] /ieee802-dot1q-cbsa-bridge/interfaces/interface/bridge-
3886 port/cbsa/cbsa-parameter-table/traffic-class

3887 [c] /ieee802-dot1q-cbsa-bridge/interfaces/interface/bridge-
3888 port/cbsa/cbsa-parameter-table/admin-idle-slope

3889 [c] /ieee802-dot1q-cbsa-bridge/interfaces/interface/bridge-
3890 port/cbsa/cbsa-parameter-table/oper-idle-slope

3891 **6.4.9.3.6 FRER**

3892 IA-stations supporting FRER according to 5.10.1 item b) or item c), shall support the
3893 ieee802-dot1cb-stream-identification and ieee802-dot1cb-frer YANG modules according to
3894 IEEE Std 802.1CBcv-2021 with the following nodes:

3895 [o] /ieee802-dot1cb-stream-identification/stream-identity/index

3896 [o] /ieee802-dot1cb-stream-identification/stream-identity/handle

3897 [o] /ieee802-dot1cb-stream-identification /stream-identity/out-
3898 facing/input-port

3899 [o] /ieee802-dot1cb-stream-identification /stream-identity/out-
3900 facing/output-port

3901 [o] /ieee802-dot1cb-stream-identification /stream-
3902 identity/parameters/null-stream-identification

3903 [o] /ieee802-dot1cb-frer/frer/sequence-generation/index

3904 [o] /ieee802-dot1cb-frer/frer/sequence-generation/stream

3905 [o] /ieee802-dot1cb-frer/frer/sequence-generation/direction-out-facing

3906 [o] /ieee802-dot1cb-frer/frer/sequence-recovery/index

3907 [o] /ieee802-dot1cb-frer/frer/sequence-recovery/stream

3908 [o] /ieee802-dot1cb-frer/frer/sequence-recovery/port

3909 [o] /ieee802-dot1cb-frer/frer/sequence-recovery/direction-out-facing

3910 [o] /ieee802-dot1cb-frer/frer/sequence-recovery/algorithm/vector

3911 [o] /ieee802-dot1cb-frer/frer/sequence-identification/port

3912 [o] /ieee802-dot1cb-frer/frer/sequence-identification/direction-out-
3913 facing

3914 [o] /ieee802-dot1cb-frer/frer/sequence-identification/stream

3915 [o] /ieee802-dot1cb-frer/frer/sequence-identification/encapsulation/r-
3916 tag

3917 [o] /ieee802-dot1cb-frer/frer/stream-split

6.4.9.4 CUC/CNC YANG

6.4.9.4.1 NETCONF Client

IA-stations with CNC or CUC functionality shall support the ietf-netconf-client YANG module according to draft-ietf-netconf-netconf-client-server, 2.1.1, with the following features:

- `tls-initiate`,
- `tls-listen`, and
- `central-netconf-client-supported`.

6.4.9.4.2 YANG module for CNC-Config

IA-stations with CNC or CUC functionality shall support the ieee802-dot1q-cnc-config YANG module according to IEEE Std 802.1Qdj-2024 with the node:
[o] `/ieee802-dot1q-cnc-config/cnc-config`

6.4.10 YANG Data Model

6.4.10.1 General

Subclause 6.4.10 specifies the YANG data model for IA-stations. YANG (IETF RFC 7950) is a data modelling language used to model configuration data and state data for remote network management protocols. The selected YANG-based remote network management protocol is NETCONF (IETF RFC 6241). A YANG module specifies the organization and rules for the management data, and a mapping from YANG to the specific encoding enables the data to be understood correctly by both client (e.g. network manager) and server (e.g. IA-stations).

6.4.10.2 YANG framework

The core of the YANG module for IEC/IEEE 60802 IA-stations consists of YANG "augment" statements, used to add members to the tree of existing YANG modules plus one new module for IEC/IEEE 60802 specific objects.

6.4.10.3 IEC/IEEE 60802 Specific managed objects

6.4.10.3.1 General

Subclause 6.4.10.3 defines the set of managed objects, and their functionality, that provides additional information about an IA-station that is required by a CNC to calculate network configurations.

IEC/IEEE 60802 specific managed objects are specified:

- per Ethernet interface, i.e. external port, in 6.4.10.3.2,
- per end station component internal or external port in 6.4.10.3.3 and 6.4.10.3.4,
- per bridge component internal or external port in 6.4.10.3.4,
- per end station component in 6.4.10.3.5 and 6.4.10.3.7,
- per bridge component in 6.4.10.3.6 and 6.4.10.3.7, and
- per IA-station in 6.4.10.3.8.

IEC/IEEE 60802 specific managed objects for CNC entities are specified in 6.4.10.3.9.

3954 **6.4.10.3.2 IEC/IEEE 60802 managed objects per Ethernet interface**

3955 **6.4.10.3.2.1 supportedMauTypes**

3956 The list of supported MAU Types including the data:

3957 a) mauType

3958 The value is the supported MAU Type derived from the list position of the corresponding
3959 dot3MauType as listed in IETF RFC 4836, Clause 5.

3960 b) preemptionSupported

3961 The Boolean value indicates if preemption is supported by the MAU Type.

3962 NOTE The operational MAU Type of an Ethernet interface is provided as leaf operational-mau-type of the
3963 ieee802-ethernet-lldp YANG module. The operational MAU Type is included in the supportedMauTypes list.

3964 **6.4.10.3.3 IEC/IEEE 60802 managed objects per end station component port**

3965 **6.4.10.3.3.1 worstCasePacketGap**

3966 The value is the worst case maximum inter-packet gap between consecutive frames in a traffic
3967 burst expressed in bit-times.

3968 NOTE Minimum interPacketGap is defined in IEEE Std 802.3-2022, 1.4.362. The worst-case-packet-gap is never
3969 less than the minimum interPacketGap.

3970 **6.4.10.3.3.2 maxBurstFrames**

3971 The value is the maximum number of frames that can be sent with minimal inter packet gap.

3972 **6.4.10.3.3.3 maxBurstBytes**

3973 The value is the maximum number of octets that can be sent with minimal inter packet gap.

3974 **6.4.10.3.3.4 committedDataRates**

3975 The list of committed data rates per traffic class and supported line speed including the data:

3976 a) committedInformationRate

3977 The value is the bandwidth limit in kbit/s.

3978 b) committedBurstSize

3979 The value is the burst size limit in octets.

3980 **6.4.10.3.4 IEC/IEEE 60802 managed objects per bridge or end station component port**

3981 **6.4.10.3.4.1 transmissionSelectionAlgorithm**

3982 The list of supported transmission selection algorithms according to IEEE Std 802.1Q-2022,
3983 8.6.8, per traffic class.

3984 **6.4.10.3.4.2 supportedResourcePools**

3985 The list of supported buffer resource pools including the data:

3986 a) resourcePoolName

3987 The value is the name of a resource pool.

3988 b) coveredTimeInterval

3989 The value specifies the covered buffering time given as rational number of seconds for the
3990 highest supported link speed.

3991 c) resourcePoolTrafficClasses

3992 The list of the traffic classes to be served by the resource pool.

3993 **6.4.10.3.4.3 minGatingTimes**

3994 The list of minimum gating times per supported line speed including the data:

3995 a) minCycleTime

3996 The value is the minimum value supported by this port of the AdminCycleTime and
3997 OperCycleTime parameters given as rational number of seconds.

3998 b) minIntervalTime

3999 The value is the minimum value supported by this port of the TimeIntervalValue parameter
4000 in nanoseconds.

4001 **6.4.10.3.5 IEC/IEEE 60802 managed objects per end station component**

4002 **6.4.10.3.5.1 frerSupported**

4003 The value indicates if FRER is supported.

4004 **6.4.10.3.5.2 maxRedundantStreams**

4005 The value is the maximum number of supported redundant streams.

4006 **6.4.10.3.6 IEC/IEEE 60802 managed objects per bridge component**

4007 **6.4.10.3.6.1 delayVariance**

4008 The value indicates variance in delay depending upon the use of a singleValue or
4009 multipleValues (see 6.4.10.3.6.2).

4010 **6.4.10.3.6.2 delayTimes**

4011 The list of minimum and maximum frame length independent and frame length dependent delay
4012 time values of frames as they pass through a bridge component. These values are given:

- 4013 • per supported MAU Type pair and traffic class, if delayVariance is singleValue, or
- 4014 • per port pair with supported MAU Types and traffic class, if delayVariance is multipleValues.

4015 The list includes the data:

4016 a) independentDelayMin

4017 The value is the minimum delay portion that is independent of frame length according to
4018 IEEE Std 802.1Q-2022, 12.32.1.1.

4019 b) independentDelayMax

4020 The value is the maximum delay portion that is independent of frame length according to
4021 IEEE Std 802.1Q-2022, 12.32.1.1.

4022 c) dependentDelayMin

4023 The value is the minimum delay portion that is dependent on frame length according to IEEE
4024 Std 802.1Q-2022, 12.32.1.2.

4025 d) dependentDelayMax

4026 The value is the maximum delay portion that is dependent on frame length according to
4027 IEEE Std 802.1Q-2022, 12.32.1.2.

4028 **6.4.10.3.6.3 latency**

4029 The PHY latency in ns.

4030 **6.4.10.3.6.4 latencyVariation**

4031 The PHY latency variation in ns.

4032 **6.4.10.3.7 IEC/IEEE 60802 managed objects per bridge or end station component**

4033 **6.4.10.3.7.1 maxFids**

4034 The value is the maximum number of supported FIDs.

4035 **6.4.10.3.7.2 maxFdbEntries**

4036 The list of the maximum number of static (6.4.10.3.7.3) and dynamic (6.4.10.3.7.4) FDB entries
4037 per FDB.

4038 **6.4.10.3.7.3 maxStaticFdbEntries**

4039 The value is the maximum number of static FDB entries.

4040 **6.4.10.3.7.4 maxDynamicFdbEntries**

4041 The value is the maximum number of dynamic FDB entries.

4042 **6.4.10.3.7.5 maxPtpInstances**

4043 The value is the maximum number of supported PTP Instances.

4044 **6.4.10.3.7.6 maxHotStandbySystems**

4045 The value is the maximum number of supported HotStandbySystem entities (see IEEE Std
4046 802.1ASdm-2024).

4047 **6.4.10.3.7.7 clockList**

4048 The list of supported application clock entities including the data:

4049 a) clockIdentity

4050 The clock identity of the application clock.

4051 b) clockTarget

4052 The Boolean value indicates if the application clock is a clock target (TRUE) or clock source
4053 (FALSE).

4054 c) arbSupported

4055 The Boolean value indicates if the application clock supports the ARB timescale.

4056 d) ptpSupported

4057 The Boolean value indicates if the application clock supports the PTP timescale.

4058 e) hotStandbySupported

4059 The Boolean value indicates if the application clock supports hot standby.

4060 f) attachedPtpInstance

4061 The value is a reference to the PTP or hot standby Instance, that is attached to the
4062 application clock.

4063 g) isSynced

4064 The Boolean value indicates if the application clock is either synchronized to the attached
4065 PTP Instance (TRUE) or to an internal/external ClockSource (FALSE).

4066 **6.4.10.3.8 IEC/IEEE 60802 managed objects per IA-station**

4067 **6.4.10.3.8.1 maxSubscriptions**

4068 The value is the maximum number of supported NETCONF Server subscriptions.

4069 **6.4.10.3.8.2 maxOnChangeSubscriptionLeaves**

4070 The value is the maximum number of supported leaves for NETCONF Server on-change
4071 subscriptions according to IETF RFC 8641.

4072 **6.4.10.3.8.3 maxPeriodicSubscriptionLeaves**

4073 The value is the maximum number of supported leaves for NETCONF Server periodic
4074 subscriptions according to IETF RFC 8641.

4075 **6.4.10.3.8.4 minPeriodicSubscriptionInterval**

4076 The value is the minimum periodic subscription interval in centiseconds (0,01 seconds) for
4077 NETCONF Server periodic subscriptions according to IETF RFC 8641.

4078 **6.4.10.3.8.5 capabilityLLDP**

4079 This Boolean value indicates if LLDP is supported.

4080 **6.4.10.3.8.6 capabilityTimesync**

4081 This Boolean value indicates if Timesync is supported.

4082 **6.4.10.3.8.7 capabilityKeystore**

4083 This Boolean value indicates if Keystore is supported.

4084 **6.4.10.3.8.8 capabilityNACM**

4085 This Boolean value indicates if NACM is supported.

4086 **6.4.10.3.8.9 capabilityTruststore**

4087 This Boolean value indicates if Truststore is supported.

4088 **6.4.10.3.8.10 capabilityYangLibrary**

4089 This Boolean value indicates if YANG library is supported.

4090 **6.4.10.3.8.11 capabilityYangPush**

4091 This Boolean value indicates if Yang Push is supported.

4092 **6.4.10.3.8.12 capabilityYangNotifications**

4093 This Boolean value indicates if YANG notifications is supported.

4094 **6.4.10.3.8.13 capabilityNetconfMonitoring**

4095 This Boolean value indicates if NETCONF Monitoring is supported.

4096 **6.4.10.3.8.14 capabilityNetconfClient**

4097 This Boolean value indicates if NETCONF client is supported.

4098 **6.4.10.3.8.15 capabilityCncConfig**

4099 This Boolean value indicates if CNC configuration is supported.

4100 **6.4.10.3.8.16 capabilitySchedTraffic**

4101 This Boolean value indicates if scheduled traffic is supported.

4102 **6.4.10.3.8.17 capabilityFramePreemption**

4103 This Boolean value indicates if frame preemption is supported.

4104 **6.4.10.3.9 IEC/IEEE 60802 managed objects per proxy station**

4105 **6.4.10.3.9.1 maxNumProxyTargets**

4106 The value indicates how many proxy targets can be handled by this proxy.

4107 **6.4.10.3.9.2 proxyType**

4108 OID of proxy targets handled by this proxy.

4109 **6.4.10.3.9.3 device**

4110 The identification of the device.

4111 **6.4.10.3.9.4 deviceType**

4112 The type of the device.

4113 **6.4.10.3.9.5 proxyAddr**

4114 The ip-address for the proxy target.

4115 **6.4.10.3.9.6 port**

4116 The port-number for the proxy target.

4117 **6.4.10.3.9.7 deviceAddr**

4118 The ip-address of the proxy target.

4119 **6.4.10.3.9.8 macAddr**

4120 The mac-address of the proxy target.

4121 **6.4.10.3.10 IEC/IEEE 60802 managed objects per proxy target**

4122 **6.4.10.3.10.1 proxyType**

4123 OID of proxy that this proxy target requires.

4124 **6.4.10.3.11 IEC/IEEE 60802 managed objects for CNC entities**

4125 **6.4.10.3.11.1 maxConfigurationDomains**

4126 The value is the maximum number of supported Configuration Domains.

4127 6.4.10.3.11.2 maxCUCs

4128 The value is the maximum number of supported CUC entities.

4129 6.4.10.3.11.3 maxIAstations

4130 The value is the maximum number of supported IA-stations.

4131 6.4.10.3.11.4 maxNetworkDiameter

4132 The value is the maximum supported network diameter.

4133 6.4.10.3.11.5 maxStreams

4134 The value is the maximum number of supported streams.

4135 6.4.10.3.11.6 maxNumSeamlessTrees

4136 The value is the maximum number of trees supported for seamless redundancy of a stream.

4137 6.4.10.3.11.7 hotStandbySupported

4138 The Boolean value indicates if hot standby is supported.

4139 6.4.10.4 RPCs and actions specific to this document**4140 6.4.10.4.1 RPC ia-factory-reset****4141 6.4.10.4.1.1 General**

4142 In contrast to the original factory-reset RPC in IETF RFC 8808, this RPC puts the device into a
4143 state where a subsequent configuration by a CNC component results in a functioning IA-station
4144 according to this document. Depending on the factory default configuration, after being reset,
4145 the device can become unreachable on the network.

4146 6.4.10.4.1.2 Input

4147 There is no input.

4148 6.4.10.4.1.3 Output

4149 There is no output.

4150 6.4.10.4.2 Action add-streams**4151 6.4.10.4.2.1 General**

4152 This Action requests a CNC to add a list of streams.

4153 6.4.10.4.2.2 Input

4154 The RPC requires the following input parameters.

- 4155 a) CUCId – The ID of the CUC for which the streams are to be added.
4156 b) StreamId – The Stream ID is a unique identifier of a Stream request and corresponding
4157 configuration.
4158 c) Container Talker – The Talker container contains:
4159 – Talker's behavior for Stream (how/when transmitted),
4160 – Talker's requirements from the network, and

- 4161 – TSN capabilities of the Talker's interface(s).
- 4162 d) List Listener – Each Listener list entry contains:
- 4163 – Listener's requirements from the network, and
- 4164 – TSN capabilities of the Listener's interface(s).

4165 **6.4.10.4.2.3 Output**

4166 Result – Status information indicating if Stream addition has been successful.

4167 **6.4.10.4.3 Action remove-listener**

4168 **6.4.10.4.3.1 General**

4169 This Action removes listeners from a stream.

4170 **6.4.10.4.3.2 Input**

4171 List Listener – A list of indices of listeners to be removed from a stream.

4172 **6.4.10.4.3.3 Output**

4173 Result – Status information indicating if Stream addition has been successful.

4174 **6.4.10.5 IEC/IEEE 60802 YANG data models**

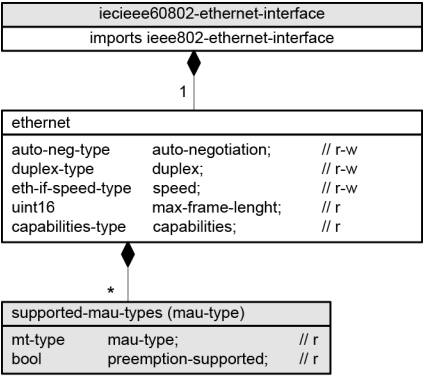
4175 A representation similar to UML® is used to provide an overview of the hierarchy of the
4176 IEC/IEEE 60802 YANG data model.

4177 A UML-like representation of the management model is provided in Figure 29, Figure 30,
4178 Figure 31, Figure 32, Figure 33, Figure 34, Figure 35, Figure 36, and Figure 37. The purpose
4179 of a UML-like diagram is to express the model design in a concise manner. The structure of the
4180 UML-like representation shows the name of the object followed by a list of properties for the
4181 object. The properties indicate their type and accessibility. It should be noted that UML-like
4182 representation is meant to express simplified semantics for the properties. It is not meant to
4183 provide the specific datatype used to encode the object in either MIB or YANG.

4184 NOTE OMG® UML® 2.5 conventions together with C++ language constructs are used as a representation to convey
4185 model structure and relationships.

4186 For all UML® figures, data that is imported from original modules is shown in white, and data
4187 that is augmented according to IEC/IEEE 60802 is shown in grey.

4188 Figure 29, Figure 30, Figure 31, Figure 32, Figure 33, Figure 34, Figure 35, Figure 36, and
4189 Figure 37 provide an overview of the IEC/IEEE 60802 augmentations.



IEC

Figure 29 – Module iecieee60802-ethernet-interface

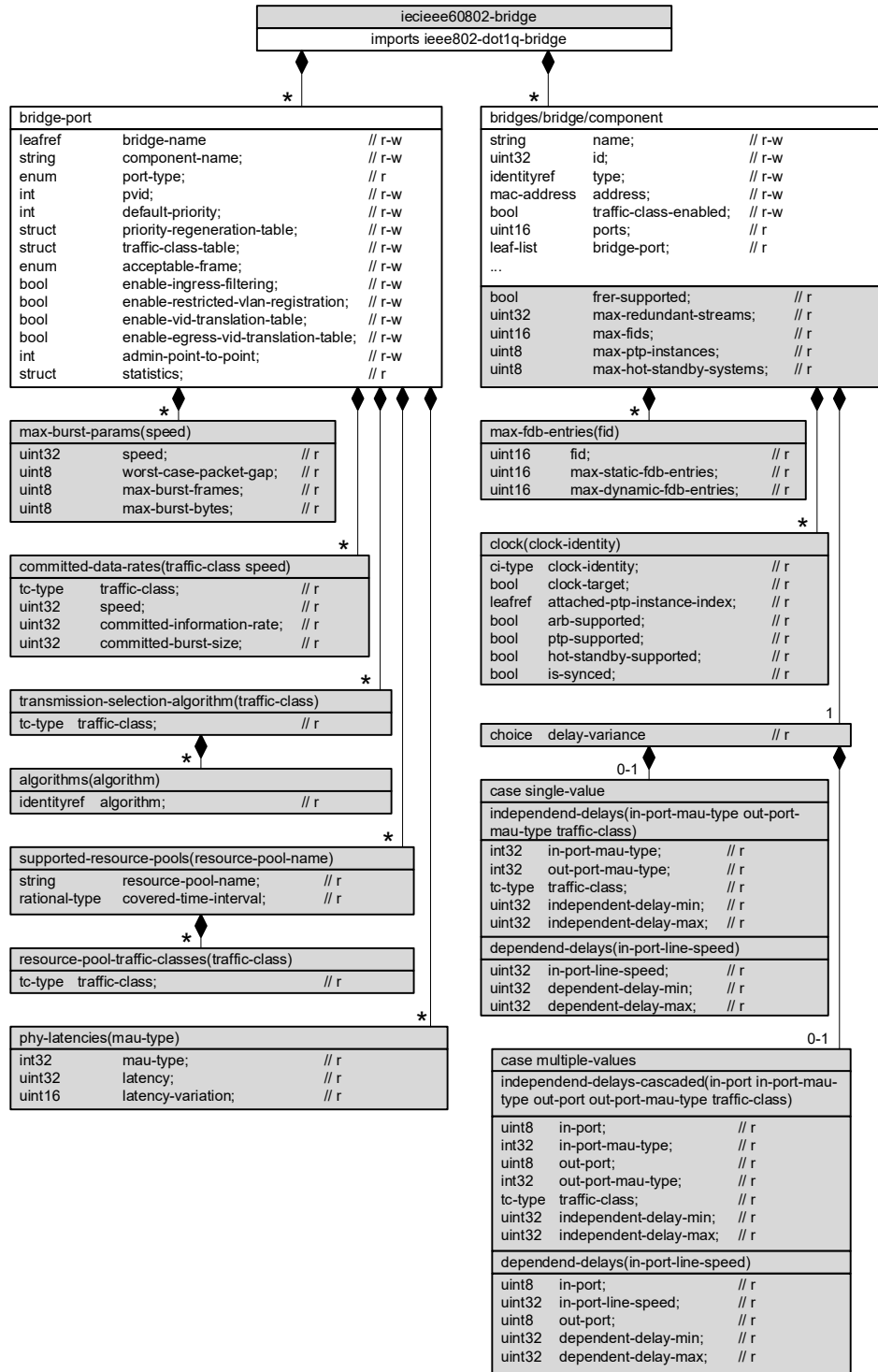


Figure 30 – Module iecieee60802-bridge

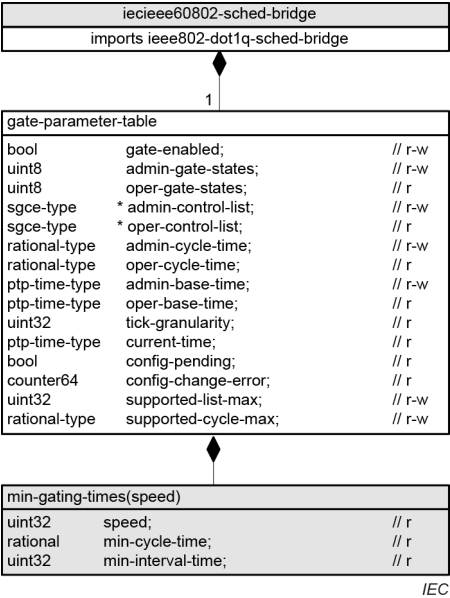


Figure 31 – Module icieeee60802-dot1-sched-bridge

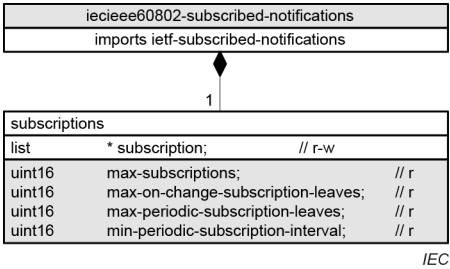


Figure 32 – Module icieeee60802-subscribed-notifications

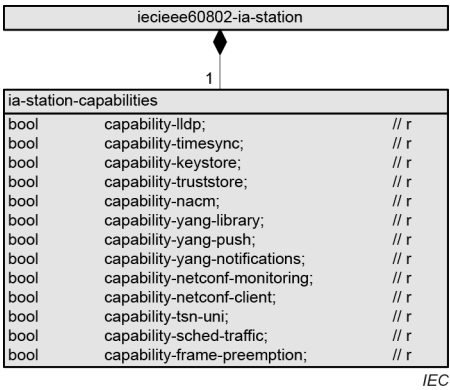


Figure 33 – Module icieeee60802-ia-station

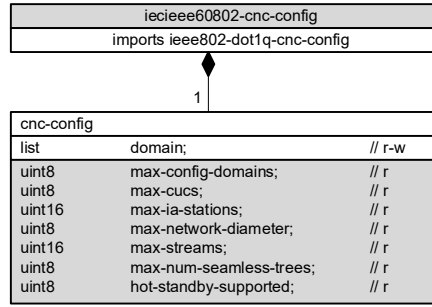


Figure 34 – Module icieeee60802-cnc-config

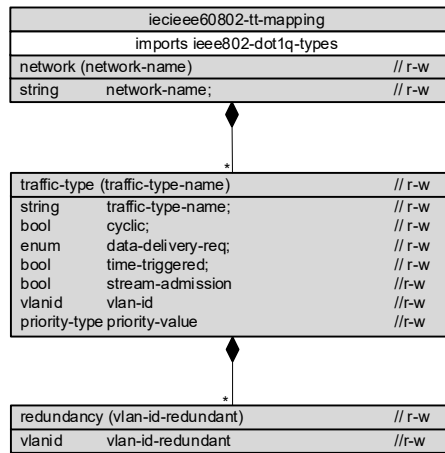


Figure 35 – Module icieeee60802-tt-mapping

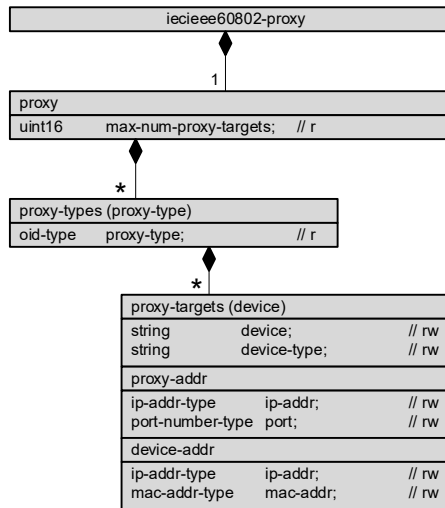


Figure 36 – Module icieeee60802-proxy

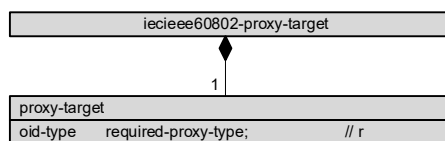


Figure 37 – icieeee60802-proxy-target

6.4.10.6 Structure of IEC/IEEE 60802 YANG data models

The YANG data models specified by this standard use the YANG modules summarized in Table 17.

In the YANG module definitions, if any discrepancy between the "description" text and the corresponding definition in any other part of this standard occurs, the definitions outside Clause 6 take precedence.

Table 17 – Summary of the YANG modules

Module	Description
ieee802-ethernet-interface	This module contains YANG definitions for configuring IEEE Std 802.3 Ethernet Interfaces.
ietf-interfaces	This module contains a collection of YANG definitions for managing network interfaces.
ieciieee60802-ethernet-interface	This module augments ieee802-ethernet-interface.
ieee802-types	This module contains a collection of generally useful derived data types for IEEE YANG data models.
ieee802-dot1q-bridge	This module describes the bridge configuration model for IEEE 802.1Q Bridges.
ieee802-dot1q-types	This module contains common types used within dot1q-bridge modules.
ieciieee60802-bridge	This module augments ieee802-dot1q-bridge.
ieee802-dot1q-sched-bridge	This module provides for management of IEEE Std 802.1Q Bridges that support Scheduled Traffic Enhancements.
ieciieee60802-dot1q-sched-bridge	This module augments ieee802-dot1q-sched-bridge.
ieee802-dot1cb-frer	This module provides management objects that control the frame replication and elimination from IEEE Std 802.1CB-2017.
ieee1588-ptp-tt	This module defines a data model for the configuration and state of IEEE Std 1588 clocks.
ietf-netconf-acm	This module provides management for the Network Configuration Access Control Model.
ieee802-dot1q-cnc-config	This module provides the Time-Sensitive Networking (TSN) User/Network Interface (UNI) for the exchange of information between CUC and CNC that are required to configure TSN Streams in a TSN network.
ieciieee60802-cnc-config	This module augments ieee802-dot1q-cnc-config.
ieciieee60802-ia-station	This module provides read-only information about the capabilities and RPCs for IEC/IEEE 60802 IA-stations.
ietf-subscribed-notifications	This module defines a YANG data model for subscribing to event records and receiving matching content in notification messages.
ieciieee60802-subscribed-notifications	This module augments ietf-subscribed-notifications.
ieciieee60802-tt-mapping	This module provides information on characteristics of traffic classes configured in a network to allow applications to map application traffic classes to them
ieciieee60802-proxy.yang	This module contains configuration and capability information for IEC/IEEE 60802 proxy stations
ieciieee60802-proxy-target.yang	This module contains capability information for IEC/IEEE 60802 proxy targets

6.4.10.7 YANG schema tree definitions

6.4.10.7.1 General

The schema tree is provided as an overview of the YANG modules. The symbols and their meaning are specified in YANG Tree Diagrams (IETF RFC 8340).

6.4.10.7.2 Module iecieeee60802-ethernet-interface

```

4220
4221 module: iecieeee60802-ethernet-interface
4222
4223   augment /if:interfaces/if:interface/eth-if:ethernet:
4224     +-ro supported-mau-types* [mau-type]
4225     +-ro mau-type             uint32
4226     +-ro preemption-supported? boolean
4227

```

6.4.10.7.3 Module iecieeee60802-bridge

```

4228
4229 module: iecieeee60802-bridge
4230
4231   augment /if:interfaces/if:interface/bridge:bridge-port:
4232     +-ro max-burst-params* [speed]
4233     | +-ro speed             uint32
4234     | +-ro worst-case-packet-gap? uint8
4235     | +-ro max-burst-frames?    uint8
4236     | +-ro max-burst-bytes?    uint8
4237     +-ro committed-data-rates* [traffic-class speed]
4238     | +-ro traffic-class      dot1q-types:traffic-class-type
4239     | +-ro speed              uint32
4240     | +-ro committed-information-rate? uint32
4241     | +-ro committed-burst-size?  uint32
4242     +-ro transmission-selection-algorithm* [traffic-class]
4243     | +-ro traffic-class      dot1q-types:traffic-class-type
4244     | +-ro algorithms* [algorithm]
4245     | +-ro algorithm          identityref
4246     +-ro supported-resource-pools* [resource-pool-name]
4247     | +-ro resource-pool-name  string
4248     | +-ro covered-time-interval
4249     | | +-ro numerator?       uint32
4250     | | +-ro denominator?    uint32
4251     | +-ro resource-pool-traffic-classes* [traffic-class]
4252     | +-ro traffic-class      dot1q-types:traffic-class-type
4253     +-ro phy-latencies* [mau-type]
4254     +-ro mau-type             uint32
4255     +-ro latency?              uint32
4256     +-ro latency-variation?    uint16
4257   augment /bridge:bridges/bridge:bridge/bridge:component:
4258     +-ro frer-supported?      boolean
4259     +-ro max-redundant-streams? uint32
4260     +-ro max-fids?            uint16
4261     +-ro max-fdb-entries* [fid]
4262     | +-ro fid                uint16
4263     | +-ro max-static-fdb-entries? uint16
4264     | +-ro max-dynamic-fdb-entries? uint16
4265     +-ro (delay-variance)?
4266     | +--:(single-value)
4267     | | +-ro independent-delays* [in-port-mau-type out-port-mau-type
4268   traffic-class]
4269     | | | +-ro in-port-mau-type    int32
4270     | | | +-ro out-port-mau-type   int32
4271     | | | +-ro traffic-class      dot1q-types:traffic-class-type
4272     | | | +-ro independent-delay-min? uint32
4273     | | | +-ro independent-delay-max? uint32
4274     | | +-ro dependent-delays* [in-port-line-speed]
4275     | | +-ro in-port-line-speed    uint32
4276     | | +-ro dependent-delay-min?  uint32
4277     | | +-ro dependent-delay-max?  uint32
4278     | +--:(multiple-values)
4279     | +-ro independent-delays-cascaded* [in-port in-port-mau-type out-
4280   port out-port-mau-type traffic-class]

```

```

4281         |         | +-ro in-port                               uint8
4282         |         | +-ro in-port-mau-type                     int32
4283         |         | +-ro out-port                              uint8
4284         |         | +-ro out-port-mau-type                     int32
4285         |         | +-ro traffic-class                         dot1q-types:traffic-class-type
4286         |         | +-ro independent-delay-min?                uint32
4287         |         | +-ro independent-delay-max?                uint32
4288         |         +-ro dependent-delays-cascaded* [in-port in-port-line-speed out-
4289 port]
4290         |         +-ro in-port                               uint8
4291         |         +-ro in-port-line-speed                   uint32
4292         |         +-ro out-port                              uint8
4293         |         +-ro dependent-delay-min?                  uint32
4294         |         +-ro dependent-delay-max?                  uint32
4295         +-ro max-ptp-instances?                               uint8
4296         +-ro max-hot-standby-systems?                         uint8
4297         +-ro clock* [clock-identity]
4298             +-ro clock-identity                             ptp-tt:clock-identity
4299             +-ro clock-target?                               boolean
4300             +-ro attached-ptp-instance-index?                -> /ptp-
4301 tt:ptp/instances/instance/instance-index
4302             +-ro arb-supported?                               boolean
4303             +-ro ptp-supported?                               boolean
4304             +-ro hot-standby-supported?                       boolean
4305             +-ro is-synced?                                   boolean

```

6.4.10.7.4 Module iecieeee60802-sched-bridge

```

4307 module: iecieeee60802-sched-bridge
4308
4309     augment /if:interfaces/if:interface/bridge:bridge-port/sched-bridge:gate-
4310 parameter-table:
4311         +-ro min-gating-times* [speed]
4312         +-ro speed                               uint32
4313         +-ro min-cycle-time
4314         | +-u ieee802:rational-grouping
4315         +-ro min-interval-time?                   uint32

```

6.4.10.7.5 Module iecieeee60802-cnc-config

```

4317 module: iecieeee60802-cnc-config
4318
4319     augment /dot1q-cnc-config:cnc-config:
4320         +-ro max-config-domains?                   uint8
4321         +-ro max-cucs?                               uint8
4322         +-ro max-ia-stations?                       uint16
4323         +-ro max-network-diameter?                 uint8
4324         +-ro max-streams?                           uint16
4325         +-ro max-num-seamless-trees?               uint8
4326         +-ro hot-standby-supported?                 uint8
4327         +-x add_streams
4328             +-w input
4329             | +-w cuc-id?                             string
4330             | +-w stream-list* [stream-id]
4331             |     +-w stream-id                       tsn-types:stream-id-type
4332             |     +-w talker
4333             |     | +-w stream-rank
4334             |     | | +-w rank?                       uint8
4335             |     | +-w end-station-interfaces* [mac-address interface-name]
4336             |     | | +-w mac-address                 string
4337             |     | | +-w interface-name              string
4338             |     | +-w data-frame-specification* [index]
4339             |     | | +-w index                               uint8
4340             |     | | +-w (field)?

```

```

4341         |         |         |         +-:(ieee802-mac-addresses)
4342         |         |         |         |   +--w ieee802-mac-addresses
4343         |         |         |         |       +--w destination-mac-address?   string
4344         |         |         |         |       +--w source-mac-address?       string
4345         |         |         |         +-:(ieee802-vlan-tag)
4346         |         |         |         |   +--w ieee802-vlan-tag
4347         |         |         |         |       +--w priority-code-point?   uint8
4348         |         |         |         |       +--w vlan-id?                 uint16
4349         |         |         |         +-:(ipv4-tuple)
4350         |         |         |         |   +--w ipv4-tuple
4351         |         |         |         |       +--w source-ip-address?         inet:ipv4-
4352 address    |         |         |         |       +--w destination-ip-address?   inet:ipv4-
4353 address    |         |         |         |       +--w destination-ip-address?   inet:ipv4-
4354 address    |         |         |         |       +--w destination-ip-address?   inet:ipv4-
4355         |         |         |         |       +--w dscp?                     uint8
4356         |         |         |         |       +--w protocol?                 uint16
4357         |         |         |         |       +--w source-port?              uint16
4358         |         |         |         |       +--w destination-port?          uint16
4359         |         |         |         +-:(ipv6-tuple)
4360         |         |         |         |   +--w ipv6-tuple
4361         |         |         |         |       +--w source-ip-address?         inet:ipv6-
4362 address    |         |         |         |       +--w destination-ip-address?   inet:ipv6-
4363 address    |         |         |         |       +--w destination-ip-address?   inet:ipv6-
4364 address    |         |         |         |       +--w destination-ip-address?   inet:ipv6-
4365         |         |         |         |       +--w dscp?                     uint8
4366         |         |         |         |       +--w protocol?                 uint16
4367         |         |         |         |       +--w source-port?              uint16
4368         |         |         |         |       +--w destination-port?          uint16
4369         |         |         |         +--w traffic-specification
4370         |         |         |         |   +--w interval
4371         |         |         |         |       +--w numerator?                 uint32
4372         |         |         |         |       +--w denominator?              uint32
4373         |         |         |         |       +--w max-frames-per-interval?    uint16
4374         |         |         |         |       +--w max-frame-size?            uint16
4375         |         |         |         |       +--w transmission-selection?    uint8
4376         |         |         |         |       +--w time-aware!
4377         |         |         |         |       +--w earliest-transmit-offset?   uint32
4378         |         |         |         |       +--w latest-transmit-offset?     uint32
4379         |         |         |         |       +--w jitter?                    uint32
4380         |         |         |         +--w user-to-network-requirements
4381         |         |         |         |   +--w num-seamless-trees?            uint8
4382         |         |         |         |   +--w max-latency?                  uint32
4383         |         |         |         +--w interface-capabilities
4384         |         |         |         |   +--w vlan-tag-capable?              boolean
4385         |         |         |         |   +--w cb-stream-iden-type-list*      uint32
4386         |         |         |         |   +--w cb-sequence-type-list*         uint32
4387         |         |         |         +--w listener* [index]
4388         |         |         |         |   +--w index                          uint32
4389         |         |         |         |   +--w end-station-interfaces* [mac-address interface-name]
4390         |         |         |         |       +--w mac-address                string
4391         |         |         |         |       +--w interface-name              string
4392         |         |         |         +--w user-to-network-requirements
4393         |         |         |         |   +--w num-seamless-trees?            uint8
4394         |         |         |         |   +--w max-latency?                  uint32
4395         |         |         |         +--w interface-capabilities
4396         |         |         |         |   +--w vlan-tag-capable?              boolean
4397         |         |         |         |   +--w cb-stream-iden-type-list*      uint32
4398         |         |         |         |   +--w cb-sequence-type-list*         uint32
4399         +-rw output
4400         +-rw result?    boolean
4401 augment /dot1q-cnc-config:cnc-config/dot1q-cnc-config:domain/dot1q-cnc-
4402 config:cuc/dot1q-cnc-config:stream:
4403         +--x remove_listener

```



```

4404     +--w input
4405     |   +--w listener* [index]
4406     |       +--w index      uint32
4407     +--rw output
4408     +--rw result?   boolean

```

4409 6.4.10.7.6 Module iecieeee60802-ia-station

```

4410 module: iecieeee60802-ia-station
4411   +--ro ia-station-capabilities
4412     +--ro capability-lldp?          boolean
4413     +--ro capability-timesync?      boolean
4414     +--ro capability-keystore?      boolean
4415     +--ro capability-truststore?    boolean
4416     +--ro capability-nacm?          boolean
4417     +--ro capability-yang-library?  boolean
4418     +--ro capability-yang-push?     boolean
4419     +--ro capability-yang-notifications? boolean
4420     +--ro capability-netconf-monitoring? boolean
4421     +--ro capability-netconf-client? boolean
4422     +--ro capability-cnc-config?    boolean
4423     +--ro capability-sched-traffic? boolean
4424     +--ro capability-frame-preemption? boolean
4425
4426   rpcs:
4427     +--x ia-factory-reset

```

4428 6.4.10.7.7 Module iecieeee60802-subscribed-notifications

```

4429 module: iecieeee60802-subscribed-notifications
4430
4431   augment /sn:subscriptions:
4432     +--ro max-subscriptions?          uint16
4433     +--ro max-on-change-subscription-leaves? uint16
4434     +--ro max-periodic-subscription-leaves? uint16
4435     +--ro min-periodic-subscription-interval? uint16

```

4436 6.4.10.7.8 Module iecieeee-tt-mapping

```

4437 module: iecieeee60802-tt-mapping
4438   +--rw tt-mapping
4439     +--rw network* [network-name]
4440       +--rw network-name      string
4441       +--rw traffic-type* [traffic-type-name]
4442         +--rw traffic-type-name string
4443         +--rw cyclic?           boolean
4444         +--rw data-delivery-req? enumeration
4445         +--rw time-triggered?   boolean
4446         +--rw stream-based?     boolean
4447         +--rw vlan-id?          dot1q-types:vlanid
4448         +--rw redundancy* [vlan-id-redundant]
4449         | +--rw vlan-id-redundant dot1q-types:vlanid
4450         +--rw priority-value?   dot1q-types:priority-type

```

4451 6.4.10.7.9 Module iecieeee60802-proxy

```

4452 module: iecieeee60802-proxy
4453   +--rw proxy!
4454     +--ro max-num-proxy-targets?  uint16
4455     +--ro proxy-types* [proxy-type]
4456       +--ro proxy-type           string
4457       +--ro proxy-targets* [device]
4458         +--ro device             string
4459         +--ro device-type?       string
4460         +--ro proxy-addr
4461         | +--ro ip-addr?         inet:ip-address

```

```

4462         |  +--ro port?          inet:port-number
4463         +--ro device-addr
4464         +--ro ip-addr?         inet:ip-address
4465         +--ro mac-addr?       ieee:mac-address

```

4466 **6.4.10.7.10 Module iecieee60802-proxy-target**

```

4467 module: iecieee60802-proxy-target
4468     +--rw proxy-target!
4469     +--ro required-proxy-type?  string

```

4470 **6.4.10.8 YANG modules^{5,6}**

4471 **6.4.10.8.1 Module iecieee60802-ethernet-interface**

```

4472 module iecieee60802-ethernet-interface {
4473     yang-version 1.1;
4474     namespace
4475         "urn:ieee:std:60802:yang:ieciieee60802-ethernet-interface";
4476     prefix ia-eth-if;
4477
4478     import ieee802-ethernet-interface {
4479         prefix eth-if;
4480     }
4481     import ietf-interfaces {
4482         prefix if;
4483     }
4484
4485     organization
4486         "IEEE 802.1 Working Group and IEC subcommittee 65C:
4487         Industrial networks, of IEC technical committee 65:
4488         Industrial-process measurement, control and automation";
4489     contact
4490         "WG-URL: http://ieee802.org/1/
4491         WG-EMail: stds-802-1-1@ieee.org
4492
4493         Contact: IEEE 802.1 Working Group Chair
4494         Postal: C/O IEEE 802.1 Working Group
4495         IEEE Standards Association
4496         445 Hoes Lane
4497         Piscataway, NJ 08854
4498         USA
4499
4500         E-mail: stds-802-1-chairs@ieee.org";
4501     description
4502         "Management objects that provide information about IEC/IEEE 60802
4503         IA-Stations as specified in IEC/IEEE 60802.
4504
4505         Copyright (C) IEC/IEEE (2025).
4506         This version of this YANG module is part of IEC/IEEE 60802;
4507         see the standard itself for full legal notices.";
4508
4509     revision 2024-02-19 {
4510         description "Published as part of IEC/IEEE 60802-2025.
4511         The following reference statement identifies each referenced
4512         IEEE Standard as updated by applicable amendments.";
4513         reference
4514             "IEC/IEEE 60802 TSN profile for industrial automation:

```

⁵ Copyright release for YANG: Users of this document may freely reproduce the YANG modules contained in this document so that they can be used for their intended purpose.

⁶ An ASCII version of each YANG module defined in this document is attached to the PDF of this document and can also be obtained from the IEEE 802.Website at <https://1.ieee802.org/yang-modules/>.

```
4515     IEC/IEEE 60802-2025.
4516     IEEE Std 802.1Q Bridges and Bridged Networks:
4517     IEEE Std 802.1Q-2022, IEEE Std 802.1Qcz-2023,
4518     IEEE Std 802.1Qcw-2023, IEEE Std 802.1Qdj-2024,
4519     IEEE Std 802.1Qdx-2024, IEEE Std 802.1Qdy-2024.";
4520 }
4521
4522 augment "/if:interfaces/if:interface/eth-if:ethernet" {
4523     description
4524         "Augment IEEE Std 802.3 ethernet.";
4525     list supported-mau-types {
4526         description
4527             "Contains a list of supported MAU parameters.";
4528         key "mau-type";
4529         config false;
4530         leaf mau-type {
4531             type int32;
4532             config false;
4533             description
4534                 "The value is the supported MAU Type derived from the list
4535                 position of the corresponding dot3MauType as listed in
4536                 Clause 5 of IETF RFC 4836.";
4537             reference
4538                 "Item a) in 6.4.10.3.2.1 of IEC/IEEE 60802";
4539         }
4540         leaf preemption-supported {
4541             type boolean;
4542             config false;
4543             description
4544                 "The Boolean value indicates if preemption is supported by
4545                 the MAU Type.";
4546             reference
4547                 "Item b) in 6.4.10.3.2.1 of IEC/IEEE 60802";
4548         }
4549     }
4550 }
4551 }
```

4552 **6.4.10.8.2 Module iecieee60802-bridge**

```
4553 module iecieee60802-bridge {
4554     yang-version 1.1;
4555     namespace "urn:ieee:std:60802:yang:ieciieee60802-bridge";
4556     prefix ia-bridge;
4557
4558     import ieee802-types {
4559         prefix ieee802;
4560     }
4561     import ieee802-dot1q-bridge {
4562         prefix bridge;
4563     }
4564     import ietf-interfaces {
4565         prefix if;
4566     }
4567     import ieee802-dot1q-types {
4568         prefix dot1q-types;
4569     }
4570     import ieee1588-ptp-tt {
4571         prefix ptp-tt;
4572     }
4573
4574     organization
4575         "IEEE 802.1 Working Group and IEC subcommittee 65C:"
```

```

4576     Industrial networks, of IEC technical committee 65:
4577     Industrial-process measurement, control and automation";
4578     contact
4579     "WG-URL: http://ieee802.org/1/
4580     WG-EMail: stds-802-1-1@ieee.org
4581
4582     Contact: IEEE 802.1 Working Group Chair
4583             Postal: C/O IEEE 802.1 Working Group
4584             IEEE Standards Association
4585             445 Hoes Lane
4586             Piscataway, NJ 08854
4587             USA
4588
4589     E-mail: stds-802-1-chairs@ieee.org";
4590     description
4591     "Management objects that provide information about
4592     IEC/IEEE 60802 IA-Stations as specified in IEC/IEEE 60802.
4593
4594     Copyright (C) IEC/IEEE (2025).
4595     This version of this YANG module is part of IEC/IEEE 60802;
4596     see the standard itself for full legal notices.";
4597
4598     revision 2025-01-09 {
4599         description
4600         "Published as part of IEC/IEEE 60802-2025.
4601         The following reference statement identifies each referenced
4602         IEEE Standard as updated by applicable amendments.";
4603         reference
4604         "IEC/IEEE 60802 TSN profile for industrial automation:
4605         IEC/IEEE 60802-2025.
4606         IEEE Std 802.1Q Bridges and Bridged Networks:
4607         IEEE Std 802.1Q-2022, IEEE Std 802.1Qcz-2023,
4608         IEEE Std 802.1Qcw-2023, IEEE Std 802.1Qdj-2024,
4609         IEEE Std 802.1Qdx-2024, IEEE Std 802.1Qdy-2024.";
4610     }
4611
4612     augment "/if:interfaces/if:interface/bridge:bridge-port" {
4613         description
4614         "Augment IEEE Std 802.1 bridge.";
4615         list max-burst-params {
4616             description
4617             "The list of maximum burst parameters per supported line
4618             speed.";
4619             key "speed";
4620             config false;
4621             leaf speed {
4622                 type uint32;
4623                 description
4624                 "This value is the line speed in Mbps.";
4625             }
4626             leaf worst-case-packet-gap {
4627                 type uint8;
4628                 config false;
4629                 description
4630                 "The value is the worst case maximum inter-packet gap
4631                 between consecutive frames in a traffic burst expressed
4632                 in bit-times.";
4633                 reference
4634                 "Item a) in 6.4.10.3.3.1 of IEC/IEEE 60802";
4635             }
4636             leaf max-burst-frames {
4637                 type uint8;
4638                 config false;

```

```
4639         description
4640             "The value is the maximum number of frames that can be sent
4641             with minimal inter packet gap.";
4642         reference
4643             "Item b) in 6.4.10.3.3.1 of IEC/IEEE 60802";
4644     }
4645     leaf max-burst-bytes {
4646         type uint8;
4647         config false;
4648         description
4649             "The value is the maximum number of octets that can be sent
4650             with minimal inter packet gap.";
4651         reference
4652             "Item c) in 6.4.10.3.3.1 of IEC/IEEE 60802";
4653     }
4654 }
4655 list committed-data-rates {
4656     description
4657         "The list of committed data rates per traffic class and
4658         supported line speed.";
4659     key "traffic-class speed";
4660     config false;
4661     leaf traffic-class {
4662         type dot1q-types:traffic-class-type;
4663         description
4664             "The traffic class of the entry (0..7).";
4665         reference
4666             "8.6.6 of IEEE Std 802.1Q";
4667     }
4668     leaf speed {
4669         type uint32;
4670         description
4671             "This value is the line speed in Mbps.";
4672     }
4673     leaf committed-information-rate {
4674         type uint32;
4675         config false;
4676         description
4677             "The value is the bandwidth limit in kbit/s.";
4678         reference
4679             "Item a) in 6.4.10.3.3.2 of IEC/IEEE 60802";
4680     }
4681     leaf committed-burst-size {
4682         type uint32;
4683         config false;
4684         description
4685             "The value is the burst size limit in bytes.";
4686         reference
4687             "Item b) in 6.4.10.3.3.2 of IEC/IEEE 60802";
4688     }
4689 }
4690 list transmission-selection-algorithm {
4691     description
4692         "The list of supported transmission selection algorithms
4693         according to 8.6.8 of IEEE Std 802.1Q per traffic class.";
4694     key "traffic-class";
4695     config false;
4696     leaf traffic-class {
4697         type dot1q-types:traffic-class-type;
4698         config false;
4699         description
4700             "Traffic class. (0..7)";
4701         reference
```

```

4702         "8.6.6 of IEEE Std 802.1Q";
4703     }
4704     list algorithms {
4705         description
4706             "The list of supported transmission selection algorithms
4707             according to 8.6.8 of IEEE Std 802.1Q for this traffic
4708             class.";
4709         key "algorithm";
4710         config false;
4711         leaf algorithm {
4712             type identityref {
4713                 base dot1q-types:transmission-selection-algorithm;
4714             }
4715             config false;
4716             description
4717                 "Transmission selection algorithm";
4718             reference
4719                 "8.6.8 of IEEE Std 802.1Q";
4720         }
4721     }
4722 }
4723 list supported-resource-pools {
4724     description
4725         "The list of supported buffer resource pools.";
4726     key "resource-pool-name";
4727     config false;
4728     leaf resource-pool-name {
4729         type string;
4730         config false;
4731         description
4732             "The value is the name of a resource pool.";
4733         reference
4734             "Item a) in 6.4.10.3.4.2 of IEC/IEEE 60802";
4735     }
4736     container covered-time-interval {
4737         config false;
4738         uses ieee802:rational-grouping;
4739         description
4740             "The value is the covered buffering time given as rational
4741             number of seconds for the highest supported link speed.";
4742         reference
4743             "Item b) in 6.4.10.3.4.2 of IEC/IEEE 60802";
4744     }
4745     list resource-pool-traffic-classes {
4746         description
4747             "The list of the traffic classes to be served by the
4748             resource pool.";
4749         reference
4750             "Item c) in 6.4.10.3.4.2 of IEC/IEEE 60802";
4751         key "traffic-class";
4752         config false;
4753         leaf traffic-class {
4754             type dot1q-types:traffic-class-type;
4755             description
4756                 "The traffic class of the entry.";
4757             reference
4758                 "8.6.6 of IEEE Std 802.1Q";
4759         }
4760     }
4761 }
4762 list phy-latencies {
4763     description
4764         "Latency information for each MAU type";

```

```
4765     key "mau-type";
4766     config false;
4767     leaf mau-type {
4768         type int32;
4769         config false;
4770         description
4771             "The MAU type";
4772     }
4773     leaf latency {
4774         type uint32;
4775         config false;
4776         description
4777             "The PHY latency in ns";
4778         reference
4779             "6.4.10.3.6.3 of IEC/IEEE 60802";
4780     }
4781     leaf latency-variation {
4782         type uint16;
4783         config false;
4784         description
4785             "The PHY latency variation in ns";
4786         reference
4787             "6.4.10.3.6.4 of IEC/IEEE 60802";
4788     }
4789 }
4790 }
4791
4792 augment "/bridge:bridges/bridge:bridge/bridge:component" {
4793     description
4794         "Augment IEEE Std 802.1 bridge component.";
4795     leaf frer-supported {
4796         type boolean;
4797         config false;
4798         description
4799             "The Boolean value indicates if FRER is supported.";
4800         reference
4801             "6.4.10.3.5.1 of IEC/IEEE 60802";
4802     }
4803     leaf max-redundant-streams {
4804         type uint32;
4805         config false;
4806         description
4807             "The value is the maximum number of supported redundant
4808             streams.";
4809         reference
4810             "6.4.10.3.5.2 of IEC/IEEE 60802";
4811     }
4812     leaf max-fids {
4813         type uint16;
4814         config false;
4815         description
4816             "The value is the maximum number of supported FIDs.";
4817         reference
4818             "6.4.10.3.7.1 of IEC/IEEE 60802";
4819     }
4820     list max-fdb-entries {
4821         config false;
4822         description
4823             "The list of the maximum number of static and dynamic
4824             FDB entries per FID.";
4825         reference
4826             "6.4.10.3.7.2 of IEC/IEEE 60802";
4827         key "fid";
```

```

4828     leaf fid {
4829         type uint16;
4830         config false;
4831         description
4832             "The FID number";
4833     }
4834     leaf max-static-fdb-entries {
4835         type uint16;
4836         config false;
4837         description
4838             "The value is the maximum number of static FDB
4839             entries.";
4840         reference
4841             "6.4.10.3.7.3 of IEC/IEEE 60802";
4842     }
4843     leaf max-dynamic-fdb-entries {
4844         type uint16;
4845         config false;
4846         description
4847             "The value is the maximum number of dynamic FDB entries.";
4848         reference
4849             "6.4.10.3.7.4 of IEC/IEEE 60802";
4850     }
4851 }
4852 choice delay-variance {
4853     config false;
4854     description
4855         "The value indicates variance in delay depending upon the use
4856         of a singleValue or multipleValues.";
4857     reference
4858         "6.4.10.3.6.1 of IEC/IEEE 60802";
4859     case single-value {
4860         list independent-delays {
4861             description
4862                 "The list of minimum and maximum frame length
4863                 independent delay time values of frames as they pass
4864                 through a bridge component.";
4865             reference
4866                 "6.4.10.3.6.2 of IEC/IEEE 60802";
4867             key "in-port-mau-type out-port-mau-type traffic-class";
4868             config false;
4869             leaf in-port-mau-type {
4870                 type int32;
4871                 config false;
4872                 description
4873                     "The MAU type of the input port";
4874             }
4875             leaf out-port-mau-type {
4876                 type int32;
4877                 config false;
4878                 description
4879                     "The MAU type of the input port";
4880             }
4881             leaf traffic-class {
4882                 type dot1q-types:traffic-class-type;
4883                 config false;
4884                 description
4885                     "The traffic class of the entry.";
4886                 reference
4887                     "8.6.6 of IEEE Std 802.1Q";
4888             }
4889             leaf independent-delay-min {
4890                 type uint32;

```



```
4891         config false;
4892         description
4893             "The value is the minimum delay portion that is
4894             independent of frame length according to 12.32.1.1.
4895             of IEEE 802.1Q";
4896         reference
4897             "Item a) in 6.4.10.3.6.2 of IEC/IEEE 60802";
4898     }
4899     leaf independent-delay-max {
4900         type uint32;
4901         config false;
4902         description
4903             "The value is the maximum delay portion that is
4904             independent of frame length according to 12.32.1.1.
4905             of IEEE 802.1Q";
4906         reference
4907             "Item b) in 6.4.10.3.6.2 of IEC/IEEE 60802";
4908     }
4909 }
4910 list dependent-delays {
4911     description
4912         "The list of minimum and maximum frame length dependent
4913         delay time values of frames as they pass through a
4914         bridge component";
4915     reference
4916         "6.4.10.3.6.2 of IEC/IEEE 60802";
4917     key "in-port-line-speed";
4918     config false;
4919     leaf in-port-line-speed {
4920         type uint32;
4921         config false;
4922         description
4923             "This value is the line speed in Mbps.";
4924     }
4925     leaf dependent-delay-min {
4926         type uint32;
4927         config false;
4928         description
4929             "The value is the minimum delay portion that is
4930             dependent on frame length according to 12.32.1.2.
4931             of IEEE 802.1Q";
4932         reference
4933             "Item c) in 6.4.10.3.6.2 of IEC/IEEE 60802";
4934     }
4935     leaf dependent-delay-max {
4936         type uint32;
4937         config false;
4938         description
4939             "The value is the maximum delay portion that is
4940             dependent on frame length according to 12.32.1.2.
4941             of IEEE 802.1Q";
4942         reference
4943             "Item d) in 6.4.10.3.6.2 of IEC/IEEE 60802";
4944     }
4945 }
4946 }
4947 case multiple-values {
4948     list independent-delays-cascaded {
4949         description
4950             "The list of minimum and maximum frame length
4951             independent delay time values of frames as they pass
4952             through a bridge component.";
4953         reference
```

```

4954         "6.4.10.3.6.2 of IEC/IEEE 60802";
4955     key "in-port in-port-mau-type out-port out-port-mau-type
4956         traffic-class";
4957     config false;
4958     leaf in-port {
4959         type uint8;
4960         config false;
4961         description
4962             "The port number of the input port";
4963     }
4964     leaf in-port-mau-type {
4965         type int32;
4966         config false;
4967         description
4968             "The MAU type of the input port";
4969     }
4970     leaf out-port {
4971         type uint8;
4972         config false;
4973         description
4974             "The port number of the output port";
4975     }
4976     leaf out-port-mau-type {
4977         type int32;
4978         config false;
4979         description
4980             "The MAU type of the input port";
4981     }
4982     leaf traffic-class {
4983         type dot1q-types:traffic-class-type;
4984         config false;
4985         description
4986             "The traffic class of the entry.";
4987         reference
4988             "8.6.6 of IEEE Std 802.1Q";
4989     }
4990     leaf independent-delay-min {
4991         type uint32;
4992         config false;
4993         description
4994             "The value is the minimum delay portion that is
4995             independent of frame length according to 12.32.1.1.
4996             of IEEE 802.1Q";
4997         reference
4998             "Item a) in 6.4.10.3.6.2 of IEC/IEEE 60802";
4999     }
5000     leaf independent-delay-max {
5001         type uint32;
5002         config false;
5003         description
5004             "The value is the maximum delay portion that is
5005             independent of frame length according to 12.32.1.1.
5006             of IEEE 802.1Q";
5007         reference
5008             "Item b) in 6.4.10.3.6.2 of IEC/IEEE 60802";
5009     }
5010 }
5011 list dependent-delays-cascaded {
5012     description
5013         "The list of minimum and maximum frame length dependent
5014         delay time values of frames as they pass through a
5015         bridge component";
5016     reference

```

```
5017         "6.4.10.3.6.2 of IEC/IEEE 60802";
5018     key "in-port in-port-line-speed out-port";
5019     config false;
5020     leaf in-port {
5021         type uint8;
5022         config false;
5023         description
5024             "The port number of the input port";
5025     }
5026     leaf in-port-line-speed {
5027         type uint32;
5028         config false;
5029         description
5030             "This value is the line speed in Mbps.";
5031     }
5032     leaf out-port {
5033         type uint8;
5034         config false;
5035         description
5036             "The port number of the output port";
5037     }
5038     leaf dependent-delay-min {
5039         type uint32;
5040         config false;
5041         description
5042             "The value is the minimum delay portion that is
5043             dependent on frame length according to 12.32.1.2.
5044             of IEEE 802.1Q";
5045         reference
5046             "Item c) in 6.4.10.3.6.2 of IEC/IEEE 60802";
5047     }
5048     leaf dependent-delay-max {
5049         type uint32;
5050         config false;
5051         description
5052             "The value is the maximum delay portion that is
5053             dependent on frame length according to 12.32.1.2.
5054             of IEEE 802.1Q";
5055         reference
5056             "Item d) in 6.4.10.3.6.2 of IEC/IEEE 60802";
5057     }
5058 }
5059 }
5060 }
5061 leaf max-ptp-instances {
5062     type uint8;
5063     config false;
5064     description
5065         "The value is the maximum number of supported PTP
5066         Instances.";
5067     reference
5068         "6.4.10.3.7.5 of IEC/IEEE 60802";
5069 }
5070 leaf max-hot-standby-systems {
5071     type uint8;
5072     config false;
5073     description
5074         "The value is the maximum number of supported
5075         HotStandbySystem entities";
5076     reference
5077         "6.4.10.3.7.6 of IEC/IEEE 60802";
5078 }
5079 list clock {
```

```

5080     description
5081         "The list of supported application clock entities.";
5082     reference
5083         "6.4.10.3.7.7 of IEC/IEEE 60802";
5084     key "clock-identity";
5085     config false;
5086     leaf clock-identity {
5087         type ptp-tt:clock-identity;
5088         config false;
5089         description
5090             "The clock identity of the application clock.";
5091         reference
5092             "Item a) in 6.4.10.3.7.7 of IEC/IEEE 60802";
5093     }
5094     leaf clock-target {
5095         type boolean;
5096         config false;
5097         description
5098             "The Boolean value indicates if the application clock is a
5099             clock target (TRUE) or clock source (FALSE).";
5100         reference
5101             "Item b) in 6.4.10.3.7.7 of IEC/IEEE 60802";
5102     }
5103     leaf attached-ptp-instance-index {
5104         type leafref {
5105             path "/ptp-tt:ptp/ptp-tt:instances/ptp-tt:instance/ptp-
5106 tt:instance-index";
5107         }
5108         config false;
5109         description
5110             "The value is a reference to the index of the PTP or hot
5111             standby Instance, that is attached to the application
5112             clock.";
5113         reference
5114             "Item f) in 6.4.10.3.7.7 of IEC/IEEE 60802";
5115     }
5116     leaf arb-supported {
5117         type boolean;
5118         config false;
5119         description
5120             "The Boolean value indicates if the application clock
5121             supports the ARB timescale.";
5122         reference
5123             "Item c) in 6.4.10.3.7.7 of IEC/IEEE 60802";
5124     }
5125     leaf ptp-supported {
5126         type boolean;
5127         config false;
5128         description
5129             "The Boolean value indicates if the application clock
5130             supports the PTP timescale.";
5131         reference
5132             "Item d) in 6.4.10.3.7.7 of IEC/IEEE 60802";
5133     }
5134     leaf hot-standby-supported {
5135         type boolean;
5136         config false;
5137         description
5138             "The Boolean value indicates if the application clock
5139             supports the hot standby.";
5140         reference
5141             "Item e) in 6.4.10.3.7.7 of IEC/IEEE 60802";
5142     }

```

```
5143     leaf is-synced {
5144         type boolean;
5145         config false;
5146         description
5147             "The Boolean value indicates if the application clock is
5148             either synchronized to the attached PTP Instance (TRUE)
5149             or to an internal/external ClockSource (FALSE).";
5150         reference
5151             "Item g) in 6.4.10.3.7.7 of IEC/IEEE 60802";
5152     }
5153 }
5154 }
5155 }
```

5156 6.4.10.8.3 Module **iecieeee60802-sched-bridge**

```
5157 module iecieeee60802-sched-bridge {
5158     yang-version 1.1;
5159     namespace "urn:ieee:std:60802:yang:iecieeee60802-sched-bridge";
5160     prefix ia-sched-bridge;
5161
5162     import ieee802-types {
5163         prefix ieee802;
5164     }
5165     import ieee802-dot1q-bridge {
5166         prefix bridge;
5167     }
5168     import ieee802-dot1q-sched-bridge {
5169         prefix sched-bridge;
5170     }
5171     import ietf-interfaces {
5172         prefix if;
5173     }
5174
5175     organization
5176         "IEEE 802.1 Working Group and IEC subcommittee 65C:
5177         Industrial networks, of IEC technical committee 65:
5178         Industrial-process measurement, control and automation";
5179     contact
5180         "WG-URL: http://ieee802.org/1/
5181         WG-EMail: stds-802-1-1@ieee.org
5182
5183         Contact: IEEE 802.1 Working Group Chair
5184         Postal: C/O IEEE 802.1 Working Group
5185         IEEE Standards Association
5186         445 Hoes Lane
5187         Piscataway, NJ 08854
5188         USA
5189
5190         E-mail: stds-802-1-chairs@ieee.org";
5191     description
5192         "Management objects that provide information about IEC/IEEE 60802
5193         IA-Stations as specified in IEC/IEEE 60802.
5194
5195         Copyright (C) IEC/IEEE (2025).
5196         This version of this YANG module is part of IEC/IEEE 60802;
5197         see the standard itself for full legal notices.";
5198
5199     revision 2024-02-19 {
5200         description "Published as part of IEC/IEEE 60802-2025.
5201         The following reference statement identifies each referenced
5202         IEEE Standard as updated by applicable amendments.";
5203         reference
```

```

5204     "IEC/IEEE 60802 TSN profile for industrial automation:
5205     IEC/IEEE 60802-2025.
5206     IEEE Std 802.1Q Bridges and Bridged Networks:
5207     IEEE Std 802.1Q-2022, IEEE Std 802.1Qcz-2023,
5208     IEEE Std 802.1Qcw-2023, IEEE Std 802.1Qdj-2024,
5209     IEEE Std 802.1Qdx-2024, IEEE Std 802.1Qdy-2024.";
5210 }
5211
5212     augment "/if:interfaces/if:interface/bridge:bridge-port/sched-bridge:gate-
5213     parameter-table" {
5214         description
5215         "Augment IEEE Std 802.1 bridge/gate-parameter-table.";
5216         list min-gating-times {
5217             description
5218             "The list of minimum gating times per supported line speed.";
5219             reference
5220             "6.4.10.3.4.3 of IEC/IEEE 60802";
5221             key "speed";
5222             config false;
5223             leaf speed {
5224                 type uint32;
5225                 config false;
5226                 description
5227                 "This value is the line speed in Mbps.";
5228             }
5229             container min-cycle-time {
5230                 uses ieee802:rational-grouping;
5231                 description
5232                 "The value is the minimum value supported by this port of
5233                 the AdminCycleTime and OperCycleTime parameters given as
5234                 rational number of seconds.";
5235                 reference
5236                 "Item a) in 6.4.10.3.4.3 of IEC/IEEE 60802";
5237             }
5238             leaf min-interval-time {
5239                 type uint32;
5240                 description
5241                 "The value is the minimum value supported by this port of
5242                 the TimeIntervalValue parameter in nanoseconds.";
5243                 reference
5244                 "Item b) in 6.4.10.3.4.3 of IEC/IEEE 60802";
5245             }
5246         }
5247     }
5248 }

```

5249 **6.4.10.8.4 Module iecieee60802-cnc-config**

```

5250 module iecieee60802-cnc-config {
5251     yang-version 1.1;
5252     namespace "urn:ieee:std:60802:yang:iecieee60802-cnc-config";
5253     prefix ia-cnc-config;
5254
5255     import ieee802-dot1q-cnc-config {
5256         prefix dot1q-cnc-config;
5257     }
5258     import ieee802-dot1q-tsn-types {
5259         prefix tsn-types;
5260     }
5261
5262     organization
5263     "IEEE 802.1 Working Group and IEC subcommittee 65C:
5264     Industrial networks, of IEC technical committee 65:

```

```
5265     Industrial-process measurement, control and automation";
5266 contact
5267     "WG-URL: http://ieee802.org/1/
5268     WG-EMail: stds-802-1-1@ieee.org
5269
5270     Contact: IEEE 802.1 Working Group Chair
5271             Postal: C/O IEEE 802.1 Working Group
5272             IEEE Standards Association
5273             445 Hoes Lane
5274             Piscataway, NJ 08854
5275             USA
5276
5277     E-mail: stds-802-1-chairs@ieee.org";
5278 description
5279     "Management objects that provide information about IEC/IEEE 60802
5280     IA-Stations as specified in IEC/IEEE 60802.
5281
5282     Copyright (C) IEC/IEEE (2025).
5283     This version of this YANG module is part of IEC/IEEE 60802;
5284     see the standard itself for full legal notices.";
5285
5286 revision 2025-01-10 {
5287     description "Published as part of IEC/IEEE 60802-2025.
5288     The following reference statement identifies each referenced
5289     IEEE Standard as updated by applicable amendments.";
5290     reference
5291         "IEC/IEEE 60802 TSN profile for industrial automation:
5292         IEC/IEEE 60802-2025.
5293         IEEE Std 802.1Q Bridges and Bridged Networks:
5294         IEEE Std 802.1Q-2022, IEEE Std 802.1Qcz-2023,
5295         IEEE Std 802.1Qcw-2023, IEEE Std 802.1Qdj-2024,
5296         IEEE Std 802.1Qdx-2024, IEEE Std 802.1Qdy-2024.";
5297 }
5298
5299 augment "/dot1q-cnc-config:cnc-config" {
5300     description
5301         "Augment main container in cnc-config.";
5302     leaf max-config-domains {
5303         type uint8;
5304         config false;
5305         description
5306             "The value is the maximum number of supported configuration
5307             domains.";
5308         reference
5309             "6.4.10.3.9.1 of IEC/IEEE 60802";
5310     }
5311     leaf max-cucs {
5312         type uint8;
5313         config false;
5314         description
5315             "The value is the maximum number of supported CUC entities.";
5316         reference
5317             "6.4.10.3.9.2 of IEC/IEEE 60802";
5318     }
5319     leaf max-ia-stations {
5320         type uint16;
5321         config false;
5322         description
5323             "The value is the maximum number of supported IA-stations.";
5324         reference
5325             "6.4.10.3.9.3 of IEC/IEEE 60802";
5326     }
5327     leaf max-network-diameter {
```

```

5328     type uint8;
5329     config false;
5330     description
5331         "The value is the maximum supported network diameter.";
5332     reference
5333         "6.4.10.3.9.4 of IEC/IEEE 60802";
5334 }
5335 leaf max-streams {
5336     type uint16;
5337     config false;
5338     description
5339         "The value is the maximum number of supported streams.";
5340     reference
5341         "6.4.10.3.9.5 of IEC/IEEE 60802";
5342 }
5343 leaf max-num-seamless-trees {
5344     type uint8;
5345     config false;
5346     description
5347         "The value is the maximum number of trees supported for
5348         seamless redundancy of a stream.";
5349     reference
5350         "6.4.10.3.9.6 of IEC/IEEE 60802";
5351 }
5352 leaf hot-standby-supported {
5353     type uint8;
5354     config false;
5355     description
5356         "The Boolean value indicates if PTP hot standby is
5357         supported.";
5358     reference
5359         "6.4.10.3.9.7 of IEC/IEEE 60802";
5360 }
5361 action add_streams {
5362     description
5363         "This Action requests a CNC to add a list of streams.";
5364     input {
5365         leaf cuc-id {
5366             type string;
5367             description
5368                 "The CUC ID where the streams are to be added";
5369         }
5370         list stream-list {
5371             key "stream-id";
5372             description
5373                 "List of Streams that should be added.";
5374             leaf stream-id {
5375                 type tsn-types:stream-id-type;
5376                 description
5377                     "The Stream ID is a unique identifier of a Stream
5378                     request and corresponding configuration. It is used to
5379                     associate a CUC's Stream request with a CNC's
5380                     corresponding response.";
5381             }
5382             container talker {
5383                 description
5384                     "The Talker container contains: - Talker's behavior for
5385                     Stream (how/when transmitted) - Talker's requirements
5386                     from the network - TSN capabilities of the Talker's
5387                     interface(s).";
5388                 uses tsn-types:group-talker;
5389             }
5390             list listener {

```



```

5391         key "index";
5392         description
5393             "Each Listener list entry contains: - Listener's
5394             requirements from the network - TSN capabilities of
5395             the Listener's interface(s).";
5396         leaf index {
5397             type uint32;
5398             description
5399                 "This index is provided in order to provide a unique
5400                 key per list entry.";
5401         }
5402         uses tsn-types:group-listener;
5403     }
5404 }
5405 }
5406 output {
5407     leaf result {
5408         type boolean;
5409         description
5410             "Returns status information indicating if Stream addition
5411             has been successful.";
5412     }
5413 }
5414 }
5415 }
5416
5417 augment "/dot1q-cnc-config:cnc-config/dot1q-cnc-config:domain/dot1q-cnc-
5418 config:cuc/dot1q-cnc-config:stream" {
5419     description
5420         "Augment stream list in tsc-config-uni.";
5421     action remove_listener {
5422         description
5423             "This Action removes listeners from a stream.";
5424         input {
5425             list listener {
5426                 key "index";
5427                 description
5428                     "Each Listener list entry contains: - Listener's
5429                     requirements from the network - TSN capabilities of the
5430                     Listener's interface(s).";
5431                 leaf index {
5432                     type uint32;
5433                     description
5434                         "This index is provided in order to provide a unique
5435                         key per list entry.";
5436                 }
5437             }
5438         }
5439         output {
5440             leaf result {
5441                 type boolean;
5442                 description
5443                     "Returns status information indicating if listene removal
5444                     has been successful.";
5445             }
5446         }
5447     }
5448 }
5449 }

```

6.4.10.8.5 Module iecieeee60802-ia-station

```

5451 module iecieeee60802-ia-station {
5452     yang-version 1.1;

```

```

5453 namespace "urn:ieee:std:60802:yang:ieciieee60802-ia-station";
5454 prefix ias;
5455
5456 import ietf-datastores {
5457     prefix ds;
5458     reference
5459         "IETF RFC 8342: Network Management Datastore Architecture
5460         (NMDA)";
5461 }
5462 import ietf-netconf-acm {
5463     prefix nacm;
5464     reference
5465         "IETF RFC 8341: Network Configuration Access Control Model";
5466 }
5467
5468 organization
5469     "IEEE 802.1 Working Group and IEC subcommittee 65C:
5470     Industrial networks, of IEC technical committee 65:
5471     Industrial-process measurement, control and automation";
5472 contact
5473     "WG-URL: http://ieee802.org/1/
5474     WG-EMail: stds-802-1-1@ieee.org
5475
5476     Contact: IEEE 802.1 Working Group Chair
5477             Postal: C/O IEEE 802.1 Working Group
5478             IEEE Standards Association
5479             445 Hoes Lane
5480             Piscataway, NJ 08854
5481             USA
5482
5483     E-mail: stds-802-1-chairs@ieee.org";
5484 description
5485     "Capability information and reset to factory defaults
5486     functionality for IEC/IEEE 60802 IA-Stations as specified in
5487     IEC/IEEE 60802.
5488
5489     Copyright (C) IEC/IEEE (2025).
5490     This version of this YANG module is part of IEC/IEEE 60802;
5491     see the standard itself for full legal notices.";
5492
5493 revision 2025-01-10 {
5494     description "Published as part of IEC/IEEE 60802-2025.
5495     The following reference statement identifies each referenced
5496     IEEE Standard as updated by applicable amendments.";
5497     reference
5498         "IEC/IEEE 60802 TSN profile for industrial automation:
5499         IEC/IEEE 60802-2025.
5500         IEEE Std 802.1Q Bridges and Bridged Networks:
5501         IEEE Std 802.1Q-2022, IEEE Std 802.1Qcz-2023,
5502         IEEE Std 802.1Qcw-2023, IEEE Std 802.1Qdj-2024,
5503         IEEE Std 802.1Qdx-2024, IEEE Std 802.1Qdy-2024.";
5504 }
5505
5506 feature ia-factory-default-datastore {
5507     description
5508         "Indicates that the factory default configuration is
5509         available as a datastore.";
5510 }
5511
5512 identity ia-factory-default {
5513     if-feature "ia-factory-default-datastore";
5514     base ds:datastore;
5515     description

```

```
5516         "This read-only datastore contains the factory default
5517         configuration for the device that will be used to replace
5518         the contents of the read-write conventional configuration
5519         datastores during a 'ia-factory-reset' RPC operation.";
5520     }
5521
5522     container ia-station-capabilities {
5523         description
5524             "This container provides read only information about an
5525             ia-station's capabilities.";
5526         reference
5527             "IEC/IEEE 60802 - YANG Data Model";
5528         config false;
5529         leaf capability-lldp {
5530             type boolean;
5531             config false;
5532             description
5533                 "The value is true if the device supports LLDP.";
5534             reference
5535                 "6.4.10.3.8.5 of IEC/IEEE 60802";
5536         }
5537         leaf capability-timesync {
5538             type boolean;
5539             config false;
5540             description
5541                 "The value is true if the device supports Timesync.";
5542             reference
5543                 "6.4.10.3.8.6 of IEC/IEEE 60802";
5544         }
5545         leaf capability-keystore {
5546             type boolean;
5547             config false;
5548             description
5549                 "The value is true if the device supports Keystore.";
5550             reference
5551                 "6.4.10.3.8.7 of IEC/IEEE 60802";
5552         }
5553         leaf capability-truststore {
5554             type boolean;
5555             config false;
5556             description
5557                 "The value is true if the device supports Truststore.";
5558             reference
5559                 "6.4.10.3.8.9 of IEC/IEEE 60802";
5560         }
5561         leaf capability-nacm {
5562             type boolean;
5563             config false;
5564             description
5565                 "The value is true if the device supports NACM.";
5566             reference
5567                 "6.4.10.3.8.8 of IEC/IEEE 60802";
5568         }
5569         leaf capability-yang-library {
5570             type boolean;
5571             config false;
5572             description
5573                 "The value is true if the device supports YANG library.";
5574             reference
5575                 "6.4.10.3.8.10 of IEC/IEEE 60802";
5576         }
5577         leaf capability-yang-push {
5578             type boolean;
```

```

5579     config false;
5580     description
5581         "The value is true if the device supports YANG push.";
5582     reference
5583         "6.4.10.3.8.11 of IEC/IEEE 60802";
5584 }
5585 leaf capability-yang-notifications {
5586     type boolean;
5587     config false;
5588     description
5589         "The value is true if the device supports YANG
5590         notifications.";
5591     reference
5592         "6.4.10.3.8.12 of IEC/IEEE 60802";
5593 }
5594 leaf capability-netconf-monitoring {
5595     type boolean;
5596     config false;
5597     description
5598         "The value is true if the device supports NETCONF
5599         monitoring.";
5600     reference
5601         "6.4.10.3.8.13 of IEC/IEEE 60802";
5602 }
5603 leaf capability-netconf-client {
5604     type boolean;
5605     config false;
5606     description
5607         "The value is true if the device supports NETCONF client.";
5608     reference
5609         "6.4.10.3.8.14 of IEC/IEEE 60802";
5610 }
5611 leaf capability-cnc-config {
5612     type boolean;
5613     config false;
5614     description
5615         "The value is true if the device supports the CNC configuration.";
5616     reference
5617         "6.4.10.3.8.15 of IEC/IEEE 60802";
5618 }
5619 leaf capability-sched-traffic {
5620     type boolean;
5621     config false;
5622     description
5623         "The value is true if the device supports scheduled
5624         traffic.";
5625     reference
5626         "6.4.10.3.8.16 of IEC/IEEE 60802";
5627 }
5628 leaf capability-frame-preemption {
5629     type boolean;
5630     config false;
5631     description
5632         "The value is true if the device supports frame preemption.";
5633     reference
5634         "6.4.10.3.8.17 of IEC/IEEE 60802";
5635 }
5636 }
5637
5638 rpc ia-factory-reset {
5639     nacm:default-deny-all;
5640     description
5641         "The server resets all datastores to their factory

```

default contents and any nonvolatile storage back to factory condition, deleting all dynamically generated files, including those containing keys, certificates, logs, and other temporary files.

Depending on the factory default configuration, after being reset, the device may become unreachable on the network.

In contrast to the original factory-reset RPC in IETF RFC 8808, this RPC puts the device into a state where a subsequent configuration by a CNC component results in a functioning 60802 IA-station";

```
}  
}
```

6.4.10.8.6 Module `iecieeee60802-subscribed-notifications`

```
module iecieeee60802-subscribed-notifications {  
  yang-version 1.1;  
  namespace  
    "urn:ieee:std:60802:yang:iecieeee60802-subscribed-notifications";  
  prefix ia-sn;  
  
  import ietf-subscribed-notifications {  
    prefix sn;  
  }  
  
  organization  
    "IEEE 802.1 Working Group and IEC subcommittee 65C:  
    Industrial networks, of IEC technical committee 65:  
    Industrial-process measurement, control and automation";  
  contact  
    "WG-URL: http://ieee802.org/1/  
    WG-EMail: stds-802-1-1@ieee.org  
  
    Contact: IEEE 802.1 Working Group Chair  
    Postal: C/O IEEE 802.1 Working Group  
    IEEE Standards Association  
    445 Hoes Lane  
    Piscataway, NJ 08854  
    USA  
  
    E-mail: stds-802-1-chairs@ieee.org";  
  description  
    "Management objects that provide information about IEC/IEEE 60802  
    IA-Stations as specified in IEC/IEEE 60802.  
  
    Copyright (C) IEC/IEEE (2025).  
    This version of this YANG module is part of IEC/IEEE 60802;  
    see the standard itself for full legal notices.";  
  
  revision 2024-02-19 {  
    description "Published as part of IEC/IEEE 60802-2025.  
    The following reference statement identifies each referenced  
    IEEE Standard as updated by applicable amendments.";  
    reference  
      "IEC/IEEE 60802 TSN profile for industrial automation:  
      IEC/IEEE 60802-2025.  
      IEEE Std 802.1Q Bridges and Bridged Networks:  
      IEEE Std 802.1Q-2022, IEEE Std 802.1Qcz-2023,  
      IEEE Std 802.1Qcw-2023, IEEE Std 802.1Qdj-2024,  
      IEEE Std 802.1Qdx-2024, IEEE Std 802.1Qdy-2024.";
```

```

5703     }
5704
5705     augment "/sn:subscriptions" {
5706         description
5707             "Augment subscriptions in ietf-subscribed-notifications.";
5708         leaf max-subscriptions {
5709             type uint16;
5710             config false;
5711             description
5712                 "The value is the maximum number of supported NETCONF Server
5713                 subscriptions.";
5714             reference
5715                 "6.4.10.3.8.1 of IEC/IEEE 60802";
5716         }
5717         leaf max-on-change-subscription-leaves {
5718             type uint16;
5719             config false;
5720             description
5721                 "The value is the maximum number of supported leaves for
5722                 NETCONF Server on-change subscriptions according to IETF
5723                 RFC 8641.";
5724             reference
5725                 "6.4.10.3.8.2 of IEC/IEEE 60802";
5726         }
5727         leaf max-periodic-subscription-leaves {
5728             type uint16;
5729             config false;
5730             description
5731                 "The value is the maximum number of supported leaves for
5732                 NETCONF Server periodic subscriptions according to IETF
5733                 RFC 8641.";
5734             reference
5735                 "6.4.10.3.8.3 of IEC/IEEE 60802";
5736         }
5737         leaf min-periodic-subscription-interval {
5738             type uint16;
5739             config false;
5740             description
5741                 "The value is the minimum periodic subscription interval in
5742                 centiseconds (0.01 seconds) for NETCONF Server periodic
5743                 subscriptions according to IETF RFC 8641.";
5744             reference
5745                 "6.4.10.3.8.4 of IEC/IEEE 60802";
5746         }
5747     }
5748 }

```

5749 **6.4.10.8.7 Module iecieee60802-tt-mapping**

```

5750 module iecieee60802-tt-mapping {
5751     yang-version "1.1";
5752     namespace "urn:ieee:std:60802:yang:iecieee60802-tt-mapping";
5753     prefix tt-mapping;
5754
5755     import ieee802-dot1q-types {
5756         prefix dot1q-types;
5757     }
5758
5759     organization
5760         "IEEE 802.1 Working Group and IEC subcommittee 65C:
5761         Industrial networks, of IEC technical committee 65:
5762         Industrial-process measurement, control and automation";
5763     contact

```

```
5764     "WG-URL: http://ieee802.org/1/
5765     WG-EMail: stds-802-1-1@ieee.org
5766
5767     Contact: IEEE 802.1 Working Group Chair
5768     Postal: C/O IEEE 802.1 Working Group
5769             IEEE Standards Association
5770             445 Hoes Lane
5771             Piscataway, NJ 08854
5772             USA
5773
5774     E-mail: stds-802-1-chairs@ieee.org";
5775 description
5776     "This module provides information on characteristics of traffic classes
5777     configured in a network to allow applications to map application
5778 traffic
5779     classes to them.
5780
5781     Copyright (C) IEC/IEEE (2025).
5782
5783     This version of this YANG module is part of IEC/IEEE 60802;
5784     see the standard itself for full legal notices.";
5785 revision 2024-12-13 {
5786     description
5787         "Published as part of IEC/IEEE 60802-2025.
5788
5789         The following reference statement identifies each referenced
5790         IEEE Standard as updated by applicable amendments.";
5791     reference
5792         "IEC/IEEE 60802 TSN profile for industrial automation:
5793         IEC/IEEE 60802-2025.
5794         IEEE Std 802.1Q Bridges and Bridged Networks:
5795         IEEE Std 802.1Q-2022, IEEE Std 802.1Qcz-2023,
5796         IEEE Std 802.1Qcw-2023, IEEE Std 802.1Qdj-2024,
5797         IEEE Std 802.1Qdx-2024, IEEE Std 802.1Qdy-2024.";
5798 }
5799
5800 container tt-mapping {
5801     description
5802         "Top-level container for the traffic class module.";
5803     list network {
5804         key "network-name";
5805         description
5806             "A list of networks, each containing one or more sets of traffic
5807 class
5808         characteristics.";
5809         leaf network-name {
5810             type string;
5811             description
5812                 "The network-name uniquely identifies a network.";
5813         }
5814         list traffic-type {
5815             key "traffic-type-name";
5816             description
5817                 "A list of traffic types in the network, each with its associated
5818                 traffic type characteristics.";
5819             leaf traffic-type-name {
5820                 type string;
5821                 description
5822                     "The traffic-type-name uniquely identifies the traffic type.";
5823             }
5824             leaf cyclic {
5825                 type boolean;
5826                 description
```

```

5827         "Indicates if cyclic transmission is mandatory (TRUE) or
5828 optional (FALSE)
5829         for this traffic type.";
5830     reference
5831         "4.7.2 of IEC/IEEE 60802-2025";
5832 }
5833 leaf data-delivery-req {
5834     type enumeration {
5835         enum frame-latency {
5836             value 0;
5837             description
5838                 "Data delivery of a frame for a given Talker-Listener pair
5839                 occurs within a bounded timespan.";
5840         }
5841         enum flow-latency {
5842             value 1;
5843             description
5844                 "Data delivery up to a certain number of frames or data
5845 size
5846                 (including bursts of frames) occurring over a defined
5847                 period.";
5848         }
5849         enum deadline {
5850             value 2;
5851             description
5852                 "Data delivery of a frame to a given Listener occurs at or
5853                 before a specific point in time.";
5854         }
5855         enum no {
5856             value 3;
5857             description
5858                 "No special data delivery requirements.";
5859         }
5860     }
5861     description
5862         "The data-delivery-req provides information on the data
5863         delivery requirements for this traffic type.";
5864     reference
5865         "4.7.2 of IEC/IEEE 60802-2025";
5866 }
5867 leaf time-triggered {
5868     type boolean;
5869     description
5870         "Indicates if it is required (TRUE) or optional (FALSE) that
5871         transmission occurs at a specific point in time based upon
5872         the Working Clock.";
5873     reference
5874         "4.7.2 of IEC/IEEE 60802-2025";
5875 }
5876 leaf stream-based {
5877     type boolean;
5878     default true;
5879     description
5880         "Indicates if traffic of this traffic type requires
5881         registering a stream, i.e., allocating dedicated
5882         resources, for the traffic, before a given Talker
5883         can start transmitting data (TRUE), or if registering
5884         a stream is not required (FALSE).";
5885     reference
5886         "4.7.2 of IEC/IEEE 60802-2025";
5887 }
5888 leaf vlan-id {
5889     type dot1q-types:vlanid;

```



```

5890         description
5891             "The VID this traffic type is mapped to.";
5892     }
5893     list redundancy {
5894         key "vlan-id-redundant";
5895         description
5896             "A list of VIDs this traffic type is mapped to for redundant
5897             transmission. There is one entry in the list per redundant
5898             transmission ";
5899         leaf vlan-id-redundant {
5900             type dot1q-types:vlanid;
5901             description
5902                 "The VID this traffic type is mapped for a redundant
5903 transmission.";
5904         }
5905     }
5906     leaf priority-value {
5907         type dot1q-types:priority-type;
5908         description
5909             "The PCP this traffic type is mapped to.";
5910     }
5911 }
5912 }
5913 }
5914 }

```

6.4.10.8.8 Module iecieee60802-proxy

```

5916 module iecieee60802-proxy {
5917     yang-version 1.1;
5918     namespace "urn:ieee:std:60802:yang:ieciieee60802-proxy";
5919     prefix pr;
5920
5921     import ieee802-types {
5922         prefix ieee;
5923     }
5924     import ietf-inet-types {
5925         prefix inet;
5926     }
5927
5928     organization
5929         "IEEE 802.1 Working Group and IEC subcommittee 65C:
5930         Industrial networks, of IEC technical committee 65:
5931         Industrial-process measurement, control and automation";
5932     contact
5933         "WG-URL: http://ieee802.org/1/
5934         WG-EMail: stds-802-1-1@ieee.org
5935
5936         Contact: IEEE 802.1 Working Group Chair
5937         Postal: C/O IEEE 802.1 Working Group
5938         IEEE Standards Association
5939         445 Hoes Lane
5940         Piscataway, NJ 08854
5941         USA
5942
5943         E-mail: stds-802-1-chairs@ieee.org";
5944     description
5945         "Configuration and capability information for IEC/IEEE 60802 proxy
5946         targets as specified in IEC/IEEE 60802.
5947
5948         Copyright (C) IEC/IEEE (2025).
5949         This version of this YANG module is part of IEC/IEEE 60802;
5950         see the standard itself for full legal notices.";

```

```

5951
5952 revision 2025-01-08 {
5953     description
5954         "Published as part of IEC/IEEE 60802-2025.
5955         The following reference statement identifies each referenced
5956         IEEE Standard as updated by applicable amendments.";
5957     reference
5958         "IEC/IEEE 60802 TSN profile for industrial automation:
5959         IEC/IEEE 60802-2025.
5960         IEEE Std 802.1Q Bridges and Bridged Networks:
5961         IEEE Std 802.1Q-2022, IEEE Std 802.1Qcz-2023,
5962         IEEE Std 802.1Qcw-2023, IEEE Std 802.1Qdj-2024,
5963         IEEE Std 802.1Qdx-2024, IEEE Std 802.1Qdy-2024.";
5964 }
5965
5966 container proxy {
5967     presence "The presence of this container indicates that this IA station
5968     provides proxy functionality for proxy targets";
5969     description
5970         "This container provides information about a
5971         60802 IA station with proxy functionality.";
5972     reference
5973         "IEC/IEEE 60802 - YANG Data Model";
5974     leaf max-num-proxy-targets {
5975         type uint16;
5976         config false;
5977         description
5978             "The value indicates how many proxy targets can be handled by this
5979 proxy component.";
5980         reference
5981             "6.4.10.3.9.1 of IEC/IEEE 60802";
5982     }
5983     list proxy-types {
5984         key "proxy-type";
5985         config false;
5986         description
5987             "Supported proxy types";
5988         leaf proxy-type {
5989             type string {
5990                 pattern '[0-9A-F]{2}(-[0-9A-F]{2}){2}';
5991             }
5992             config false;
5993             description
5994                 "OID of proxy targets handled by this proxy.";
5995             reference
5996                 "6.4.10.3.9.2 of IEC/IEEE 60802";
5997         }
5998     list proxy-targets {
5999         key "device";
6000         description
6001             "List of proxy targets that are connected to this proxy";
6002         leaf device {
6003             type string;
6004             description
6005                 "The identification of the device.";
6006             reference
6007                 "6.4.10.3.9.3 of IEC/IEEE 60802";
6008         }
6009         leaf device-type {
6010             type string;
6011             description
6012                 "The type of the device.";
6013             reference

```

```
6014         "6.4.10.3.9.4 of IEC/IEEE 60802";
6015     }
6016     container proxy-addr {
6017         leaf ip-addr {
6018             type inet:ip-address;
6019             description
6020                 "The ip-address for the proxy target.";
6021             reference
6022                 "6.4.10.3.9.5 of IEC/IEEE 60802";
6023         }
6024         leaf port {
6025             type inet:port-number;
6026             description
6027                 "The port-number for the proxy-target.";
6028             reference
6029                 "6.4.10.3.9.6 of IEC/IEEE 60802";
6030         }
6031     }
6032     container device-addr {
6033         leaf ip-addr {
6034             type inet:ip-address;
6035             description
6036                 "The ip-address of the proxy target.";
6037             reference
6038                 "6.4.10.3.9.7 of IEC/IEEE 60802";
6039         }
6040         leaf mac-addr {
6041             type ieee:mac-address;
6042             description
6043                 "The mac-address of the proxy target.";
6044             reference
6045                 "6.4.10.3.9.8 of IEC/IEEE 60802";
6046         }
6047     }
6048 }
6049 }
6050 }
6051 }
```

6052 6.4.10.8.9 Module **iecieeee60802-proxy-target**

```
6053 module iecieeee60802-proxy-target {
6054     yang-version 1.1;
6055     namespace "urn:ieee:std:60802:yang:iecieeee60802-proxy-target";
6056     prefix cd;
6057
6058     organization
6059         "IEEE 802.1 Working Group and IEC subcommittee 65C:
6060         Industrial networks, of IEC technical committee 65:
6061         Industrial-process measurement, control and automation";
6062     contact
6063         "WG-URL: http://ieee802.org/1/
6064         WG-EMail: stds-802-1-1@ieee.org
6065
6066         Contact: IEEE 802.1 Working Group Chair
6067         Postal: C/O IEEE 802.1 Working Group
6068         IEEE Standards Association
6069         445 Hoes Lane
6070         Piscataway, NJ 08854
6071         USA
6072
6073         E-mail: stds-802-1-chairs@ieee.org";
6074     description
```

```

6075     "Capability information for IEC/IEEE 60802 proxy targets as
6076         specified in IEC/IEEE 60802.
6077
6078     Copyright (C) IEC/IEEE (2025).
6079     This version of this YANG module is part of IEC/IEEE 60802;
6080     see the standard itself for full legal notices.";
6081
6082     revision 2025-01-08 {
6083         description
6084             "Published as part of IEC/IEEE 60802-2025.
6085             The following reference statement identifies each referenced
6086             IEEE Standard as updated by applicable amendments.";
6087         reference
6088             "IEC/IEEE 60802 TSN profile for industrial automation:
6089             IEC/IEEE 60802-2025.
6090             IEEE Std 802.1Q Bridges and Bridged Networks:
6091             IEEE Std 802.1Q-2022, IEEE Std 802.1Qcz-2023,
6092             IEEE Std 802.1Qcw-2023, IEEE Std 802.1Qdj-2024,
6093             IEEE Std 802.1Qdx-2024, IEEE Std 802.1Qdy-2024.";
6094     }
6095
6096     container proxy-target {
6097         presence "The presence of this container indicates that an IA station
6098             is a proxy target that requires a proxy device";
6099         description
6100             "This container provides information about a
6101             60802 proxy target device.";
6102         reference
6103             "IEC/IEEE 60802 - YANG Data Model";
6104         leaf required-proxy-type {
6105             type string {
6106                 pattern '[0-9A-F]{2}(-[0-9A-F]{2}){2}';
6107             }
6108             config false;
6109             description
6110                 "OID of proxy that this proxy target requires.";
6111             reference
6112                 "6.4.10.3.10.1 of IEC/IEEE 60802";
6113         }
6114     }
6115 }

```

6116 6.5 Topology discovery and verification

6117 6.5.1 Topology discovery and verification requirements

6118 Electrical engineering of machines with multiple IA-stations includes the definition of the
6119 machine internal network topology (i.e. the engineered topology).

6120 The machine internal network topology includes type specific data of IA-stations (for example
6121 model name or manufacturer name) as well as instance specific data (for example IP addresses
6122 or DNS names).

6123 The electrical engineering data of the network topology is used:

- 6124 • during commissioning so that machine planning and installation are identical;
- 6125 • by the TDE during operation to verify that the actual topology of the Configuration Domain
6126 matches the engineered topology;
- 6127 • by maintenance staff during repair to easily identify failed IA-stations, ports, or links to be
6128 replaced.

Repair and replacement of an IA-station do not require verification of the updated engineered topology so that the TDE does not produce a verification error.

IA-stations are necessarily pre-configured when they are repaired or replaced. IA-stations report type and instance data as described in 6.5.3.

6.5.2 Topology discovery overview

6.5.2.1 General

LLDP enables the discovery of IA-stations, their external ports, and their external connectivity. A Topology Discovery Entity can query LLDP data by remote management to derive the physical network topology.

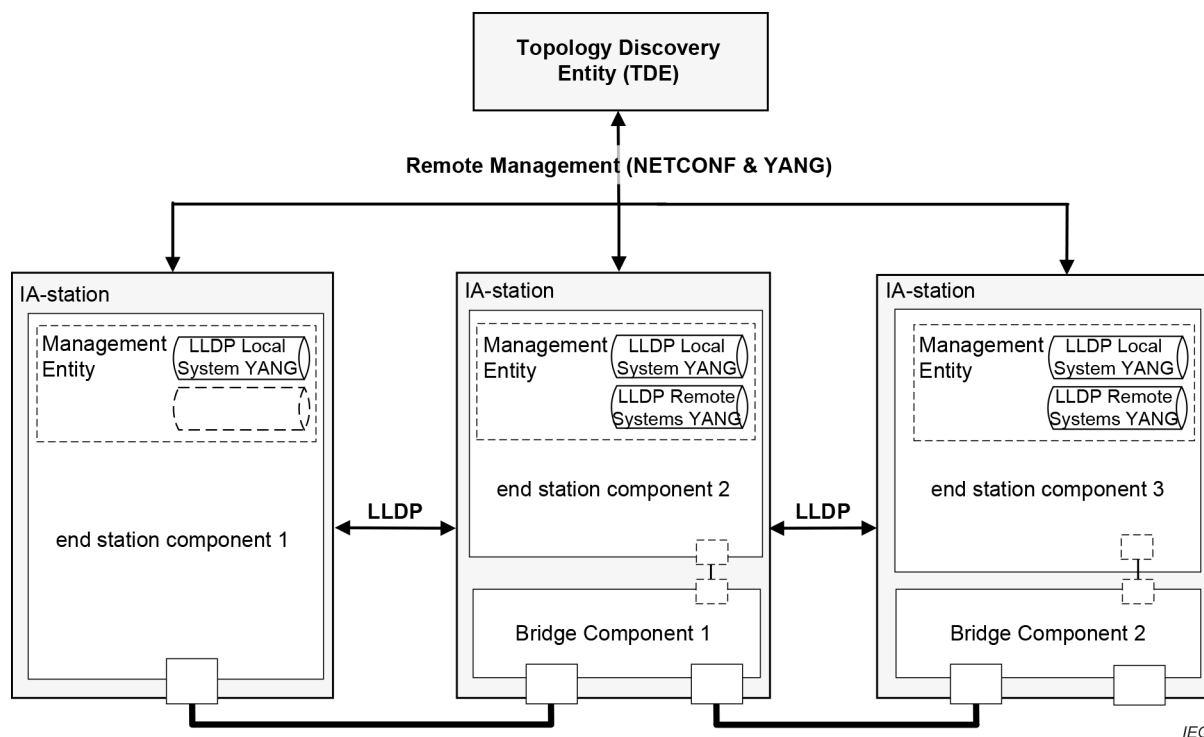


Figure 38 – Usage example of LLDP

Figure 38 illustrates a network showing the LLDP agent implementations in an IA-station consisting of a single end station component and two IA-stations with end station and Bridge components (see 4.3). The LLDP protocol is used to convey neighborhood information among peers, and NETCONF is used between the TDE and the IA-stations to query this neighborhood information from the IA-stations. This information allows the TDE to discover IA-stations and the physical network topology.

NOTE A Topology Discovery Entity (TDE) can be run from anywhere in the network with reachability to the to-be-discovered devices.

IA-stations announce themselves via LLDP to support discovery by the TDE. Announcements contain the management address (see 6.5.2.4.6) and system capabilities (see 6.5.2.4.5) for the discovery operation. The announced system capabilities information enables the TDE to identify IA-stations with multiple end station and Bridge components. The TDE can use the definitions in 6.4.3 for the discovery of the internal structure of such IA-stations.

To allow for operational behavior and exchanged information, IA-stations support the local system YANG (see 6.4.9.2.2). IA-stations that include a Bridge component additionally support

6155 the processing of received LLDP messages and support the remote systems YANG (see
6156 6.4.9.2.2).

6157 **6.5.2.2 LLDP operational control parameters**

6158 LLDP defines several operational parameters that control the protocol behavior (see IEEE Std
6159 802.1AB-2016, 10.5.1). These parameter definitions apply to all external ports of an IA-station.

6160 NOTE According to IEEE Std 802.1AB-2016, 9.1.1 c), changes to the local system that impact information
6161 exchanged via LLDP immediately trigger the transmission of an LLDPDU to communicate the local changes as quickly
6162 as possible to any neighboring systems.

6163 An IA-station shall support LLDP transmit mode (adminStatus enabledTxOnly) on an external
6164 end station component port and may support transmit and receive mode (adminStatus
6165 enabledRxTx) on that port (see IEEE Std 802.1AB-2016, 10.5.1).

6166 An IA-station shall support LLDP transmit and receive mode (adminStatus enabledRxTx) on an
6167 external Bridge component port (see IEEE Std 802.1AB-2016, 10.5.1).

6168 **6.5.2.3 LLDPDU transmission, reception, and addressing**

6169 The destination address to be used for LLDPDU transmission (dest-mac-address) shall be the
6170 nearest bridge group MAC address, i.e., 01-80-C2-00-00-0E, on all ports to limit the scope of
6171 LLDPDU propagation to a single physical link (see IEEE Std 802.1AB-2016, 7.1 a).

6172 NOTE IEEE Std 802.1AB-2016 defines LLDPDUs to be transmitted untagged, i.e. frames do not carry priority
6173 information for traffic class selection. At the same time, IEEE Std 802.1AB-2016 neither specifies a well-defined
6174 device-internal priority nor management capabilities for the configuration of the traffic class to be used for the
6175 transmission of LLDPDUs. It is the user's responsibility to prevent LLDPDUs from interfering with the transmission
6176 of time-critical control data.

6177 **6.5.2.4 LLDP TLV selection**

6178 **6.5.2.4.1 General**

6179 An IA-station transmitting LLDPDUs shall include the LLDP TLVs selected in 6.5.2.4 and may
6180 include additional TLVs (tlvs-tx-enable). An IA-station receiving LLDPDUs shall process
6181 LLDPDUs.

6182 Each LLDPDU shall contain the following LLDP TLVs specified in IEEE Std 802.1AB-2016, 8.5:

- 6183 • exactly one Chassis ID TLV according to 6.5.2.4.2,
- 6184 • exactly one Port ID TLV according to 6.5.2.4.3,
- 6185 • exactly one Time To Live TLV according to 6.5.2.4.4,
- 6186 • exactly one System Capabilities TLV according to 6.5.2.4.5, and
- 6187 • one or more Management Address TLVs according to 6.5.2.4.6.

6188 NOTE The concatenation of the Chassis ID and Port ID fields enables the recipient of an LLDPDU to identify the
6189 sending LLDP agent/port.

6190 **6.5.2.4.2 Chassis ID TLV**

6191 The Chassis ID field shall contain the same value for all transmitted LLDPDUs independent
6192 from the transmitting port of the IA-station, i.e. be a non-volatile identifier which is unique within
6193 the context of the administrative domain.

6194 The Chassis ID subtype field (chassis-id-subtype) should contain subtype 4, indicating that the
6195 Chassis ID field (chassis-id) contains a MAC address to achieve the Chassis ID's desired
6196 uniqueness. For IA-stations with multiple unique MAC addresses, any one of the IA-station's
6197 MAC addresses may be used and shall be the same for all external ports of that IA-station.

6198 6.5.2.4.3 Port ID TLV

6199 The Port ID field shall contain the same value for all transmitted LLDPDUs for a given external
6200 port, i.e. be a non-volatile, IA-station-unique identifier of the LLDPDU-transmitting port.

6201 The Port ID subtype field (port-id-subtype) should contain subtype 5, indicating that the Port ID
6202 field contains the port interface name (name) according to IETF RFC 8343.

6203 IA-stations should restrict the system-defined port ID to read-only access and a maximum name
6204 length of 255 characters. The names should match the port names printed on the chassis.

6205 6.5.2.4.4 Time To Live TLV

6206 The Time To Live value shall be set according to IEEE Std 802.1AB-2016, 8.5.4.

6207 6.5.2.4.5 System capabilities TLV

6208 An IA-station consisting of a single end station component shall set the system capabilities and
6209 enabled capabilities fields (system-capabilities-supported, system-capabilities-enabled) to
6210 Station Only (i.e. bit 8 set to 1) for all transmitted LLDPDUs.

6211 An IA-station consisting of at least one end station component and at least one Bridge
6212 component shall set the system capabilities and enabled capabilities fields to Station Only (i.e.
6213 bit 8 set to 1) and C-VLAN component (i.e. bit 9 set to 1) for all transmitted LLDPDUs.

6214 NOTE The combination of the Station Only and C-VLAN component flags is used as a marker indicating to the TDE
6215 that the internal structure of the IA-station consists of multiple components. This is a deliberate deviation from IEEE
6216 Std 802.1AB-2016, Table 8-4, which states in a footnote: "The Station Only capability is intended for devices that
6217 implement only an end station capability, and for which none of the other capabilities in the table apply. Bit 8 therefore
6218 is not set in conjunction with any other bits."

6219 6.5.2.4.6 Management address TLV

6220 An IA-station shall announce at least one IPv4 address by which its Management entity (see
6221 4.3) can be reached (management-address-tx-port).

6222 6.5.2.5 LLDP remote systems data

6223 An IA-station supporting the remote systems YANG shall be able to store information from at
6224 least one neighbor per external port.

6225 Receiving LLDPDUs from more neighbors than supported on a given port shall result in the last
6226 one received being saved to the remote systems YANG as described in
6227 IEEE Std 802.1AB-2016, 9.2.7.7.5.

6228 6.5.3 Topology verification overview

6229 Topology verification checks discovered topologies against engineered topologies. Topology
6230 verification data includes for every IA-station:

- 6231 • model name,
- 6232 • manufacturer name, and
- 6233 • management address.

6234 Topology verification data includes for every external port of an IA-station:

- 6235 • port name, and
- 6236 • remote connection (i.e. management address and port name of connected IA-station).

6237 To support topology verification IA-stations shall support LLDP YANG data as specified in
6238 6.4.9.2.2 and Hardware Management YANG data as specified in 6.4.9.2.5.8.

6239 IA-station hardware instance specific data like MAC addresses or serial numbers are not
6240 considered for topology verification. This kind of data changes after a repair and replacement
6241 operation and thus, induces a topology verification error.

6242 **6.6 CNC**

6243 **6.6.1 General**

6244 Subclause 6.6 describes stream destination MAC address handling at the CNC.

6245 **6.6.2 Stream destination MAC address range**

6246 A CNC manages the destination MAC address for requested streams. This destination MAC
6247 address together with the VID identifies the path used for these streams. Thus, a stream
6248 destination MAC address is unique together with the VID in a Configuration Domain.

6249 Figure 39 shows the possible selections of a CNC for a contiguous address range. The CNC
6250 selects an OUI and an offset of the address range for the stream destination MAC addresses.

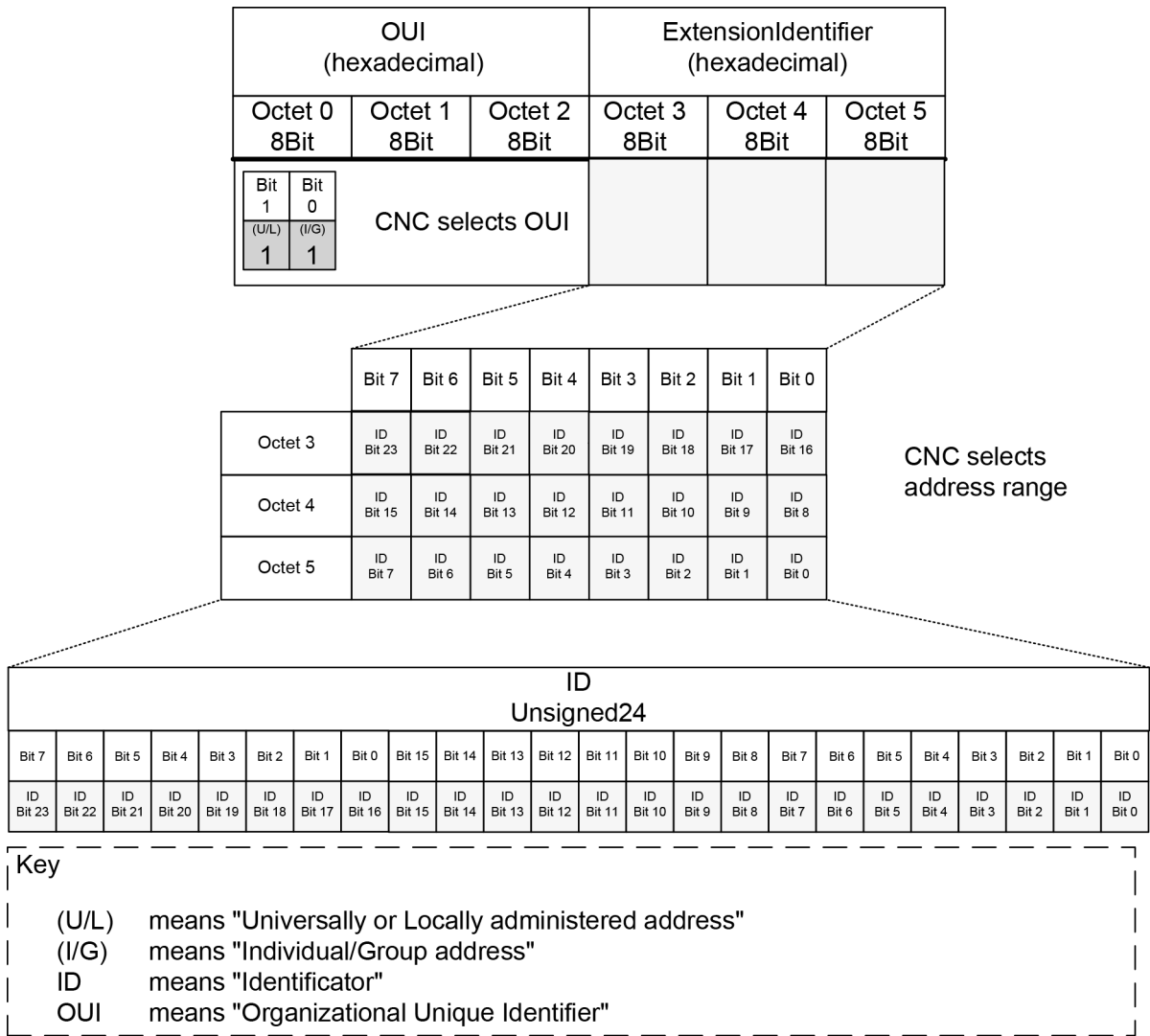
6251 An address range of 2 048 stream destination MAC addresses allows together with a VID the
6252 usage of 2 048 streams. Each additional VID used for streams allows an additional 2 048
6253 streams.

6254 **EXAMPLE**

6255 CNC selected OUI:= 00-80-C2

6256 CNC selected address range:= 0.2047

6257 CNC selected VID:= 101



6258

IEC

6259

Figure 39 – Stream destination MAC Address

Annex A (normative)

PCS proforma – Time-sensitive networking profile for industrial automation

A.1 General⁷

The supplier of an implementation that is claimed to conform to the profile specified in this document shall complete the corresponding Profile Conformance Statement (PCS) proforma, which is presented in a tabular format based on the format used for Protocol Implementation Conformance Statement (PICS) proformas.

The tables do not contain an exhaustive list of all requirements that are stated in the referenced standards; for example, if a row in a table asks whether the implementation is conformant to Standard X, and the answer "Yes" is chosen, then it is assumed that it is possible, for that implementation, to fill out the PCS proforma specified in Standard X to show that the implementation is conformant; however, the tables in this document only further refine those elements of conformance to Standard X where particular answers are required for the profiles specified here.

A completed PCS proforma is the PCS for the implementation in question. The PCS is a statement of which capabilities and options of the protocol have been implemented. The PCS can have several uses, including use by the following.

- a) Protocol implementer, as a checklist to reduce the risk of failure to conform to the document through oversight.
 - b) Supplier and acquirer, or potential acquirer, of the implementation, as a detailed indication of the capabilities of the implementation, stated relative to the common basis for understanding provided by the standard PCS proforma.
 - c) User, or potential user, of the implementation, as a basis for initially checking the possibility of interworking with another implementation.
- NOTE While interworking can never be guaranteed, failure to interwork can often be predicted from incompatible PCS.
- d) Protocol tester, as the basis for selecting appropriate tests against which to assess the claim for conformance of the implementation.
 - e) The user, to verify whether the IA-station, as described by the PCS, fulfills use-case requirements.

A.2 Abbreviated terms and special symbols

A.2.1 Status symbols

- M mandatory
- O optional
- O.n optional, but support of at least one of the group of options labeled by the same numeral n is required
- X prohibited
- pred conditional-item symbol, including predicate identification (see A.3.4)
- ¬ logical negation, applied to a conditional item's predicate

⁷ Copyright release for the PCS: Users of this document may freely reproduce the PCS contained in this document so that they can be used for their intended purpose.

A.2.2 General abbreviated terms

N/A not applicable

PCS Profile Conformance Statement

A.3 Instructions for completing the PCS proforma**A.3.1 General structure of the PCS proforma**

The first part of the PCS proforma, implementation identification and protocol summary, is to be completed as indicated with the information necessary to identify fully both the supplier and the implementation.

The main part of the PCS proforma is a fixed-format questionnaire, divided into several subclauses, each containing a number of individual items. Answers to the questionnaire items are to be provided in the rightmost column, either by simply marking an answer to indicate a restricted choice (usually Yes or No) or by entering a value or a set or range of values. There are some items where two or more choices from a set of possible answers can apply; all relevant choices are to be marked. Each item is identified by an item reference in the first column. The second column contains the question to be answered; the third column records the status of the item – whether support is mandatory, optional, or conditional (see also A.3.4). The fourth column contains the reference or references to the material that specifies the item in the main body of this document, and the fifth column provides the space for the answers.

The PCS indicates support of one of the conformance classes, ccA or ccB, per bridge and end-station component, specified in this profile.

A single IA-station can incorporate the functionality of one or more of the functions listed in this PCS. For example, an IA-station could have both an end station component and a Bridge component.

A supplier can also provide (or be required to provide) further information, categorized as either additional information (see A.3.2) or exception information (see A.3.3). When present, each kind of further information is to be provided in a further subclause of items labeled Ai or Xi, respectively, for cross-referencing purposes, where (i) is any unambiguous identification for the item (for example, simply a numeral). There are no other restrictions on its format and presentation.

A completed PCS proforma, including any Additional Information and Exception Information, is the Protocol Implementation Conformance Statement for the implementation in question.

NOTE Where an implementation is capable of being configured in more than one way, a single PCS can be used to describe all such configurations. However, the supplier has the choice of providing more than one PCS, each covering some subset of the implementation's configuration capabilities, in case that makes for easier and clearer presentation of the information.

A.3.2 Additional information

Items of Additional Information allow a supplier to provide further information intended to assist the interpretation of the PCS. It is not intended or expected that a large quantity be supplied, and a PCS can be considered complete without any such information. Examples can be an outline of the ways in which a (single) implementation can be set up to operate in a variety of environments and configurations, or information about aspects of the implementation that are outside the scope of this document but that have a bearing on the answers to some items.

References to items of Additional Information can be entered next to any answer in the questionnaire and can be included in items of Exception Information.

A.3.3 Exception information

It can occasionally happen that a supplier wishes to answer an item with mandatory status (after any conditions have been applied) in a way that conflicts with the indicated requirement. No preprinted answer will be found in the Support column for this item. Instead, the supplier shall write the missing answer into the Support column, together with an Xi reference to an item of Exception Information and shall provide the appropriate rationale in the Exception item itself.

An implementation for which an Exception item is required in this way does not conform to this document.

NOTE A possible reason for the situation described previously is that a defect in this document has been reported, a correction for which is expected to change the requirement not met by the implementation.

A.3.4 Conditional status

A.3.4.1 Conditional items

The PCS proforma contains a number of conditional items. These are items for which both the applicability of the item itself, and its status if it does apply (mandatory or optional) are dependent on whether certain other items are supported.

Where a group of items is subject to the same condition for applicability, a separate preliminary question about the condition appears at the head of the group, with an instruction to skip to a later point in the questionnaire if the "Not Applicable" (N/A) answer is selected. Otherwise, individual conditional items are indicated by a conditional symbol in the Status column.

A conditional symbol is of the form "pred: S" where pred is a predicate as described in A.3.4.2, and S is a status symbol, M or O.

If the value of the predicate is true (see A.3.4.2), the conditional item is applicable, and its status is indicated by the status symbol following the predicate: the answer column is to be marked in the usual way. If the value of the predicate is false, the "Not Applicable" (N/A) answer is to be marked.

A.3.4.2 Predicates

A predicate is one of the following:

- a) an item-reference for an item in the PCS proforma: the value of the predicate is true if the item is marked as supported and is false otherwise,
- b) a predicate-name, for a predicate defined as a Boolean expression constructed by combining item-references using the Boolean operator OR: the value of the predicate is true if one or more of the items is marked as supported, or
- c) the logical negation symbol "¬" prefixed to an item-reference or predicate-name: the value of the predicate is true if the value of the predicate formed by omitting the "¬" symbol is false, and vice versa.

Each item whose reference is used in a predicate or predicate definition, or in a preliminary question for grouped conditional items, is indicated by an asterisk in the Item column.

A.4 Common requirements

A.4.1 Instructions

One instance of Clause A.4 shall be filled out per IA-station.

A.4.2 Implementation identification

The entire PCS proforma is a form that shall be filled out by a supplier according to Table A.1.

Table A.1 – Implementation identification template

Supplier	
Contact point for queries about the PCS	
Implementation Name(s) and Version(s)	
Other information necessary for full identification, for example, name(s) and version(s) of machines and operating system names	

Only the first three items are required for all implementations; other information can be completed as appropriate in meeting the requirement for full identification. The terms "Name" and "Version" should be interpreted appropriately to correspond with a supplier's terminology (for example, Type, Series, Model).

A.4.3 Profile summary

Table A.2 shows the profile summary template.

Table A.2 – Profile summary template

Identification of profile specification	IEC/IEEE 60802 – Time-Sensitive Networking profile for industrial automation			
Identification of amendments (Amd) and corrigenda (Corr) to the PCS proforma that have been completed as part of the PCS	Amd	:	Corr	:
	Amd	:	Corr	:
Have any Exception items been required? (See A.3.3: the answer "Yes" means that the implementation does not conform to IEC/IEEE 60802)	No	[]	Yes	[]
Date of Statement				

A.4.4 Implementation summary

The form in Table A.3 is used to indicate the type of system that the PCS describes.

Table A.3 – Implementation type

Item	Feature	Status	References	Support
BC-CCA-N	State the number of Conformance Class A bridge components implemented by the IA-station.	O	5.7.2, 5.8.2	Number _____
BC-CCB-N	State the number of Conformance Class B bridge components implemented by the IA-station.	O	5.7.3, 5.8.3	Number _____
ESC-CCA-N	State the number of Conformance Class A end station components implemented by the IA-station.	O.1	5.9.2, 5.10.2	Number _____
ESC-CCB-N	State the number of Conformance Class B end station components implemented by the IA-station.	O.1	5.9.3, 5.10.3	Number _____
CNC	Does the IA-station include a CNC?	O	5.5.4.1 m)	Yes [] No []
CUC	Does the IA-station include a CUC?	O	5.5.4.1 n)	Yes [] No []
PRX	Does the IA-station support proxy functionality?	O	5.5.4.1 o)	Yes [] No []

A.5 IA-station requirements and options

A.5.1 Instructions

One instance of Clause A.5 shall be filled out for an IA-station.

A.5.2 IA-station requirements

The form in Table A.4 is used to indicate the IA-station requirements.

Table A.4 – IA-station requirements

Item	Feature	Status	References	Support
IASTA-1	Does the IA-station support PHY and MAC requirements for external ports?	M	5.5.1	Yes []
IASTA-2	Does the IA-station support topology discovery requirements?	M	5.5.2	Yes []
IASTA-3	Does the IA-station support requirements for time synchronization?	M	5.5.3	Yes []
IASTA-4	Does the IA-station provide a digital data sheet?	M	5.5.4.3	Yes []

A.5.3 Management Requirements for non-proxied IA-stations

The form in Table A.5 is used to indicate management requirements for IA-stations containing conformance class A components

Table A.5 – Additional requirements for non-proxied IA-stations

Item	Feature	Status	References	Support
IASTA-5	Does the IA-station support requirements for Secure management exchanges?	M	5.5.4.1 a) through l)	Yes []
IASTA-6	Number of of Dynamic Subscriptions to YANG Events and Datastores over NETCONF	M	5.5.4.1 h)	Number _____
IASTA-7	Does the IA-station support management YANG modules?	M	5.5.4.2	Yes []

A.5.4 IA-station PHY and MAC options for external ports

The form in Table A.6 is used to indicate PHY and MAC options for external ports.

Table A.6 – IA-station PHY and MAC options

Item	Feature	Status	References	Support
DOT3-1	Does the IA-station support PoE over 2 pairs?	O	5.6.1 a)	Yes [] No [] N/A []
DOT3-2	Does the IA-station support Power Interfaces?	O	5.6.1 b)	Yes [] No [] N/A []
DOT3-3	Does the IA-station support PoE?	O	5.6.1 c)	Yes [] No [] N/A []

A.5.5 IA-station options for time synchronization

The form in Table A.7 is used to indicate options for time synchronization.

6419

Table A.7 – IA-station time synchronization options

Item	Feature	Status	References	Support
PTP-1	Does the IA-station support media-independent TimeTransmitter capability according to IEEE Std 802.1AS-2020, 5.4.2 item b) as amended by IEEE Std 802.1ASdr-2024?	O	5.6.2 a)	Yes [] No []
PTP-2	Does the IA-station support Grandmaster PTP Instance capability Grandmaster PTP Instance capability according to IEEE Std 802.1AS-2020 as amended by IEEE Std 802.1ASdr-2024 and IEEE Std 802.1ASdm-2024, 5.4.2 item c)?	O	5.6.2 b)	Yes [] No []
PTP-3	Does the IA-station support more than one PTP port as a PTP Relay Instance according to IEEE Std 802.1AS-2020, 5.4.2 item d)?	O	5.6.2 c)	Yes [] No []
PTP-4	Does the IA-station support hot standby redundancy requirements according to IEEE Std 802.1AS-2020 as amended by IEEE Std 802.1ASdm-2024, 5.4.2, item m)?	O	5.6.2 d)	Yes [] No []

6420

A.5.6 IA-station secure management exchange options

6422 The form in Table A.8 is used to indicate options for secure management exchange.

6423

Table A.8 – IA-station secure management exchange options

Item	Feature	Status	References	Support
SECMGMT-5	Does the IA-station support Writable-Running capability?	O	5.6.3 a)	Yes [] No []
SECMGMT-6	Does the IA-station support Confirmed Commit capability?	O	5.6.3 b)	Yes [] No []
SECMGMT-7	Does the IA-station support Distinct Startup capability?	O	5.6.3 c)	Yes [] No []
SECMGMT-8	Does the IA-station support URL capability?	O	5.6.3 d)	Yes [] No []
SECMGMT-9	Does the IA-station support XPath capability?	O	5.6.3 e)	Yes [] No []
SECMGMT-10	Does the IA-station support NETCONF-over-TLS server supporting TLS version 1.3	O	5.6.3 f)	Yes [] No []

6424

A.5.7 Proxy target requirements

6426 The form in Table A.9 is used to indicate requirements for proxy targets.

6427

Table A.9 – Proxy target requirements

Item	Feature	Status	References	Support
PT-1	Does the IA-station support Secure management exchanges?	O	5.6.4 a)	Yes [] No []
PT-1	Does the IA-station support Secure management exchanges using a management proxy?	O	5.6.4 b)	Yes [] No []

6428

A.6 Bridge component**A.6.1 Bridge component requirements**

6431 The form in Table A.10 is used to indicate bridge component requirements.

Table A.10 – Bridge component requirements

Item	Feature	Status	References	Support
BC-1	Does the bridge component support the common bridge component requirements?	M	5.7.1	Yes []
BC-2	Does the bridge component support ccA bridge component requirements?	O.2	5.7.2	Yes [] No []
BC-3	Does the bridge component support ccB bridge component requirements?	O.2	5.7.3	Yes [] No []

A.6.2 Instructions

One instance of Clause A.6 shall be filled out per bridge component implemented by an IA-station.

A.6.3 Bridge Common bridge component options

The form in Table A.11 is used to indicate bridge component options.

Table A.11 – Common bridge component options

Item	Feature	Status	References	Support
BC-4	Does the bridge component support the operation of the credit-based shaper algorithm?	O	5.8.1	Yes [] No []

A.6.4 Conformance class A bridge component options

The form in Table A.12 is used to indicate options for bridge components conforming to conformance class A.

Table A.12 – Conformance class A bridge component options

Item	Feature	Status	References	Support
CCA-BC-1	Does the bridge component support any of the common bridge component options?	O	5.8.2 a)	Yes [] No [] N/A []
CCA-BC-2	Does the bridge component support more than 2 PTP instances?	O	5.8.2 b)	Yes [] No [] N/A []
CCA-BC-3	State the number of PTP instances supported by the bridge component.	CCA-BC-2:M	5.8.2 b)	Number ____
CCA-BC-4	Does the bridge component support enhancements for scheduled traffic for the 10 Mb/s, 2,5 Gb/s, 5 Gb/s, or 10 Gb/s data rates?	O	5.8.2 c)	Yes [] No [] N/A []
CCA-BC-5	Does the bridge component support frame preemption for the 10 Mb/s, 2,5 Gb/s, 5 Gb/s, or 10 Gb/s data rates?	O	5.8.2 d)	Yes [] No [] N/A []

A.6.5 Conformance class B bridge component options

The form in Table A.13 is used to indicate options for bridge components conforming to conformance class B.

Table A.13 – Conformance class B bridge component options

Item	Feature	Status	References	Support
CCB-BC-1	Does the bridge component support any of the common bridge component options?	O	5.8.3 a)	Yes [] No [] N/A []
CCB-BC-2	Does the bridge component support more than 4 but not more than 8 egress queues?	O	5.8.3 b)	Yes [] No [] N/A []

CCB-BC-3	State the number of egress queues supported by the bridge component.	CCB-BC-2:M	5.8.3 b)	Number ____
CCB-BC-4	Does the bridge component support more than 1 PTP Instance?	O	5.8.3 c)	Yes [] No [] N/A []
CCB-BC-5	State the number of PTP Instances supported by the bridge component.	CCB-BC-4:M	5.8.3 c)	Number ____
CCB-BC-6	Does the bridge component support enhancements for scheduled traffic?	O	5.8.3 d)	Yes [] No [] N/A []
CCB-BC-7	Does the bridge component support frame preemption?	O	5.8.3 e)	Yes [] No [] N/A []

6450

6451 **A.7 End station component**6452 **A.7.1 Instructions**

6453 One instance of Clause A.7 shall be filled out per end station component implemented by an
 6454 IA-station.

6455 **A.7.2 Common end station component requirements**

6456 The form in Table A.14 is used to indicate common requirements for end station components.

6457 **Table A.14 – Common end station component requirements**

Item	Feature	Status	References	Support
ESC-1	Does the end station component support the common end station component requirements?	M	5.9.1	Yes []
ESC-2	Does the end station component support the ccA end station component requirements?	O.3	5.9.2	Yes [] No []
ESC-3	Does the end station component support the ccB end station component requirements?	O.3	5.9.3	Yes [] No []

6458

6459 **A.7.3 Common end station component options**

6460 The form in Table A.15 is used to indicate options for end station components.

6461 **Table A.15 – Common end station component options**

Item	Feature	Status	References	Support
ESC-4	Does the end station component support the operation of the credit-based shaper?	O	5.10.1 a)	Yes [] No []
ESC-5	Does the end station component support talker end system behaviors?	O	5.10.1 b)	Yes [] No []
ESC-6	Does the end station component support listener end system behaviors?	O	5.10.1 c)	Yes [] No []

6462

6463 **A.7.4 Conformance class A end station component options**

6464 The form in Table A.16 is used to indicate options for end station components conforming to
 6465 conformance class A.

6466 **Table A.16 – Conformance class A end station component options**

Item	Feature	Status	References	Support
CCA-ESC-1	Does the end station component support any of the common end station component options?	O	5.10.2 a)	Yes [] No [] N/A []

CCA-ESC-2	Does the end station component support more than 2 PTP Instances?	O	5.10.2 b)	Yes [] No [] N/A []
CCA-ESC-3	State the number of PTP Instances supported by the end-station component.	CCA-ESC-2:M	5.10.2 b)	Number _____
CCA-ESC-4	Does the end station component support enhancements for scheduled traffic for data rates 10 Mb/s, 2,5 Gb/s, 5 Gb/s, or 10 Gb/s?	O	5.10.2 c)	Yes [] No [] N/A []
CCA-ESC-5	Does the end station component support requirements for frame preemption for data rates 10 Mb/s, 2,5 Gb/s, 5 Gb/s, or 10 Gb/s?	O	5.10.2 d)	Yes [] No [] N/A []

6467

6468 **A.7.5 Conformance class B end station component options**

6469 The form in Table A.17 is used to indicate options for end station components conforming to
6470 conformance class B.

6471 **Table A.17 – Conformance class B end station component options**

Item	Feature	Status	References	Support
CCB-ESC-1	Does the end station component support any of the common end station component options?	O	5.10.3 a)	Yes [] No [] N/A []
CCB-ESC-2	Does the end station component support one or more PTP Instances?	O	5.10.3 b)	Yes [] No [] N/A []
CCB-ESC-3	State the number of PTP Instances supported by the end-station component.	CCB-ESC-2:M	5.10.3 b)	Number _____
CCB-ESC-4	Does the end station component support enhancements for scheduled traffic?	O	5.10.3 c)	Yes [] No [] N/A []
CCB-ESC-5	Does the end station component support requirements for frame preemption?	O	5.10.3 d)	Yes [] No [] N/A []

6472

Annex B (informative)

Representative Configuration Domain

The following quantities are representative of what can be supported in a single Configuration Domain.

- IA-stations: 1 024.
- Network diameter: 64.
- Streams per IA-controller for IA-controller to IA-device (C2D) communication:
 - 512 Talker and ≥ 512 Listener streams, and
 - 1 024 Talker and $\geq 1\,024$ Listener streams in case of seamless redundancy.
- Streams per IA-controller for IA-controller to IA-controller (C2C) communication:
 - 64 Talker and ≥ 64 Listener streams, and
 - 128 Talker and ≥ 128 Listener streams in case of seamless redundancy.
- Streams per IA-device for IA-device-to-IA-device (D2D) communication:
 - 2 Talker and 2 Listener streams, and
 - 4 Talker and 4 Listener streams in case of seamless redundancy.
- Example calculation of data flow quantities for eight PLCs – without seamless redundancy:
 - $8 \times 512 \times 2 = 8\,192$ streams for C2D communication,
 - $8 \times 64 \times 2 = 1\,024$ streams for C2C communication, and
 - $(8\,192 + 1\,024) \times 2\,000 = 18\,432\,000$ octets data for all streams.

Annex C (informative)

Description of clock control system

C.1 Clock control system introduction

Annex C provides an introductory discussion of a basic clock control system. For more detailed information, see the Bibliography References ITU-T Recommendation G.8260 and ITU-T Series G Supplement 65.

Figure C.1 shows a basic control system model that uses a proportional plus integral (PI) controller. This is meant to be reference model, i.e. it is not meant to specify an implementation. Requirements for the clock control system can be expressed using parameters (e.g. 3 dB bandwidth, gain peaking, frequency response) that are based on this reference model. Any implementation whose parameters are within the requirements is considered to be acceptable. For example, the model of Figure C.1 is expressed in the analog domain (i.e. s-domain), and is shown to be second order. An actual implementation can be digital, and can be higher order, as long as it meets the respective requirements.

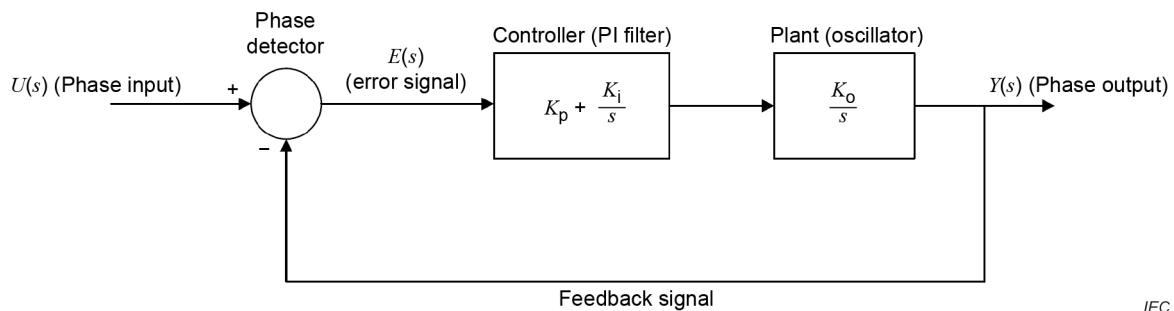


Figure C.1 – Reference model for clock control system

In Figure C.1, the plant, i.e. the entity being controlled, represents the clock oscillator. It is desired that the phase output, $y(t)$, of the oscillator follows the phase input, $u(t)$, as closely as possible (the signals are shown in the frequency domain (i.e. as Laplace Transforms) in Figure C.1; however, they can equivalently be expressed in the time domain, with t representing time). The parameter K_o is the oscillator gain; the oscillator frequency is equal to the oscillator input multiplied by K_o . In some implementations the input signal to the oscillator is a voltage, and the oscillator is referred to as a voltage-controlled oscillator (VCO). However, other implementations are possible, e.g. digital implementations, where the oscillator is a digital controlled oscillator (DCO). Since the input to the oscillator depends on the implementation, it is not labeled in Figure C.1.

The control system of Figure C.1 uses negative feedback to enable the phase output to follow the phase input. The phase detector computes the difference between the input and output signals to produce the error signal $e(t)$. The error signal is then filtered by the PI filter to produce the input to the oscillator. The filter is referred to as a PI filter because its output is the sum of the proportional gain, K_p , multiplied by the error signal and the integral gain, K_i , multiplied by the integral of the error signal. The gains K_o , K_p , and K_i must be chosen such that the performance of the control system is acceptable, i.e. the time-domain behavior of the output with respect to the input is acceptable. However, an alternative set of parameters, which are more convenient, can be defined in terms of K_o , K_p , and K_i ; this is done in Clause C.2.

C.2 Transfer function for control system

From the block diagram of Figure C.1, the input and output are related by:

$$Y(s) = \left(K_p + \frac{K_i}{s} \right) \left(\frac{K_o}{s} \right) (U(s) - Y(s)) \quad (\text{C.1})$$

where

$Y(s)$ is the phase output, expressed in the s-domain;

$U(s)$ is the phase input, in the s-domain;

K_p is the proportional gain;

K_i is the integral gain;

K_o is the oscillator gain.

or

$$Y(s) = \frac{\left(K_p + \frac{K_i}{s} \right) \left(\frac{K_o}{s} \right)}{1 + \left(K_p + \frac{K_i}{s} \right) \left(\frac{K_o}{s} \right)} U(s) \quad (\text{C.2})$$

This can be simplified by multiplying the numerator and denominator by s^2 to produce:

$$Y(s) = H(s)U(s) \quad (\text{C.3})$$

where the transfer function $H(s)$ is given by:

$$H(s) = \frac{K_p K_o s + K_i K_o}{s^2 + K_p K_o s + K_i K_o} \quad (\text{C.4})$$

In Formula (C.4), the parameter K_o does not appear independently of K_p and K_i ; rather, only the products $K_p K_o$ and $K_i K_o$ appear. The plant and PI filter could have been combined in the model of Figure C.1; this is consistent with the fact that the exact nature of the signal between the PI filter and plant is unimportant in this reference model. The units of $K_p K_o$ are (time)⁻¹ and the units of $K_i K_o$ are (time)⁻². The frequency units are the same as the units of s , e.g. if s has units rad/s, then $K_p K_o$ has units rad/s and $K_i K_o$ has units (rad/s)². The integration operation in the plant results in the transfer function being dimensionless, which is consistent with the fact that the input and output of the control system both have units of phase.

6553 The transfer function can be expressed in an equivalent form by defining the undamped natural
6554 frequency, ω_n , and damping ratio, ζ :

$$H(s) = \frac{2\zeta\omega_n s + \omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \quad (\text{C.5})$$

6555

6556 where

6557 ζ is the damping ratio;

6558 ω_n is the undamped natural frequency expressed in rad/s.

6559 And, ζ and ω_n are given by:

$$\omega_n = \sqrt{K_i K_o} \quad (\text{C.6})$$

$$\zeta = \frac{K_p K_o}{2\sqrt{K_i K_o}} = \frac{K_p}{2} \sqrt{\frac{K_o}{K_i}} \quad (\text{C.7})$$

6560

6561 In the Formula (C.7), the first form shows explicitly that ζ depends only on the products $K_p K_o$
6562 and $K_i K_o$.

6563 C.3 Frequency response for control system

6564 The frequency response is obtained by setting $s = j\omega$ in Formula (C.5) and taking the absolute
6565 value (here j rather than i is used for $\sqrt{-1}$ to avoid confusion with other uses of i), where ω is
6566 the frequency in rad/s. The result is:

$$|H(j\omega)| = \left| \frac{2\zeta\omega_n \omega j + \omega_n^2}{-\omega^2 + \omega_n^2 + 2\zeta\omega_n \omega j} \right| = \left(\frac{4\zeta^2 \omega_n^2 \omega^2 + \omega_n^4}{(\omega_n^2 - \omega^2)^2 + 4\zeta^2 \omega_n^2 \omega^2} \right)^{1/2} \quad (\text{C.8})$$

6567

6568 Dividing the numerator and denominator of Formula (C.7) by ω_n^4 and defining the dimensionless
6569 frequency $x = \omega/\omega_n$ produces:

$$|H(j\omega)| = \left(\frac{4\zeta^2 x^2 + 1}{(1 - x^2)^2 + 4\zeta^2 x^2} \right)^{1/2} \quad (\text{C.9})$$

6570

where

ζ is the damping ratio;

ω is the frequency expressed in rad/s;

$x = \omega/\omega_n$.

Figure C.2 contains plots of frequency response (Formula (C.9)) versus dimensionless frequency x , on a log-log scale, for damping ratio ζ equal to 0,3, 0,5, 0,707, 1,0, 2,0, 3,0, 4,0, and 5,0. It is seen that the frequency response is very close to 1 for values of dimensionless frequency much less than 1 (i.e. for $\omega \ll \omega_n$). The frequency response increases as the frequency approaches the undamped natural frequency (i.e. as dimensionless frequency approaches 1) and reaches a peak for dimensionless frequency slightly less than 1. The frequency response then decreases, eventually having a slope (i.e. roll-off) of 20 dB/decade (i.e. frequency response decreases by a factor of 10 for every factor of 10 increase in x for $x \gg 1$). Figure C.3 shows the detail of frequency response for x in the range 0,1 to 10.

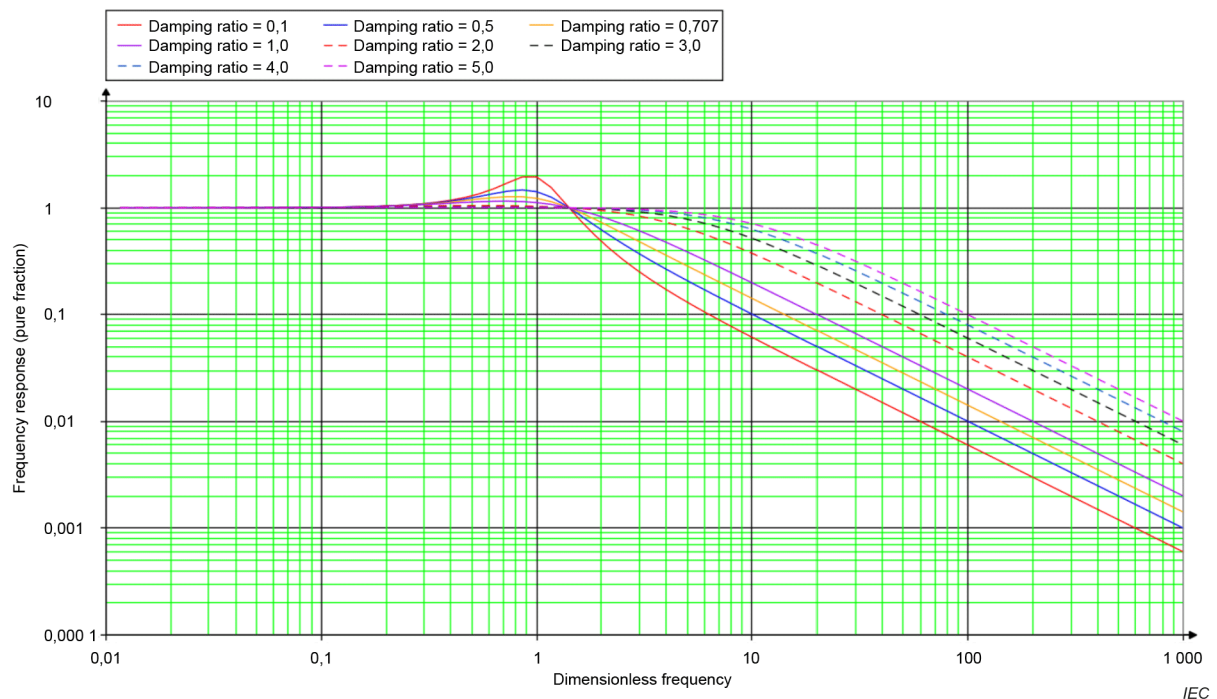


Figure C.2 – Frequency response for the control system of Figure C.1

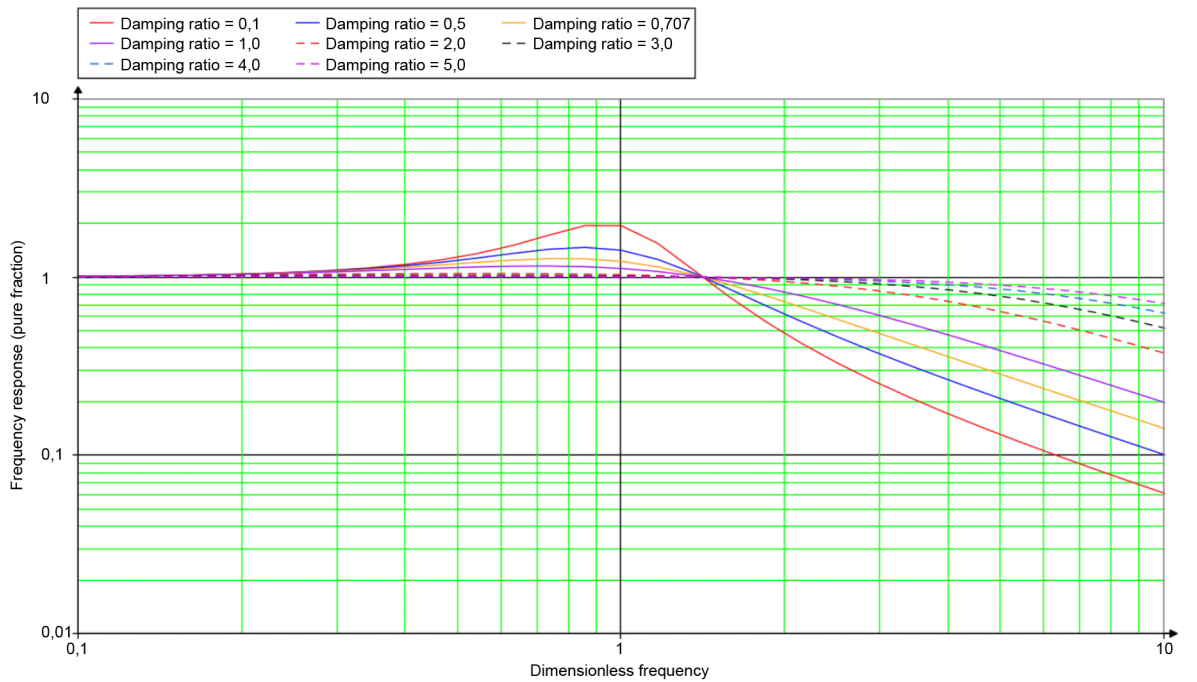


Figure C.3 – Detail of frequency response for the control system of Figure C.1 for dimensionless frequency in the range 0,1 to 10

In addition to undamped natural frequency ω_n and damping ratio ζ , the parameters 3 dB bandwidth and gain peaking are often used when specifying clock performance. The 3 dB bandwidth is defined as the value of frequency for which the frequency response is equal to -3 dB. Since dB is given by 10 multiplied by the logarithm to base 10 of the power ratio, which is 20 multiplied by the logarithm to base 10 of the amplitude ratio, -3 dB corresponds to the value $10^{-3/20}$. The 3 dB bandwidth can be computed by setting Formula (C.8) equal to $10^{-3/20}$ and solving for x in terms of ζ . This is equivalent to setting the quantity in parentheses (i.e. inside the square root) in Formula (C.8) equal to $10^{-3/10}$ and solving for x . Now, $10^{-3/10}$ is approximately equal to 0,501 2, i.e. it is very close to $\frac{1}{2}$. Then the 3dB bandwidth can be obtained by solving the following formula for x in terms of ζ :

$$\frac{4\zeta^2 x^2 + 1}{(1-x^2)^2 + 4\zeta^2 x^2} = \frac{1}{2} \quad (\text{C.10})$$

or

$$x^4 - 2(2\zeta^2 + 1)x^2 - 1 = 0 \quad (\text{C.11})$$

The result is:

$$x = \left[2\zeta^2 + 1 + \sqrt{(2\zeta^2 + 1)^2 + 1} \right]^{1/2} \quad (\text{C.12})$$

6603 or

$$\omega_{3\text{dB}} = \omega_n \left[2\zeta^2 + 1 + \sqrt{(2\zeta^2 + 1)^2 + 1} \right]^{1/2} \quad (\text{C.13})$$

6604

6605 The gain peaking is the maximum value of the frequency response, in dB. It is computed by
6606 differentiating Formula (C.8) with respect to x , setting the result to zero, solving for x , and then
6607 substituting this value of x into Formula (C.8) to obtain the maximum. The result is:

$$H_p = \left[1 - 2\alpha - 2\alpha^2 + 2\alpha(2\alpha + \alpha^2)^{1/2} \right]^{-1/2} \quad (\text{C.14})$$

6608

6609 where α is related to damping ratio by:

$$\alpha = \frac{1}{4\zeta^2} \quad (\text{C.15})$$

6610

6611 and H_p is the gain peaking expressed as a pure fraction. The gain peaking in dB is equal to
6612 $20 \cdot \log_{10} H_p$. In some cases, it is necessary to compute damping ratio from gain peaking. The
6613 result for this is:

$$\alpha = \frac{(1-q)(1+\sqrt{1-q})}{2q} \quad (\text{C.16})$$

6614

6615 where

$$q = \frac{1}{H_p^2} \quad (\text{C.17})$$

6616

6617 Damping ratio is obtained from α using Formula (C.15).

6618 If 3 dB bandwidth and gain peaking are given, damping ratio can be obtained using Formulas
6619 (C.15), (C.16) and (C.17). Undamped natural frequency can then be obtained using Formula
6620 (C.13).

6621 Figure C.4 shows gain peaking, expressed as a pure fraction, as a function of damping ratio.
6622 Figure C.5 shows gain peaking in dB as a function of damping ratio.

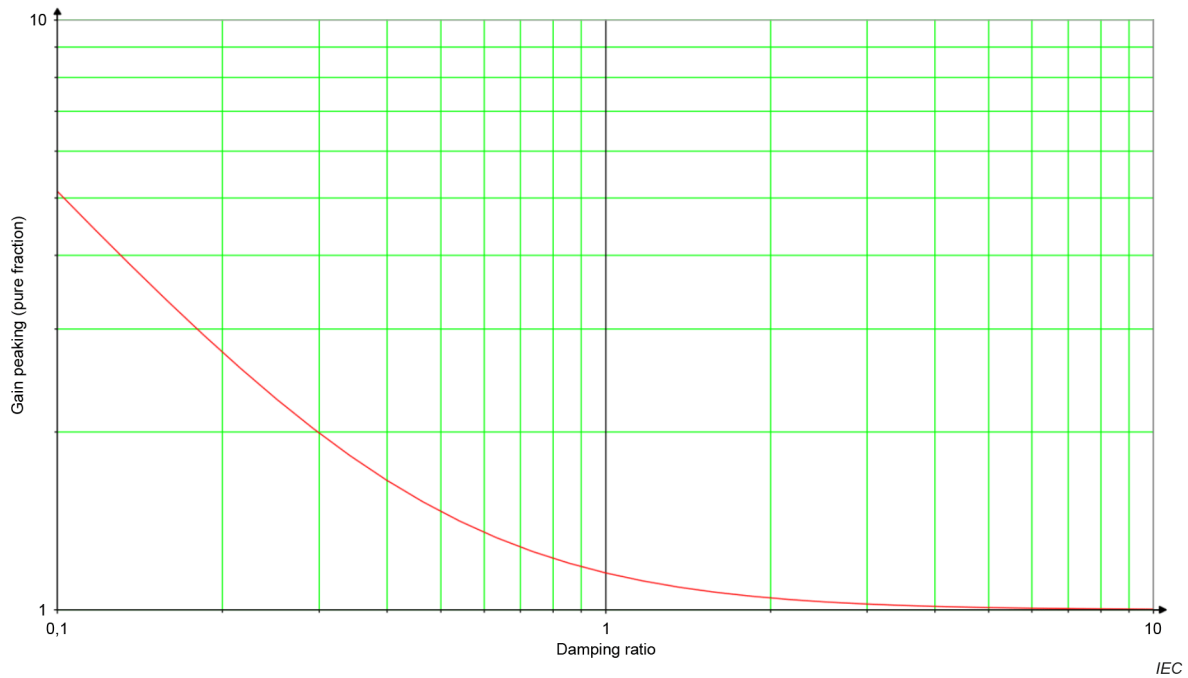


Figure C.4 – Gain peaking (pure fraction) as a function of damping ratio

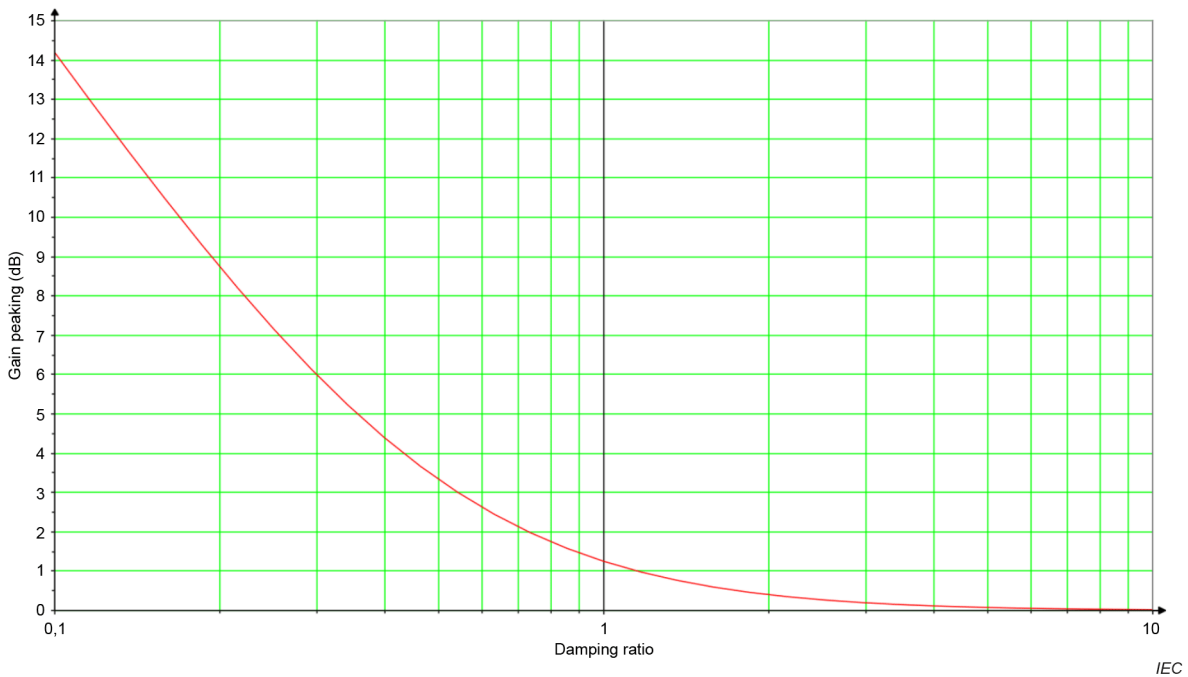


Figure C.5 – Gain peaking in dB as a function of damping ratio

The performance of the clock control system can be described using the frequency response as follows:

- a) Maximum 3 dB bandwidth in Hz,
- b) Maximum gain peaking in dB, and
- c) Frequency response plot (mask) corresponding to (a) and (b) that is not to be exceeded.

C.4 Example

Consider a clock control system with $K_p K_o = 4,23 \text{ rad/s}$ and $K_i K_o = 9,62 (\text{rad/s})^2$. The undamped natural frequency and damping ratio are:

$$\omega_n = \sqrt{K_i K_o} = \sqrt{9,62 (\text{rad/s})^2} = 3,10 \text{ rad/s} \quad (\text{C.18})$$

$$\zeta = \frac{K_p K_o}{2\sqrt{K_i K_o}} = \frac{4,23 \text{ rad/s}}{2\sqrt{9,62 (\text{rad/s})^2}} = 0,682 \quad (\text{C.19})$$

The gain peaking is obtained from:

$$\alpha = \frac{1}{4(0,682)^2} = 0,537 \quad (\text{C.20})$$

$$H_p (\text{purefraction}) = \left[1 - 2(0,537) - 2(0,537)^2 + 2(0,537) \sqrt{2(0,537) + (0,537)^2} \right]^{\frac{1}{2}} = (\text{C.21})$$

$$H_p (\text{dB}) = 20 \log_{10}(1,29) \text{ dB} = 2,2 \text{ dB} \quad (\text{C.22})$$

The 3dB bandwidth is:

$$f_{3\text{dB}} (\text{Hz}) = \frac{\omega_n}{2\pi} \left[1 + 2\zeta^2 + \sqrt{(1 + 2\zeta^2)^2 + 1} \right]^{1/2} \quad (\text{C.23})$$

$$= \frac{3,10}{2\pi} \left[1 + 2(0,682)^2 + \sqrt{(1 + 2(0,682)^2)^2 + 1} \right]^{1/2} = 1,0 \text{ Hz}$$

The frequency response is shown in Figure C.6.

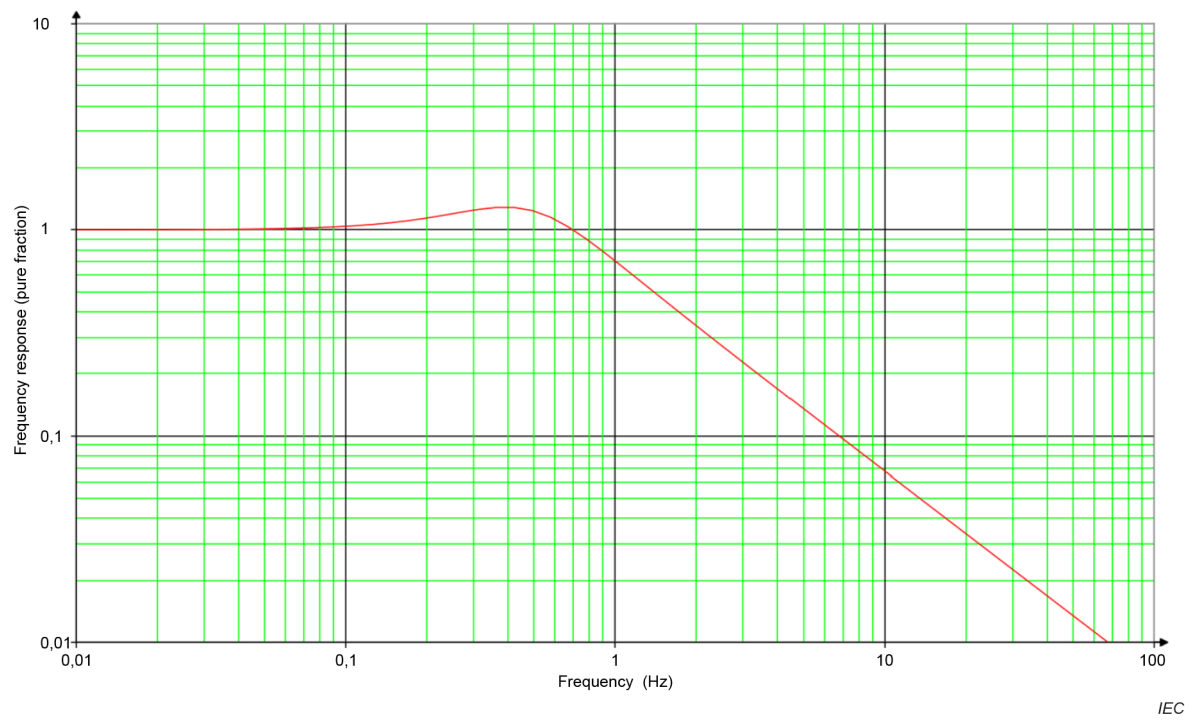


Figure C.6 – Example frequency response

Annex D
(informative)

Time synchronization annex

D.1 Overview

Annex D describes how a network of compliant devices can achieve a time synchronization accuracy, at the application level, of $\pm 1\ \mu\text{s}$, relative to the Clock Source at the Grandmaster, over 100 network hops. To achieve this, it allocates the overall error budget of 1 000 ns as described in Table D.1.

Table D.1 – Time synchronization error budget

Network Aspect	Error Type	Network-Level Error Budget (ns)
All PTP Instances	Constant time error	200
	Dynamic time error	600
All PTP links	Constant time error	200
	Dynamic time error	

A chain of 1 Grandmaster PTP Instance, 99 PTP Relay Instances and 1 PTP End Instance (100 network hops) that all comply with the normative requirements of 6.2.2, 6.2.3, 6.2.4, and 6.2.5 generates a network-level Time Error at or below the Error Budget for All PTP Instances.

Clause D.2 describes the principles of operation this document assumes.

Clause D.3 provides additional information on specific normative requirements.

The principles of operation include the use of crystal oscillators (XOs) as opposed to more accurate, stable, and costly options such as temperature-compensated crystal oscillators (TCXOs).

Clause D.4 describes a potential approach to testing the normative requirements. It is not a test specification, but rather a high-level overview of one potential approach that can be adopted by a full test specification.

The use of XOs means that some of the normative requirements are difficult or impossible to meet without employing algorithms that track Neighbor Rate Ratio drift and Rate Ratio drift and compensate for consequent errors in calculating Rate Ratio and Correction Field.

Clause D.5 provides examples of algorithms that can be used for this purpose, and which have been shown to enable compliance with the normative requirements.

Implementations that employ TCXOs or other more accurate, stable oscillators can still find some of the normative requirements difficult or impossible to meet without employing algorithms to track and compensate for errors due to clock drift. This is because other PTP Instances that use XOs can still cause the implementation to experience Neighbor Rate Ratio drift, Rate Ratio drift or both.

There is no normative requirement to use the algorithms described in Clause D.5; an implementation can employ alternative algorithms provided the normative requirements are met. Clause D.5 describes the potential risks of deploying a network whose instances employ a mix of different algorithms. It is the responsibility of implementers to mitigate the risks and ensure alternative algorithms deliver the desired network-level performance.

This document does not include normative requirements for PTP Links. Annex D.2.4 describes PTP Link characteristics that influence achieving 1 μ s time synchronization accuracy. It includes some examples using common PTP Link characteristics.

This document's normative requirements regarding instance-level error generation are intended to ensure not just an overall level of dTE generation at each node, but also the performance of drift tracking and error compensation algorithms and the amount of dTE generation due to timestamp error versus clock drift. The algorithms are employed to mitigate errors due to clock drift but cannot mitigate timestamp errors.

D.2 Principles of operation

D.2.1 General

Achieving ± 1 μ s time synchronization accuracy across 100 network hops involves managing the accumulation of errors in the preciseOriginTimestamp plus correctionField and the Rate Ratio as they are passed, via Sync or Follow_Up messages, down the chain of PTP Instances and are then used by the PTP End Instance to keep its ClockTarget in line with the ClockSource at the Grandmaster PTP Instance. The majority of significant errors can ultimately be traced back to one of three sources: timestamp error, clock drift, or path delay asymmetry. The selection of PTP protocol parameters often involves trading off one source of error against the other. This document requires specific PTP protocol configurations, and assumes the use of mechanisms (algorithms), that reduce dTE due to timestamp error but would also – without additional measures – increase dTE due to clock drift to the point where the latter exceeds the allocated error budget. However, this document also assumes additional measures to minimize some sources of dTE due to clock drift, and mechanisms and to track and compensate for errors from other sources to a sufficient degree that the error budget is not exceeded.

The specific protocol configurations and other measures, along with their intended effects, are described in Table D.2.

Table D.2 – Protocol configurations & other measures to achieve dTE budget

Configuration or measure	Description and intended effect(s)
Sync Interval 125 ms	Effects: 1) Calibrates the balance between dTE from timestamp error vs error due to clock drift. Larger intervals lead to less timestamp error and more error due to clock drift. 2) Keeps below acceptable limits the impact of errors in Rate Ratio and Rate Ratio Drift estimation when keeping ClockTarget in line with ClockSource between arrival of Sync messages. Larger intervals increase the impact of any errors.
Drift_Tracking TLV – syncEgressTimestamp	Effect: Enables calculation of NRR using Sync message timestamps, which eliminates error due to NRR clock drift that would otherwise occur between calculation of NRR using Pdelay_Resp messages and use during Sync message processing (i.e. calculation of Rate Ratio and output Correction Field values).

Configuration or measure	Description and intended effect(s)
NRR Smoothing	Description: Algorithm to use timestamps from multiple past Sync messages when calculating NRR. Effect: Reduces the amount of error in the estimate of NRR due to timestamp error while increasing the amount of error due to clock drift.
NRR Drift Tracking & Compensation	Description: Algorithm to use timestamps from multiple past Sync messages to estimate NRR drift rate and then apply compensation to correct for consequent errors in NRR calculation. Effect: Mitigates the effect of errors due to clock drift when calculating and using the estimated NRR.
Drift_Tracking TLV – rateRatioDrift	Description: Carries estimate of Rate Ratio drift rate from one node to the next. Effect: Allows each node to estimate its own Rate Ratio drift rate by combining the incoming Rate Ratio drift rate with the local estimate of NRR drift rate.
RR Drift Compensation	Description: Algorithm that uses the estimate of RR drift rate to compensate for that drift, adjusting the estimated RR over time according to the drift rate. Effect: For PTP Relay Instances, minimizes errors in the Correction Field caused by Rate Ratio drift. For PTP End Instances, a similar approach can reduce errors in keeping ClockTarget in line with ClockSource between arrival of Sync messages, but is outside the scope of this document.
Pdelay Interval Consistency	Description: This document requires tighter control of the interval between Pdelay messages generated at the Grandmaster PTP Instance than the defaults in IEEE Std 802.1AS-2020. Effect: This document requires the use of Sync messages to calculate NRR (see above). However, when a sufficient number of Sync messages are not available, for example on startup or after a reconfiguration, Pdelay_Resp messages can be used instead. In such cases, errors due to clock drift at Relay Instances have a tendency to cancel out. A clock drift that generates a positive error in NRR measurement on receipt of a Pdelay_Resp message generates a negative error in NRR measurement at the next node. The degree of cancellation depends on the consistency of the intervals over which NRR is measured at neighboring nodes. Tighter control of the Pdelay Interval increases the consistency of the measurement interval and thus decreases the amount of error.
Mean Residence Time	Description: This document defines a mean Residence Time requirement, which is significantly lower than the default maximum Residence Time in IEEE Std 802.1AS-2020. Effect: The amount of error in the Correction Field at the PTP End Instance due to clock drift is proportional to the cumulative meanLinkDelay and residenceTime experienced by a Sync message during transit from the Grandmaster PTP Instance to the PTP End Instance. Specifying a lower mean residenceTime reduces this source of error.

D.2.2 Grandmaster PTP Instance implementation

Depending on implementation, a Grandmaster PTP Instance can:

- a) contain a single oscillator used for both Local Clock and Clock Source,
- b) contain separate oscillators for Local Clock and Clock Source, or
- c) contain only an oscillator for Local Clock and accept an external input for Clock Source.

In some cases, a Grandmaster PTP Instance can support more than one mode of operation and transition between them depending on changes in network configuration (see Splitting, Joining and Aligning Time Domains).

In the first case, the rateRatio transmitted by the Grandmaster PTP Instance is one (i.e. 0 ppm) the rateRatioDrift is 0 ppm/s, reflecting the fact there is no difference between the Local Clock and Clock Source frequencies.

In the second and third cases there can be differences between the Local Clock and Clock Source frequencies. Any differences are reflected in the rateRatio and rateRatioDrift fields transmitted by the Grandmaster PTP Instance. This means that the Grandmaster PTP Instance tracks rateRatio over time in order to calculate rateRatioDrift, similarly to PTP Relay Instances and PTP End Instances. The exact implementation can vary.

D.2.3 Splitting, joining and aligning time domains

Modular machines or production cells can allow the splitting and combining of machines if this is required by the production process. When separate, the ClockSources of two machines run separately, each with its own time domain. If both ClockSources are traceable to the same PTP timescale, the difference between the ClockSources can be relatively small. If traceable to different timescales, especially if one or both are ARB timescales, there can be a very large difference between the ClockSources.

When two machines are joined, the first machine's time domain remains unaffected, and it can continue operation without disruption. There are two typical approaches to how the second machine behaves. In the first case, at a time of the end user's choosing, the second machine's time domain ceases to exist, with its PTP Instances becoming part of the first machine's time domain. In the second case, the second machine's time domain is gradually aligned with the first machine's time domain such that control loop cycles are coordinated. In the first case the second machine's time domain is unaffected, and it can continue production even if the machines are accidentally connected, until the end user chooses to join the time domains. In the second case the second machine can continue production while its time domain is being aligned.

D.2.3.1 Joining machines with single time domain

In the first case, where the second machine's time domain ceases to exist, a discontinuity in timing for the second machine's PTP Instances can occur, as they switch to use the first machine's Grandmaster. Some implementations implement measures to limit such timing discontinuities, but these measures are outside the scope of this document. Typically, in this case, the second machine is not operational while it is joined to the first. It resumes operation once its PTP Instances have synchronized with the first machine's Grandmaster.

D.2.3.2 Joining machines with multiple coordinated time domains

In the second case, where the second machine's time domain is gradually aligned with the first machine's time domain, this typically requires both machines to be implementing the same control loop cycle time. The goal is that, once coordinated, each control loop cycle of the first machine is aligned with the start of a control loop cycle of the second machine, even though the two machines maintain separate time domains and there can be a large time difference between their Clock Sources.

6756 In this case, after being joined together, the first machine effectively drives the second
6757 machine's Clock Source faster or slower, during an alignment period, until coordination is
6758 achieved. During the alignment period, this drive from the first machine can result in the second
6759 machine's Clock Source temporarily exceeding the usual normative requirement on range of
6760 fractional frequency offset relative to the nominal frequency of ± 50 ppm. The usual normative
6761 requirement on range of rate of change of fractional frequency offset of ± 1 ppm/s, applicable
6762 when split (i.e. independent) or coordinated (i.e. joined and stable, after the alignment period),
6763 may also be temporarily exceeded. However, if the value stays within the range ± 3 ppm/s, the
6764 network-level performance of 1 μ s time synchronization accuracy can be maintained. For this
6765 reason, this document specifies a separate normative requirement for temporary, externally
6766 driven, rate of rate of change of fractional frequency offset.

6767 Since the second machine experiences no time discontinuities and the network-level
6768 performance is maintained the second machine can, if desired, continue operation during the
6769 alignment period.

6770 Once coordinated, the first machine continues to drive the Clock Source of the second machine
6771 to maintain coordination. In this stable, coordinated mode of operation the normal range of
6772 ± 1 ppm/s is not exceeded.

6773 The mechanism by which the first machine drives the Clock Source of the second machine is
6774 not addressed in this document.

6775 **D.2.3.3 Splitting machines**

6776 In the first case, where the second machine's time domain ceased to exist while joined to the
6777 first, splitting machines means that the second machine must create its own time domain again.
6778 The second machine's Clock Source typically starts at the PTP Grandmaster Instance's last,
6779 best estimate of the first machine's Clock Source. The goal is for no discontinuities in time sync
6780 to occur; however, depending on implementation, it can take some time to before the time
6781 synchronization accuracy of all the second machine's PTP Instances relative to its Grandmaster
6782 can be relied upon. For this reason, it is possible the second machine is not operational during
6783 the split. Hot Standby can be employed to mitigate this transition time, but the details of how to
6784 do so are out of scope for this document.

6785 In the second case, where the second machine maintains its time domain while joined to the
6786 first, splitting machines means that the first machine ceases driving the second machine's Clock
6787 Source to maintain coordination of control loop cycle times. Without this drive, the two time
6788 domains can drift relative to each other resulting in loss of coordination. Time synchronization
6789 performance within the second machine is maintained during the split and the second machine
6790 can, if desired, continue operation throughout the process.

6791 **D.2.4 PTP link characteristics**

6792 A vast majority of time synchronization error due to PTP link characteristics is caused by
6793 asymmetrical path delay in one direction versus the other. The mechanism to measure path
6794 delay assumes the link is symmetrical and cannot detect asymmetry, thus asymmetry causes
6795 an error. The potential maximum asymmetry and thus error typically scales linearly with physical
6796 path length.

6797 The error budget due to PTP link characteristics for an entire network is 200 ns. In any specific
6798 network this budget can be allocated as required with some links allocated a higher budget
6799 (typically longer length) than others.

6800 A typical specified maximum delay skew for Category 6 Ethernet cables is 50 ns per 100 m. If
6801 such cables are used, a maximum total cable length between Clock Source and Clock Target
6802 with 99 PTP Relay Instances between them (i.e. 100 network hops) is 400 m. Extending the
6803 cable length beyond 400 m without jeopardizing network-level performance would require the
6804 use of cables with less delay skew or asymmetry compensation for delay skew.

6805 It is possible for the delay skew in one section of cable to cancel all or part of a delay skew in
6806 the opposite direction from prior section but, depending on how cables are manufactured and
6807 deployed, it is feasible for the delay skews of every cable segment between a Grandmaster
6808 PTP Instance and a PTP End Instance to be additive.

6809 **D.3 Notes on normative requirements**

6810 **D.3.1 Oscillator requirements**

6811 Clock drift at the Grandmaster PTP Instance causes greater dTE than the same amount of clock
6812 drift at a PTP Relay Instance or the PTP End Instance. This document therefore requires tighter
6813 limits on maximum fractional frequency offset for an oscillator at the Grandmaster PTP Instance
6814 than at other instances.

6815 This document does not place requirements on operational temperature range or other
6816 environmental factors. The required oscillator behavior is delivered for the operational
6817 conditions across which a device claims it is compliant. These conditions typically include
6818 temperature range but can also include rate of change of ambient temperature, supply voltage
6819 stability, amount of vibration and others.

6820 **D.3.2 Timestamp Granularity Error**

6821 Timestamp Granularity Error (TSGE) is the error in timestamping each incoming and outgoing
6822 message due to the maximum timestamp resolution of which an implementation is capable. It
6823 is typically directly related to an implementation's clock rate.

6824 For example: a clock rate of 125 MHz typically results in a maximum resolution of 8 ns.
6825 Depending on implementation the consequent TSGE range can be –8 ns to 0 ns, 0 ns to 8 ns,
6826 or anything in between. In some implementations, offsets are applied to ensure the average
6827 TSGE is 0 ns with, assuming uniform error distribution, a range of –4 ns to +4 ns

6828 Similarly, a clock rate of 500 MHz results in a maximum resolution of 2 ns; a consequent TSGE
6829 range between –2 ns to 0 ns and 0 ns to 2 ns; and, if a suitable offset is applied to ensure a
6830 TSGE average of 0 ns, a range of –1 ns to +1 ns.

6831 A minimum resolution of 8 ns, i.e. minimum clock rate of 125 MHz is assumed. It is further
6832 assumed that TSGE for the sum of the preciseOriginTimestamp and followUpCorrectionField at
6833 the Grandmaster PTP Instance (see IEEE Std 802.1AS-2020 as amended by IEEE Std
6834 802.1ASdm-2024, 10.2.9.2.1) has an average of 0 ns and that the TSGE averages for other
6835 timestamps are stable and consistent across all a PTP Instance's ports. No assumption
6836 regarding the value of the TSGE average for these other timestamps is necessary as they are
6837 always used to measure intervals such that any stable, consistent offset cancels out.

6838 **D.3.3 Dynamic Timestamp Error**

6839 Dynamic Timestamp Error (DTSE) is the, effectively random, error in timestamping each
6840 incoming and outgoing event message due to an implementation's inherent inaccuracies,
6841 excluding TSGE. It is assumed to vary between a minimum of –6 ns and a maximum of + 6 ns
6842 with an average of 0 ns. Lower levels of DTSE are better.

6843 If an implementation timestamps an incoming or outgoing message at a point other than the
6844 PHY, any variability in delay between that point and the PHY (PHY delay) translates to DTSE.
6845 Some common implementations were not designed to limit this variability. If care is not taken
6846 to avoid implementations with high variability, the assumed DTSE range is easily exceeded.
6847 Such implementations can find some of the normative requirements difficult or impossible to
6848 meet.

D.3.4 Grandmaster PTP Instance error generation

Table 12 sets normative requirements for error generation at a Grandmaster PTP Instance that ensure the relevant fields in the Sync and Follow_Up messages it transmits are sufficiently accurate to deliver the network-level performance. Table D.3 describes how the normative requirements align with major sources of error.

Table D.3 – Main sources of error affecting normative requirements for Grandmaster PTP Instance

Item	Normative requirement	Main sources of error
1	preciseOriginTimestamp + correctionField vs Direct measurement of Working Clock at Grandmaster (acting as a Clock Source)	Timestamp Error relative to Clock Source plus accuracy measuring any internal delay between generation of the preciseOriginTimestamp and Sync message transmission.
2	rateRatio vs Direct measurement of Rate Ratio of Clock Source vs Local Clock	Accuracy of internal mechanism to measure Rate Ratio of Clock Source vs. Local Clock, potentially including algorithms that track RateRatioDrift and modify Rate Ratio accordingly ^a
3	syncEgressTimestamp vs Direct measurement of Local Clock	Timestamp Error relative to Local Clock
4	(responseOriginTimestamp – requestReceiptTimestamp) vs Direct measurement of Local Clock interval between reception of Pdelay_Req and transmission of Pdelay_Resp	Timestamp error relative to Local Clock
^a Only applicable if Clock Source and Local Clock are not locked to the same frequency by the implementation. If they are locked, then rateRatio is 0 ppm and rateRatioDrift is 0 ppm/s.		

D.3.5 PTP Relay Instance error generation

Table 13 sets normative requirements for error generation at a PTP Relay Instance that ensure the relevant fields in the Sync and Follow_Up messages it transmits as part of Sync processing are sufficiently accurate to deliver the network-level time sync performance. The requirements include the ability to mitigate errors in rateRatio and rateRatio drift that would otherwise occur due to clock drift at the current PTP Relay Instance, an adjacent PTP Relay Instance, or the Grandmaster PTP Instance. Table D.4 describes how the normative requirements align with major sources of error.

Table D.4 – Main sources of error affecting normative requirements for PTP Relay Instance

Item	Normative requirement	Clock drifts	Main sources of error
1	preciseOriginTimestamp + correctionField vs Direct measurement of Clock Source at Grandmaster PTP Instance	None	Timestamp Errors relative to Local Clock when measuring Residence Time, i.e. Sync message ingress and egress. Accuracy of meanLinkDelay measurement. Errors in Rate Ratio used when translating Residence Time measured in terms of Local Clock to Residence time in terms of Clock Source, although these are typically orders of magnitude smaller than those from Timestamp Errors.
2	rateRatio vs	None	Timestamp Error affecting measurement of NRR when there is no NRR Drift. The effect should be low. This normative

Item	Normative requirement	Clock drifts	Main sources of error
	Direct measurement of Rate Ratio of Clock Source vs Local Clock		requirement is a baseline for the next two requirements.
3		Clock Source (RR Drift)	Accuracy of measurement of NRR when there is no NRR Drift (as above). Accuracy of calculation of rateRatio, including algorithms for RR Drift tracking & error compensation.
4		Clock Source and Local Clock at previous PTP Instance (RR Drift & NRR drift)	Accuracy of measurement of NRR when there is NRR Drift, including algorithms for NRR Drift tracking & error compensation Accuracy of calculation of rateRatio, including algorithms for RR Drift tracking & error compensation. Combined with test 3 this effectively requires a level of performance regarding NRR Drift tracking & error compensation, whether the source of the NRR drift is the Local Clock of the current PTP Instance or the previous PTP Instance.
5	rateRatioDrift vs Direct measurement of Rate Ratio Drift of Clock Source vs Local Clock	None	Timestamp Error affecting measurement of NRR Drift when there is no NRR Drift. The effect should be low. This normative requirement is a baseline for the next two requirements.
6		Clock Source (RR Drift)	Accuracy of measurement of NRR Drift when there is no NRR Drift (as above). Accuracy of calculation of rateRatioDrift, including algorithms for RR Drift tracking & error compensation.
7		Clock Source and Local Clock at previous PTP Instance (RR Drift & NRR drift)	Accuracy of measurement of NRR Drift when there is NRR Drift, including algorithms for NRR Drift tracking & error compensation. Accuracy of calculation of rateRatioDrift, including algorithms for RR Drift tracking & error compensation. Combined with test 6 this effectively requires a level of performance regarding NRR Drift tracking & error compensation, whether the source of the NRR drift is the Local Clock of the current PTP Instance or the previous PTP Instance.
8	syncEgressTimestamp vs Direct measurement of Local Clock	None	Timestamp Error relative to Local Clock
9	(responseOriginTimestamp – requestReceiptTimestamp) vs Direct measurement of Local Clock interval between reception of Pdelay_Req and transmission of Pdelay_Resp	None	Timestamp Error relative to Local Clock
10	Mean Link Delay	None	Timestamp error affecting measurement of meanLinkDelay. Each Pdelay measurement can be subject to relatively large errors from both the local and neighboring node. This normative requirement ensures a suitable algorithm mitigates the impact of any such single Pdelay measurement.

D.3.6 PTP End Instance error generation

Table 14 sets normative requirements for error generation at a PTP End Instance that ensure the ClockTarget it generates from incoming Sync and Follow_Up messages is sufficiently accurate to deliver the network-level time sync performance. Table D.5 describes how the normative requirements align with major sources of error.

Table D.5 – Main sources of error affecting normative requirements for PTP End Instance

Item	Normative requirement	Clock drifts	Main sources of error
1	ClockTarget vs ClockSource	None	Timestamp Error affecting measurement of NRR Drift when there is no NRR Drift. The effect should be low. This normative requirement is a baseline for the next two tests.
2		Clock Source (RR Drift)	Accuracy of measurement of NRR Drift when there is no NRR Drift (as above). Accuracy of calculation of rateRatioDrift, including algorithms for RR Drift tracking & error compensation.
3		Clock Source and Local Clock at previous PTP Instance (RR Drift & NRR drift)	Accuracy of measurement of NRR Drift when there is NRR Drift, including algorithms for NRR Drift tracking & error compensation. Accuracy of calculation of rateRatioDrift, including algorithms for RR Drift tracking & error compensation. Combined with test 2 this effectively requires a level of performance regarding NRR Drift tracking & error compensation, whether the source of the NRR drift is the Local Clock of the current PTP Instance or the previous PTP Instance.
4	Mean Link Delay	None	Timestamp error affecting measurement of meanLinkDelay. Each Pdelay measurement can be subject to relatively large errors from both the local and neighboring node. This normative requirement ensures a suitable algorithm mitigates the impact of any such single Pdelay measurement.

D.3.7 Explanation for the asymmetric normative requirements for the allowable range of dTE in Table 14, rows 2 and 3

The PTP End Instance requirements for dTE in Table 13, rows 2 and 3, are asymmetric, i.e., the allowable range in both cases is -145 ns to +20 ns. These tests are performed with the fractional frequency offset of the Working Clock (acting as the ClockSource) at the Grandmaster PTP Instance increasing at the rate of 1 ppm/s. dTE is measured at the output of the PTP End Instance clock control system after steady-state is reached, i.e., after any transients due to initiating the 1 ppm/s frequency drift rate have decayed. The clock control system is a second-order filter, with bandwidth and gain peaking requirements given in Table 10. Table 10 also indicates that the filter has a minimum roll-off of 20 dB/decade. The worst-case dTE response can be analyzed using the second-order filter model described in Annex C. The transfer function for this filter is given by Eq. (C.5). The example in C.4 shows that the damping ratio, ζ , for a filter whose gain peaking is 2.2 dB, i.e., the maximum allowed in Table 11, is 0.682 (see Eq. (C.19)). The allowable range for filter 3dB bandwidth in Table 10 is 0.9 Hz – 1.0 Hz. The example in C.4 indicates that the undamped natural frequency, ω_n corresponding to the maximum bandwidth of 1.0 Hz is 3.10 rad/s (see Eq. (C.18)).

6892 It is well-known that if a frequency drift is applied to a linear, second-order filter with 20
 6893 dB/decade roll-off, the steady-state output has a time offset. This means that if the input to the
 6894 PTP End Instance from the ClockSource (see Figure D.5) has a frequency drift of 1 ppm/s, the
 6895 measured dTE output of the ClockTarget has a steady-state offset. Any errors due to other
 6896 sources (e.g., timestamp granularity, dynamic timestamp error, clock noise) are in addition to
 6897 this. If these other errors are symmetric about zero, the addition of the steady-state offset
 6898 results in an asymmetric range of dTE.

6899 The requirement in Table 13 is on the Working Clock (acting as ClockTarget) at the PTP End
 6900 Instance minus the Working Clock (acting as ClockSource) at the Grandmaster. This difference,
 6901 which is the error between the input and output and is denoted by $E(s)$ in the s-domain, is given
 6902 by:

$$\begin{aligned} E(s) &= Y(s) - U(s) \\ &= H(s)U(s) - U(s) \\ &= (H(s) - 1)U(s) \\ &= H_e(s)U(s) \end{aligned} \quad (D.24)$$

6903 where $Y(s)$ and $U(s)$ are the Laplace transforms of the output and input, respectively, $H(s)$ is the
 6904 transfer function between the input and output and is given by Eq. (C.5), and $H_e(s)$ is the transfer
 6905 function between the input and error. Using Eq. (C.5), the error transfer function is given by:

$$H_e(s) = -\frac{s^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \quad (D.25)$$

6906 The phase drift corresponding to an $A = 1$ ppm/s = 1000 ns/s² frequency drift is $0.5At^2$, where
 6907 t is the time in seconds and the phase drift is in ns if A is in ns/s². The Laplace transform of this
 6908 waveform is $U(s) = A/s^3$. The steady-state value of the filter output, $y_{\text{steady-state}}$, due to this drift
 6909 is obtained from the final value theorem:

$$y_{\text{steady-state}} = \lim_{s \rightarrow 0} sH(s)U(s) = \lim_{s \rightarrow 0} s \cdot \left(-\frac{s^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \right) \cdot \frac{A}{s^3} = -\frac{A}{\omega_n^2} \quad (D.26)$$

6910 Since the 3dB bandwidth is proportional to the undamped natural frequency (see Eq. (C.23)),
 6911 Eq. (D.26) indicates that the worst case, i.e., largest steady-state error occurs for the minimum
 6912 3dB bandwidth, i.e., 0.9 Hz in Table 10. In the example of C.4, the undamped natural frequency
 6913 of 3.10 rad/s and damping ratio 0,682 correspond to a 3dB bandwidth of 1,0 Hz. Then, due to
 6914 the direct proportionality between 3dB bandwidth and undamped natural frequency (for the
 6915 same damping ratio), a 3dB bandwidth of 0.9 Hz corresponds to an undamped natural frequency
 6916 of:

$$\omega_n = \left(\frac{0.9 \text{ Hz}}{1.0 \text{ Hz}} \right) (3.10 \text{ rad/s}) = 2.79 \text{ rad/s} \quad (D.27)$$

6917 Then, using Eq. (D.26), the steady-state error corresponding to a frequency drift of 1 ppm/s is:

$$y_{\text{steady-state}} = -\frac{1000 \text{ ns/s}}{(2.79 \text{ rad/s})^2} = -128 \text{ ns} \quad (D.28)$$

6918 Then, assuming the error without the filter is 15 ns as given in Table 14, row 1, the lower end
 6919 of the range of dTE when the filter is present is:

$$dTE = -128\text{ ns} - 15\text{ ns} = -143\text{ ns} \approx -145\text{ ns}$$

(D.29)

D.4 Approach to testing normative requirements

D.4.1 General

This document does not specify tests to ensure conformance with the requirements. However, it is important that the requirements are, in principle, testable. Clause D.4 describes, at a high level, approaches a test specification can take to testing conformance with some of the requirements related to time synchronization.

It is assumed that test equipment can precisely measure the output of the ClockSource (at a Grandmaster PTP Instance), ClockTarget (at a PTP End Instance) and Local Clock (at any PTP Instance) to ensure conformance with frequency offset and frequency offset drift requirements. This might be via a Pulse per Second (PPS) plus Time-of-Day information or another mechanism.

It is also assumed that test equipment can generate sequences of PTP messages with precise timing and content (for testing PTP Relay Instances and PTP End Instances) and receive, log, and process sequences of PTP messages with precise timing measurement (e.g. of message arrival).

D.4.2 Testing Grandmaster PTP Instance

Figure D.1 illustrates an approach to testing the three normative requirements discussed in D.3.4.

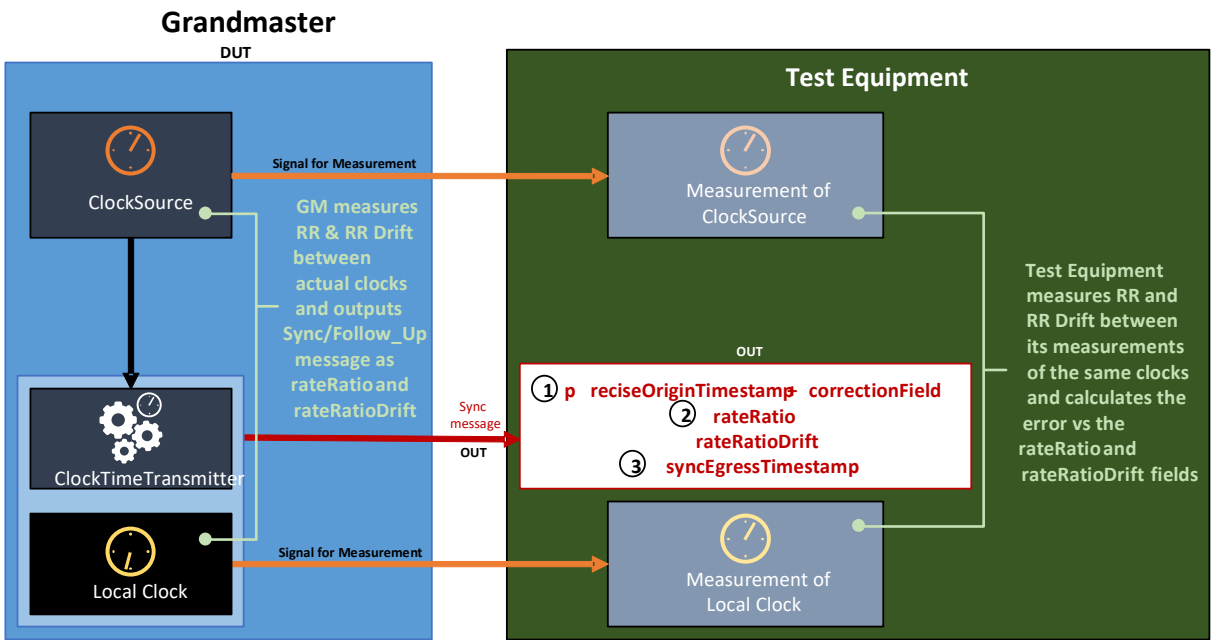


Figure D.1 – Approach to testing normative requirements for Grandmaster PTP Instance

The test equipment can calculate the time the Sync message is output at the DUT by subtracting the link delay from the measured arrival time at the test equipment.

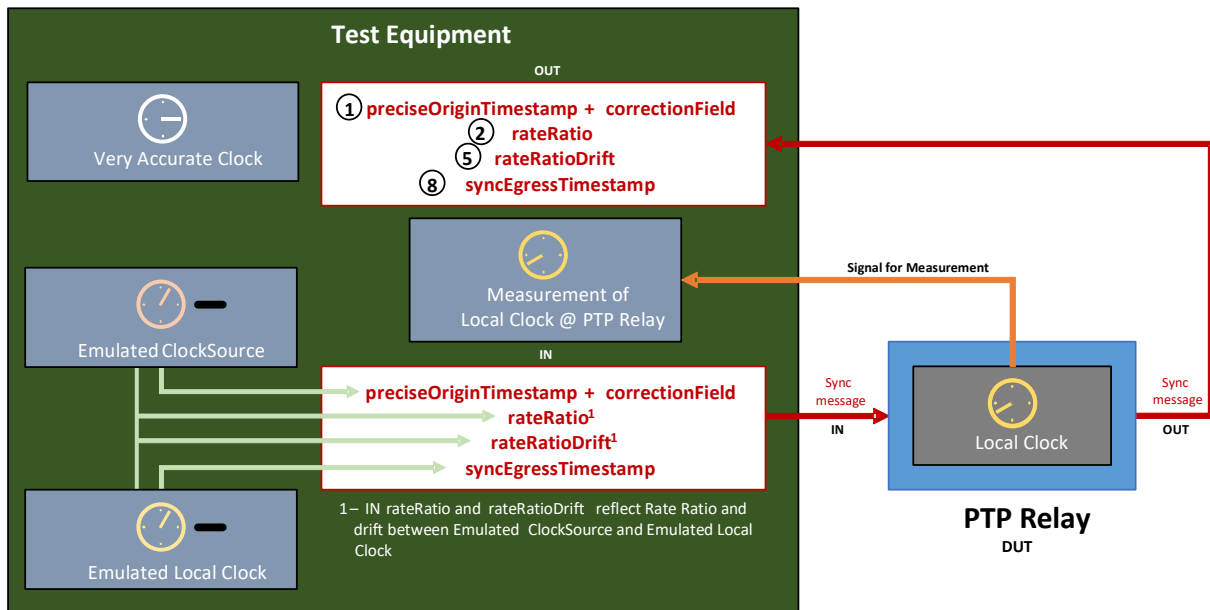
For test 1, the test equipment compares the value of the **preciseOriginTimestamp** + **correctionField** against its measurement of the **ClockSource**.

6945 For test 2, the test equipment compares the value in the rateRatio field with its calculation of
6946 the equivalent value based on its measurement of the ClockSource.

6947 For test 3, the test equipment compares the value of the syncEgressTimestamp against its
6948 measurement of the Local Clock.

6949 D.4.3 Testing PTP Relay Instance

6950 Figure D.2 illustrates an approach to testing normative requirements 1, 2, 5 and 8 discussed in
6951 D.3.5.



6952

6953 **Figure D.2 – Approach to testing normative requirements for PTP Relay Instance - 1**

6954 The test equipment can compare the DUT's output Sync message to the expected result given
6955 the measurement of the Local Clock and the timing of the input PTP timing message
6956 transmission and output PTP timing message reception.

6957 For these four tests, the Emulated ClockSource and Emulated Local Clock are stable and in
6958 sync. In practice, both can be equal to the test equipment's Very Accurate Clock. In the input
6959 Follow_Up information TLV, rateRatio is 0 ppm, and in the input Drift_Tracking TLV
6960 rateRatioDrift is 0 ppm/s. If the Local Clock of the PTP Relay Instance is also stable, it measures
6961 NRR of 0 ppm and NRR Drift of 0 ppm/s.

6962 The test equipment can calculate the time the output Sync message is output at the DUT by
6963 subtracting the link delay from the measured arrival time at the test equipment.

6964 For test 1, the test equipment can compare the increase in the value of the correctionField to
6965 the measured meanLinkDelay (from the test equipment to the DUT) plus residenceTime. The
6966 test equipment accounts for the additional delay between the PTP Relay Instance's
6967 transmission of the input PTP timing message and its reception by the test equipment.

6968 For tests 2 and 5, the test equipment can compare the rateRatio and rateRatioDrift fields in the
6969 output PTP timing message with the equivalent calculated values between the measured Local
6970 Clock and the Emulated ClockSource.

6971 For test 8, the test equipment can compare syncEgressTimestamp value in the output PTP
6972 timing message with its measurement of the Local Clock.

Figure D.3 illustrates an approach to testing normative requirements 3 and 6 discussed in D.3.5.

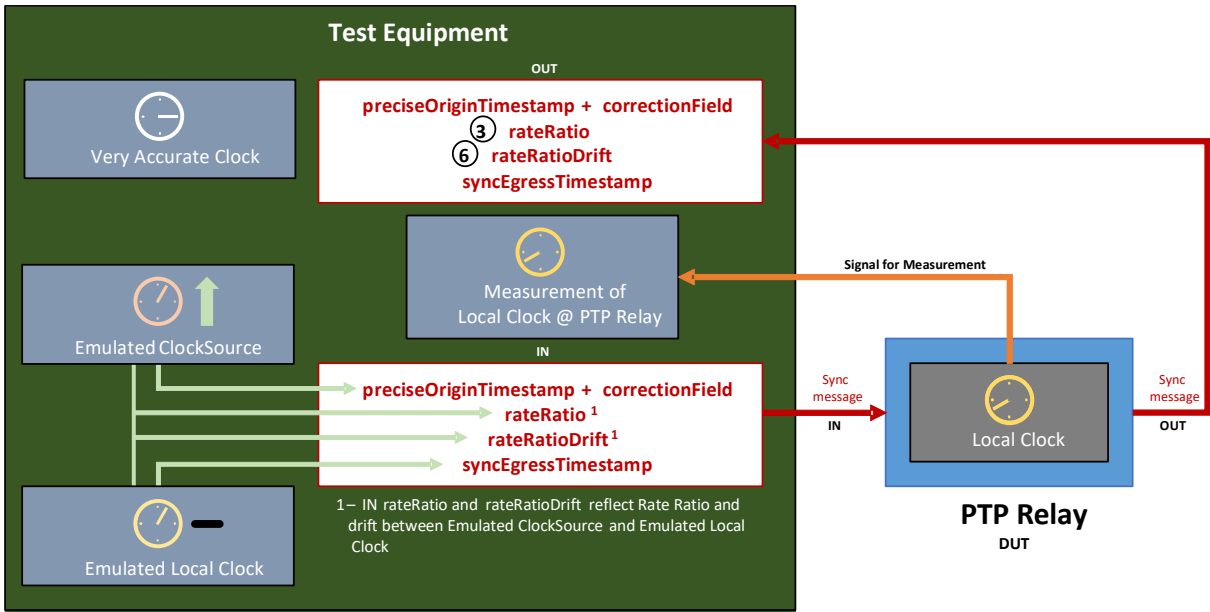


Figure D.3 – Approach to testing normative requirements for PTP Relay Instance – 2

For these two tests, the fractional frequency offset of the Emulated ClockSource is increasing at a defined ppm/s rate relative to the Very Accurate Clock. The Emulated Local Clock is stable; in practice, it can be equal to the test equipment's Very Accurate Clock. In the output Follow_Up information TLV, the rateRatio field increases over time, and in the output Drift_Tracking TLV, the rateRatioDrift field maintains a matching positive value. If the Local Clock of the PTP Relay Instance is also stable, it measures a NRR of 0 ppm and a NRR Drift of 0 ppm/s.

For tests 3 and 6, the test equipment can compare the rateRatio and rateRatioDrift fields in the output Follow_Up information TLV and Drift_Tracking TLV respectively with the equivalent calculated values between the measured Local Clock and the Emulated ClockSource.

Figure D.4 illustrates an approach to testing normative requirements 4 and 7 discussed in D.3.5.

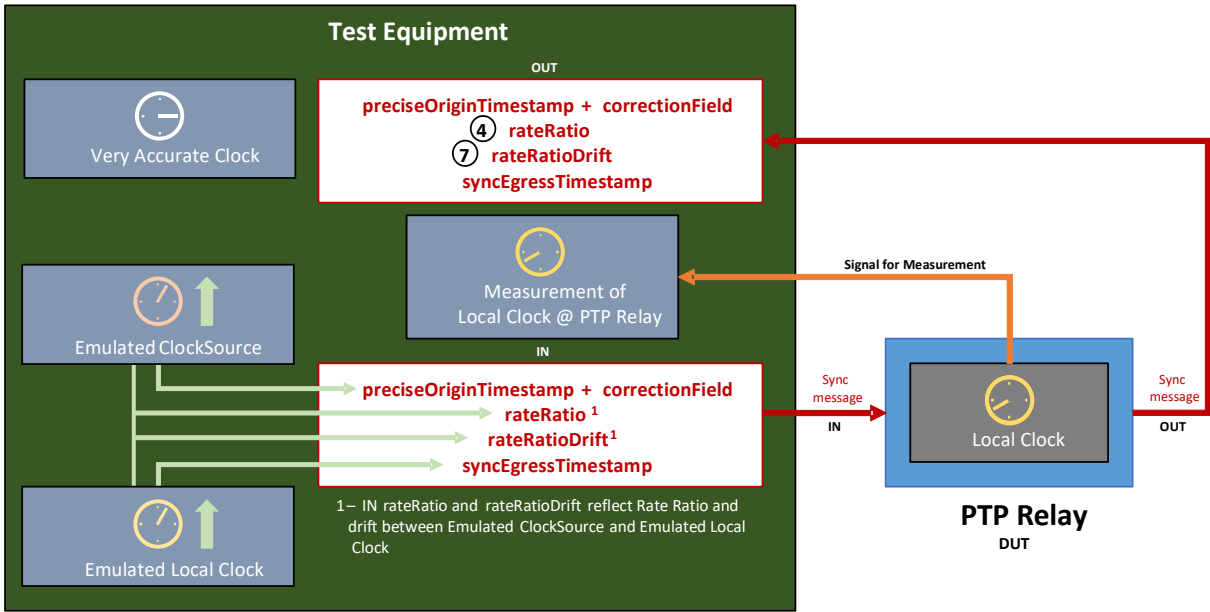


Figure D.4 – Approach to testing normative requirements for PTP Relay Instance – 3

For these two tests, the fractional frequency offsets of the Emulated ClockSource and the Emulated Local Clock are equal and increasing at a defined ppm/s rate relative to the Very Accurate Clock. In the output Follow_Up information TLV, the rateRatio field is 0 ppm, and in the output Drift_Tracking TLV the rateRatioDrift field is 0 ppm/s. If the Local Clock of the PTP Relay Instance is stable, the NRR it measures increases over time and the NRR Drift it measures maintains a matching positive value.

For tests 4 and 7, the test equipment can compare the rateRatio and rateRatioDrift fields in the output Follow_Up information TLV and Drift_Tracking TLV respectively with the equivalent calculated values between the measured Local Clock and the Emulated ClockSource.

D.4.4 Testing PTP End Instance

Figure D.5 illustrates an approach to testing the three normative requirements discussed in D.3.6.

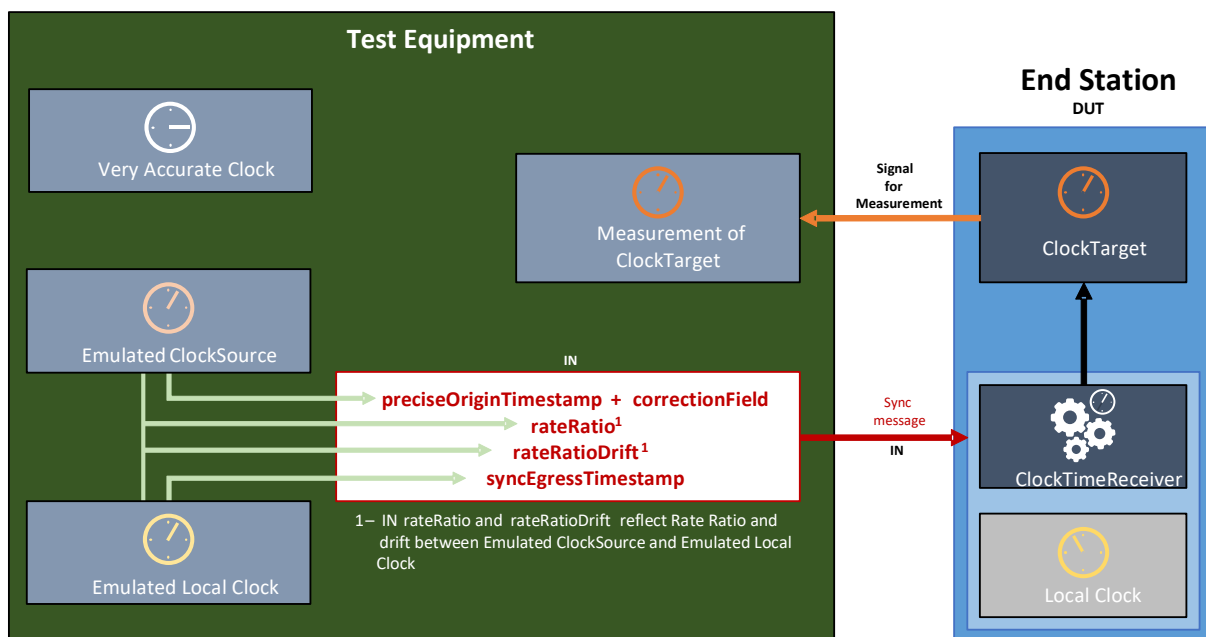


Figure D.5 – Approach to testing normative requirements for PTP End Instance

The test equipment can compare its measurement of the DUT's ClockTarget to the Emulated ClockSource. It accounts for the additional delay between its transmission of the input Sync message and the reception of the message by the DUT.

For test 1, the Emulated ClockSource and Emulated Local Clock are stable and in sync. In practice, both can be equal to the test equipment's Very Accurate Clock. In the input Follow_Up information TLV, rateRatio is 0 ppm, and in the input Drift_Tracking TLV, rateRatioDrift is 0 ppm/s. If the Local Clock of the PTP End Instance is also stable, it measures NRR of 0 ppm and NRR Drift of 0 ppm/s.

For test 2, the fractional frequency offset of the Emulated ClockSource is increasing at a defined ppm/s rate relative to the Very Accurate Clock. The Emulated Local Clock is stable; in practice, it can be equal to the test equipment's Very Accurate Clock. In the output Follow_Up information TLV, the rateRatio field increases over time, and the output Drift_Tracking TLV, the rateRatioDrift field maintains a matching positive value. If the Local Clock of the PTP Relay Instance is also stable, it measures NRR of 0 ppm and NRR Drift of 0 ppm/s.

For test 3, the fractional frequency offsets of the Emulated ClockSource and the Emulated Local Clock are equal and increasing at a defined ppm/s rate relative to the Very Accurate Clock. In the output Follow_Up information TLV, the rateRatio field is 0 ppm, and in the output output Drift_Tracking TLV, the rateRatioDrift field is 0 ppm/s. If the Local Clock of the PTP Relay Instance is stable, the NRR it measures increases over time and the NRR Drift it measures maintains a matching positive value.

D.4.5 Testing meanLinkDelay Accuracy

D.3.5 and D.3.6 both discuss a normative requirement to test the accuracy of meanLinkDelay measurements for PTP Relay Instances and PTP End Instances respectively. This can be achieved by using a cable of known length and asymmetry to connect the test equipment to the DUT. The test equipment then responds to the DUT's Pdelay_Req messages with Pdelay_Resp messages, allowing the DUT to estimate the path delay.

The DUT's estimate of the path delay can be accessed via the meanLinkDelay global variable (see IEEE 802.1AS-2020, Clause 14.8.8).

The error in a single path delay measurement via a Pdelay_Req / Pdelay_Resp exchange can be large relative to the normative requirement, but an estimate with significantly less error can be generated by considering more than one measurement, hence the "mean" in "meanLinkDelay". Two consequences of this are noteworthy.

Firstly, since errors can be introduced by Timestamp Errors at both the DUT and a neighboring node, and the DUT is responsible for mitigating both, the test equipment simulates typical errors on its timestamps for receiving Pdelay_Req and transmitting Pdelay_Resp messages to provide a complete test.

Secondly, since the meanLinkDelay measurement can take some time to stabilize based on multiple Pdelay_Req / Pdelay_Resp exchanges, leaving sufficient time for these multiple exchanges before assessing meanLinkDelay accuracy results in a more accurate test.

D.5 Example algorithms

D.5.1 General

This document does not place normative requirements on the use of specific algorithms. However, the normative requirements assume the use of algorithms to reduce the effect of errors in meanLinkDelay and to track clock drift and compensate for consequent errors. PTP Instances that do not implement algorithms can find it difficult or impossible to meet the normative requirements.

D.5 provides examples of algorithms that can be used for:

- tracking NRR drift;
- correcting for errors in measured NRR (mNRR) due to NRR drift;
- calculating RR drift;
- correcting for errors in measured RR (mRR) due to RR drift;
- reducing the effect of errors in meanLinkDelay.

D.5.2 Algorithm for tracking NRR drift

For measured NRR, measured RR, and meanLinkDelay, an example for how startup behavior can be handled is provided.

7057 NRR Drift Tracking and Error Correction is carried out for each network hop, i.e. at every node
7058 other than the Grandmaster. It is based on pairs of timestamps with each pair associated with
7059 a Sync message transmitted from the previous node ($n-1$) to the current node (n).

- 7060 • t_{s1outP} – Timestamp of the Sync message egress from the previous node ($n-1$), timestamped
7061 by that node's Local Clock. Unit: ns.
- 7062 • t_{s2in} – Timestamp of the Sync message ingress to the current node (n), timestamped by that
7063 node's Local Clock. Unit: ns.

7064 All timestamps are affected by Timestamp Errors.

7065 The algorithm uses information from the 32 most recent Sync messages. However, a node only
7066 keeps track of the 9 most recent pairs of timestamps from the most recent (x) to the 9th most
7067 recent ($x-8$) Sync message. The algorithm generates one measurement of NRR using the prior
7068 2 s of Sync message data (on average, based on a nominal Sync Interval of 125 ms), and a
7069 second measure based on the 2 s of Sync message data prior to that. It then uses the difference
7070 in the two measurements over the interval between the effective measurement points to
7071 calculate the NRR drift rate.

7072 On arrival of a new Sync message (x), or Follow_Up in the case of two-step time transport, a
7073 node executes a NRR calculation:

$$NRRcalc(x) = \left(\frac{t_{s1outP}(x) - t_{s1outP}(x-8)}{t_{s2in}(x) - t_{s2in}(x-8)} - 1 \right) \times 10^6 \quad (D.1)$$

7074
7075 where

7076 $NRRcalc$ is the calculated Neighbor Rate Ratio, expressed in ppm;
7077 x is the most recent Sync message;
7078 t_{s1outP} is the timestamp of the Sync message egress from the previous node ($n-1$),
7079 timestamped by that node's Local Clock, expressed in ns;
7080 t_{s2in} is the Timestamp of the Sync message ingress to the current node (n), timestamped
7081 by that node's Local Clock, expressed in ns.

7082 with an associated effective measurement point:

$$NRRcalcT(x) = \frac{t_{s2in}(x) + t_{s2in}(x-8)}{2} \quad (D.2)$$

7083
7084 where

7085 $NRRcalcT$ is the effective measurement point, expressed in ns;
7086 t_{s1outP} is the timestamp of the Sync message egress from the previous node ($n-1$),
7087 timestamped by that node's Local Clock, expressed in ns;
7088 t_{s2in} is the Timestamp of the Sync message ingress to the current node (n), timestamped
7089 by that node's Local Clock, expressed in ns.

7090 A node keeps track of the 24 most recent NRR calculations and effective measurement points,
7091 from the most recent (x) to the 24th most recent ($x-23$).

7092 After of a new most-recent NRR calculation, a node calculates an NRR drift rate:

$$NRRaverageA = \sum_{i=x-7}^x \frac{mNRRcalc(i)}{8} \quad (D.3)$$

7093

7094 where

7095 *NRRaverageA* is the average of the 8 most recent Neighbor Rate Ratio calculations, expressed
7096 in ppm.

$$NRRaverageB = \sum_{i=x-23}^{x-16} \frac{mNRRcalc(i)}{8} \quad (D.4)$$

7097

7098 where

7099 *NRRaverageB* is the average of the 8 least recent Neighbor Rate Ratio calculations, expressed
7100 in ppm.

$$NRRdriftInterval = \sum_{i=x-7}^x \frac{mNRRcalcT(i)}{8} - \sum_{i=x-23}^{x-16} \frac{mNRRcalcT(i)}{8} \quad (D.5)$$

7101

7102 where

7103 *NRRdriftInterval* is the period across which Neighbor Rate Ratio drift is measured, expressed
7104 in ns.

$$NRRdriftRate(n) = \left(\frac{NRRaverageA - NRRaverageB}{NRRdriftInterval} \right) \times 10^9 \quad (D.6)$$

7105

7106 where

7107 *NRRdriftRate(n)* is the the NRR drift rate for the current Node *n*, expressed in ppm/s.

7108 **D.5.3 Algorithm to compensate for errors in measured NRR due to clock drift**

7109 **D.5.3.1 General**

7110 The algorithm to measure NRR uses data from the previous 1 s of Sync message data,
7111 combined with the NRR drift estimate from the previous step. This smaller amount of data (vs
7112 that used for either of the NRR measurements in the previous step) is employed as it improves
7113 responsiveness to sudden changes in NRR drift with minimal loss of accuracy.

7114 On arrival of a Sync message (*x*), or Follow_Up in the case of two-step time transport, a node
7115 executes a NRR calculation:

$$mNRRcalc(x) = \left(\frac{t_{s1outP}(x) - t_{s1outP}(x-4)}{t_{s2in}(x) - t_{s2in}(x-4)} - 1 \right) \times 10^6 \quad (D.7)$$

7116

7117 with an associated effective measurement point:

$$mNRRcalcT(x) = \frac{t_{s2in}(x) + t_{s2in}(x-4)}{2} \quad (D.8)$$

7118

7119 A node keeps track of the 4 most recent mNRR calculations and effective measurement points,
7120 from the most recent (x) to the 4th most recent ($x-3$). The mNRR calculations use information
7121 from the 5 most recent Sync messages, but the node is already keeping track of information
7122 from the 9 most recent Sync messages for the NRR drift tracking algorithm.

7123 The node then calculates an error corrected measured NRR value.

7124 For $i = x$ to $(x-3)$

$$mNRRcorrected(i) = mNRRcalc(i) + \left(NRRdriftRate(n) \times \frac{(t_{s2in}(x) - mNRRcalcT(i))}{10^9} \right) \quad (D.9)$$

7125

7126 where

7127 $mNRRcorrected(i)$ is the error-corrected measured NRR value at message i , expressed in ppm.

$$mNRR(n) = \sum_{i=x-3}^x \frac{mNRRcorrected(i)}{4} \quad (D.10)$$

7128

7129 where

7130 $mNRR$ is the error-corrected measured NRR value, expressed in ppm.

7131 The result is a measured NRR value, error-corrected to the time when the most recent Sync
7132 message was received.

7133 D.5.3.2 Measured NRR algorithm – Startup behavior

7134 NRR is used when calculating meanLinkDelay and output Sync/Follow_Up message fields. The
7135 first NRR drift calculation is only be available after receipt of 32 Sync/Follow_Up messages, i.e.
7136 after approximately 4 seconds of operation given the 125 ms Sync Interval. During this time
7137 meanLinkDelay and output Sync/Follow_Up message fields have not been calculated, so an
7138 alternative must be used, even if it cannot deliver the same assurances regarding network-level
7139 performance.

7140 If measured NRR from Sync/Follow_Up message information is unavailable but equivalent
7141 information from Pdelay_Resp messages is available, it may be substituted for Sync/Follow_Up
7142 message information. However, measuring NRR using Pdelay_Resp messages is vulnerable to
7143 additional error due to clock drift between the time NRR is measured, on receipt of the latest
7144 Pdelay_Resp message, and use of the measurement during Sync message processing. This is
7145 the reason using Sync/Follow_Up message information is preferable. It also means that a switch
7146 to using Sync/Follow_Up message information as soon as possible is desirable. It is technically
7147 possible to calculate a NRR using a combination of Pdelay_Resp and Sync messages but this
7148 can be risky due to the potential for very short intervals between messages and resulting high
7149 error due to timestamp errors, so it not recommended.

7150 It is the responsibility of implementers to decide whether and when to use Pdelay_Resp
 7151 message information and when to switch to using Sync/Follow_Up message information. The
 7152 normative requirements in this document are for operation after 32 Sync/Follow_Up messages
 7153 have been received and assume use of the algorithms in D.5.2 and D.5.3, or more effective
 7154 algorithms. Implementations that continue to use Pdelay_Resp message information to
 7155 calculate NRR after 32 messages have been received can find some of the normative
 7156 requirements difficult or impossible to meet.

7157 The following describes potential startup behavior applicable to either Sync/Follow_Up or
 7158 Pdelay_Resp message information.

- 7159 a) At least two messages must be received before calculating a NRR value.
 7160 b) Prior to two messages being received, $NRR = 1$ (i.e. 0 ppm) should be used.
 7161 c) Once two messages have been received, NRR should be calculated using the formula:

$$2^{\text{nd}} \text{ message: } mNRR = \left(\left(\frac{t_3(x) - t_3(x-1)}{t_4(x) - t_4(x-1)} \right) - 1 \right) \times 10^6 \quad (\text{D.11})$$

7162
 7163 where

7164 $t_3(i)$ is the timestamp of the i^{th} Pdelay_Resp message on egress from the previous node ($n-1$),
 7165 timestamped by that node's Local Clock, expressed in ns;

7166 $t_4(i)$ is the timestamp of the Pdelay_Resp message on ingress to the current node (n),
 7167 timestamped by that node's Local Clock, expressed in ns.

- 7168 d) When three to four messages have been received, NRR should be calculated using the
 7169 following formula:

$$3^{\text{rd}} \text{ message: } mNRR = \left(\left(\frac{t_{1\text{outP}}(x) - t_{1\text{outP}}(x-2)}{t_{2\text{in}}(x) - t_{2\text{in}}(x-2)} \right) - 1 \right) \times 10^6 \quad (\text{D.12})$$

$$4^{\text{th}} \text{ message: } mNRR = \left(\left(\frac{t_{1\text{outP}}(x) - t_{1\text{outP}}(x-3)}{t_{2\text{in}}(x) - t_{2\text{in}}(x-3)} \right) - 1 \right) \times 10^6 \quad (\text{D.13})$$

- 7170
 7171 e) On arrival of the 5th Sync/Follow_Up message the first mNRRcalc and mNRRcalcT
 7172 calculations can take place and should be used for NRR:

$$mNRRcalc(x) = \left(\frac{t_{s1\text{outP}}(x) - t_{s1\text{outP}}(x-4)}{t_{s2\text{in}}(x) - t_{s2\text{in}}(x-4)} - 1 \right) \times 10^6 \quad (\text{D.14})$$

$$mNRRcalcT(x) = \frac{t_{s2\text{in}}(x) + t_{s2\text{in}}(x-4)}{2} \quad (\text{D.15})$$

$$5^{\text{th}} \text{ Sync message: } mNRR = mNRRcalc(x) \quad (\text{D.16})$$

7174 f) As the 6th, 7th and 8th messages arrive an average can be taken and used for NRR, so:

$$6^{\text{th}} \text{ message: } mNRR = \sum_{i=x-1}^x \frac{mNRR_{calc}(i)}{2} \quad (\text{D.17})$$

$$7^{\text{th}} \text{ message: } mNRR = \sum_{i=x-2}^x \frac{mNRR_{calc}(i)}{3} \quad (\text{D.18})$$

$$8^{\text{th}} \text{ message: } mNRR = \sum_{i=x-3}^x \frac{mNRR_{calc}(i)}{4} \quad (\text{D.19})$$

7175 g) For the 9th to the 31st message, the Formula (D.19) can be used.

7176 h) Once the 32nd message arrives, the regular formulas with NRR drift tracking and error
7177 correction can be used.

7178 **D.5.4 Algorithm for tracking RR Drift**

7179 A Sync or Follow_Up message carries the rateRatio field, which informs each node of the
7180 previous node's estimate of its (the previous node's) Rate Ratio. This document also requires
7181 support for the Drift_Tracking TLV, which carries the rateRatioDrift field and informs each node
7182 of the previous node's estimate of its (the previous node's) Rate Ratio Drift.

7183 If the implementation of the Grandmaster PTP Instance means the ClockSource and Local Clock
7184 (at the Grandmaster PTP Instance) are linked such that the two are always operating at the
7185 same frequency, the rateRatio field received by the first node (Node 1) is always 0 ppm and the
7186 rateRatioDrift field is always 0 ppm/s. Thus, at Node 1, RR equals NRR, RR Drift equals NRR
7187 Drift, and therefore D.5.3 and D.5.3.2 describe how to calculate RR and RR Drift at Node 1.

7188 If the implementation of the Grandmaster PTP Instance means the ClockSource and Local Clock
7189 (at the Grandmaster PTP Instance) can operate at different frequencies, the implementation
7190 populates the rateRatio and rateRatioDrift field with values reflecting those differences.

7191 In either case all PTP Instances, other than the Grandmaster PTP Instance, calculate an
7192 estimate of the local Rate Ratio Drift when the latest Sync/Follow_Up Message is received,
7193 based on the received rateRatioDrift field and the local measure of NRR Drift. The Rate Ratio
7194 Drift Rate from the previous node is in ppm/s relative to the timebase of its Local Clock (i.e. the
7195 "s" in "ppm/s"). For highest precision, this can be converted to the timebase of the current
7196 node's Local Clock.

$$rateRatioDrift(n) = \frac{rateRatioDrift(n-1)}{\left(1 + \frac{mNRR(n)}{10^6}\right)} + NRRdriftRate(n) \quad (\text{D.20})$$

7197
7198 where

7199 *rateRatioDrift* is the Rate Ratio drift rate, expressed in ppm/s.

7200 However, given that adding ppm/s already lacks the precision of multiplying actual ratios, this
7201 simplification delivers similarly accurate results.

$$rateRatioDrift(n) = rateRatioDrift(n-1) + NRRdriftRate(n) \quad (D.21)$$

7202

7203 D.5.5 Algorithm to compensate for errors in measured RR due to clock drift

7204 On receipt of a Sync or Follow_Up message, all PTP Relay Instances estimate a measured RR
 7205 ($mRR(n)$) based on the received $rateRatio$ field ($mRR(n-1)$) and the local measure of NRR
 7206 ($mNRR(n)$). An $mRR(n)$ value is used to translate the sum of $meanLinkDelay$ and $residenceTime$
 7207 from Local Clock timebase into Grandmaster timebase. An $mRR(n)$ value is also passed in the
 7208 transmitted Sync or Follow_Up message's $rateRatio$ field to the next node. Errors in these
 7209 estimates due to clock drift can be reduced by taking account of RR Drift. Since the optimal
 7210 point in time for each estimate is different, the amount of applicable RR Drift is different, and
 7211 hence the estimates are different.

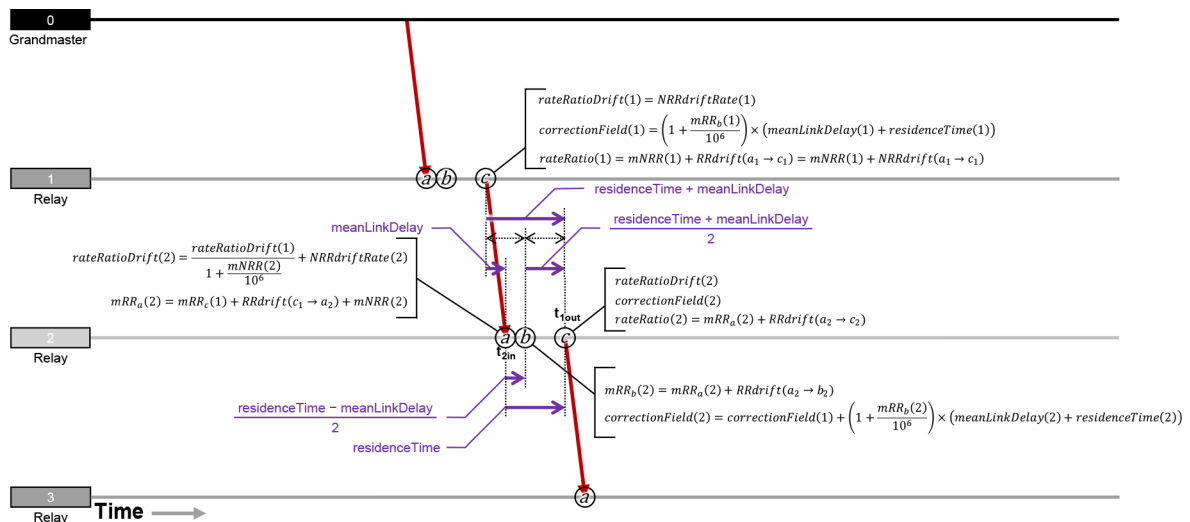
7212 (For discussion of how different Grandmaster PTP Implementations affect the behavior of a
 7213 PTP Relay Instance at Node 1 – or not – see D.5.4.)

7214 A PTP End Instance is similar in that it estimates $mRR(n)$ on receipt of a Sync message,
 7215 subsequently uses an $mRR(n)$ value, and errors in the latter due to clock drift can be mitigated
 7216 by taking account of RR Drift. However, unlike a PTP Relay Instance, the $mRR(n)$ value is used
 7217 to keep the ClockTarget in line with the ClockSource and transmitting a $rateRatio$ field to a
 7218 subsequent node is unnecessary.

7219 For a PTP Relay Instance there are three points in time of interest:

- 7220 • Point a: receipt of the Sync Message by the current node (Node n),
- 7221 • Point b: mid-point between transmission of the Sync message by the previous node (Node
 7222 $n-1$) and transmission of the consequent Sync message by the current node (Node n), and
- 7223 • Point c: transmission of the Sync Message by the current node (Node n).

7224 Figure D.6 illustrates these points and the associated calculations.



7225

IEC

7226 **Figure D.6 – RR Drift tracking and error compensation calculations –**
 7227 **PTP Relay Instance**

7228 The estimate of RR when the Sync message arrives can be calculated as follows:

$$mRR_a(n) = rateRatio(n-1) + RRdrift(c_{n-1} \rightarrow a_n) + mNRR(n). \quad (D.22)$$

$$7229 = rateRatio(n-1) + \left(rateRatioDrift(n-1) \times \left(1 + \frac{mNRR(n)}{10^6} \right) \times meanLinkDelay(n) \right) + mNRR(n)$$

7230 where

7231 mRR_a is the estimate of RR when the Sync message arrives at node n , expressed in ppm;

7232 $RRdrift(c_{n-1} \rightarrow a_n)$ is the amount $rateRatio(n-1)$ drifts between transmission of the Sync
7233 message at Node $n-1$ and reception at Node n , expressed in ppm/s.

7234 $RRdrift(c_{n-1} \rightarrow a_n)$ is equivalent to $rateRatioDrift(n-1)$ multiplied by meanLinkDelay but,
7235 since meanLinkDelay is measured in terms of Node n 's Local Clock and rateRatioDrift is in
7236 terms of Node $n-1$'s Local Clock the former should be multiplied by the NRR at Node n for the
7237 highest accuracy.

7238 However, given that adding ppm/s already lacks the precision of multiplying actual ratios, the
7239 following simplification delivers similarly accurate results:

$$mRR_a(n) = rateRatio(n-1) + (rateRatioDrift(n-1) \times meanLinkDelay(n)) + mNRR(n) \quad (D.23)$$

7240

7241 Once the time when Node n transmits the consequent Sync message is known, the
7242 correctionField value can be calculated.

$$mRR_b(n) = mRR_a(n) + RRdrift_n(a \rightarrow b) \quad (D.24)$$

$$7243 = mRR_a(n) + \left(rateRatioDrift(n) \times \frac{residenceTime(n) - meanLinkDelay(n)}{2} \right)$$

7244 where

7245 $mRR_b(n)$ is the estimate of RR when Node n transmits the consequent Sync message,
7246 expressed in ppm.

7247 The correctionField is calculated as follows:

$$correctionField(n) = correctionField(n-1) + \left(1 + \frac{mRR_b(n)}{10^6} \right) \quad (D.25)$$

$$\times (meanLinkDelay(n) + residenceTime(n))$$

7248

7249 where

7250 *correctionField*(*n*) is the value of the correction field transmitted by node *n*, expressed in
 7251 ns.

7252

7253 The rateRatio field is calculated as follows:

$$\begin{aligned}
 7254 \quad \text{rateRatio}(n) &= mRR_a(n) + RRdrift_n(a \rightarrow c) \\
 &= mRR_a(n) + (RRdriftRate(n) \times residenceTime(n))
 \end{aligned}
 \tag{D.26}$$

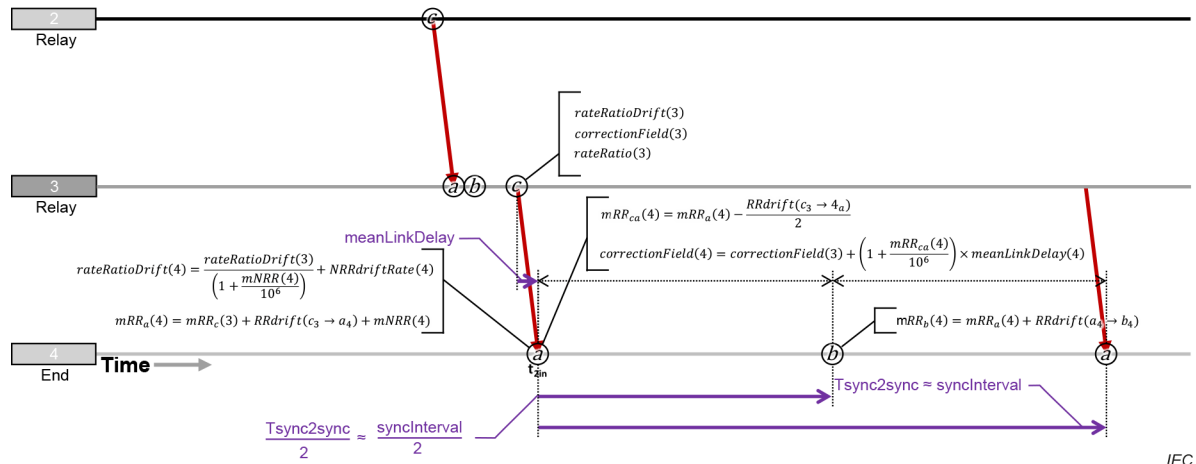
7255

7256 where

7257 *rateRatio*(*n*) is the value of the rateRatio field transmitted by node *n*, expressed in ppm.

7258 **D.5.6 Algorithm to compensate for errors in measured RR due to clock drift at PTP** 7259 **End Instance**

7260 Figure D.7 illustrates a possible approach to applying similar RR drift tracking and error
 7261 compensation at a PTP End Instance.



7262

IEC

7263 **Figure D.7 – RR Drift tracking and error compensation calculations –**
 7264 **PTP End Instance**

7265 The initial calculations for rateRatioDrift and mRR_a are exactly the same as for a PTP Relay
 7266 Instance. Instead of using RR to translate meanLinkDelay + residenceTime from the Local Clock
 7267 timebase to the Grandmaster timebase – as is done at a PTP Relay Instance – a PTP End
 7268 Instance uses mRR to translate meanLinkDelay to the Grandmaster timebase (there is no
 7269 residenceTime at an End Instance), adding the result to the incoming correctionField to obtain
 7270 an estimate of the ClockSource at the time the Sync message arrives, then uses mRR to keep
 7271 its ClockTarget in line with the ClockSource until arrival of the next Sync message. The optimal
 7272 mRR value for translating meanLinkDelay is halfway between meanLinkDelay's transmission
 7273 (at c_{n-1} , i.e. point C at the previous node) and reception (at a_n , i.e. point A at the current node);
 7274 in the formulas below, this value is referred to as mRR_{ca} .

$$7275 \quad mRR_{ca}(n) = mRR_a(n) - \frac{RRdrift(c_{n-1} \rightarrow a_n)}{2}$$

$$= mRR_a(n) - \left(rateRatioDrift(n) \times \frac{meanLinkDelay(n)}{2} \right) \quad (D.27)$$

7276

7277 where

7278 $mRR_{ca}(n)$ is the estimate of RR calculated based upon one-half of the meanLinkDelay,
 7279 expressed in ppm.

7280 The correction field at the PTP End Instance is given by:

$$correctionField(n) = correctionField(n-1) + mRR_{ca}(n) \times meanlinkDelay(n) \quad (D.28)$$

7281

7282 The optimal value of mRR for keeping the ClockTarget in line with the Clock Source is mRR_b ,
 7283 where Point B is halfway between the most recently received Sync message and the next Sync
 7284 message.

7285 Of course, the exact interval until the next Sync message's arrival ($T_{sync2sync}$ in Figure D.7)
 7286 can't be known before it happens, but the Rate Ratio value is required as soon as possible after
 7287 arrival of the most recent Sync message. The solution is to use the nominal value of the interval,
 7288 i.e. $syncInterval$, which is 125 ms.

7289

$$mRR_b(n) = mRR_a(n) + RRdrift_n(a \rightarrow b)$$

7290

$$= mRR_a(n) + \left(rateRatioDrift(n) \times \frac{syncInterval}{2} \right)$$

$$= mRR_a(n) + (rateRatioDrift(n) \times 0,0625) \quad (D.29)$$

7291

7292 where

7293 $syncInterval$ is the nominal value of the interval between sync messaged, expressed in ms.

7294 It is also possible to use more complex algorithms that repeatedly or continuously adjust the
 7295 mRR value between Sync messages, but such an approach is not addressed in this document.

7296 **D.5.7 Mean Link Delay averaging**

7297 The actual Path Delay from one node to the next – for a wired connection – is very stable and
 7298 errors measuring it due to Timestamp Error average to zero. Thus, taking a long average or
 7299 applying a low-pass filter with a low bandwidth is an effective way to reduce error in
 7300 meanLinkDelay. Care is taken during system startup or after any other initialization of the
 7301 algorithm, to quickly converge on a stable value.

7302 The basic Pdelay calculation, used by the Common Mean Link Delay service, remains the same.
 7303 Figure D.8 illustrates it.

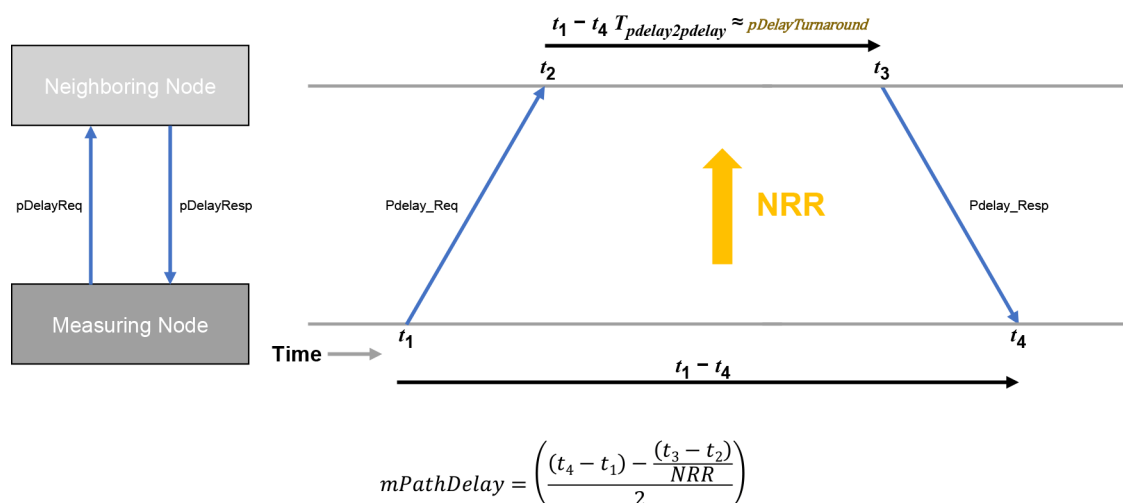


Figure D.8 – Signals and timestamps to measure path delay

Following each Pdelay_Req – Pdelay_Resp exchange, the measured path delay (mPathDelay) is calculated.

For the x^{th} message after initialisation

$$mPathDelay(x) = \frac{(t_4 - t_1) - \frac{(t_3 - t_2)}{NRR}}{2} \quad (D.30)$$

where

mPathDelay is the measured path delay between the measuring node and the neighboring node expressed in ns;

t_1 is a measurement point as defined in Figure D.8, expressed in ns;

t_2 is a measurement point as defined in Figure D.8, expressed in ns;

t_2 is a measurement point as defined in Figure D.8, expressed in ns;

t_i is a measurement point as defined in Figure D.8, expressed in ns.

The meanLinkDelay is then updated via an IIR (Infinite Impulse Response) filter. For the first measurement, the filter is initialized:

$$meanLinkDelay(x) = mPathDelay(x) \quad (D.31)$$

For the next few minutes after initialization (when $x < 1\,000$) the filter is in startup mode. It then transitions to steady-state mode.

If $x < 1\,000$ then $f = x$ else $f = 1\,000$

$$meanLinkDelay(x) = \frac{(meanLinkDelay(x-1) \times (f-1)) + mPathDelay(x)}{f} \quad (D.32)$$

7323 For example:

$$meanLinkDelay(100) = \frac{(meanLinkDelay(99) \times 99) + pDelay(x)}{100} \quad (D.33)$$

7324

$$meanLinkDelay(5\ 836) = \frac{(meanLinkDelay(5\ 835) \times 999) + pDelay(x)}{1\ 000} \quad (D.34)$$

7325

7326 It is possible to automatically reinitialize the algorithm if an *mPathDelay* value, or series of
7327 values, deviates too much from the *meanLinkDelay*, but the details are outside the scope of this
7328 document.

7329 The behavior of timestamp error means that, for shorter actual link delays, *mPathDelay* can be
7330 a negative value. It can seem tempting to reject negative values, since a negative delay is
7331 impossible. However, at a device level, including negative values of *mPathDelay* in the input to
7332 the IRR filter results in a more accurate filter output, i.e. *meanLinkDelay* value; values lower
7333 than the actual delay, even when negative, are balanced by values higher than the actual delay.

7334 Similarly, at a network level, using negative values of *meanLinkDelay*, i.e. the output of the IIR
7335 filter, results in a more accurate *correctionField* calculation at the PTP End Instance when there
7336 are many networking hops between it and the Grandmaster PTP Instance.

Annex E

(normative)

60802 constrained device proxy TLV definition

E.1 60802 constrained device proxy TLV

E.1.1 General

In the TLV definition that follows, any fields that are labelled as “Reserved” are transmitted as zero and ignored on receipt.

The 60802 device info TLV is a mandatory TLV that indicates a device’s configuration capabilities.

Figure E.1 shows the 60802 device info TLV.

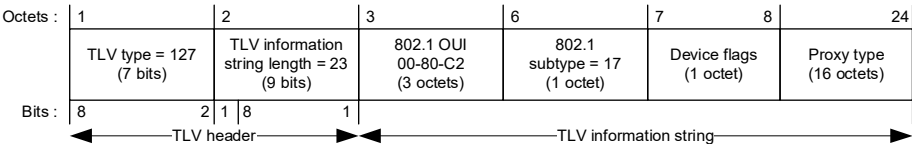


Figure E.1 – 60802 constrained device proxy TLV

E.1.2 Device flags

The Device flags field shall contain a bit map indicating the device configuration capabilities as defined in Table E.1.

Table E.1 – Device flags

Bit	Value/meaning	Function
0	1 = Proxy required 0 = no Proxy required	Proxy required
1	1 = supported 0 = not supported	NETCONF configuration supported
2-7	(set to zero)	Reserved for future standardization

E.1.3 Proxy type

The proxy type field is conditionally mandatory if the device flags indicate that the device requires a proxy. It indicates the proxy type that is required to configure the device. The format is a human readable string, that is specific for a specific proxy and its associated devices.

Annex F (informative)

MAC and PHY delay

F.1 Accumulated delays in industrial automation

F.1.1 General

MAC and PHY delays affect the performance of industrial communication. These delays impact forwarding delay, the accuracy of time-triggered transmission, and synchronization accuracy. These delays result from:

- the clock system between the MAC application, the MAC, the Media Independent Interface (MII), and the PHY,
- the processing of the frame, and
- the transmission of bits from the MAC application over MAC, MII and PHY to the MDI.

These delays consist of two components:

- the propagation delay, and
- the propagation delay variance (often called jitter).

F.1.2 Forwarding delay

Linear topologies, as commonly used in industrial communication, result in cumulative MAC and PHY delays. With 64 hops, there are 130 MAC and PHY delays (64 hops x 2 + 2x end stations) in end-to-end transmission.

For example, a frame sent from the end station at the beginning of a linear topology (64 Hops) to an end station at the end of the line is delayed by 26 μ s / 65 μ s / 130 μ s solely due to a PHY delay of 200 ns / 500 ns / 1 μ s. This means that the lower the PHY delay, the shorter the achievable update interval and thus, the roundtrip delay.

F.1.3 Time-triggered transmission

The reference point for time-triggered transmission is at the MDI. However, the frames are handed over from the transmission control to the MAC in a time-controlled manner. To transmit a frame at the correct time at the reference point, the MAC and PHY delay are compensated by the transmission control.

For example, the magnitude of the propagation delay reduces the computation time available for the application to calculate new process values, as the frames are handed over from the transmission control to the MAC earlier so they can be sent at the correct time at the reference point. Additionally, the frames are also received later by the application, which further reduces the computation time (see 4.1).

The magnitude of the propagation delay variance affects the safety margin of the windows in TAS and is a consideration when parameterizing policers. Additionally, it increases the required TAS window size in linear topologies to ensure that all planned frames can pass through their assigned TAS windows.

F.1.4 Synchronization

The reference point for synchronization is also at the MDI. The actual point at which the transmitted or received synchronization frames are timestamped depends on the implementation and can be in the PHY, on the MII, or behind the MAC. Depending on the

implementation, part or all the MAC and PHY delay is compensated to determine the time at which the synchronization frame was at the reference point.

For example, during synchronization, the mean link delay and residence time are determined. The propagation delay variance may be different for the transmission of pdelay frames and synchronization frames. This results in an uncompensated delay error for the sync frames, which accumulates in linear topologies.

F.1.5 Recommended delays

Ethernet hardware used for industrial communication typically considers the requirements for MAC and MII propagation delay and propagation delay variance, minimizing these values during development. PHYs, if not integrated, are appropriately developed or selected.

Table F.1 shows the recommended values for PHYs in industrial communication.

Table F.1 – Recommended PHY delay

Link speed	Recommended Propagation delay	Recommended Propagation delay variance
10 Mbit/s	maximum < 5 μ s minimum < 500 ns	maximum < 16 ns minimum < 4 ns
100 Mbit/s – 10 Gbit/s	maximum < 1 μ s minimum < 500 ns	maximum < 16 ns minimum < 4 ns

7414

7415

Bibliography

7416 IEEE Std 802-2014, *IEEE Standard for Local and Metropolitan Area Networks: Overview and*
7417 *Architecture*

7418 IETF RFC 4210, Adams, C., Farrell, S., Kause, T., and Mononen, T., *Internet X.509 Public Key*
7419 *Infrastructure Certificate Management Protocol (CMP)*, September 2005, available at
7420 <https://www.rfc-editor.org/info/rfc4210>

7421 IETF RFC 6020, Bjorklund, M., *YANG: A Data Modeling Language for the Network Configuration*
7422 *Protocol (NETCONF)*, October 2010, available at <https://www.rfc-editor.org/info/rfc6020>

7423 IETF RFC 6242, Wasserman, M., *Using the NETCONF Protocol over Secure Shell (SSH)*, June
7424 2011, available at <https://www.rfc-editor.org/info/rfc6242>

7425 IETF RFC 7224, Bjorklund, M., *IANA Interface Type YANG Module*, May 2014, available at
7426 <https://www.rfc-editor.org/info/rfc7224>

7427 IETF RFC 8340, Bjorklund, M. and Berger, L., *YANG Tree Diagrams*, March 2018, available at
7428 <https://www.rfc-editor.org/info/rfc8340>

7429 IETF RFC 8808, Wu, Q., Lengyel, B., and Niu, Y., *A YANG Data Model for Factory Default*
7430 *Settings*, August 2020, available at <https://www.rfc-editor.org/info/rfc8808>

7431 IETF RFC 8995, Pritikin, M., Richardson, M., Eckert, T., Behringer, M., and Watsen, K.,
7432 *Bootstrapping Remote Secure Key Infrastructure (BRSKI)*, May 2021, available at
7433 <https://www.rfc-editor.org/info/rfc8995>

7434 ITU-T Recommendation G.8260, *Definitions and terminology for synchronization in packet*
7435 *networks*

7436 ITU-T Series G Supplement 65, *Simulations of transport of time over packet networks*, Geneva,
7437 October 2018.

7438

7439
